

NPTEL ONLINE CERTIFICATION COURSES

NPTEL

Management Information Systems

Prof. Surojit Mookherjee
VGSoM, IIT KHARAGPUR

Week 01: Introduction to MIS

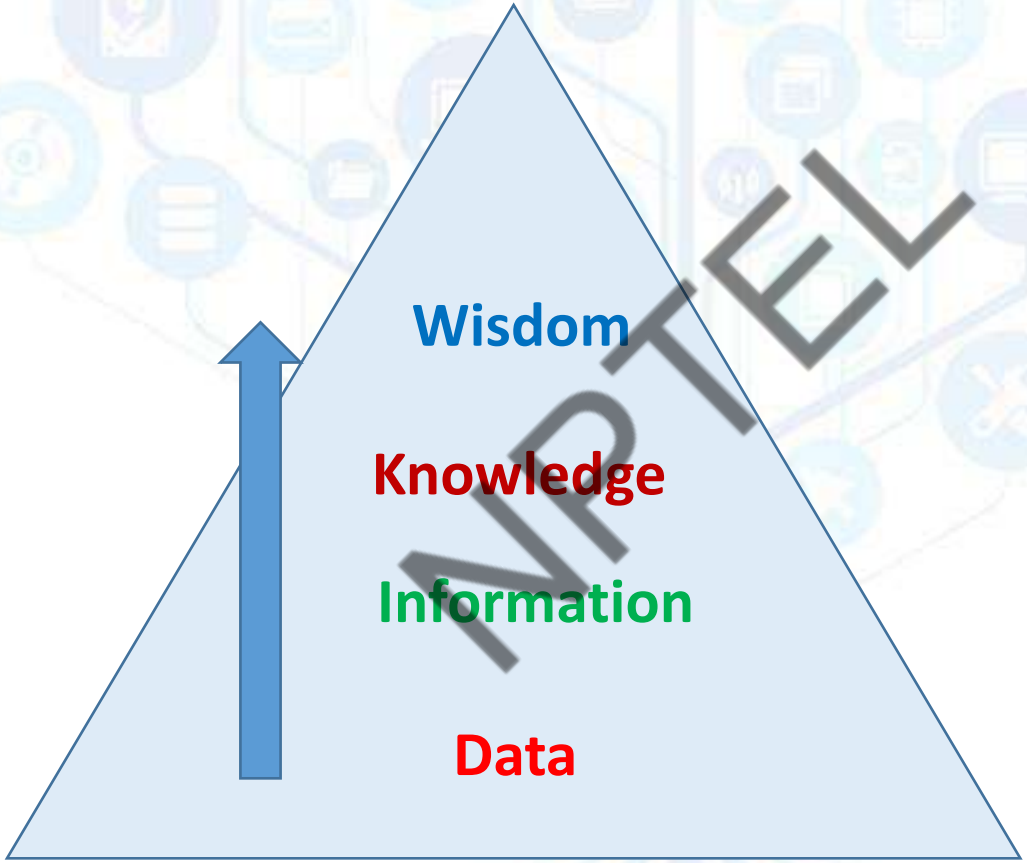
Lecture 01 : Introduction – Part 1 of 2

Learning Objectives

- How are information systems transforming business, and why are they so essential for running and managing a business today?
- What is an information system? How does it work? What are its management, organization, and technology components? Why are complementary assets essential for ensuring that information systems provide genuine value for organizations?
- What academic disciplines are used to study information systems, and how does each contribute to an understanding of information systems?

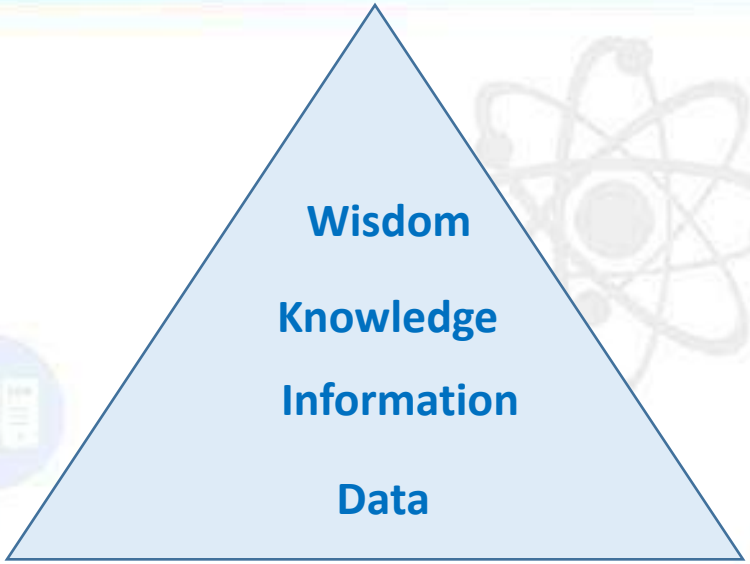
NPTEL

Goals of MIS



NPTEL

What Is an Information System? (1 of 3)



- Information system
  - Set of interrelated components
  - Collect, process, store, and distribute information
  - Support decision making, coordination, and control
- Information vs. data
  - Data are streams of raw facts
  - Information is data shaped into meaningful form
  - Information is an ordered set of data that you can understand and act on.

NPTEL

Data – Information - Decision

Data

Region	Sales in Rs.
North	50 lacs
West	40 lacs
South	22 lacs
East	10 lacs

Information

Sales is best in North region and worst in the South and East region, where the target of 40 lacs has been missed.

Decision

1. Appoint more sales engineers in South and East region
2. Increase advertisement budget
3. Offer more discounts to dealers

NPTEL

What Is an Information System? (2 of 3)

- Three activities of information systems produce information organizations need
  - Input: Captures raw data from organization or external environment
  - Processing: Converts raw data into meaningful form
  - Output: Transfers processed information to people or activities that use it

NPTEL

What Is an Information System? (3 of 3)

- Feedback
  - Output is returned to appropriate members of organization to help evaluate or correct input stage
- Computer/computer program vs. information system
  - Computers and software are technical foundation and tools, similar to the material and tools used to build a house , but are not complete by themselves.
  - These are the foundation or the base for building any Information system.

NPTEL

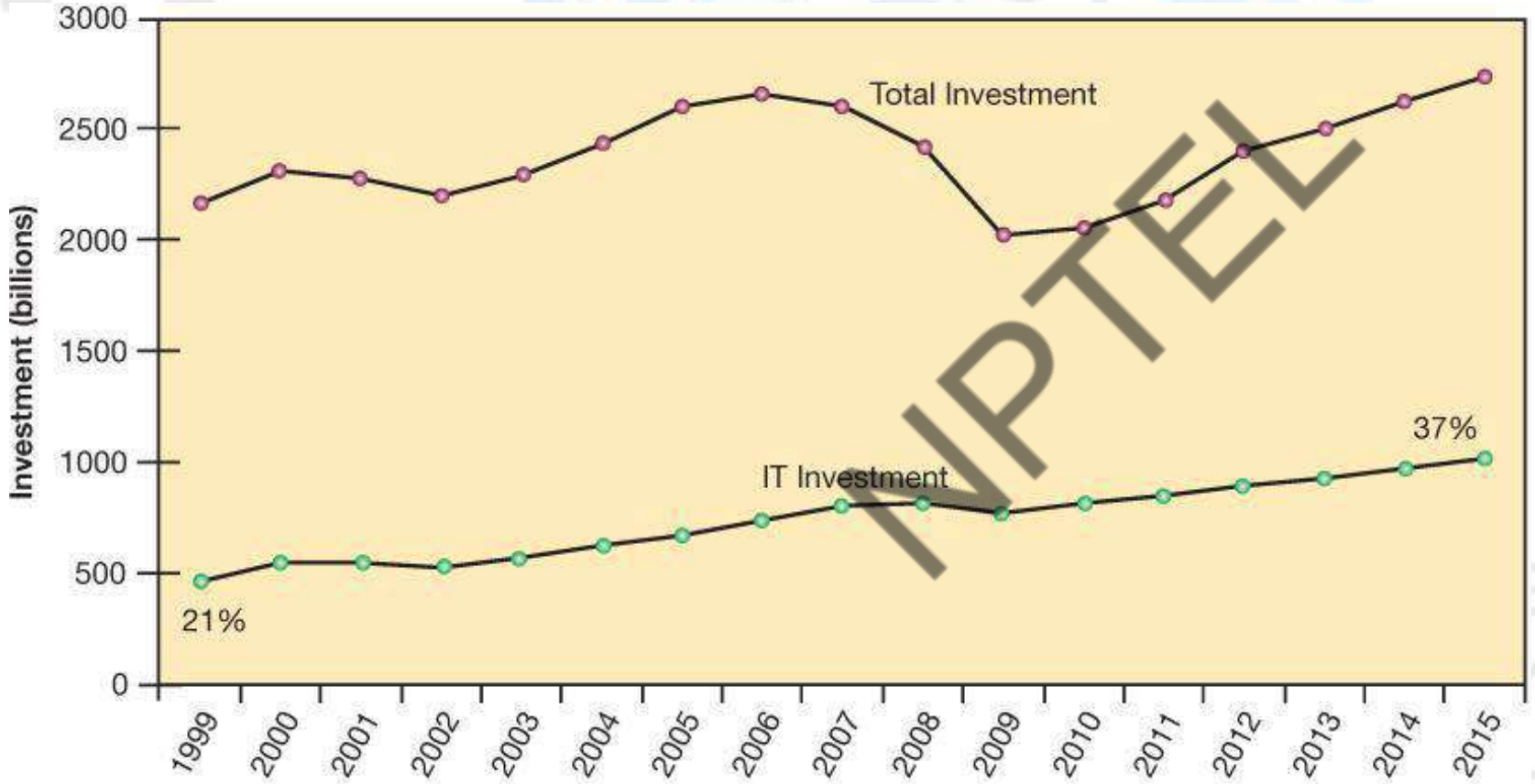
How Information Systems Are Transforming Business

- Global networking on real-time
- Mobile digital platform
- Systems used to improve customer experience, respond to customer demand, reduce inventories, and more
- Growing online learning and readership
- Expanding e-Commerce and Internet advertising
- Banking , Finance and Stock Markets
- New federal security and accounting laws

  (companies to store e-mails for 5 yrs)

NPTEL

Figure 1.1. Information Technology Capital Investment



NPTEL

What's New In Management Information Systems (1 of 2)

- Technology
  - Cloud computing
  - Big Data and the Internet of Things (IoT)
  - Mobile digital platform
- Management
  - Online collaboration and social networking software
  - Business intelligence
  - Virtual meetings (Post Covid-19 has become essential)

NPTEL

What's New In Management Information Systems (2 of 2)

- Organizations
  - Globally connected enterprises
  - Changing business models driven by new technology (e.g Uber / Airbnb)
  - Rapid change in technologies
  - Time / Speed to market (e.g Vaccines and drugs for Covid 19)
  - Social business (education sector)

NPTEL

The Emerging Digital Firm

- In a fully digital firm:
  - Significant business relationships are digitally enabled and mediated
  - Core business processes are accomplished through digital networks
  - Key corporate assets are managed digitally
- Digital firms offer greater flexibility in organization and management
  - Time shifting (multiple time zones),
  - Space shifting (multiple geo locations)

  – Work from Home

NPTEL

Strategic Business Objectives of Information Systems (1 of 2)

- Growing interdependence between:
  - Ability to use information technology and
  - Ability to implement corporate strategies and achieve corporate goals

Class Q - (For instance, Speed to Market is very important to firms introducing new products. How can IT help achieve that objective?)

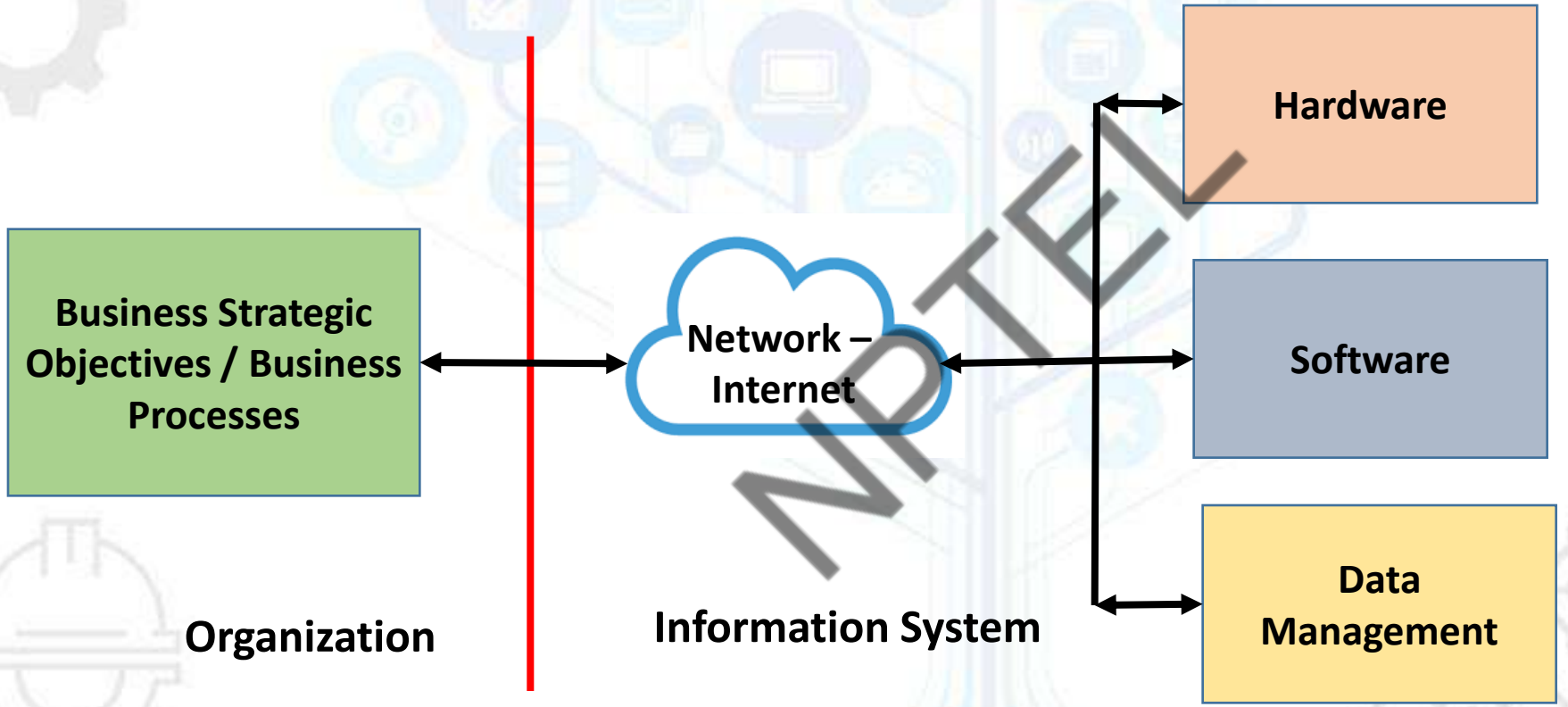
NPTEL

Strategic Business Objectives of Information Systems (2 of 2)

- Firms invest heavily in information systems to achieve six strategic business objectives:
  1. Operational excellence
  2. New products, services, and business models
  3. Customer and Supplier intimacy
  4. Improved decision making
  5. Competitive advantage
  6. Survival

NPTEL

The Interdependence Between Organizations and Information Systems



NPTEL

REFERENCES

- **The World is Flat : Thomas L Friedman**
- **Management Information Systems: Managing the Digital Firm - Kenneth C. Laudon & Jane P. Laudon**

NPTEL



NPTEL

THANK YOU !



NPTEL ONLINE CERTIFICATION COURSES

NPTEL

Management Information Systems

Prof. Surojit Mookherjee
VGSoM, IIT KHARAGPUR

Week 01: Introduction to MIS

Lecture 02 : Introduction – Part 2 of 2

Learning Objectives

- How are information systems transforming business, and why are they so essential for running and managing a business today?
- What is an information system? How does it work? What are its management, organization, and technology components? Why are complementary assets essential for ensuring that information systems provide genuine value for organizations?
- What academic disciplines are used to study information systems, and how does each contribute to an understanding of information systems?

NPTEL

Strategic Business Objectives - 1.Operational Intelligence (Excellence)

- Improvement of efficiency to attain higher profitability
- Information Systems Technology is an important tool in achieving greater efficiency and productivity
  - Take a look at Amazon / Flipkart / Alibaba etc. from an Operational excellence perspective.

NPTEL

Strategic Business Objectives – 2.New Products, Services, and Business Models

- Business model: describes how company produces, delivers, and sells product or service to create wealth (Uber / Airbnb / Amazon)
- Information systems and technology is a major enabling tool for new products, services, business models
  - Examples: Apple's iPad, Google's Android OS, and Netflix

Class Q - (What is IT's role in development of Electric / Hybrid cars / Green technologies like Wind and Solar power generation ?)

NPTEL

Strategic Business Objectives – 3.Customer and Supplier Intimacy

- Serving customers well leads them to return, increasing revenue and profits
  - Example: High-end hotels that use computers to track customer preferences and then monitor and customize the environment
- Intimacy with suppliers allows them to provide vital inputs, which lowers costs

Class Q – (Online sites achieving a high degree of Customer intimacy – Amazon / Swiggy etc. How are they doing this ?)

NPTEL

Strategic Business Objectives 4.Improved Decision Making

- Without accurate information:
  - Managers must use forecasts, best guesses, luck
  - Results in:
    - Overproduction, underproduction
    - Misallocation of resources
    - Poor response times
  - Poor outcomes raise costs, lose customers

NPTEL

Strategic Business Objectives 5.Competitive Advantage

- Delivering better performance
- Charging less for superior products
- Responding to customers and suppliers in real time
- Examples: Apple, Walmart, UPS , Airbnb , Uber, OYO

NPTEL

Strategic Business Objectives - 6.Survival

- Information technologies as necessity of business
- Governmental regulations requiring record-keeping

  Examples: Toxic Substances Control Act, Sarbanes-Oxley Act

  Compliances – Financial , Statutory , Environmental , Income-Tax , GST records etc.
- Current Pandemic situations forcing Organizations to work differently (e.g Working from Home)

NPTEL

Information System and Organization Strategy

NPTEL

Information technology and Organizations influence one another

Following are the major influencers in an Organization.

- ✓ Structure
- ✓ Business processes
- ✓ Politics
- ✓ Culture
- ✓ Environment, and
- ✓ Management decisions

NPTEL

Features of an Organization

- Use of hierarchical structure
- Accountability, authority in system of impartial decision making
- Adherence to principle of efficiency
- Decision making and business processes
- Organizational politics, culture and environment
- Ethical practices

NPTEL

Major economic impacts of Information Systems / Technology

- IT changes relative costs of capital and the costs of information
- Information systems technology is a factor of production, like capital and labor
- IT affects the cost and quality of information and changes economics of information
- Information technology helps firms contract in size because it can reduce transaction costs (the cost of participating in markets)
- Outsourcing

NPTEL

Organizational and Behavioral impacts

- IT flattens organizations
  - Decision making pushed to lower levels
  - Fewer management levels / managers needed (IT enables faster decision making and increases span of control)
- To days organizations
  - Organizations flatten because in post industrial societies, authority increasingly relies on knowledge and competence rather than formal positions

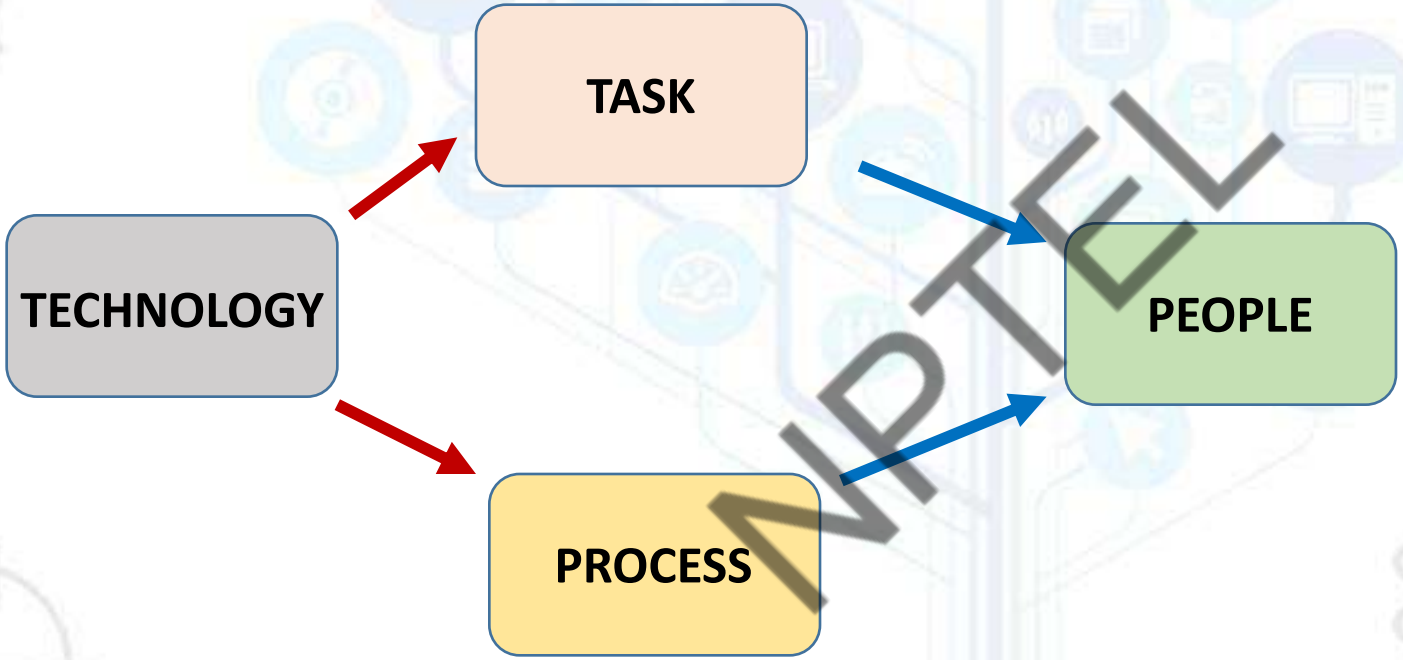
NPTEL

Organizational resistance to change…

- Information systems become bound up in organizational politics because they influence access to a key resource – **information**
- Information systems potentially change an organization's structure, culture, politics, and work
- Most common reason for failure of large projects is due to organizational and political **resistance to change**

NPTEL

Organizational resistance to change…



NPTEL

Michael Porter's Competitive Forces Model

Five competitive forces shape fate of firm

1. Traditional competitors
2. New market entrants
3. Substitute products and services
4. Customers
5. Suppliers

NPTEL

IT enabled Strategies for managing competitive forces

- Low-cost leadership – e.g Walmart
- Product differentiation – e.g. Apple , Google
- Focus on market niche – e.g. Uber , Airbnb
- Strengthen customer and supplier intimacy – e.g. Netflix , Amazon

Michael Porter's Competitive Forces Model

1. Traditional competitors
2. New market entrants
3. Substitute products and services
4. Customers
5. Suppliers

NPTEL

Internet's impact on Competitive Advantage

- Transformation, destruction, threat to some industries - e.g. travel agency, printed encyclopedia, newspaper
- Competitive forces still at work, but rivalry more intense – e.g. Booking.com
- Universal standards allow new rivals, entrants to market - e.g. Amazon competing with Microsoft & IBM with Cloud technology
- New opportunities for building brands and loyal customer bases – e.g. Uber , Airbnb

NPTEL

REFERENCES

- **The World is Flat (book) : Thomas L Friedman**
- **Management Information Systems: Managing the Digital Firm - Kenneth C. Laudon & Jane P. Laudon**

NPTEL



NPTEL

THANK YOU !



NPTEL ONLINE CERTIFICATION COURSES

NPTEL

Management Information Systems

Prof. Surojit Mookherjee
VGSoM, IIT KHARAGPUR

Week 01: Introduction to MIS

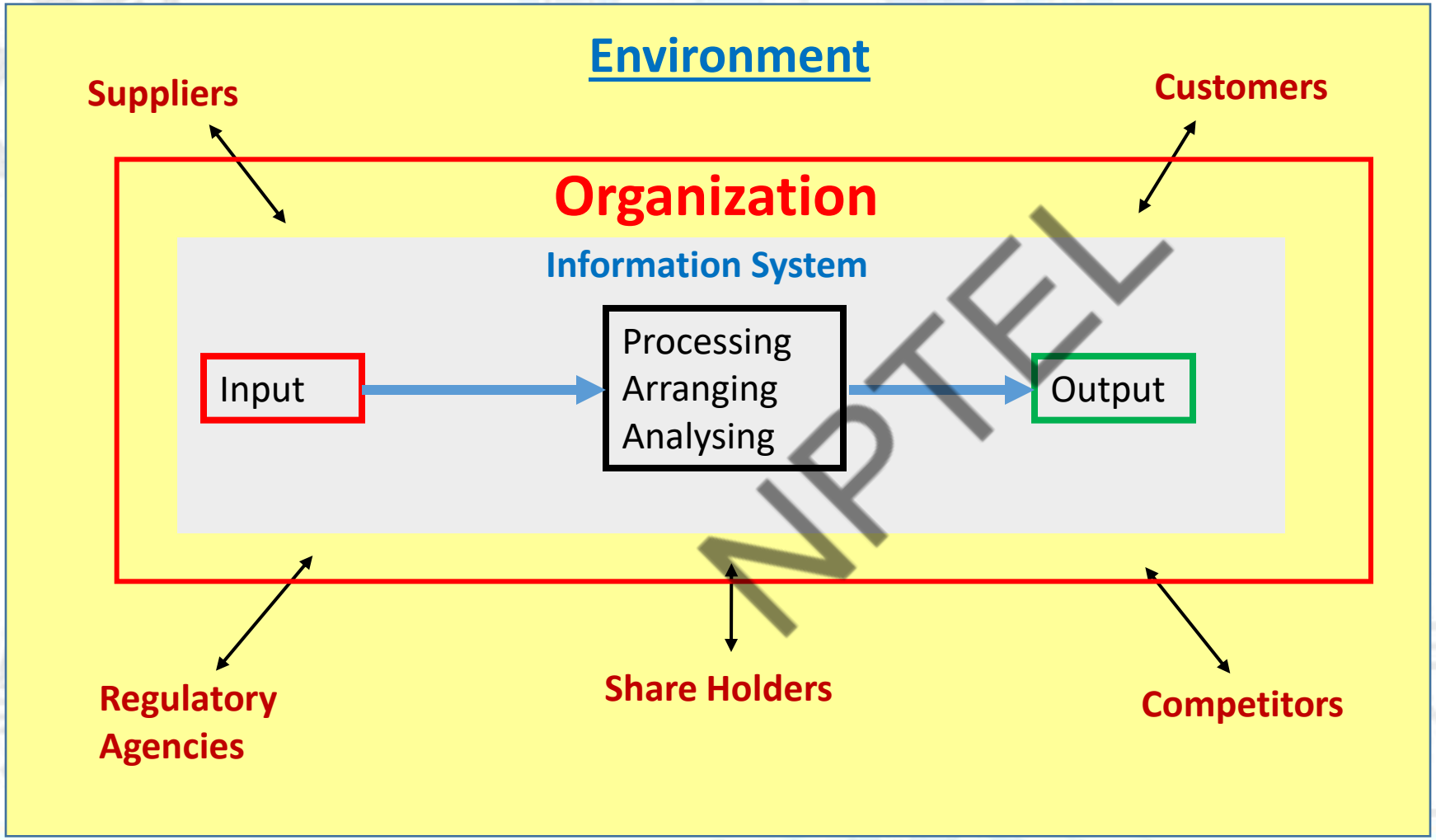
Lecture 03 : Dimensions of Information System

Learning Objectives

- How are information systems transforming business, and why are they so essential for running and managing a business today?
- What is an information system? How does it work? What are its management, organization, and technology components? Why are complementary assets essential for ensuring that information systems provide genuine value for organizations?
- What academic disciplines are used to study information systems, and how does each contribute to an understanding of information systems?

NPTEL

Information System – Eco System

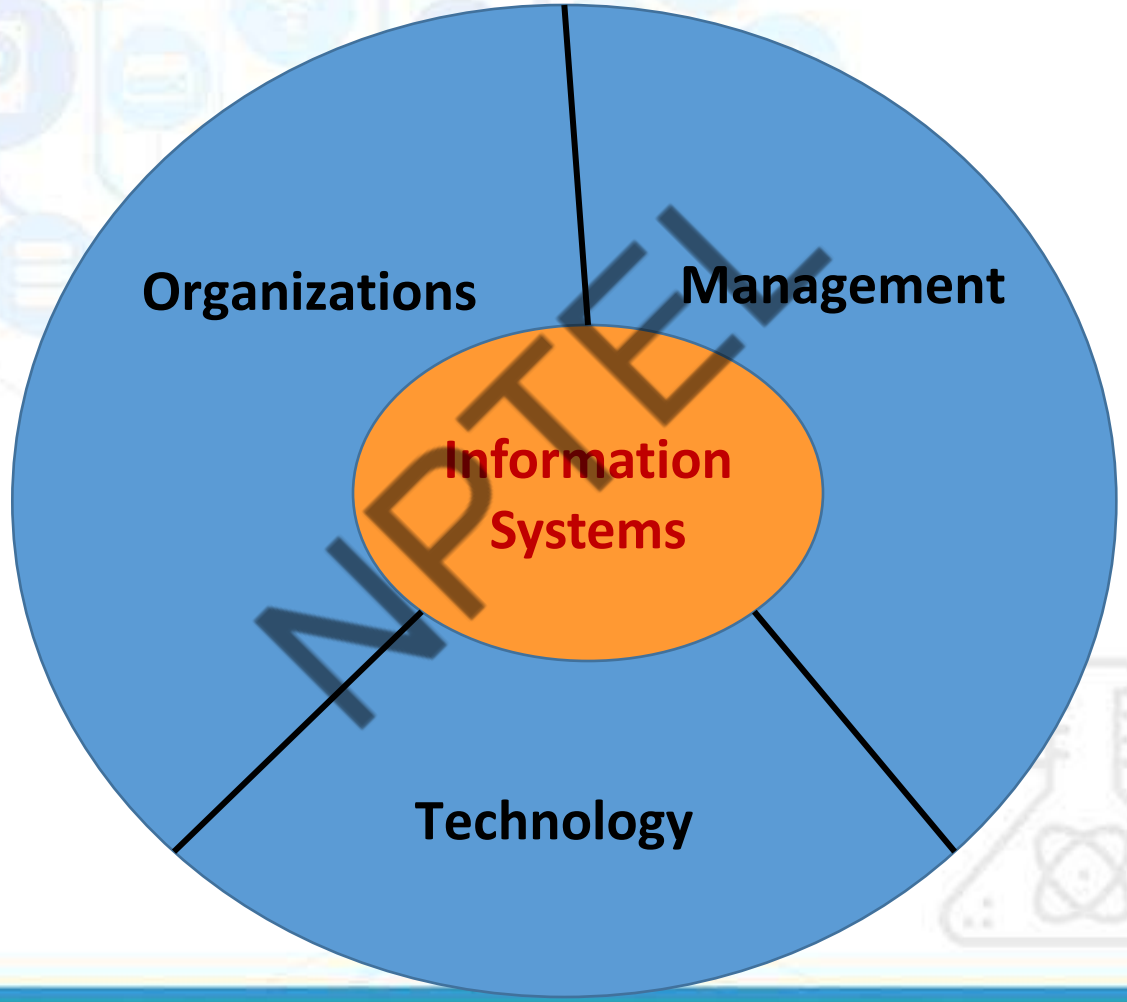


NPTEL

Dimensions of Information Systems

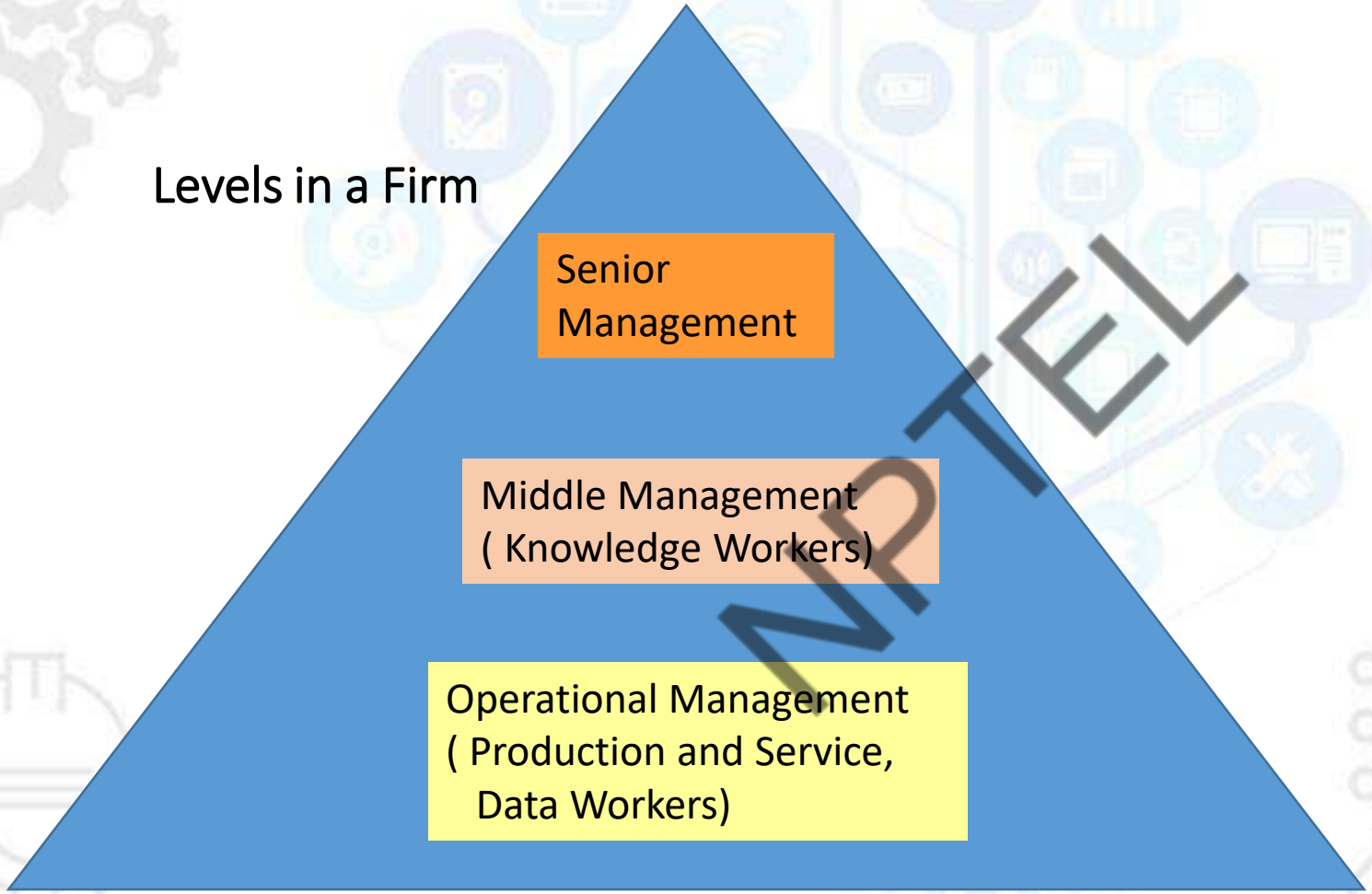
- Organizations
- Management
- Technology

Not just Computers



NPTEL

Dimensions of Information Systems: Organizations (1 of 2)



NPTEL

Dimensions of Information Systems: Organizations (2 of 2)

- Separation of business functions
  - Sales and Marketing
  - Human resources
  - Finance and Accounting
  - Manufacturing and Production

"Every business is different."

Class Exercise – Study the "Dabbawala " business of food delivery systems in Mumbai. What type of Information System is practiced for achieving 'Zero' defect service round the year.

NPTEL

Dimensions of Information Systems: Management

- Managers set Organizational Strategy for responding to business challenges
- In addition, managers must act creatively
  - Creation of new products and services
  - Occasionally re-creating the organization (e.g. Mergers , acquisitions , hiving-off , Joint Venture , Collaboration etc.)

Class Discussion – MIS issues related to the recent merger of Public Sector Banks

NPTEL

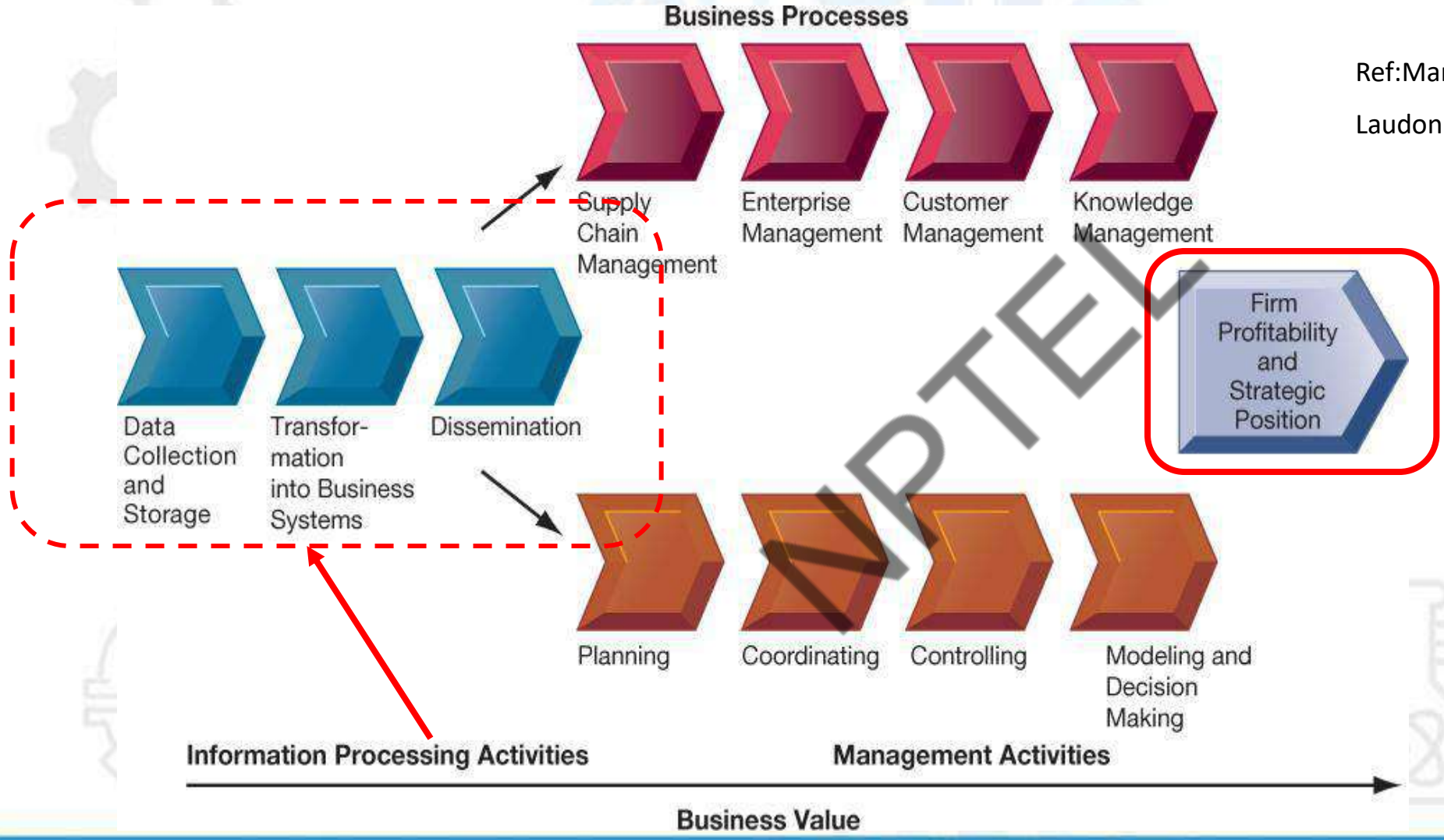
Dimensions of Information Systems: Technology

- Computer hardware and software
- Data management technology
- Networking and Telecommunications technology
  - Networks, the Internet, intranets and extranets, World Wide Web
- IT infrastructure: provides the platform on which the system is built on

Information technology is at the heart of information systems.

NPTEL

The Business Information Value Chain



Ref:Management Information Systems - Kenneth C. Laudon & Jane P. Laudon

NPTEL

Complementary Assets: Organizational Capital and the Right Business Model (1 of 2)

- Assets required to derive value from a primary investment
- Firms supporting technology investments with investment in complementary assets receive superior returns
- Example: Invest in technology and the people to make it work properly

e.g – Petroleum Companies having Retail business in Petrol Pumps

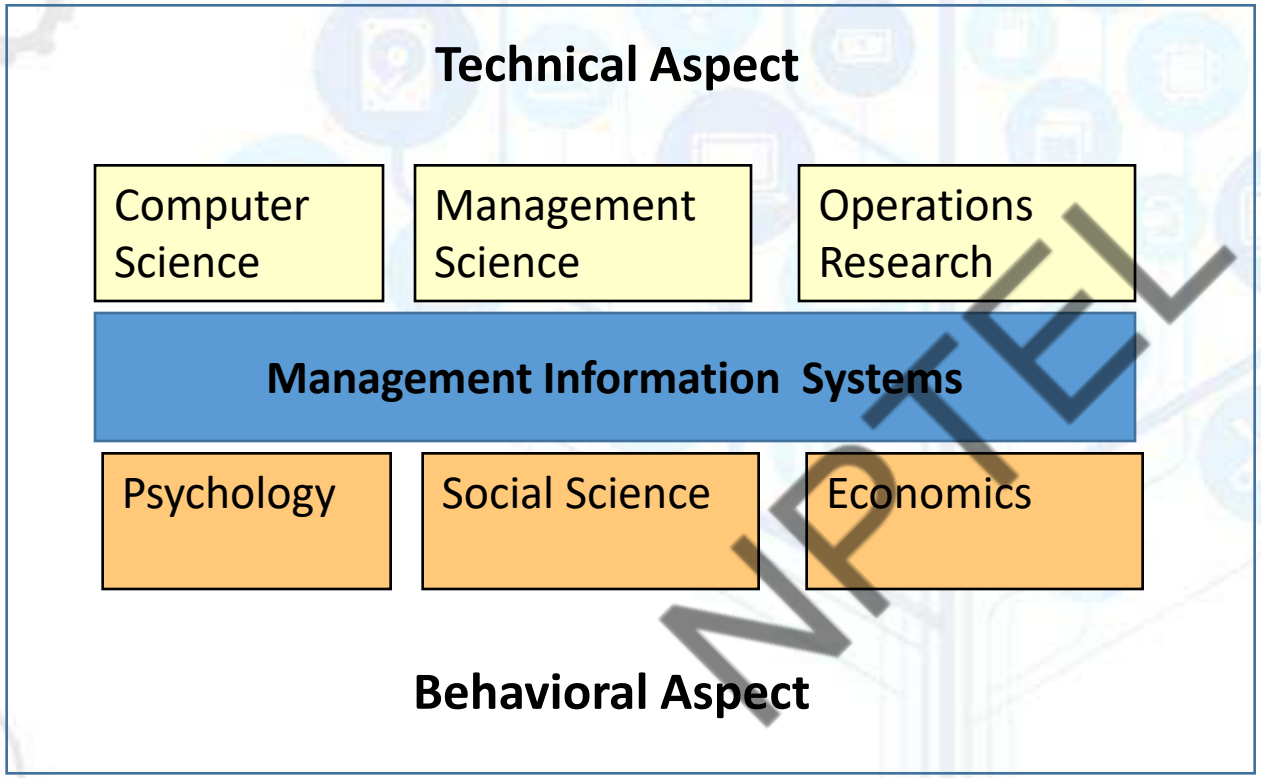
NPTEL

Complementary Assets: Organizational Capital and the Right Business Model (2 of 2)

- Complementary assets
  - Examples of organizational assets
    - Appropriate business model
    - Efficient business processes
  - Examples of managerial assets
    - Incentives for management innovation
    - Teamwork and collaborative work environments
  - Examples of social assets
    - The Internet and telecommunications infrastructure
    - Technology standards

NPTEL

Contemporary Approaches to Information Systems



NPTEL

The Three Categories of IT

IT Category	Examples
Functional IT	Spreadsheets, computer-aided design, and statistical software
Network IT	E-mail, instant messaging, wikis, blogs
Enterprise IT	Software for Enterprise resource planning, Customer resource management, and Supply chain management

NPTEL

The IT Dialogue….

Functional IT

Will any of the new software on the market enable our engineers, scientists, analysts, and other workers to do their jobs more efficiently? Do we need to invest in software.

Are our function technologies outdated? If so, why? What has changed?

NPTEL

The IT Dialogue….

Network IT

- How do our people collaborate? Do we know what technologies they're using?
- If we wanted to get broad feedback on an important topic, how would we do it?
- How do we know what our people are working on and what they think the hot topics are?

NPTEL

The IT Dialogue

Enterprise IT

- In what ways are our current processes not supporting the needs of the business? Which ones need to be redesigned? Which ones should be extended to our customers and suppliers?
- Are there important business activities, events, or trends that we should monitor?
- Are the data unavailable or stored across so many systems that the information is difficult to assemble?

NPTEL

REFERENCES

- **The World is Flat (book) : Thomas L Friedman**
- **Management Information Systems - Kenneth C. Laudon & Jane P. Laudon**

NPTEL



NPTEL

THANK YOU !



NPTEL ONLINE CERTIFICATION COURSES

NPTEL

Management Information Systems

Prof. Surojit Mookherjee
VGSoM, IIT KHARAGPUR

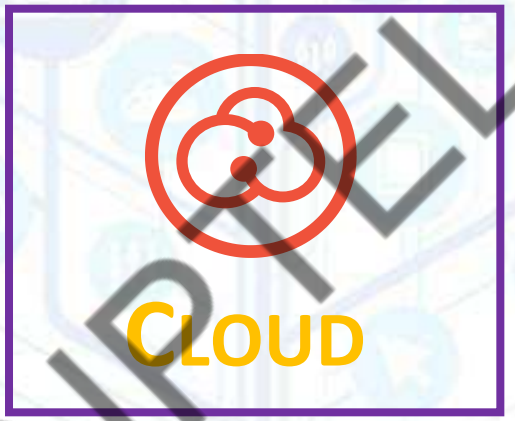
Week 01: Introduction to MIS

Lecture 04 : Information Management in the Digital World

Learning Objectives

- How are information systems transforming business, and why are they so essential for running and managing a business today?
- What is an information system? How does it work? What are its management, organization, and technology components? Why are complementary assets essential for ensuring that information systems provide genuine value for organizations?
- What academic disciplines are used to study information systems, and how does each contribute to an understanding of information systems?

NPTEL



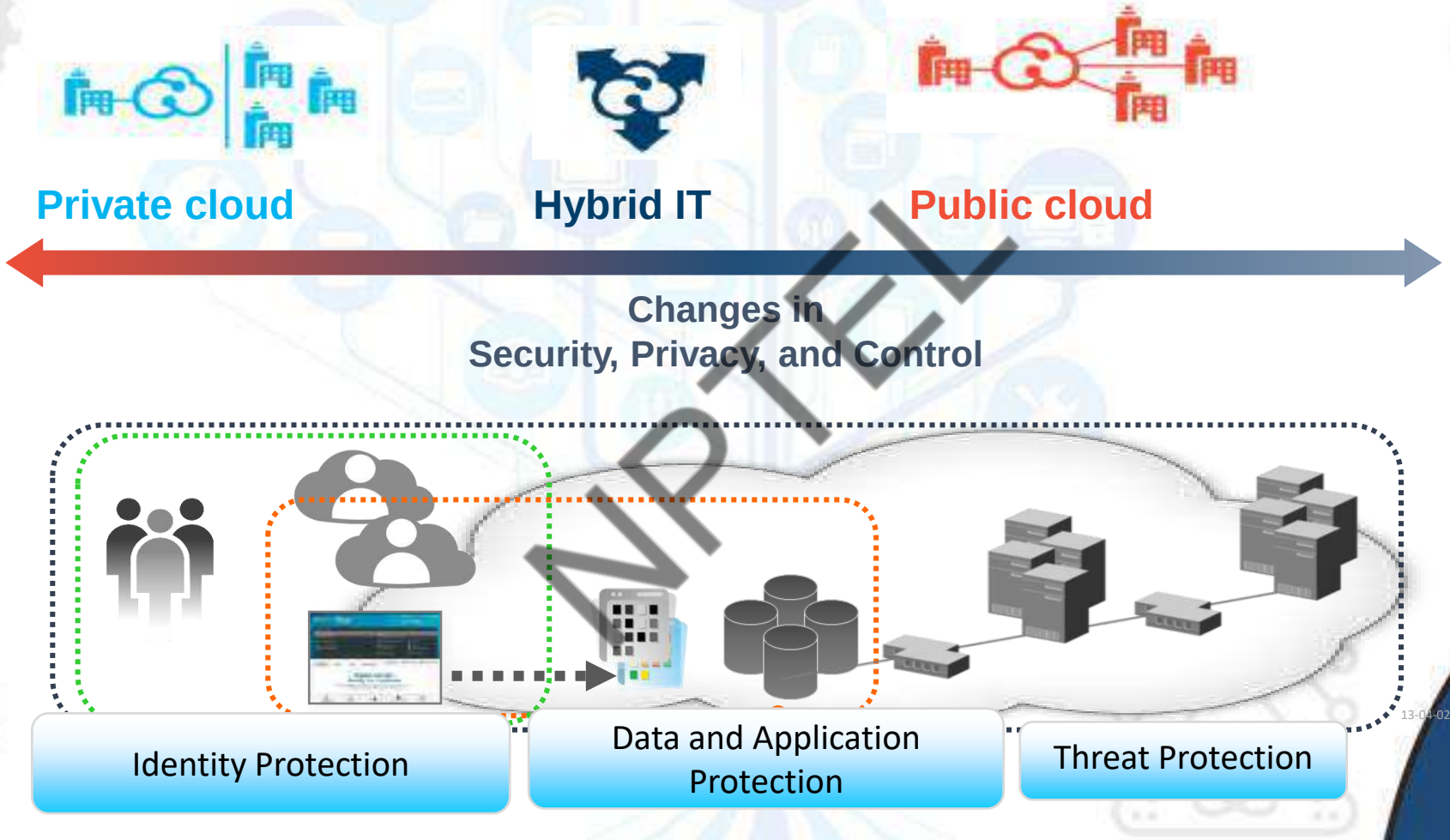
NPTEL

Cloud computing: Four key service categories of Cloud offerings

Business Process As A Service	Sales Force.com , Workday.com
Software As A Service	MS Office , Google Docs
Platform As A Service	Amazon Web Service , Azure
Infrastructure As A Service	Data Centers

NPTEL

Cloud computing changes the way we think about security



NPTEL



NPTEL

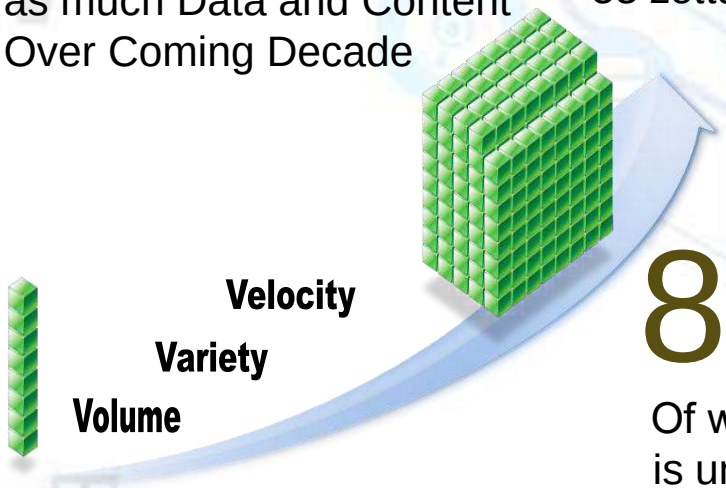
Organizations are thirsty, despite oceans and streams of data

Need for business analytics was never so much …

44x

as much Data and Content Over Coming Decade

2020
35 zettabytes



2009
800,000 petabytes

80%

Of world's data is unstructured

90%

Of world's data has been generated in the last TWO years

… They need deeper insights!

1 in 3 Business leaders frequently make decisions based on information they don't trust, or don't have

1 in 2 Business leaders say they don't have access to the information they need to do their jobs

83% of CIOs cited "Business intelligence and analytics" as part of their visionary plans to enhance competitiveness

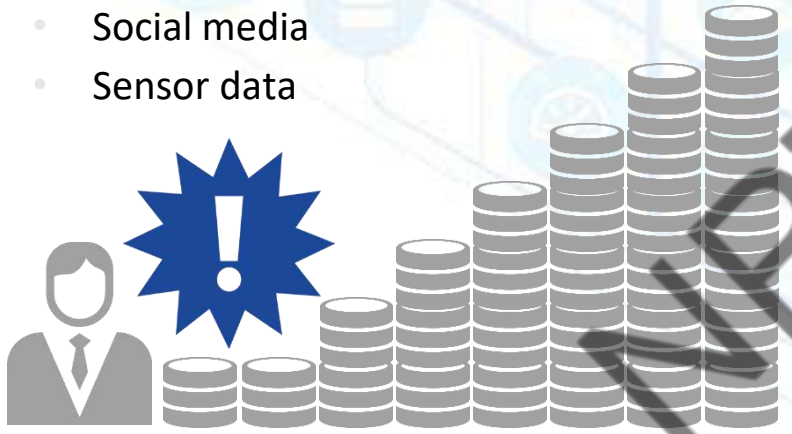
60% of CEOs need to do a better job capturing and understanding information rapidly in order to make swift business decisions

NPTEL

Analytics presents security risks and opportunities

Big data explosion

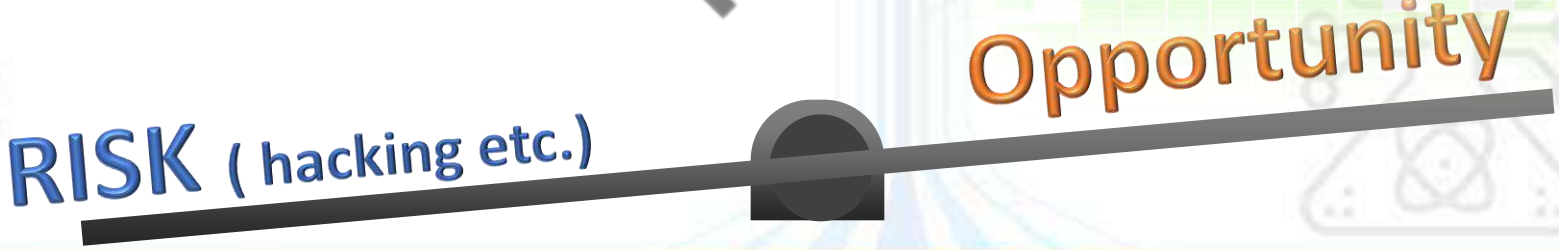
- Personally identifiable
- Credit card data
- Health data
- Intellectual property
- Social media
- Sensor data



Analytic insights for security

- Larger more diverse data sets
- Faster analysis
- Deeper insights
- Predictive models





NPTEL



NPTEL

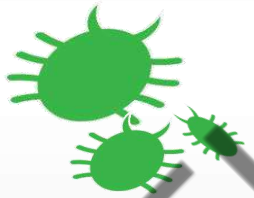
As mobile grows, so do security threats



In 2014 the number of cell phones (7.3 billion) will exceed the number of people on the planet (7 billion).



Mobile downloads increase to 108 billion in 2017.



Mobile malware is growing. Malicious code is infecting more than 11.6 million mobile devices at any given time.



Mobile devices and the apps we rely on are under attack. 90% of the top mobile apps have been hacked.

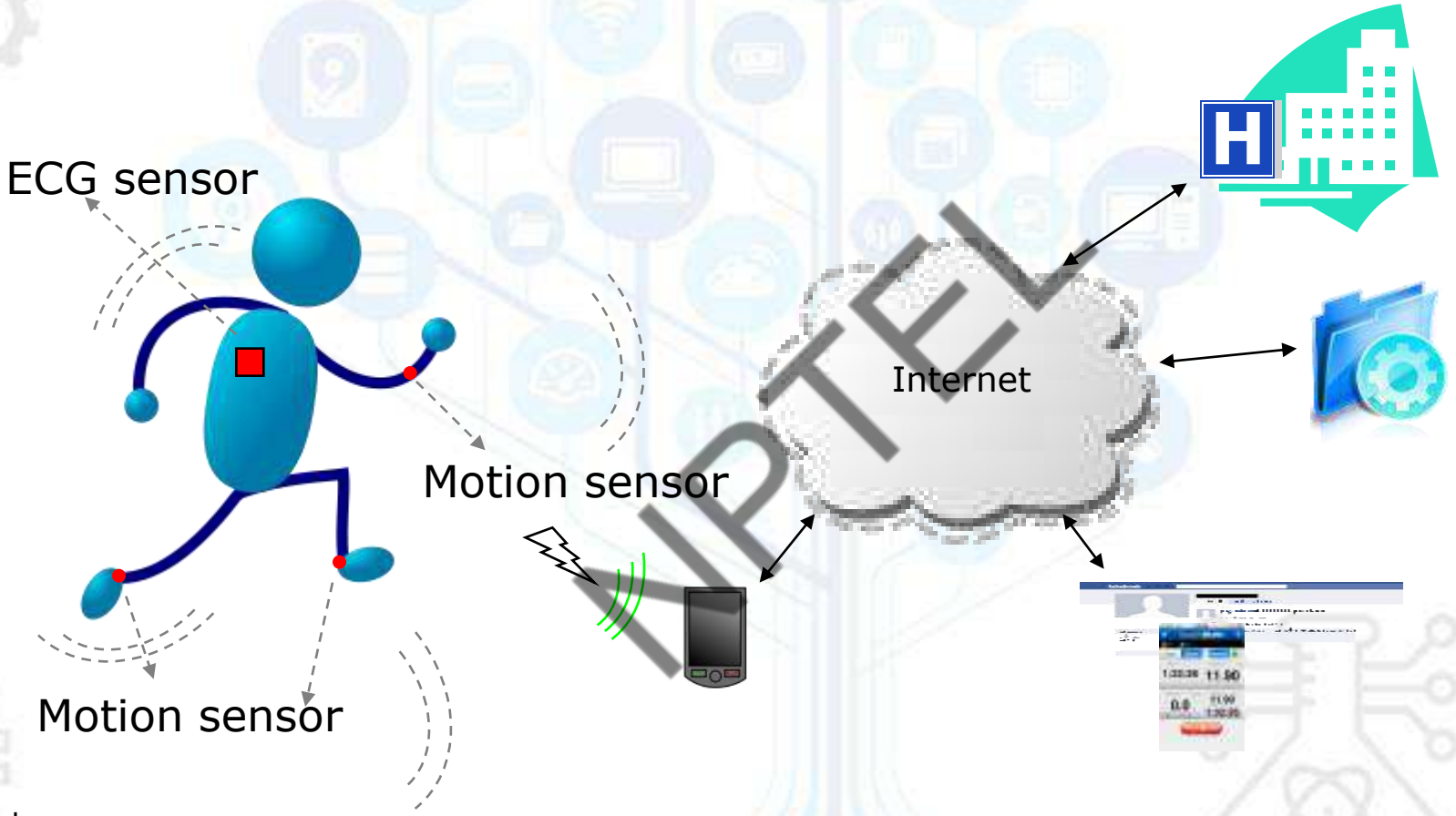
NPTEL



Internet of Things
Industry 4.0

NPTEL

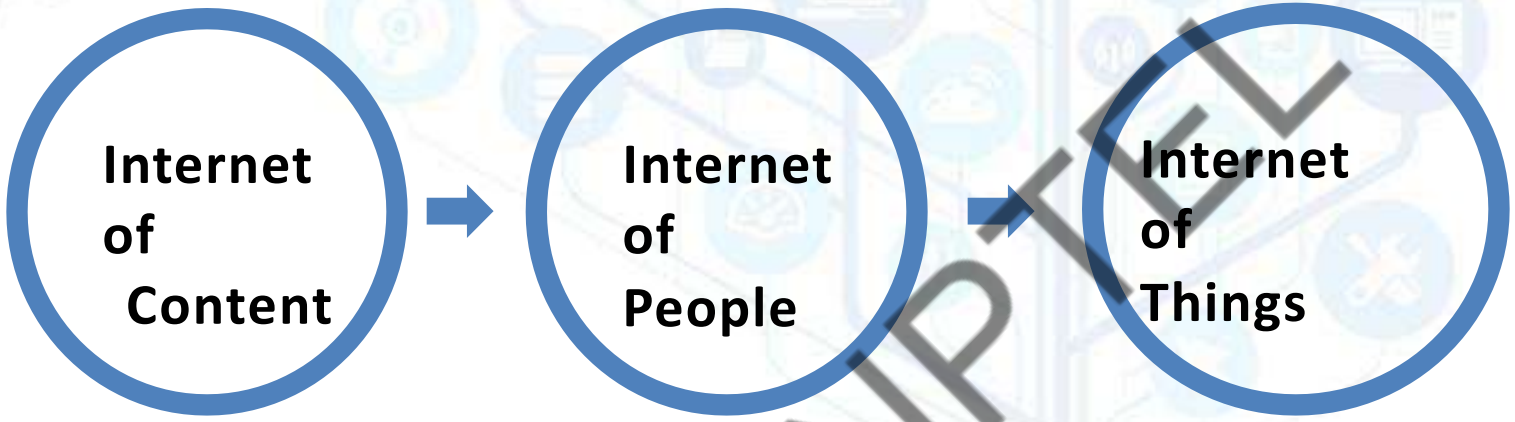
People Connecting to Things



Ref:www.kayarvizhy.com

NPTEL

Connecting information, people, and things is greatest resource ever to drive insightful action

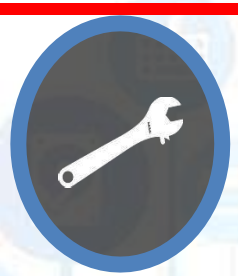


NPTEL

Manufacturing domain



Responsive supply chains



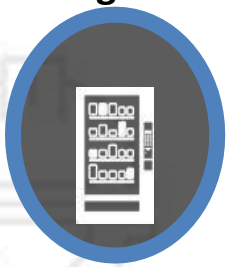
Predictive maintenance



Connected asset management



Connected logistics



Connected retail



Connected Factory

Human Society's Ecosystem



Connected home



Connected cities



Connected car



Connected healthcare

IoT Everywhere

NPTEL

Industry 4.0

Industry 4.0 has been defined as “a name for the current trend of **automation** and **data exchange** in manufacturing technologies, including **cyber-physical systems**, the **Internet of things**, **cloud computing** and **cognitive computing** and creating the **smart factory**”.

NPTEL

IoT and “Datafication”

In today’s world , we are seeing a rapid acceleration of Sensors , Actuators and Devices.

This will lead to a deluge of data being generated from all of these connected devices. This is what is known a s “Datafication” of IoT.

This Datafication will have a similar effect of pervasiveness as that of ‘Electricity’ in the 20th Century and ‘availability of Data’ will be as taken for granted as we do for Electricity.

NPTEL

Key impacts of IoT on Industry

Automation: Connecting machines, sensors, and actuators to computing systems enables a large degree of process automation. E.g.- jet engines.

Integration: Integrating the data from a machine with data from other sources,(e.g ERP, CRM systems) greatly enhances the value derived

“Servitization”: Combination of automation and integration help organizations move from product-centered business models to service-oriented business models .

NPTEL

Business Needs of IoT -



Wearable Tech



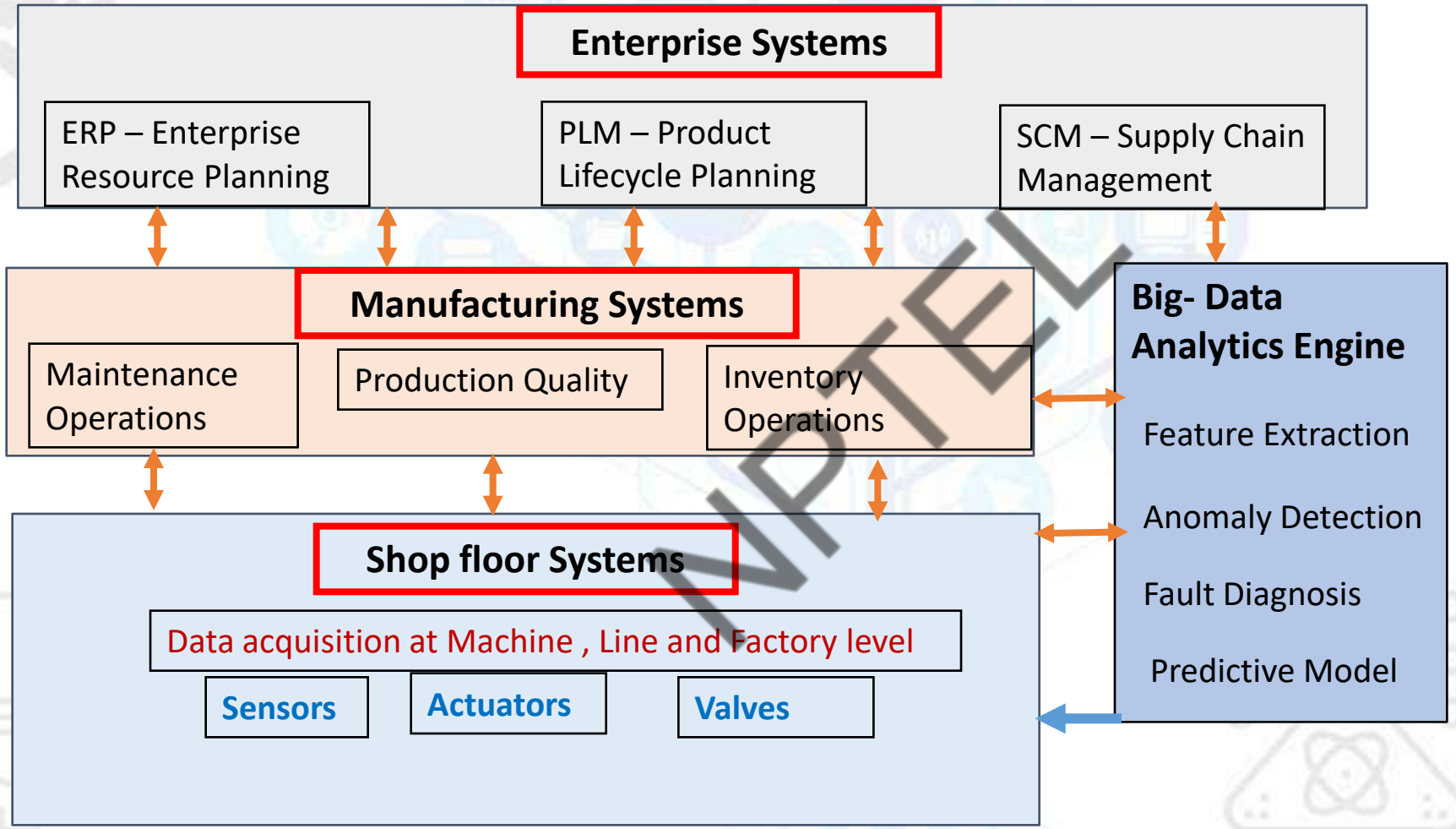
Healthcare

Smart Appliances



NPTEL

Manufacturing Analytics- How it is integrated



NPTEL



NPTEL

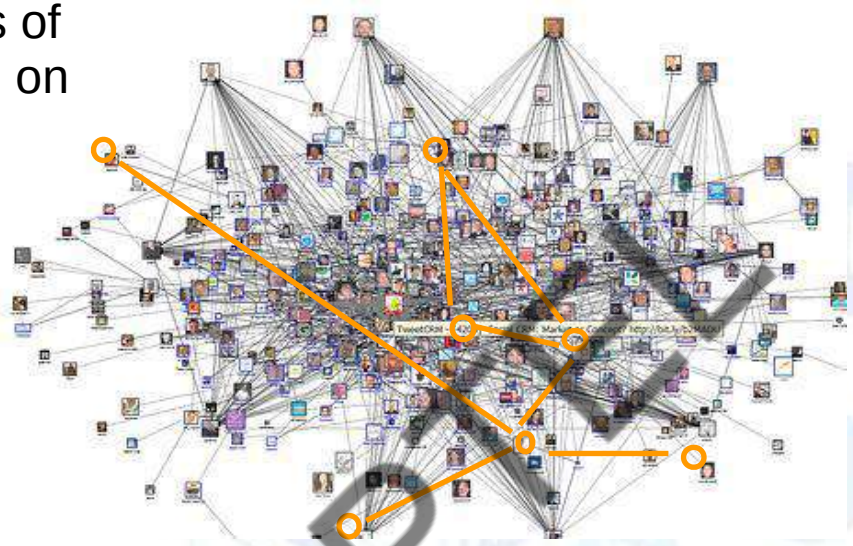
2018 This Is What Happens In An Internet Minute



NPTEL

You are what you share, the social transformation

30 billion pieces of content are shared on **Facebook** each month



More companies now use social internally than externally

66% of top financially performing companies leverage social in their business processes

Social technologies raise the productivity of interaction workers by 20-25%

NPTEL

Security Implications

Information is exposed in new ways that can pose security concerns

Mixing of personal and corporate data on social platforms creates risk. **Your privacy is now limited**.

REFERENCES

- **The World is Flat (book) : Thomas L Friedman**
- **Management Information Systems - Kenneth C. Laudon & Jane P. Laudon**

NPTEL



NPTEL

THANK YOU !



NPTEL ONLINE CERTIFICATION COURSES

NPTEL

Management Information Systems

Prof. Surojit Mookherjee
VGSoM, IIT KHARAGPUR

Week 01: Introduction to MIS

Lecture 05 : Class Discussions and Conclusion

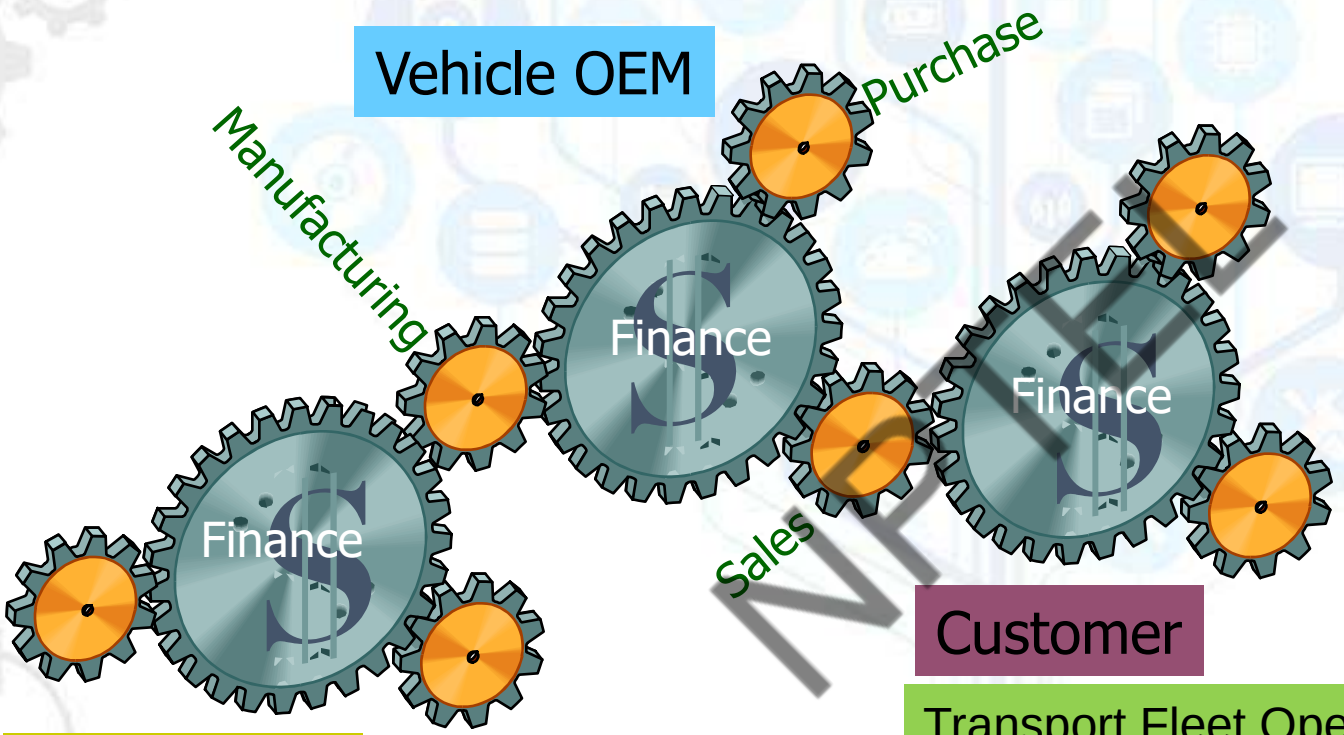
Learning Objectives

- How are information systems transforming business, and why are they so essential for running and managing a business today?
- What is an information system? How does it work? What are its management, organization, and technology components? Why are complementary assets essential for ensuring that information systems provide genuine value for organizations?
- What academic disciplines are used to study information systems, and how does each contribute to an understanding of information systems?

NPTEL

Enterprise Resource Planning

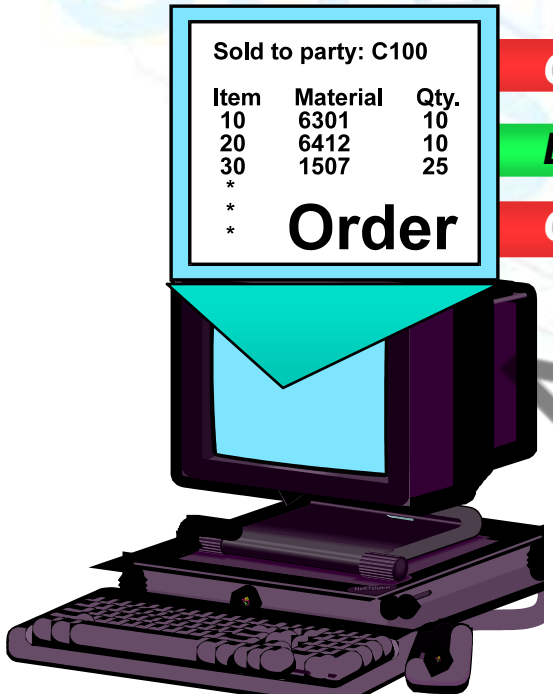
… extends to the complete value chain…

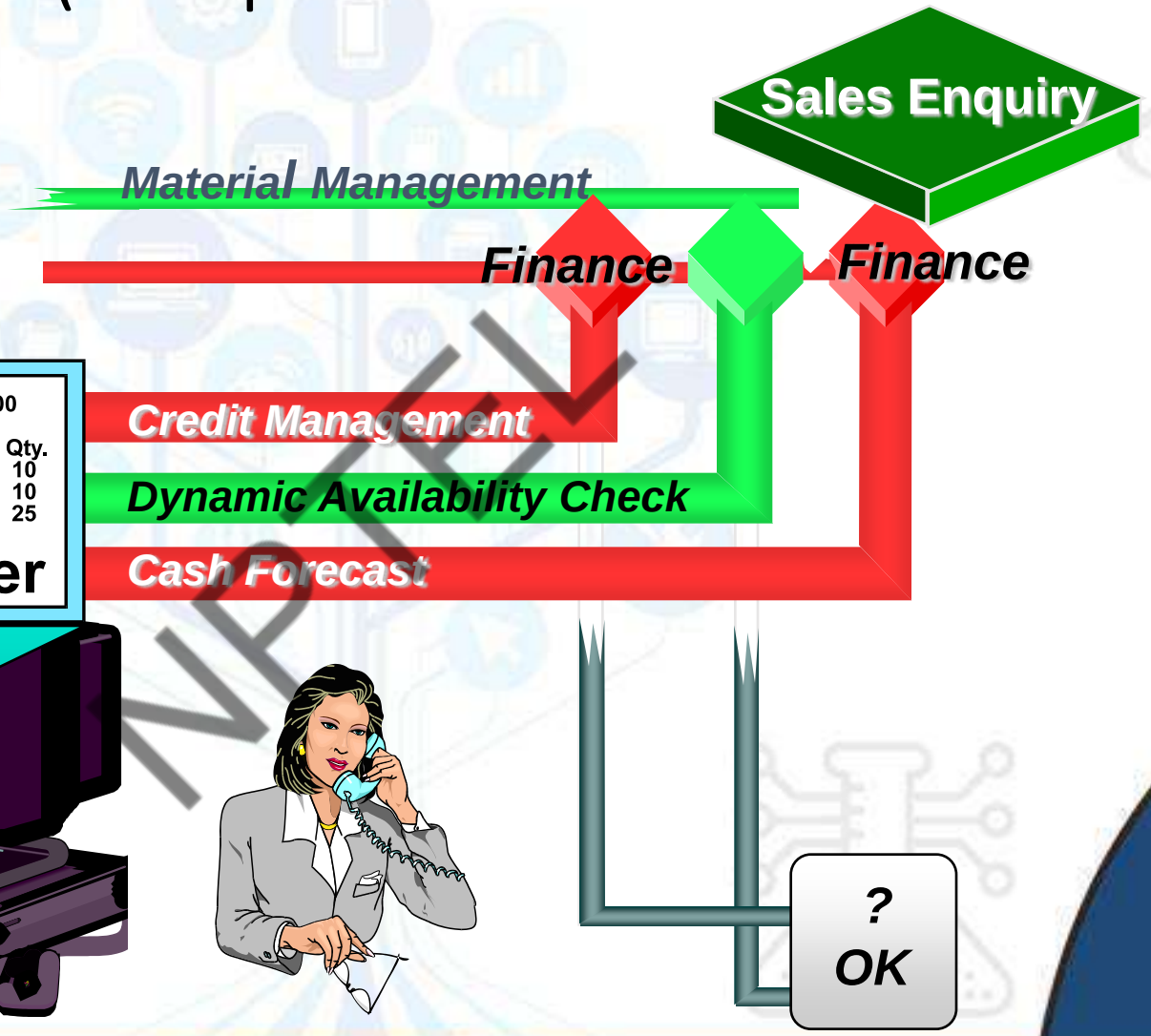


The beauty & power of MIS (Enterprise Resource Planning)…



Request from a Customer





NPTEL

IIT Kharagpur

Globalization Challenges and Opportunities: A Flattened World

Class Discussion :

- **Internet** has drastically reduced costs of operating on global scale
- Increases in foreign trade, outsourcing
- Presents both challenges and opportunities

Class Q – Airbnb / Uber / Booking.com etc.
How are they managing their entire business only using Internet and Information Systems ? They donot have any other material assets.

NPTEL

Customer and Supplier Intimacy

- Serving customers well leads them to return, increasing revenue and profits
  - Example: High-end hotels that use computers to track customer preferences and then monitor and customize the environment
- Intimacy with suppliers allows them to provide vital inputs, which lowers costs

Class Q – (Online sites achieving a high degree of Customer intimacy – Amazon / Swiggy etc. How are they doing this ?)

Pointers :

1. Return policy gives confidence to customers
2. Royalty bonus
3. Offers and Discounts
4. Are you aware that your data / browsing history / preferences are sold to marketing agencies .
5. Your every click may mean revenue for someone else.

NPTEL

Interactive Session: Management- The Mobile Pocket Office

- Class Discussion
  - What kinds of businesses are most likely to benefit from equipping their employees with mobile digital devices such as iPhones and iPads?

e.g. – Point of Sales Order and Supply information at the Retailers

Customer satisfaction survey

Marketing data collection / questionnaire

Payment banks in rural areas.

NPTEL

Conclusion… (1 of 2)

- Management information systems
  - Combines computer science, management science, operations research, and practical orientation with behavioral issues
- Four main actors
  - Suppliers of hardware and software
  - Business firms
  - Managers and employees
  - Firm's environment (Customer Satisfaction , legal, social and cultural context)

NPTEL

Conclusion (2 of 2)

In a socio-technical perspective, the performance of a system is optimized when both the technology and the organization mutually adjust (mesh) to one another until a satisfactory fit is obtained



It is extremely important for the business leaders (specially the CIO) to understand which technology would be best suited for their organization. Follow the Jones's often can lead to wrong investments.

NPTEL

REFERENCES

- **The World is Flat (book) : Thomas L Friedman**
- **Management Information Systems: Managing the Digital Firm - Kenneth C. Laudon & Jane P. Laudon**

NPTEL



NPTEL

THANK YOU !

Management Information System

Saini Das
Vinod Gupta School of Management, IIT Kharagpur

NPTEL

Module 02: Foundations of Business Analytics

Lecture 01 : Databases & Information Management

File organization concepts

Computer system uses hierarchies

- **Bit :** Smallest unit of data a computer can handle
- **Byte:** Group of bits, represents a single character
- **Field:** Group of bytes put into a word
- **Record:** Group of related fields
- **File:** Group of records of same type
- **Database:** Group of related files

NPTEL

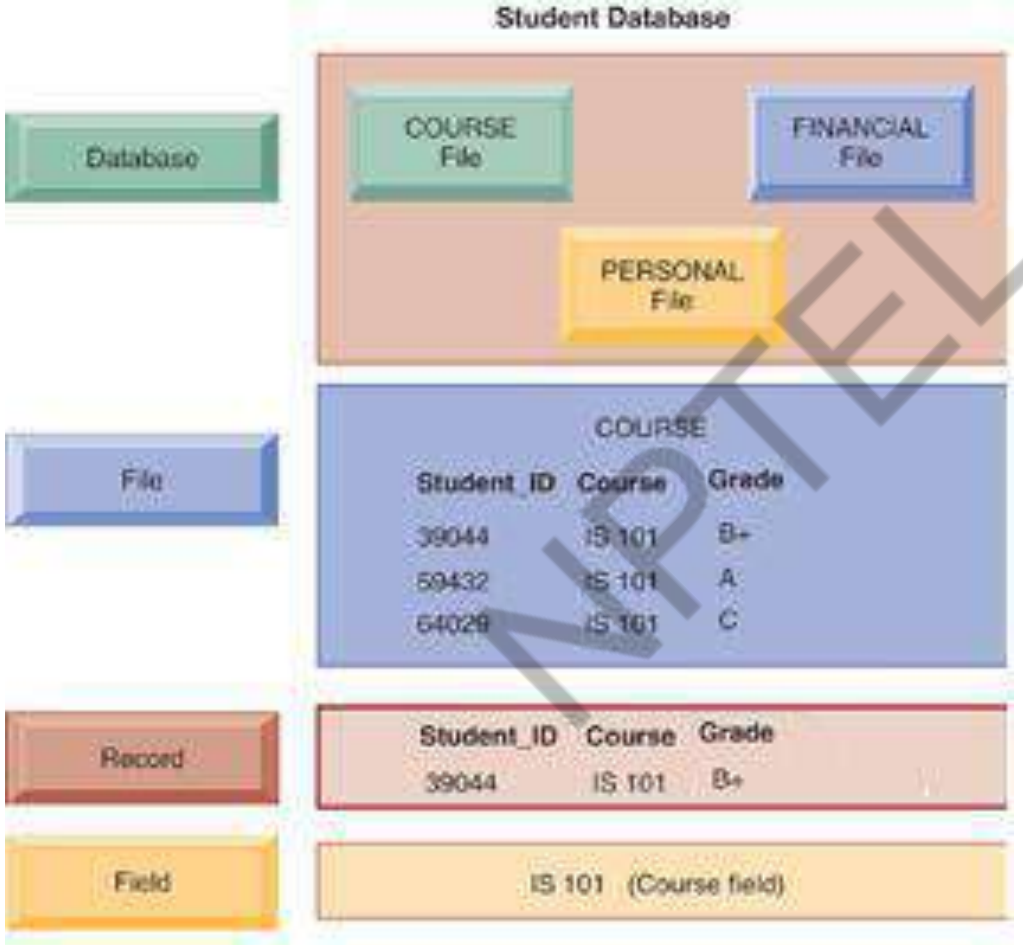
File organization concepts (contd..)

- **Record**: Describes an entity

- **Entity**: Person, place, thing about which we store information
  - **Attribute**: Each characteristic, or quality, describing entity
    - E.g. Attribute **Course and Grade** belong to entity **STUDENT**

NPTEL

The Data Hierarchy



NPTEL

Problems with the traditional files

- **Data redundancy and inconsistency**
  - **Data redundancy:** Presence of duplicate data in multiple files
  - **Data inconsistency:** Same attribute has different values
- **Poor security**
- **Lack of data sharing and availability**

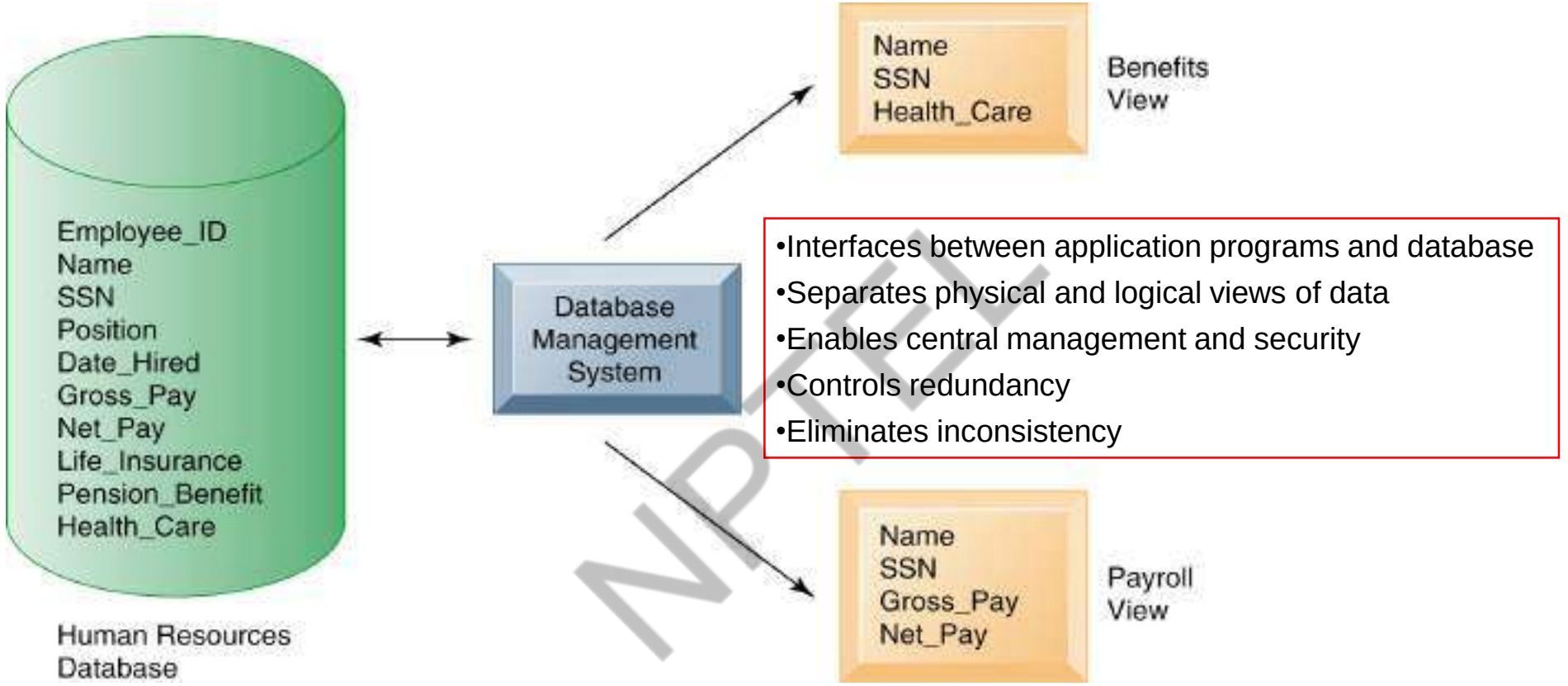
NPTEL

The Database Approach to Data Management

- **Database**
  - Collection of data organized to serve many applications by centralizing data and controlling redundant data
- **Database management system**
  - Interfaces between application programs and database
  - Separates logical and physical views of data
  - Solves problems of traditional file environment
    - Controls redundancy
    - Eliminates inconsistency
    - Enables central management and security

NPTEL

Human Resources Database with Multiple Views



- Interfaces between application programs and database
- Separates physical and logical views of data
- Enables central management and security
- Controls redundancy
- Eliminates inconsistency

A single human resources database provides many different views of data, depending on the information requirements of the user. Illustrated here are two possible views, one of interest to a benefits specialist and one of interest to a member of the company's payroll department.

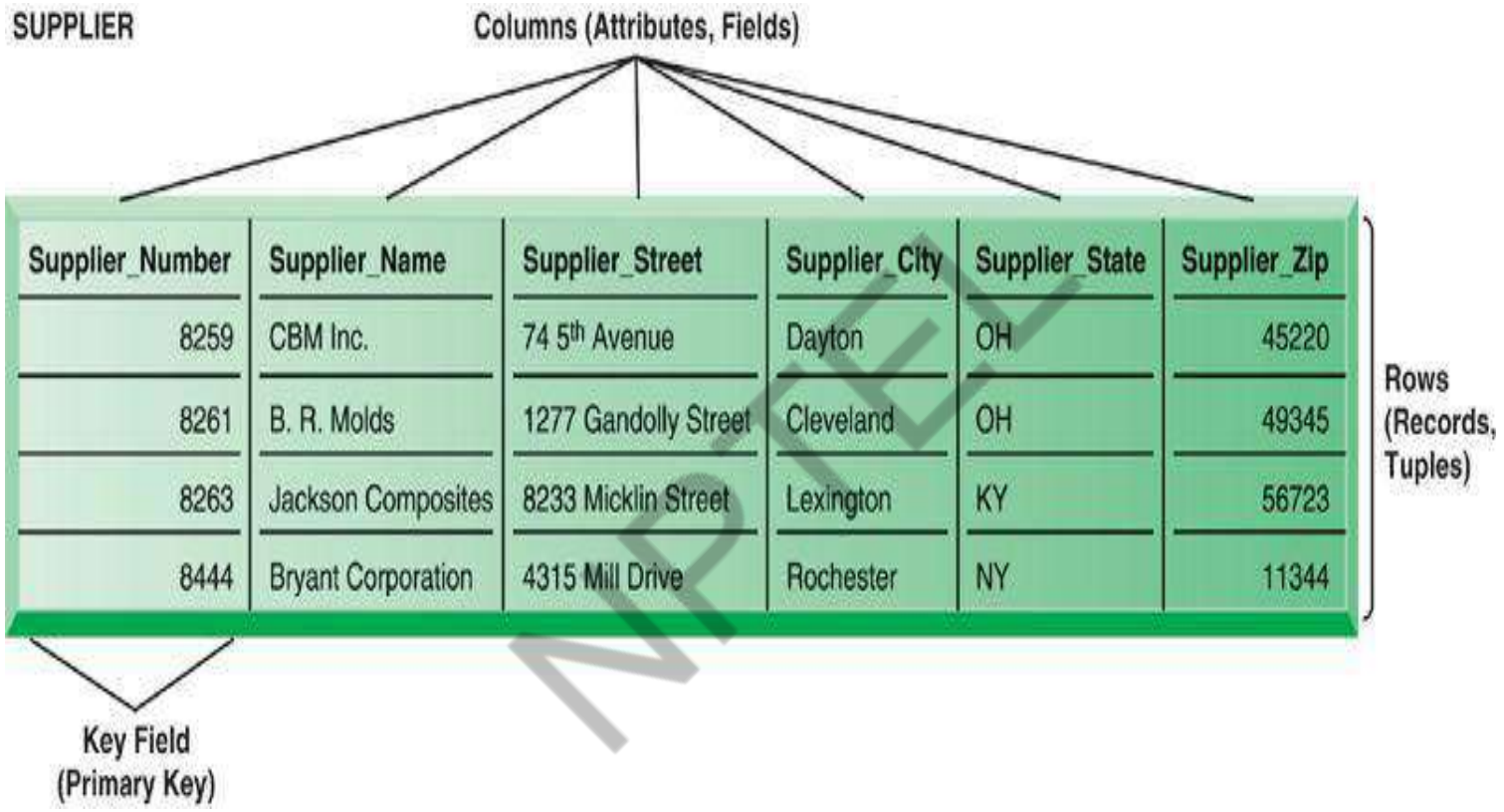
A single human resources database provides many different views of data, depending on the information requirements of the user. Illustrated here are two possible views, one of interest to a benefits specialist and one of interest to a member of the company's payroll department.

Relational DBMS

- **Relational DBMS**
  - Represent data as two-dimensional tables called relations or files
  - Each table contains data on entity and attributes
  - Examples: Microsoft Access, Oracle Database, MySQL, Microsoft SQL Server.
- **Table**: Grid of columns and rows
  - **Rows (tuples):** Records for different entities
  - **Fields (columns):** Represents attribute for entity
  - **Primary key**: Field in table used for key fields or unique identification
  - **Foreign key**: Primary key used in second table as look-up field to identify records from original table

NPTEL

Relational Database Tables



A relational database organizes data in the form of two-dimensional tables. Illustrated here are tables for the entities SUPPLIER and PART showing how they represent each entity and its attributes. Supplier_Number is a primary key for the SUPPLIER table and a foreign key for the PART table.

Relational Database Tables ….

PART

Part_Number	Part_Name	Unit_Price	Supplier_Number
137	Door latch	22.00	8259
145	Side mirror	12.00	8444
150	Door molding	6.00	8263
152	Door lock	31.00	8259
155	Compressor	54.00	8261
178	Door handle	10.00	8259

Primary Key

Foreign Key

NPTEL

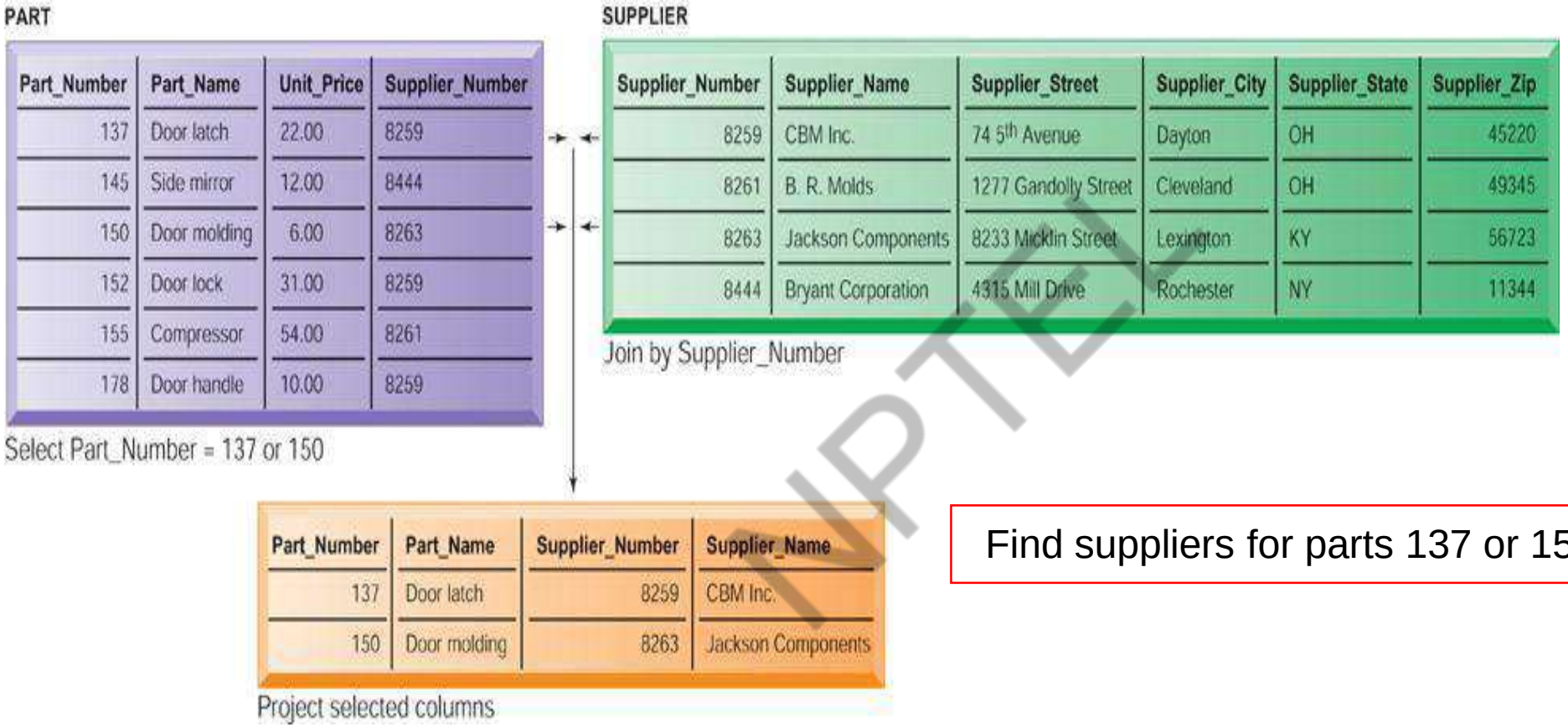
Operations of a Relational DBMS

Three basic operations used to develop useful sets of data

- **SELECT**: Creates subset of data of all records that meet stated criteria
- **JOIN**: Combines relational tables to provide user with more information than available in individual tables
- **PROJECT**: Creates subset of columns in table, creating tables with only the information specified

NPTEL

The Three Basic Operations of a Relational DBMS



Find suppliers for parts 137 or 150

The select, project, and join operations enable data from two different tables to be combined and only selected attributes to be displayed.

Capabilities of DBMS

- **Data manipulation language:** Used to add, change, delete, retrieve data from database
  - Structured Query Language (SQL)
  - Microsoft Access user tools

- Many DBMS have **report generation capabilities** for creating polished reports

NPTEL

Example of an SQL Query

SELECT: Lists the desired fields that have to be included in the query
FROM: Lists the tables from where the data has to be drawn
WHERE: Specifies the values of the fields that have to be included or the conditions that have to be met to include the field.

```
SELECT PART.Part_Number, PART.Part_Name, SUPPLIER.Supplier_Number,
SUPPLIER.Supplier_Name
FROM PART, SUPPLIER
WHERE PART.Supplier_Number = SUPPLIER.Supplier_Number AND
Part_Number = 137 OR Part_Number = 150;
```

Illustrated here are the SQL statements for a query to select suppliers for parts 137 or 150

NPTEL

References

- K. Laudon and J. Laudon (2016). Management Information Systems Publisher: Pearson. Edition 14e.

- R. De. (2018). MIS Managing Information Systems in Business, Government and Society. Publisher: Wiley. Second Edition.

NPTEL

NPTEL

Management Information System

Saini Das
Vinod Gupta School of Management, IIT Kharagpur

NPTEL

Module 02: Foundations of Business Analytics

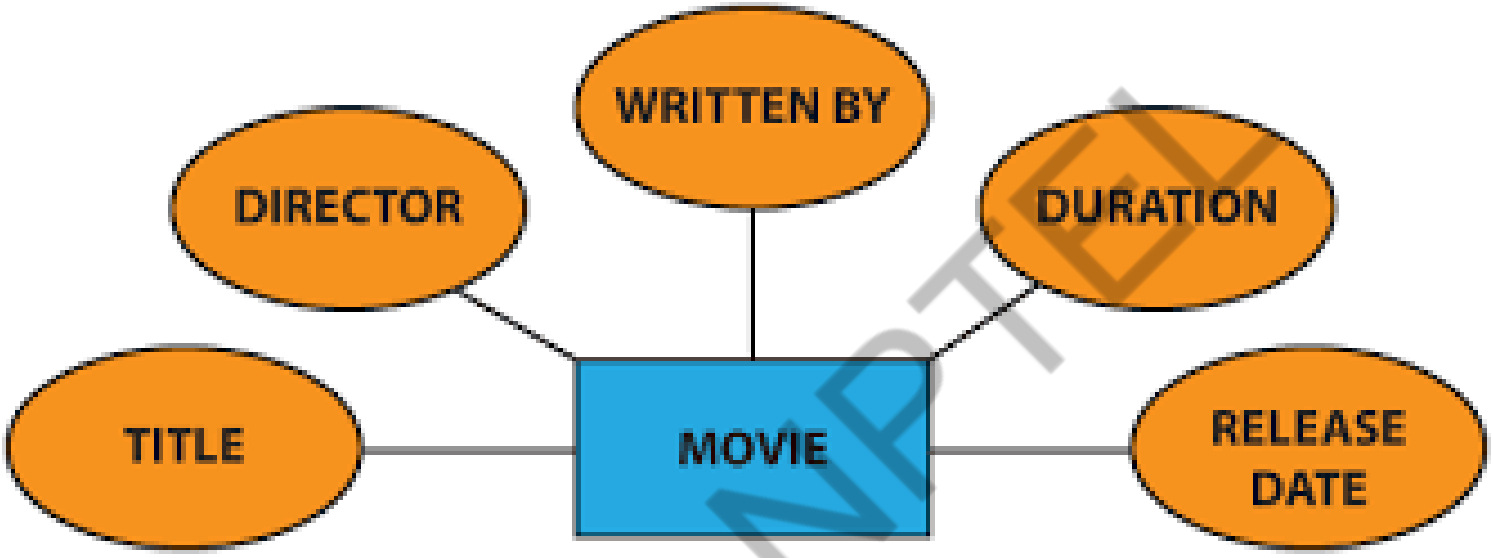
Data Warehouses & Business Intelligence

E-R (Entity-Relationship Diagrams)

- E-R Models are descriptions of the business requirements of data from the user perspective.
- They are often the first step in database design.
- A set of diagrammatic tools used to create these models are called E-R diagrams.
- An E-R diagram consists of data *entities*, *relationships* between these entities and *attributes* that describe the entities and relationships.
- Relationships identify natural links or associations between entities.

NPTEL

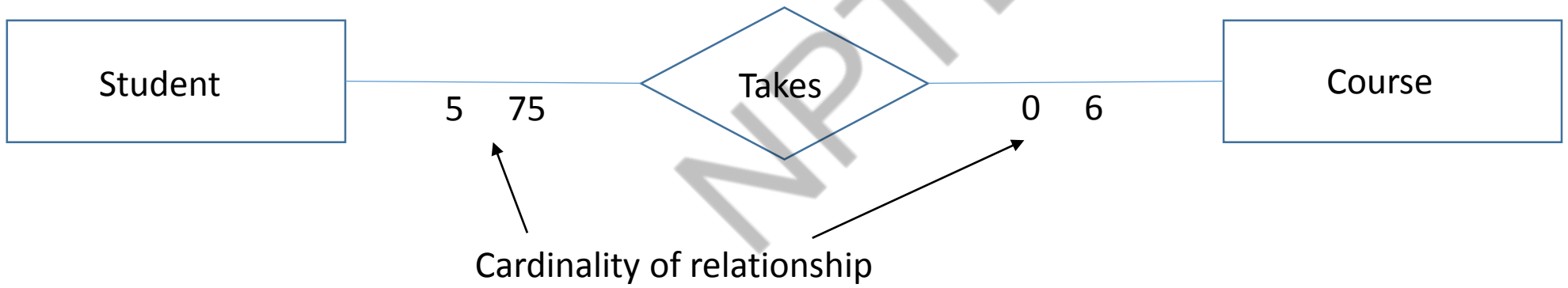
E-R (Entity-Relationship Diagrams) contd…



Movie entity with attributes

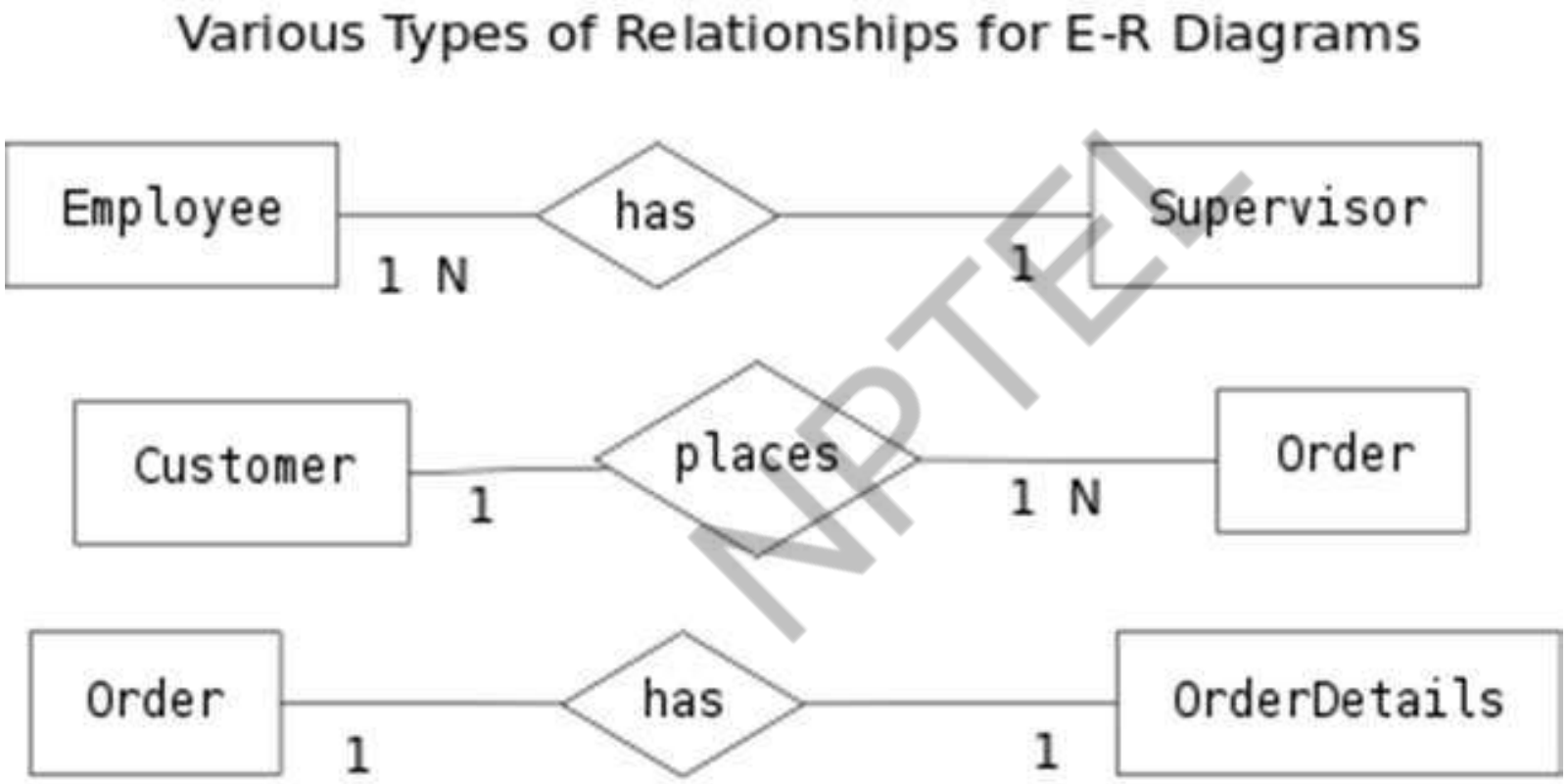
E-R (Entity-Relationship Diagrams) contd..

- The number of entities in a relationship represent the *cardinality* of a relationship.



NPTEL

E-R (Entity-Relationship Diagrams) contd..



NPTEL

Normalization

Streamlining complex groupings of data to minimize redundant data elements.

Students Engg. Mechanics

Roll No.	Name	Dept.	HoD	Dept. Contact no.
101	Sachin	Electrical	Prof. X	1234567
102	Rahul	Mechanical	Prof. Y	4567899
103	Saurav	Electronics	Prof. Z	6789048
104	Virat	Mechanical	Prof. Y	4567899
105	Dhoni	Electrical	Prof. X	1234567
106	Anil	Mechanical	Prof. Y	4567899

NPTEL

Normalization (contd..)

Students Engg. Mechanics

Roll No.	Name	Dept. Id
101	Sachin	001
102	Rahul	002
103	Saurav	003
104	Virat	002
105	Dhoni	001
106	Anil	002

Department

Dept. Id	Dept. Id	HoD	Dept. Contact no.
001	Electrical	Prof. X	1234567
002	Mechanical	Prof. Y	4567899
003	Electronics	Prof. Z	6789048

NPTEL

Using Databases to Improve Business Performance & Decision Making

- Very large organizations with huge amounts of data felt the need for:
  - Consolidating much of the data from various databases into a whole that could be understood clearly.
  - Focusing on the use of data for decision making, as opposed to simply for running transactions.

NPTEL

Using Databases to Improve Business Performance & Decision Making (contd..)

- This gave rise to special capabilities and tools required for analyzing large quantities of data:
  - Data warehouses
  - Data marts
  - Data mining

NPTEL

Data Warehouses

- Stores current and historical data from many core operational transaction systems
- Consolidates and standardizes information for use across enterprise, but data cannot be altered.
- To create a data warehouse, data is extracted from transactional tables, pre-processed to remove unwanted data types and then loaded into tables in the data warehouse.
- Warehouses differ from transaction databases as users can run complex queries on them.
- Data warehouses provide querying, analysis, and reporting tools.
-

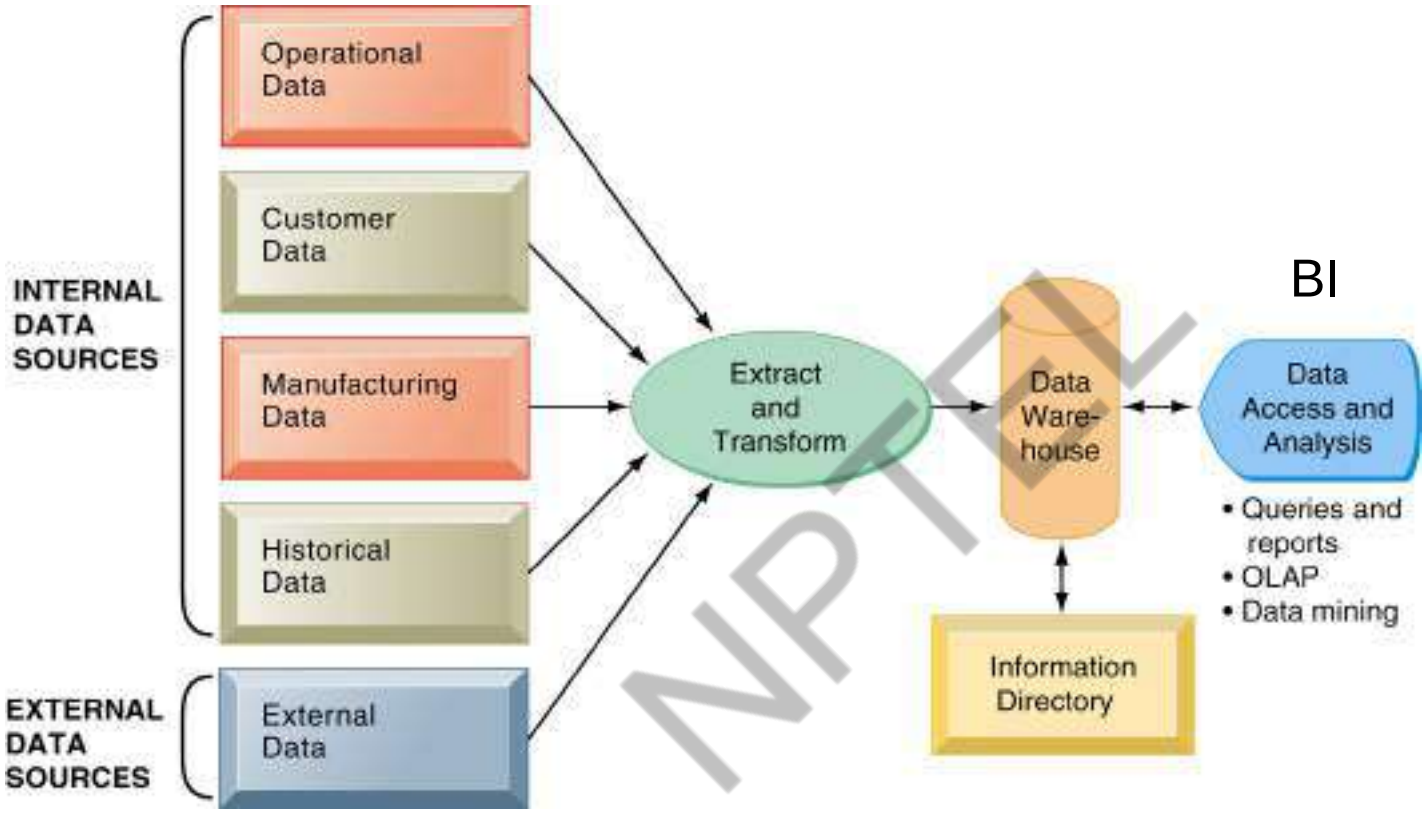
NPTEL

Data Mart

- Subset of data warehouse with summarized or highly focused portion of firm's data for use by specific population of users.
- Typically focuses on single subject or line of business.
- Used to identify problems and find solutions pertaining to a particular domain.
- Example: Sales data mart.

NPTEL

Components of a Data Warehouse



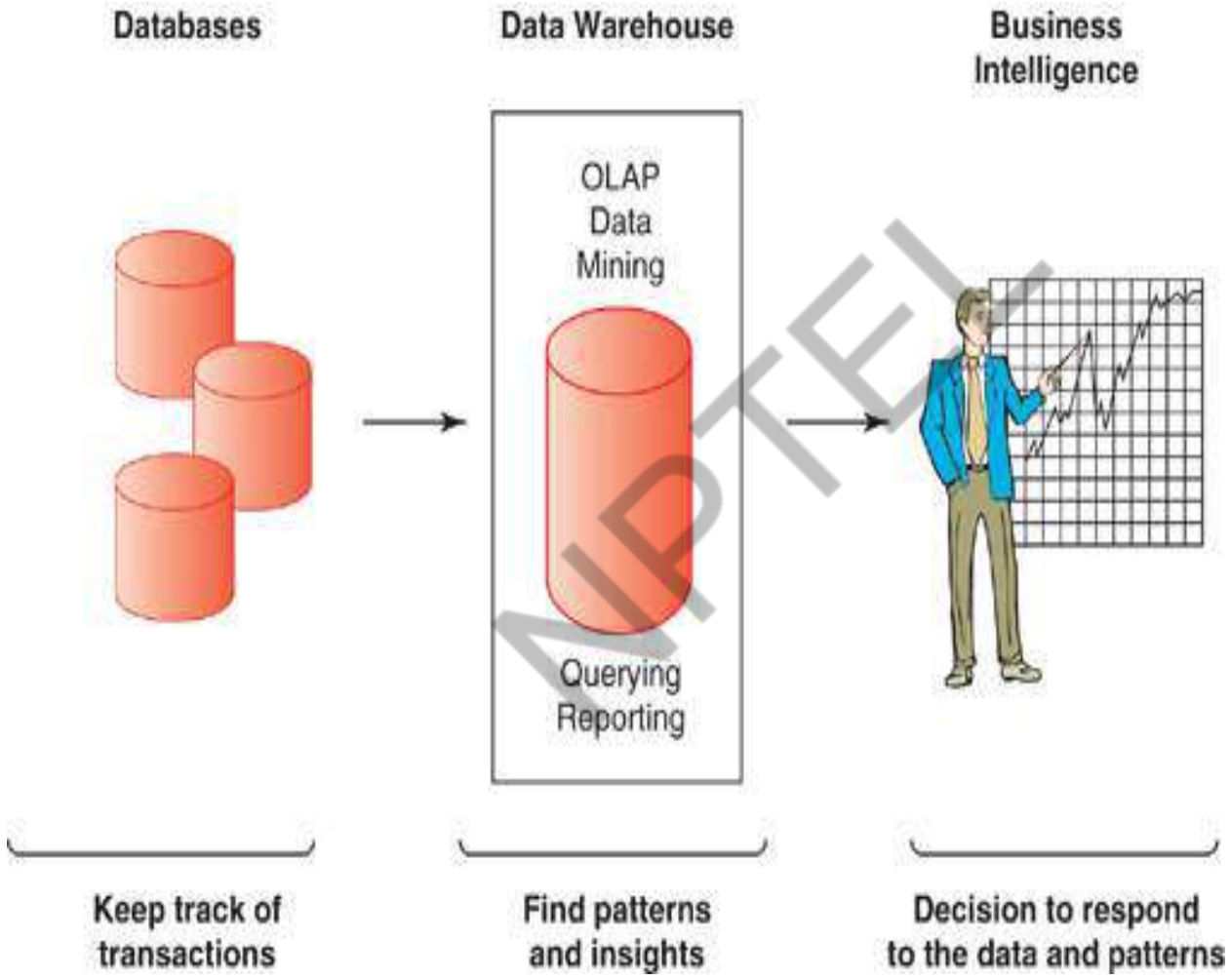
The data warehouse extracts current and historical data from multiple operational systems inside the organization. These data are combined with data from external sources and reorganized into a central database designed for management reporting and analysis. The information directory provides users with information about the data available in the warehouse.

Business Intelligence

- Tools for consolidating, analyzing, and providing access to vast amounts of data to help users make better business decisions
  - E.g. Harrah's Entertainment analyzes customers data to develop gambling profiles and identify most profitable customers
- Principle tools to derive BI from data in a warehouse include:
  - Software for database querying and reporting
  - Online analytical processing (OLAP)
  - Data mining

NPTEL

What drives Business Intelligence?



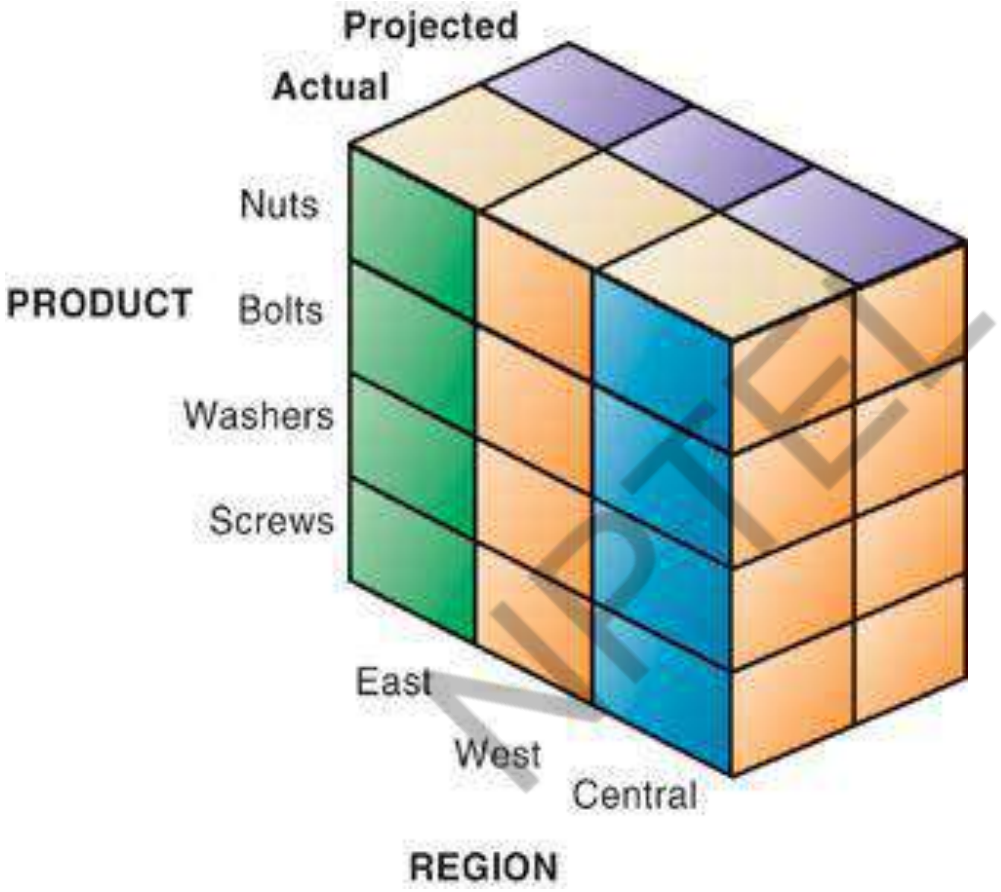
NPTEL

Online analytical processing (OLAP)

- Supports multidimensional data analysis
- Enables viewing data using multiple dimensions
  - Each aspect of information (product, pricing, cost, region, time period) is different dimension
  - E.g. How many cycles were sold in Eastern India in June?
- OLAP enables rapid, online answers to ad hoc queries

NPTEL

Multidimensional Data Model



- What were the actual sales of Bolts in Central India?

References

- K. Laudon and J. Laudon (2016). Management Information Systems Publisher: Pearson. Edition 14e.

- R. De. (2018). MIS Managing Information Systems in Business, Government and Society. Publisher: Wiley. Second Edition.

NPTEL

NPTEL

Management Information System

Saini Das
Vinod Gupta School of Management, IIT Kharagpur

NPTEL

Module 02: Foundations of Business Analytics

Introduction to Data Mining



- W. Edwards Deming

“There is a striking correlation between an organization's analytics sophistication and its competitive performance.”

10 Insights: A first look at the new intelligent enterprise survey on winning with data, MIT Sloan Management Review, Vol 52, No 1, 2010

NPTEL

“Data Scientists will be the sexiest job of 21st century”

NPTEL

Harvard Business Review 2012

Why mine data?

- Huge amounts of data being collected and warehoused
  - Walmart records 20 million items in transactions per day
  - Health care transactions: multi-gigabyte databases
- Source of competitive advantage

NPTEL

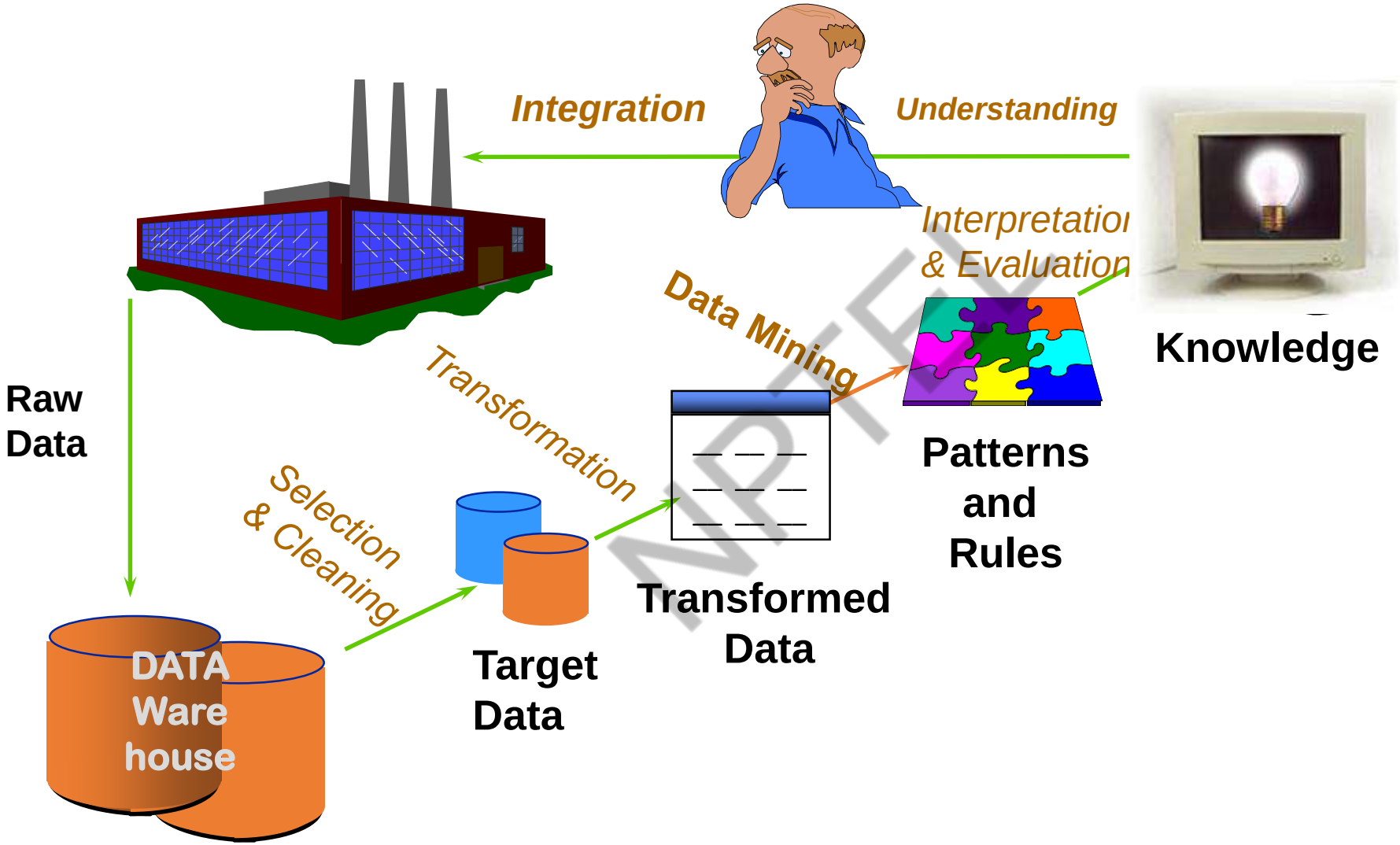
What is data mining and KDD?

- Knowledge discovery in databases (KDD) is the non-trivial process of identifying **valid**, potentially **useful, understandable** and ultimately **actionable patterns** in data.

- Data mining is a step in the KDD process of applying data analytics and discovery algorithms

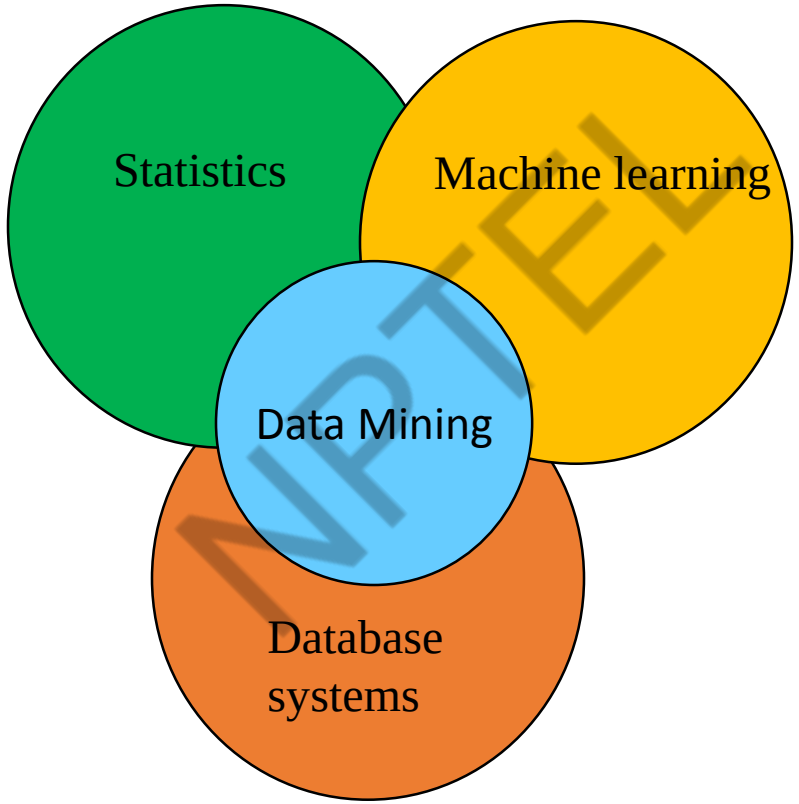
NPTEL

Knowledge Discovery Process



NPTEL

DM is multidisciplinary



NPTEL

Data Mining Applications

Typical Applications

Customer Segmentation

Propensity to Buy

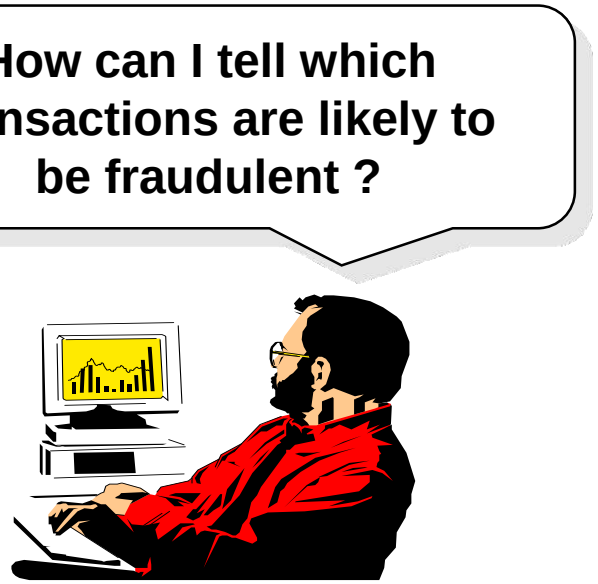
Profitability Modeling & Profiling

Customer Attrition

Channel Optimization

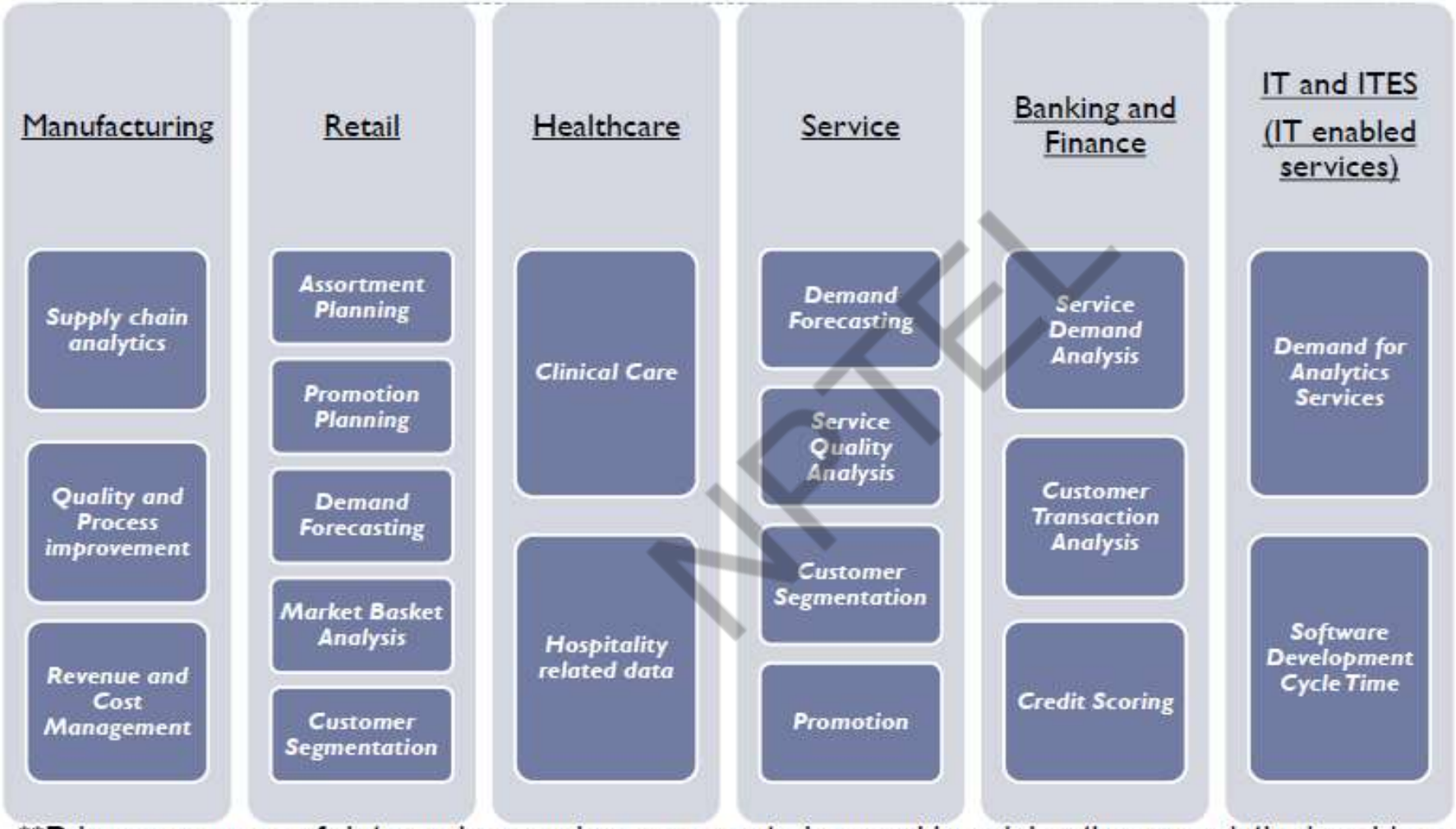
Fraud Detection

NPTEL



Detect and prevent fraud to minimize loss.

Industry wide applications of analytics



**Primary sources of data and secondary sources to be used in solving these analytical problems

NPTEL

Some other applications….

- **Marketing**
  - Which customers are likely to respond to this campaign?
  - Which customers are likely to be profitable ?
  - Who might want to buy this product ? (‘Cross-selling’)
  - Which web-pages are customers visiting before buying products or before leaving our site ? Which types of customers visit which pages ?
- **Telecommunications**
  - Which customers are vulnerable to attrition (at risk of churning) ?
  - Based on these symptoms, where are problems located in the network ?
- **Finance and Insurance**
  - Which customers are credit risks / insurance risks ?
  - Which claims or credit transactions are fraudulent ?
  - Which stocks are likely to perform well in the next 3 months, and why ? When should we buy and sell, given the likely performance and transaction costs ?

NPTEL

11

Some other applications (Contd..)

- **Healthcare**
  - Which patients may take longer to recover ?
  - What is the likely cause of this illness ?
  - Which patients are at risk of disease (and might benefit from medication)? Pfizer pharmaceuticals used data mining to construct a predictive model that was then embedded in their online cholesterol health risk assessment, which tells patients their cholesterol risk score. High risk patients can consult their doctors and request Lipitor, Pfizer's cholesterol medication.

- **Retail**
  - Which products do customers buy together (or in sequence) & which do they not buy together ? ('Category management'.)
  - What characterizes customers at various stores ?
  - What items are bought for cash, on credit, or by check ?
  - What type of customer buys this item, or this product type ?

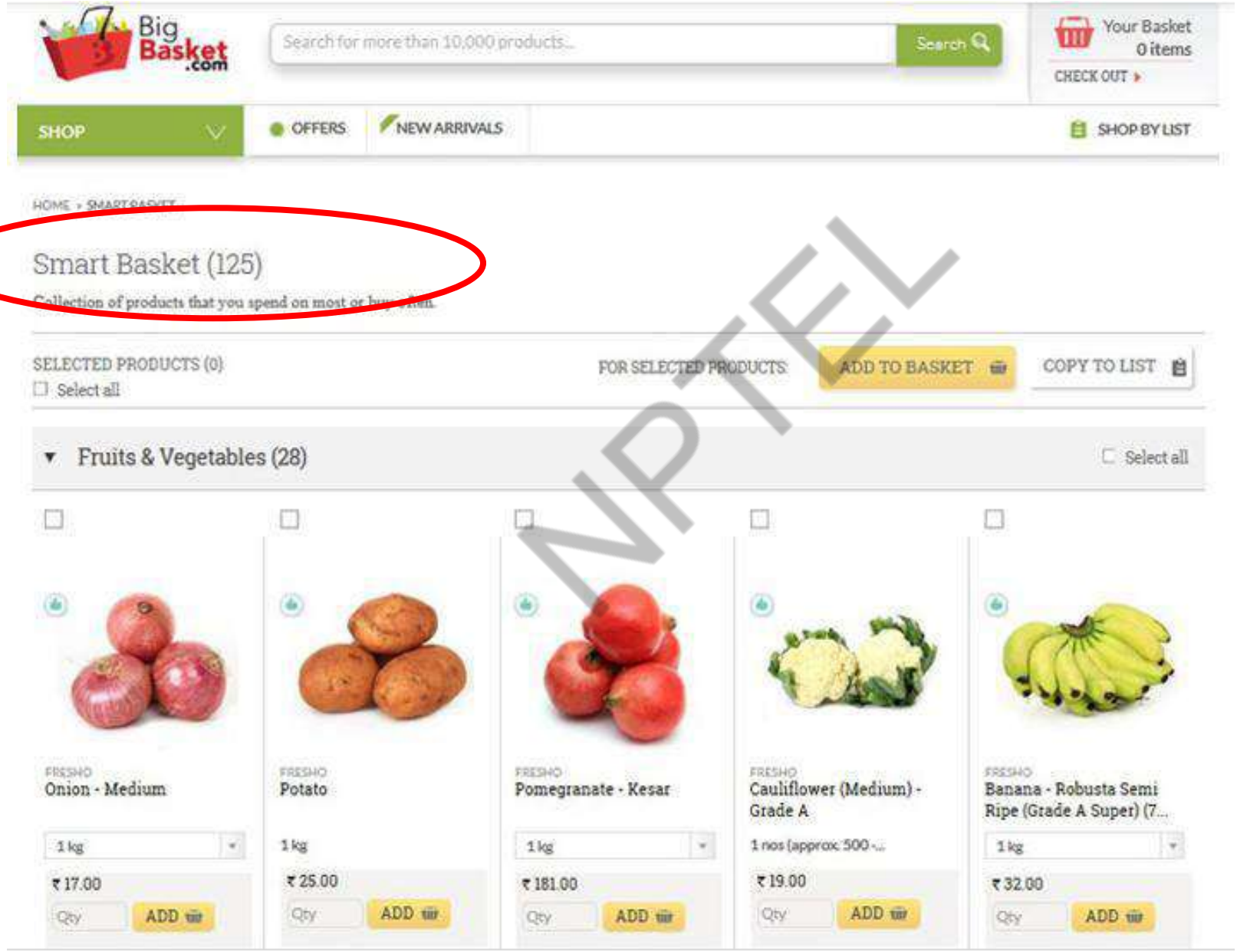
NPTEL

Data Mining Applications (Contd..)

- **Quality Control**
  - Which shipments are high-risk and need to be inspected ?
- **Customer Support**
  - Which tasks schedule (ordering) is optimal (or good enough) ?
  - Which customer service representative should I assign to a task ?
  - What documents or people are likely to be helpful to the customer in solving their problem ?

NPTEL

Real-life Applications…



NPTEL


flipkart.com
flipkart
flipkart.com

At Flipkart…

- Forecast demand for each SKU.
- Predict customer cancellations and returns.
- Predict what a customer is likely to purchase in the future?
- How to optimize the delivery system?

NPTEL

Diaper-beer syndrome

IT'S PART OF the folklore of data processing. A retail chain put all its checkout-counter data into a giant digital warehouse and set the disk drives spinning.

Out popped a most unexpected correlation: sales of diapers and beer.

Evidently, young fathers would make a late-night run to the store to pick up Pampers and get some Bud Light while they were there.



NPTEL

NPTEL

Management Information System

Saini Das
Vinod Gupta School of Management, IIT Kharagpur

NPTEL

Module 02: Foundations of Business Analytics

Data Analytics Tools and Techniques

Divorce360.com



Target predicts customer pregnancy

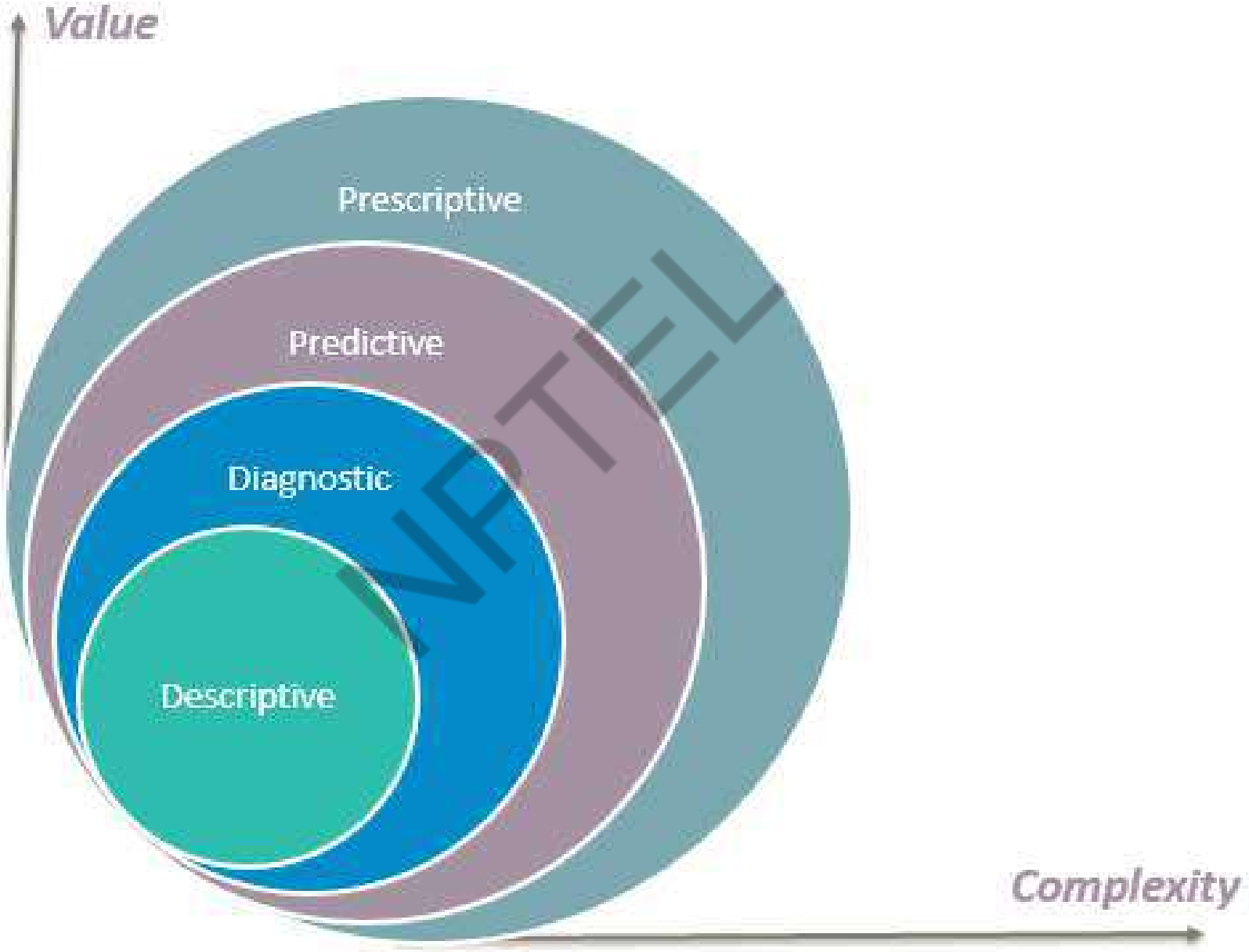


https://www.youtube.com/watch?v=XH1wQEgROg4

Categories of Data Analytics



NPTEL


Value
Prescriptive
Predictive
Diagnostic
Descriptive
Complexity

NPTEL

Descriptive Analytics Applications

- Most shoppers turn towards right when they enter the a retail store.
- Conversion rate of women shoppers is higher than male shoppers among electronic gadgets purchasers (Radio Shack).
- Strawberry pop-tarts sell 7 times more during hurricane compared to regular period (Wal Mart).
- Women car buyers prefer women sales person.

NPTEL

Diagnostic Analytics Applications

- Why customers liked your social media campaign or why they didn’t?
- Why certain products were popular at a certain time, at a certain place?

NPTEL

Predictive Analytics Applications

- Which product the customer is likely to buy in his next purchase (recommender system).
- Which customer is likely to default in his/her loan payment.
- Who is likely to cancel the product that was ordered through e-commerce portal.

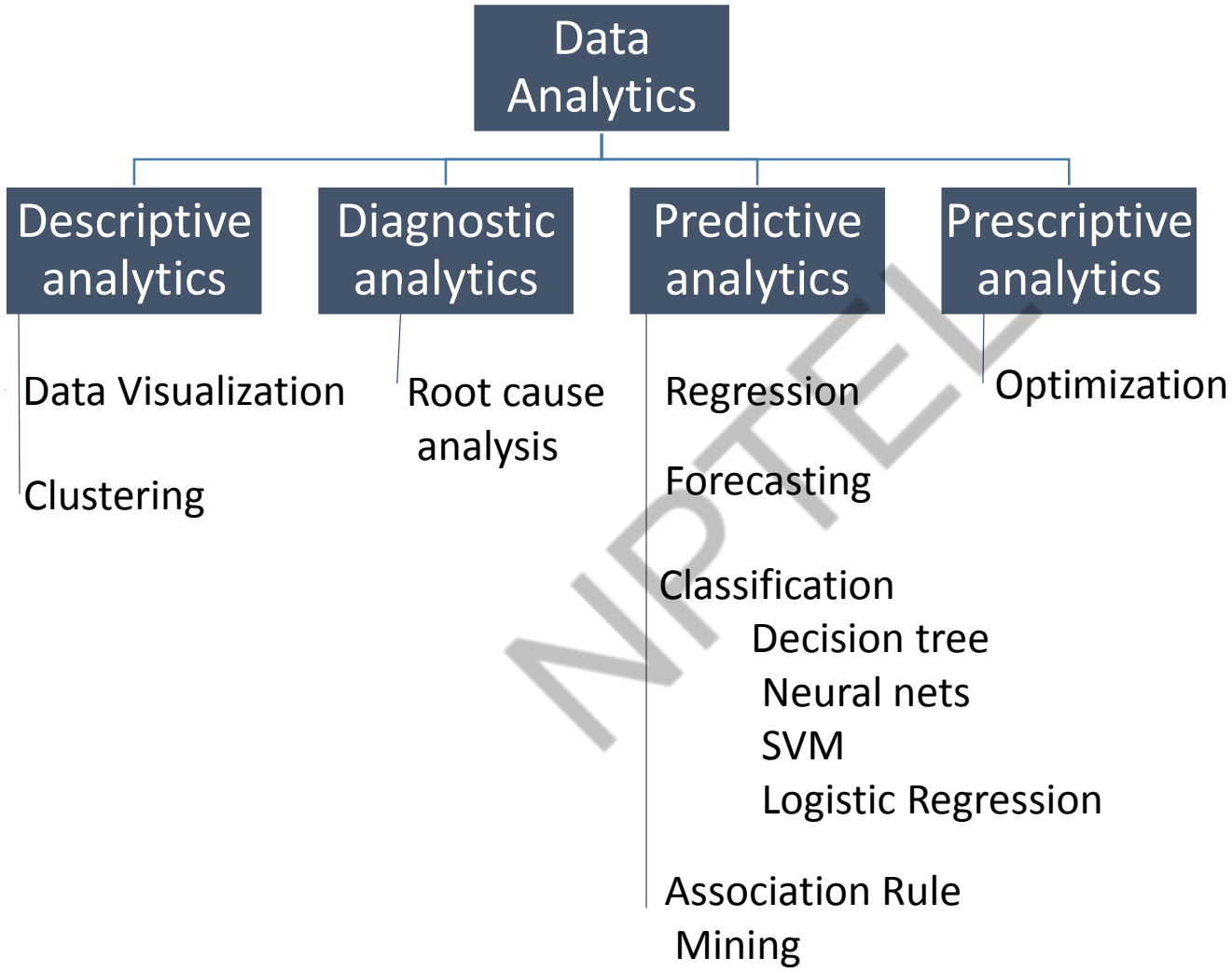
NPTEL

Prescriptive Analytics Applications

- What is the optimal product mix?
- What is the optimal route for a delivery truck.
- Best markdown pricing for fashion products.
- Optimal assignment of aircraft to flight.
- How to manage the fleet of vehicles owned by a company for employee drop and pick up?

NPTEL

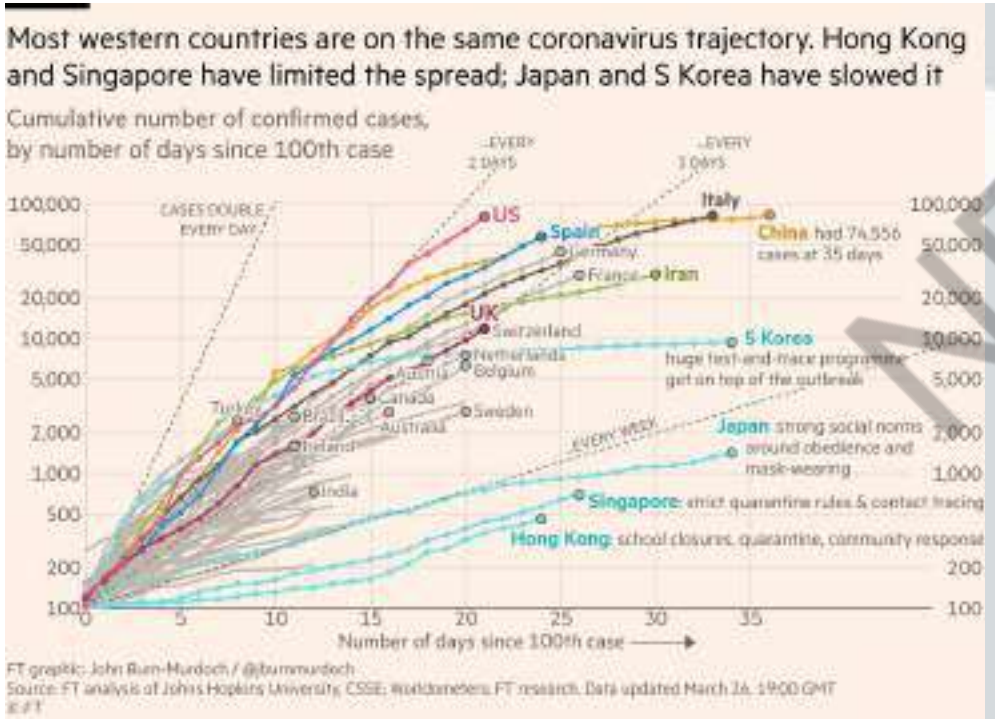
Analytics Tools & Techniques



NPTEL

Descriptive Analytics Techniques

- **Data Visualization:** Graphical representation of **data**. By using visual elements like charts, graphs, and maps, **data visualization** tools provide an accessible way to see and understand trends, outliers, and patterns in **data**.
- Software for Data Visualization: MS Excel; Power BI, Tableau; QlikView

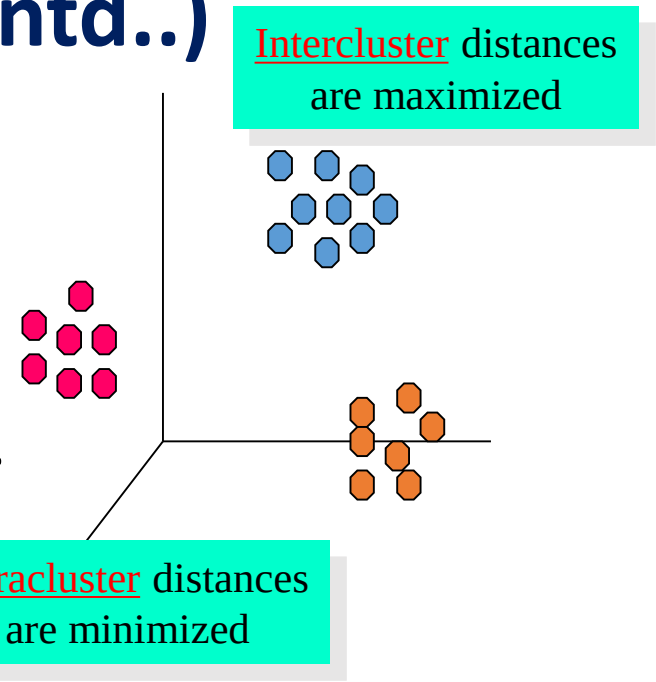




NPTEL

Descriptive Analytics Techniques (contd..)

- **Clustering:** Given a set of data points, each having a set of attributes, find clusters such that:
  - data points in one cluster are more similar to one another
  - data points in separate clusters are less similar to one another.

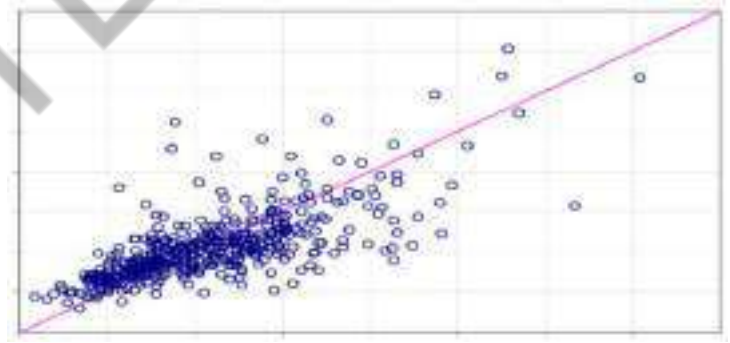


- Application in Market Segmentation: Subdivide a market into distinct subsets of customers based on their geographical and lifestyle related information where any subset may conceivably be selected as a market target to be reached with a distinct marketing mix.

NPTEL

Predictive Analytics Techniques

- **Regression analysis:** Regression analysis is a statistical technique used to describe relationships among variables. The simplest case to examine is one in which a variable Y , referred to as the dependent or target variable, may be related to one variable X, called an independent or explanatory variable.



NPTEL

- **Applications:**
- Predicting events that are yet to occur:
  - Demand analysis or number of units consumers will purchase in the next quarter based on past trends
  - Number of shoppers who will watch a particular advertisement.
  - Number of policy holders who will be involved in accidents in the next year.

Predictive Analytics Techniques (contd..)

- **Classification:** Data defined in terms of attributes, one of which is the class or target and others are predictor variables.
- Find a model for class/target attribute as a function of the values of other(predictor) attributes, such that previously unseen records can be assigned a class as accurately as possible. Given data is usually divided into training and test sets.
- Training Data: used to build the model
- Test data: used to validate the model (determine accuracy of the model).
- Classification Techniques:
  - Decision tree
  - Neural Networks
  - Support Vector Machines
  - Bayesian Classifier

NPTEL

Predictive Analytics Techniques (contd..)

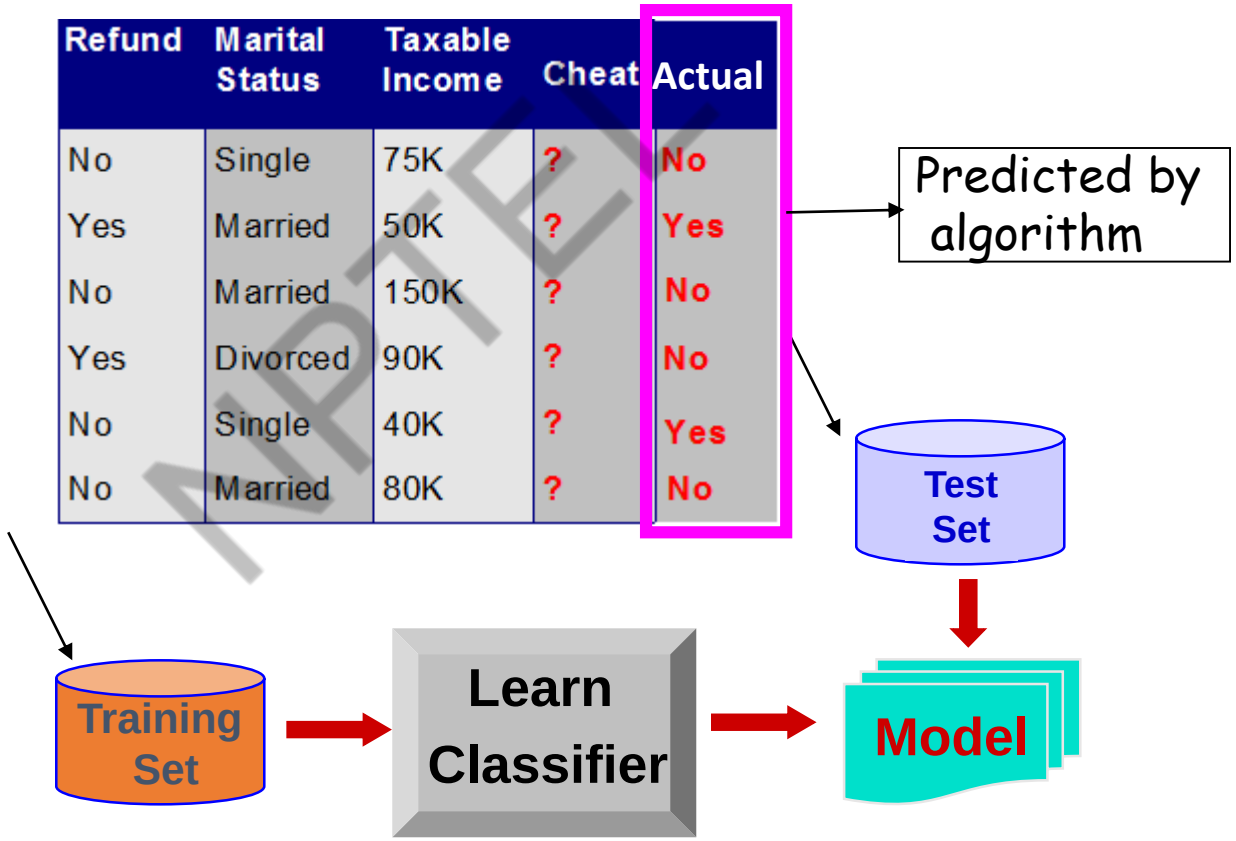
- **Application 1:** Given old data about fraudulent customers and their background, predict whether new customer will commit fraud or not.

Predictor attributes

Class/ Target

Tid	Refund	Marital Status	Taxable Income	Cheat
1	Yes	Single	125K	No
2	No	Married	100K	No
3	No	Single	70K	No
4	Yes	Married	120K	No
5	No	Divorced	95K	Yes
6	No	Married	60K	No
7	Yes	Divorced	220K	No
8	No	Single	85K	Yes
9	No	Married	75K	No
10	No	Single	90K	Yes

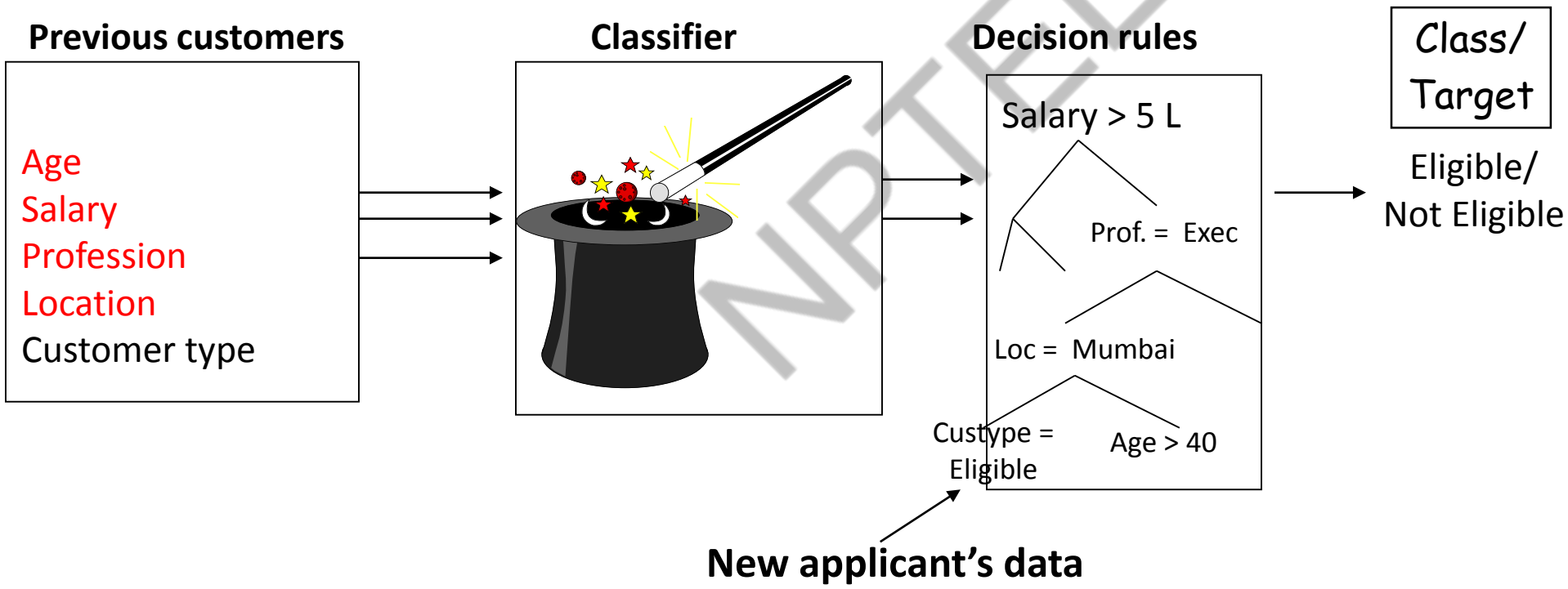
Refund	Marital Status	Taxable Income	Cheat	Actual
No	Single	75K	?	No
Yes	Married	50K	?	Yes
No	Married	150K	?	No
Yes	Divorced	90K	?	No
No	Single	40K	?	Yes
No	Married	80K	?	No



APTEX

Predictive Analytics Techniques (contd..)

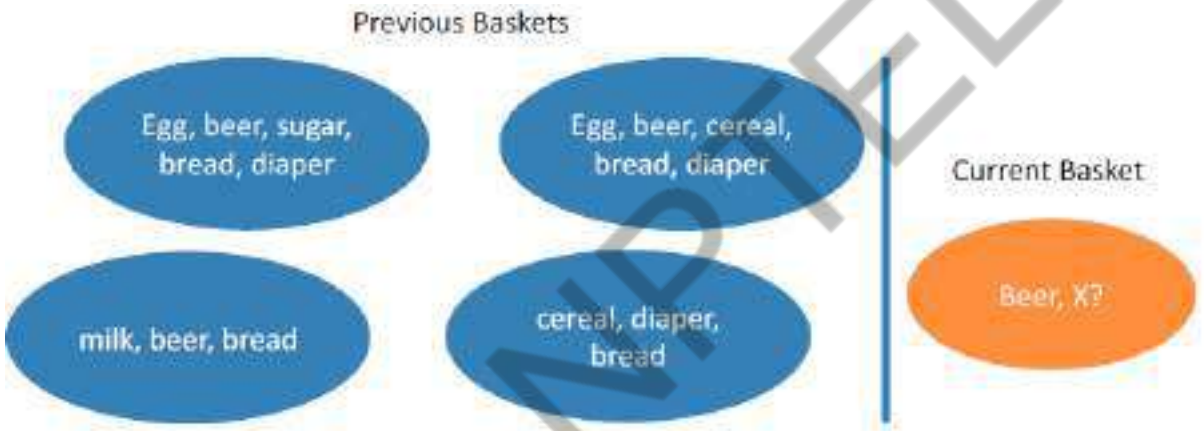
- **Application 2:** Given old data about customers and payments, predict new applicant's loan eligibility.



NPTEL

Predictive Analytics Techniques (contd..)

- **Association Rule Mining:** Finding frequent patterns:
  - Analysing item sets in customers basket or transactions and identifying frequently occurring item sets, which can be basis for recommendations.
  - Known as **Market Basket Analysis** in Retail.



NPTEL

- Applications:
  - Store layout: Place the frequently occurring items either in close proximity or as far apart as possible; give combo offers.
  - Warehousing and inventory management.

THANK YOU!

NPTEL



NPTEL ONLINE CERTIFICATION COURSES

NPTEL

Management Information Systems

Prof. Surojit Mookherjee
VGSoM, IIT KHARAGPUR

Week 03: ERP

Lecture 01 : What is ERP ?

Course Expectations…..

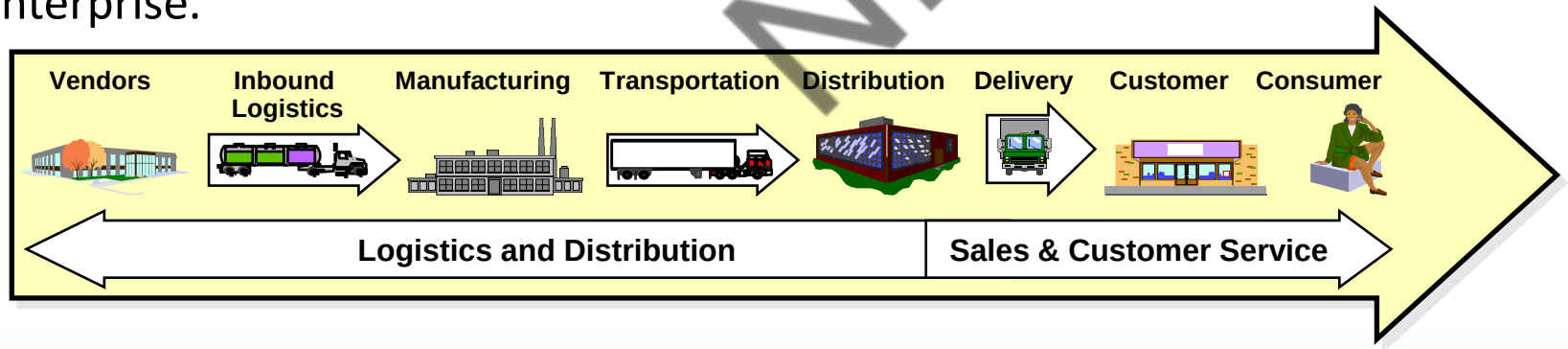
- To make you understand ‘WHY ERP’.
- To help you understand some of the key product features at a very HIGH LEVEL
- How to take the BEST from an ERP.
- To help you understand the REQUIREMENTS from an organization in IMPLEMENTING an ERP
- This course will NOT make you a SAP/ERP consultant

NPTEL

ERP Overview – Defining ERP…..

- **Enterprise**: Any organization that has a set of common goals
- **Resource**: Assets of the enterprise in the form of Man, Machine, Material
- **Planning**: Planning for effective utilization of resources

ERP is integrated information system built on a centralized database and common computing platform helps in effective usage of enterprise's resources, facilitates the flow of information between all business functions of the enterprise.



NPTEL

ERP Overview – Defining ERP

Enterprise resource planning (ERP) is a method of efficiently utilizing **people**, **hardware** and **software to increase productivity and profit**.

Thus simplifying a company's business processes.

ERP may include many software applications or a single (but more complex) software package that smoothly disseminates data required by two or more unique business departments.

NPTEL

ERP Overview – Why ERP….

The need for enterprise resource planning (ERP) software grew with big business' mandate for a centralized solution to manage all information system requirements.
An ERP may consist of many different business modules, including:

- Manufacturing / Planning
- Human Resources / Payroll
- Sales
- Materials / Inventory / Purchase
- Supply Chain / Partners
- Finance and Accounting

NPTEL

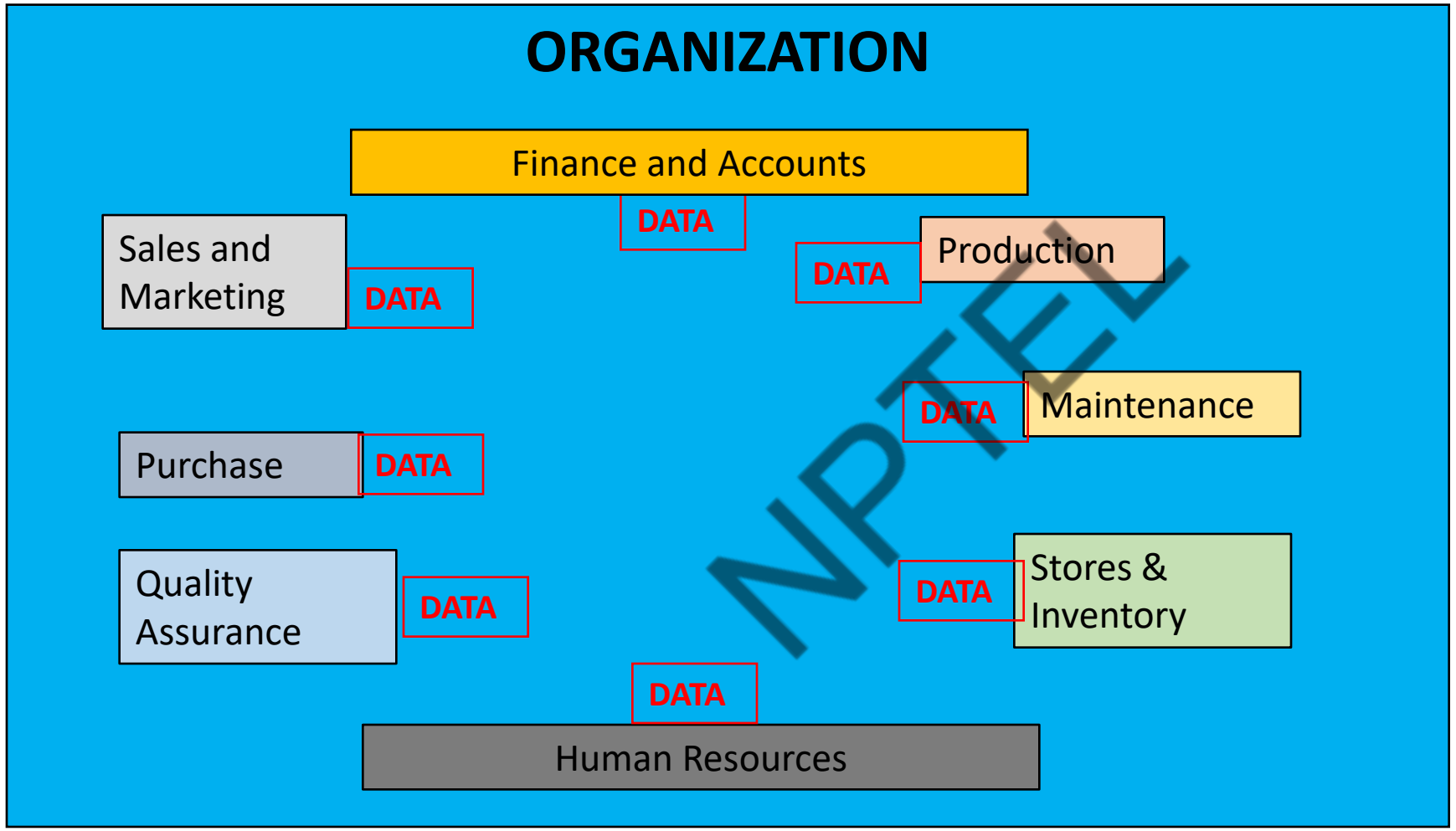
ERP Overview – Why ERP

The three Major Drivers:

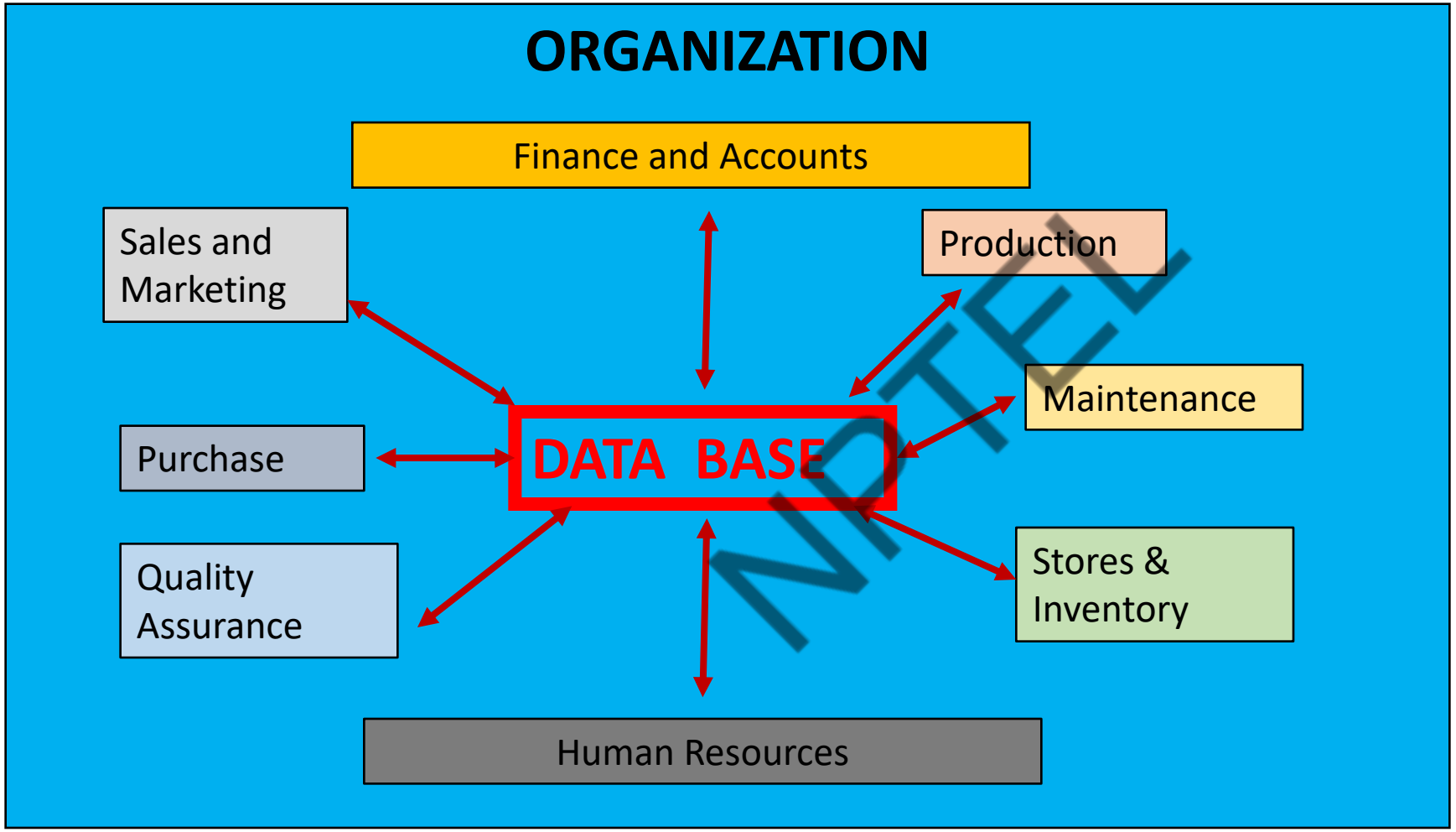
- From Department To Enterprise
- From Function To Process (e.g. Order to Cash)
- From Functional Silos To Integrated Information System

NPTEL

Stand Alone Solutions – Islands……. PAST



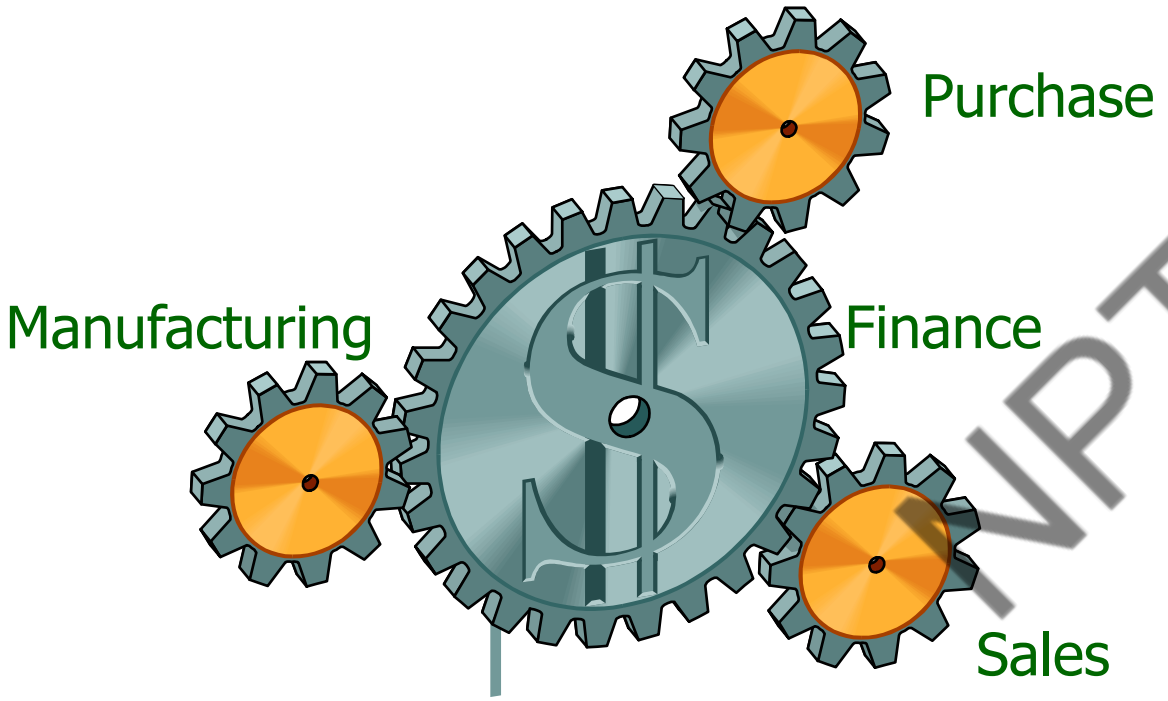
Integrated Platform……. Present



NPTEL

Enterprise Resource Planning…..present

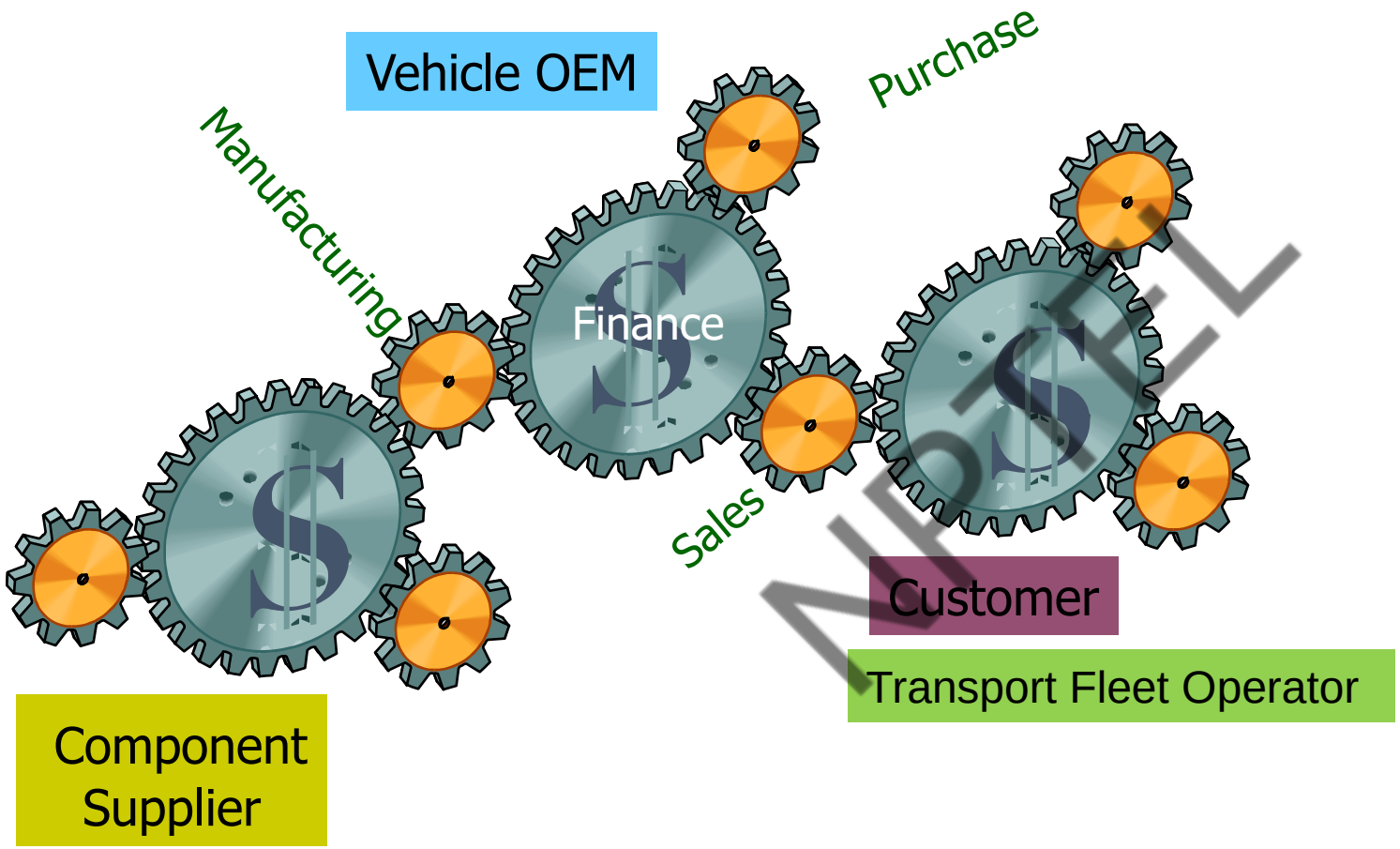
… synchronization of Company functions…



Question – When , Manufacturing happens , how do Purchase , Sales and Finance get impacted ?

NPTEL

ERP… and extended to the complete value chain…



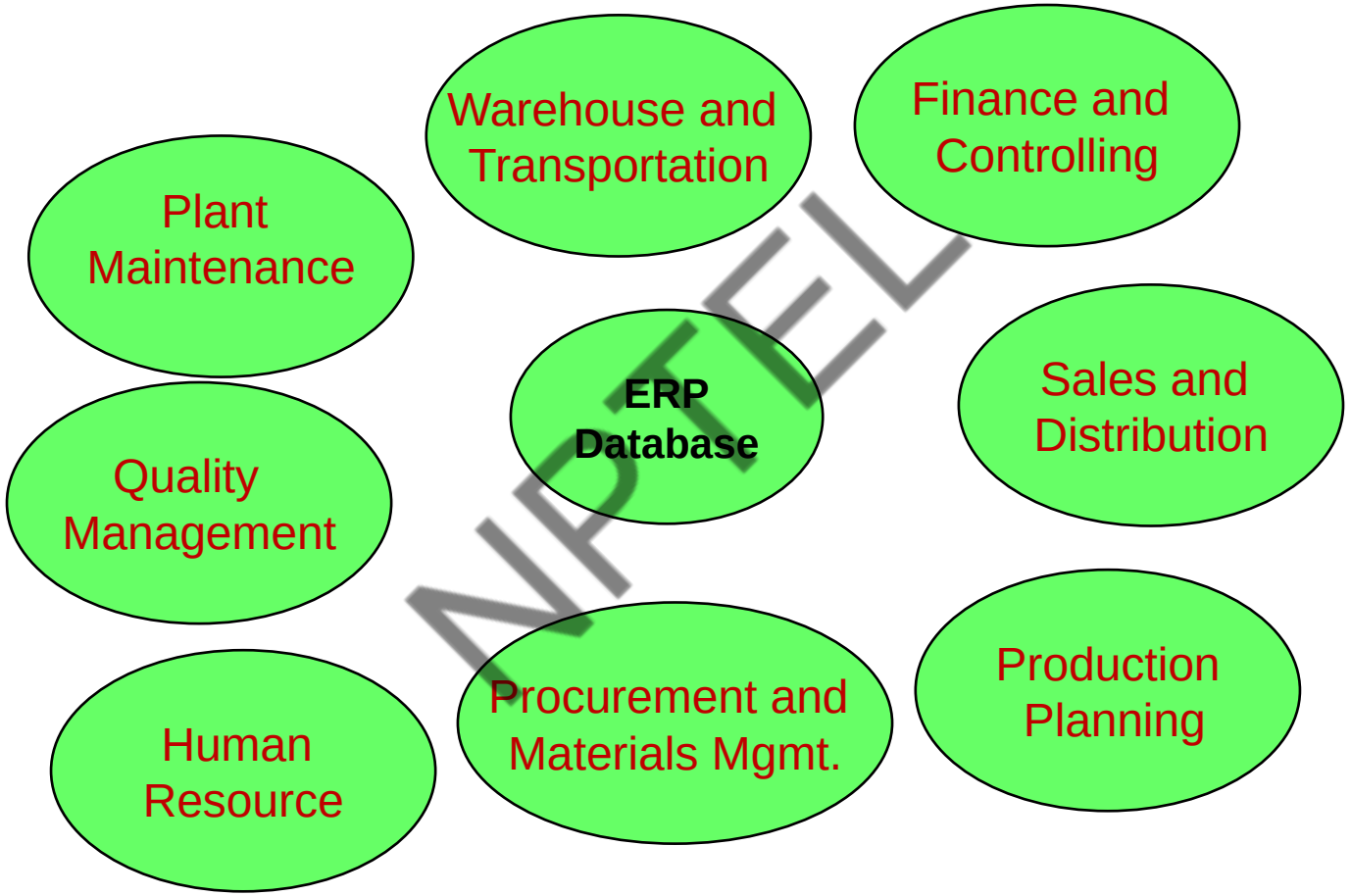
NPTEL

Problems with Information Silos

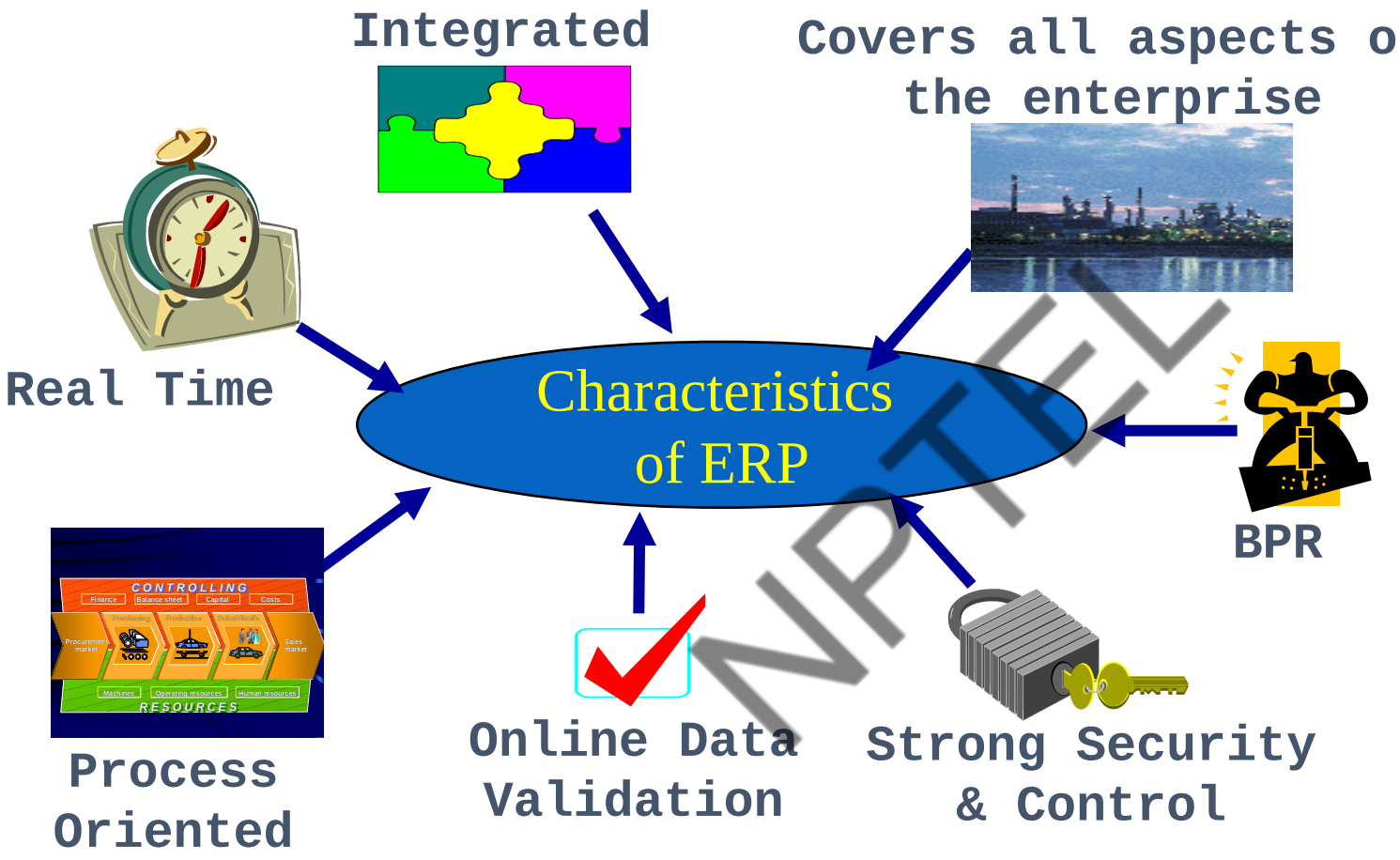
- ✓ Duplication Of Data and chances of making mistakes
- ✓ No Integration between different Systems
- ✓ No online updation of information
- ✓ No real – time availability of information
- ✓ Multiple data entry , increase possibility of making mistakes
- ✓ Different ‘naming’ conventions in different departments.

NPTEL

ERPs bring Central Database and Integrated Information System



NPTEL


Integrated
Covers all aspects of the enterprise
Real Time
Characteristics of ERP
BPR
CONTROLLING
Finance
Balance sheet
Capital
Costs
Procurement market
Sales market
Machines
Operating resources
Human resources
RESOURCES
Process Oriented
Online Data Validation
Strong Security & Control

NPTEL

REFERENCES

- **Enterprise Resource Planning : Rajesh Ray**
- **Enterprise Resource Planning Systems : O'Leary**

NPTEL



NPTEL

THANK YOU !



NPTEL ONLINE CERTIFICATION COURSES

NPTEL

Management Information Systems

Prof. Surojit Mookherjee
VGSoM, IIT KHARAGPUR

Week 03: ERP – Evolution and Benefits

Lecture 02 : Part 2 of 5

ERP Evolution - MRP

ERP systems evolved from MRP (Material Requirement Planning) systems of 60's and 70's.

Advantages

- Provides information on which items to buy or manufacture, how much quantity and on which date.

Disadvantages

- Did not look at available capacity while planning
- Could not replan quickly
- Not integrated with other organizational processes.

NPTEL

ERP Evolution – Closed Loop MRP

Closed Loop MRP was the next evolution after MRP which partially solved some of the MRP issues.

Advantages

- Does a manual feasibility check of MRP plan.
- Provides alerts / triggers if the plan is not feasible.

Disadvantages

- Could not replan quickly
- Not integrated with other organizational processes.

NPTEL

ERP Evolution – MRP 2

MRP 2 systems were much more stronger compared to MRP / Closed Loop MRP in terms of functionality. These systems were well integrated with several organizational business processes and had multiple modules.

Advantages

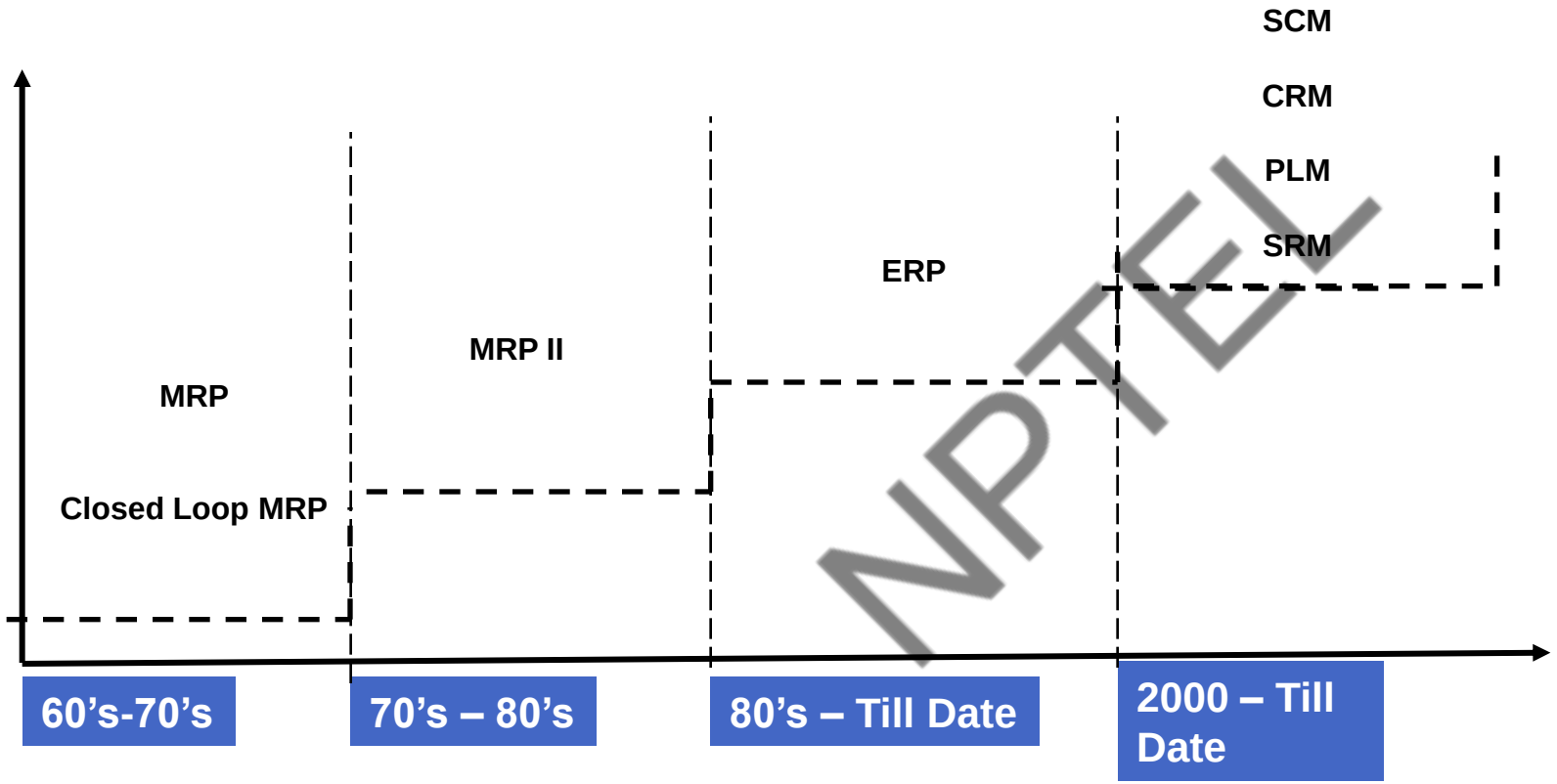
- Include processes for Sales planning, Demand management etc.
- Helped in several supply chain execution processes (like Purchase and Production order execution, Sales execution etc.).

Disadvantages

- Not fully integrated with all organization processes (like: HR)

NPTEL

History of ERP and Enterprise Applications



NPTEL

Enterprise Applications = ERP, CRM, SRM, PLM, APS, EAM

ERP – Some Early History of few Leading ERPs

- 1972: Five engineers (IBM) in Mannheim , Germany started the company SAP, with the idea of producing and marketing standard software for integrated business solutions.
- 1975: Richard Lawson, Bill Lawson, and John Cerullo started Lawson Software, a pre-packaged enterprise technology solutions as an alternative to customized business software applications.
- 1977: Jack Thompson (J), Dan Gregory (D), and Ed McVaney (ED) formed JD Edwards. (merged with Oracle ERP)
- 1978 Jan Baan begins The Baan Corporation that offered leading Baan ERP for several years. It is currently known as Infor.
- 1987: PeopleSoft was founded by Dave Duffield and Ken Morris.

JD Edwards and PeopleSoft have been taken over by Oracle and merged with Oracle ERP

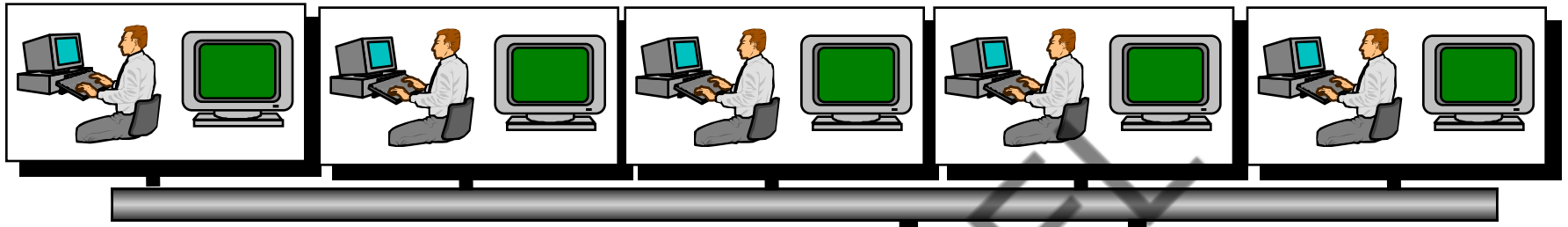
NPTEL

SAP - Systems Applications & Products in Data Processing

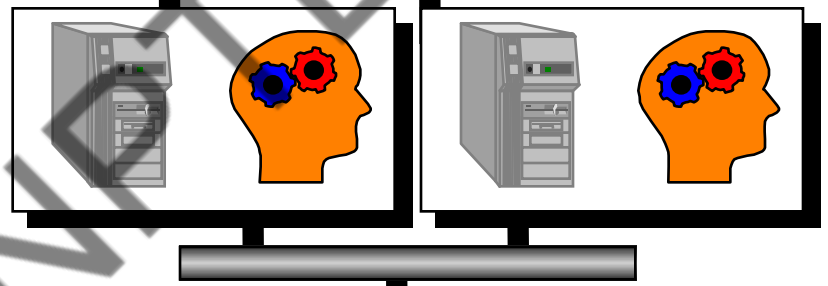


SAP R/3 : 3 Tier Architecture

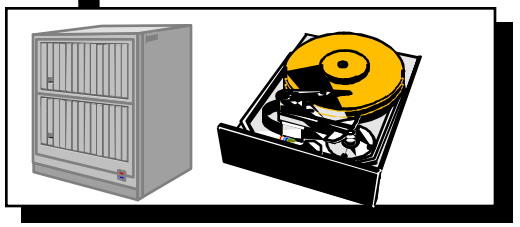
Presentation Layer (GUI) components are installed across many PCs.



The Application Layer components are installed across one or more high-end servers.



The Database Layer components are installed on high-end database server.



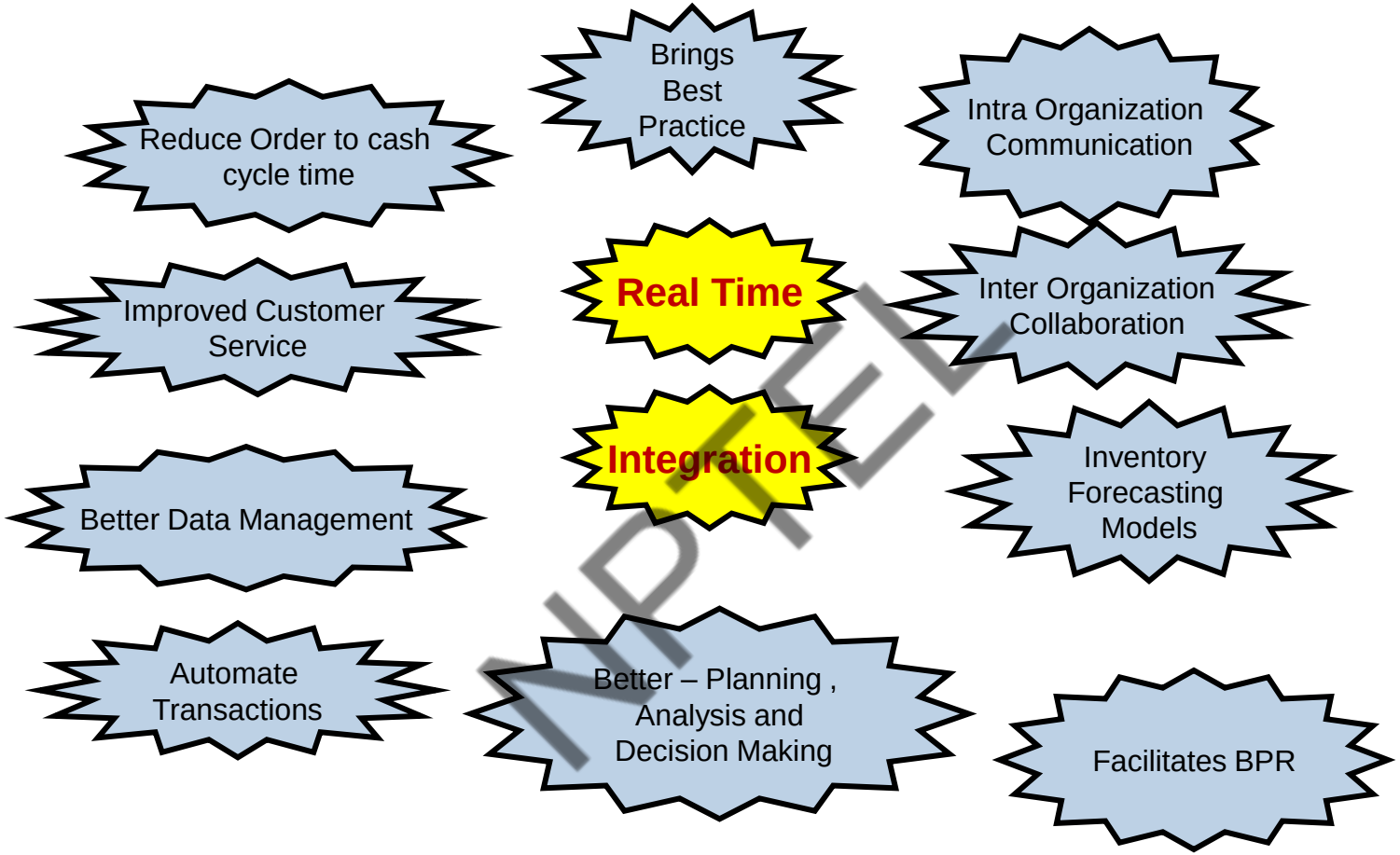
NPTEL

Business Process benefits clients expect from an ERP project

- Global process/ Product management
- Integrated Supply Chain Management
- Leverage purchasing and vendor management
- Order cycle time/ customer service improvement
- Inventory reductions
- Reduced information systems costs on an ongoing basis

NPTEL

ERP Benefits



NPTEL

Getting the best out of an ERP – Maximize ROI…..

Need to have a comprehensive or thorough business case for the executives and business stake holders

- **It is not merely an IT project**
- Provides with an excellent opportunity to transform business
- It entails significant organizational change
- Without support from business, the ERP implementation falters and IT often has a hard sell for keeping it going.

NPTEL

Contd…

Getting the best out of an ERP – Maximize ROI….

- **Know your ERP and reduce your old applications**
  - The standardized information and processes of these solutions will streamline your operations so that they are more efficient.
  - The industry best practices will prepare you to handle external pressures, such as competition and compliance.
  - Customization will ensure that your particular needs are being met.
  - Avoid too many customization to make it cheaper with less bugs , better system performance (response time) and easier and less expensive to upgrade versions.

NPTEL

Contd…

Getting the best out of an ERP – Maximize ROI…..

Cross the IT-Business divide

- ***ERP Implementation is NOT an IT program but a Business Program***
- ERP applications form the information backbone of your business. Increasingly, business executives, the ones making decisions and trying to increase profits, are using these applications and need the information it can provide.
- Business professionals and not IT experts should be part of the process of defining functional requirements for the application.

NPTEL

Contd…

Getting the best out of an ERP – Maximize ROI

- **Conclusion**
  - ERP software can be an expensive and time-consuming IT investment
  - ERP applications facilitate the standardization and acceleration of business processes and contribute to the visibility necessary to run your business well and grow revenue and profits.

NPTEL

REFERENCES

- **Enterprise Resource Planning : Rajesh Ray**
- **Enterprise Resource Planning Systems : O'Leary**

NPTEL



NPTEL

THANK YOU !



NPTEL ONLINE CERTIFICATION COURSES

NPTEL

Management Information Systems

Prof. Surojit Mookherjee
VGSoM, IIT KHARAGPUR

Week 03: ERP

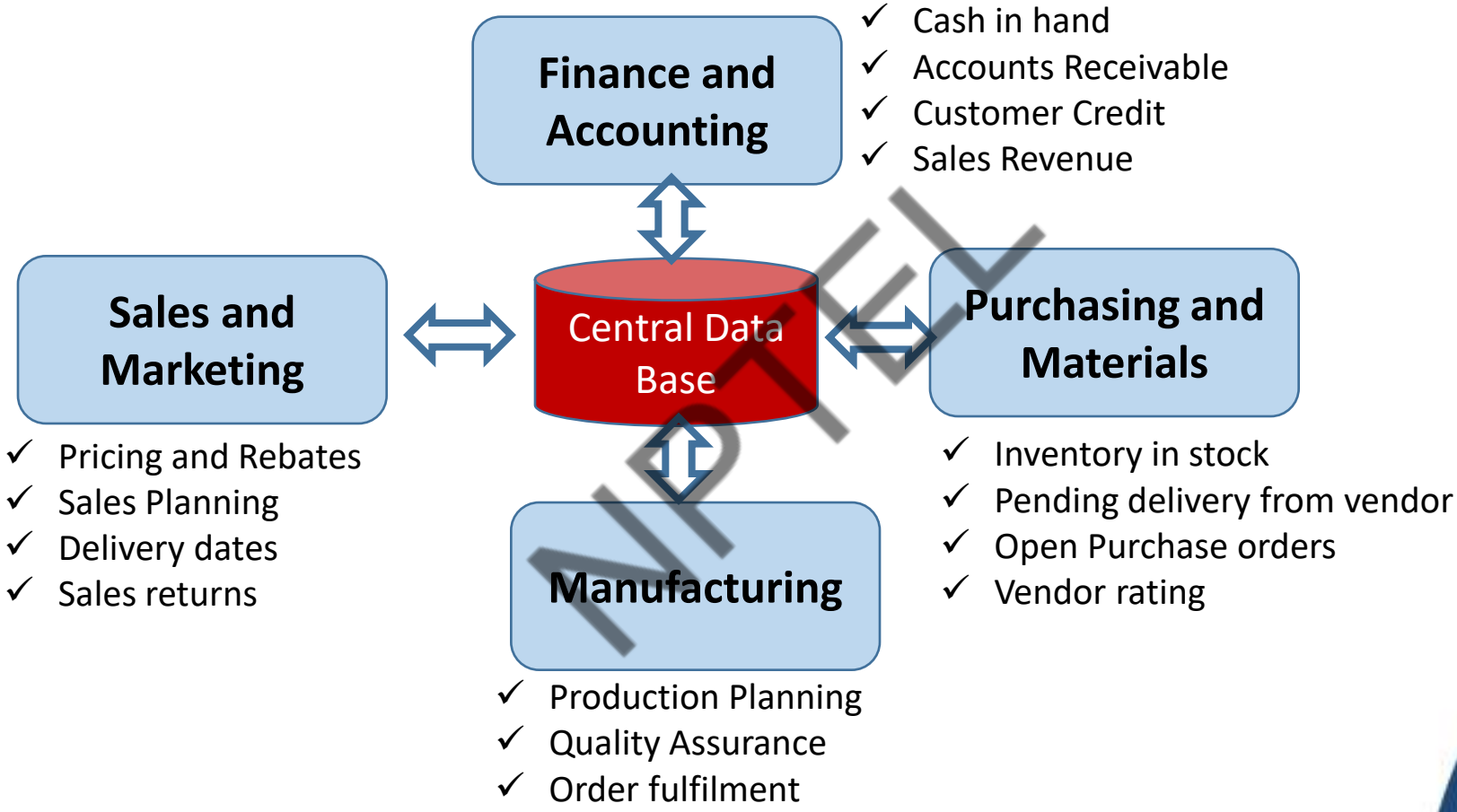
Lecture 03 : Advantages of ERP

Some of the impacts of ERP implementation

- **ERP will impact**
  - Communication and information domain
  - Leadership and authority
  - Business Processes
  - Best Practices
  - Company policies
  - Jobs and rewards
  - Company structure and culture
  - Skills and knowledge of employees

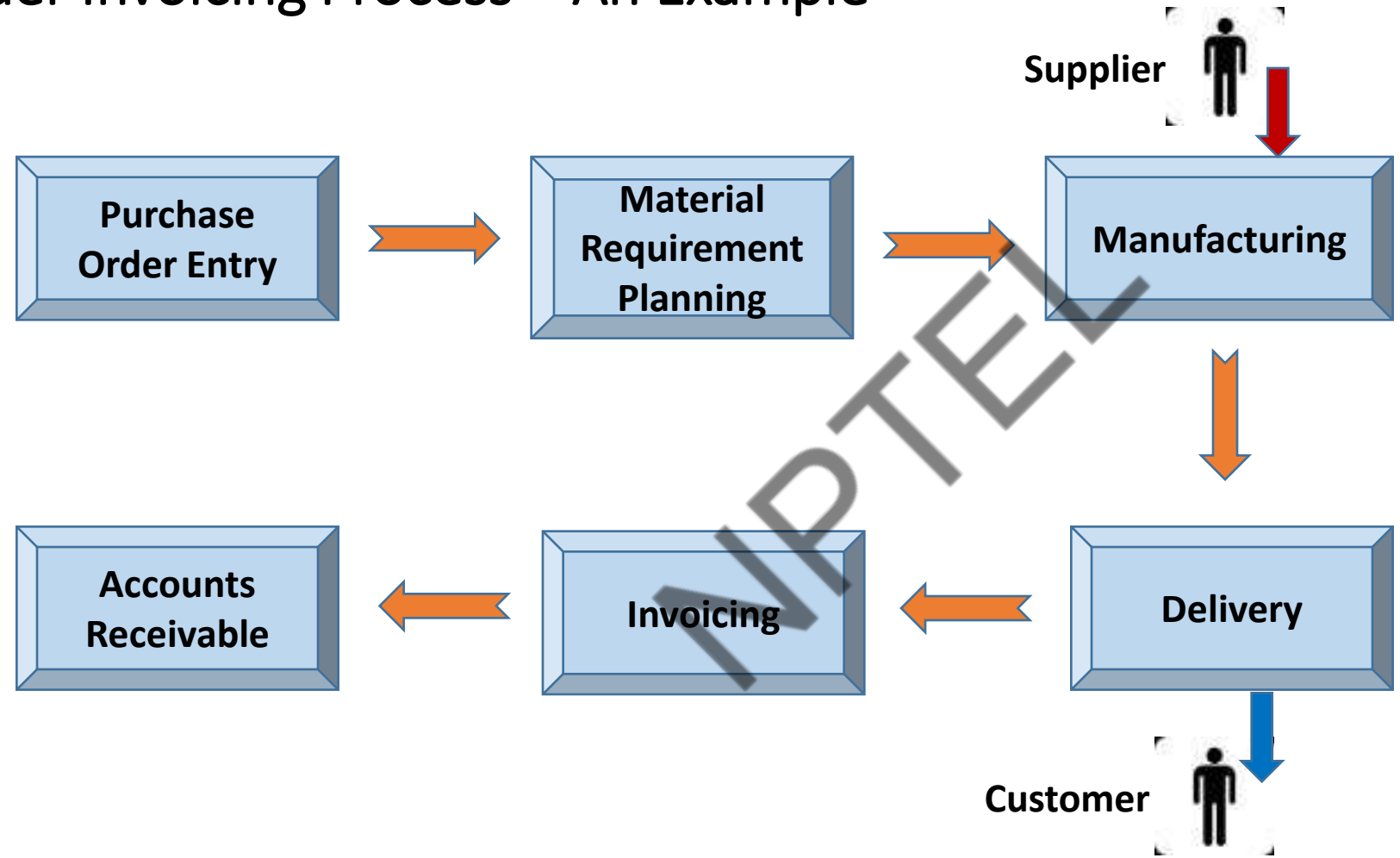
NPTEL

How the Organization uses ERP ?



NPTEL

Order Invoicing Process – An Example



NPTEL

Sales Order Enquiry Process Flow (Retail sales)

Request from a Customer

SD

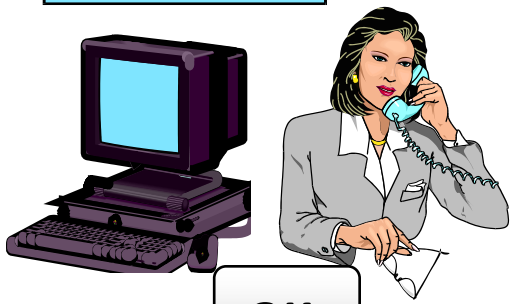
Credit Management

FI

Sold to party: C100

Item	Material	Qty.
10	6301	10
20	6412	10
30	1507	25

Order



OK

Availability Check – FG Stock

MM

Shop Floor status

PP

Goods in Transit

WM

Conforming Order- Price & Date

SD

Cash Forecast

FI

NPTEL

Impact of a Goods Receipt Transaction…..



A simple Goods Receipt transaction can trigger several activities , e.g.

- Stock and Value of inventory get updated
- Inspection Lot gets created to trigger a quality inspection
- Stock and Consumption account updated
- Purchase order history gets updated
- A transfer request is created to move the goods to the warehouse
- Vendor evaluation process gets initiated

NPTEL

Areas of performance improvement through ERP (1 of 4)

Revenue Generation

- ✓ Ability to Service new market segments.
- ✓ Increased asset utilization and capacity through better planning
- ✓ Redeploying workforce into higher value - added roles
- ✓ Broader Product Offering

NPTEL

Areas of performance improvement through ERP (2 of 4)

Cost Reduction

- ✓ Reduced working capital requirements
- ✓ Better Supplier Management
- ✓ Global centralized purchase using Strategic Sourcing with leveraged procurement spend.
- ✓ Shared Services in Purchase , HR and other general functions.

NPTEL

Areas of performance improvement through ERP (3 of 4)

Strategic Positioning

- ✓ Better information to run the business (Real Time and Decision Support Systems)
- ✓ Strong and flexible foundation to support future growth (seamless IT Platform)
- ✓ Improved supply chain integration (with Vendors and Customers' ERP systems)
- ✓ Standardized processes across divisions. This is very important for global / multi-national organizations (commonality of processes).
- ✓ Ability to more easily operate as a "single" business.
- ✓ Better employee productivity
- ✓ Better employee satisfaction through job-enrichment

NPTEL

Areas of performance improvement through ERP (4 of 4)

Improved Customer Satisfaction

- ✓ Reduced contract cycle times
- ✓ More accurate availability and delivery info
- ✓ Single face to the customer for sales, status and service
- ✓ Quicker response to contract and order changes
- ✓ ATP – Availability to Promise. Products can be booked for specific customers

NPTEL

REFERENCES

- **Enterprise Resource Planning : Rajesh Ray**
- **Enterprise Resource Planning Systems : O'Leary**

NPTEL



NPTEL

THANK YOU !



NPTEL ONLINE CERTIFICATION COURSES

NPTEL

Management Information Systems

Prof. Surojit Mookherjee
VGSoM, IIT KHARAGPUR

Week 03: ERP

Lecture 04 : Risks and Challenges of ERP Projects

ERP Project Management - Challenges and Best Practices

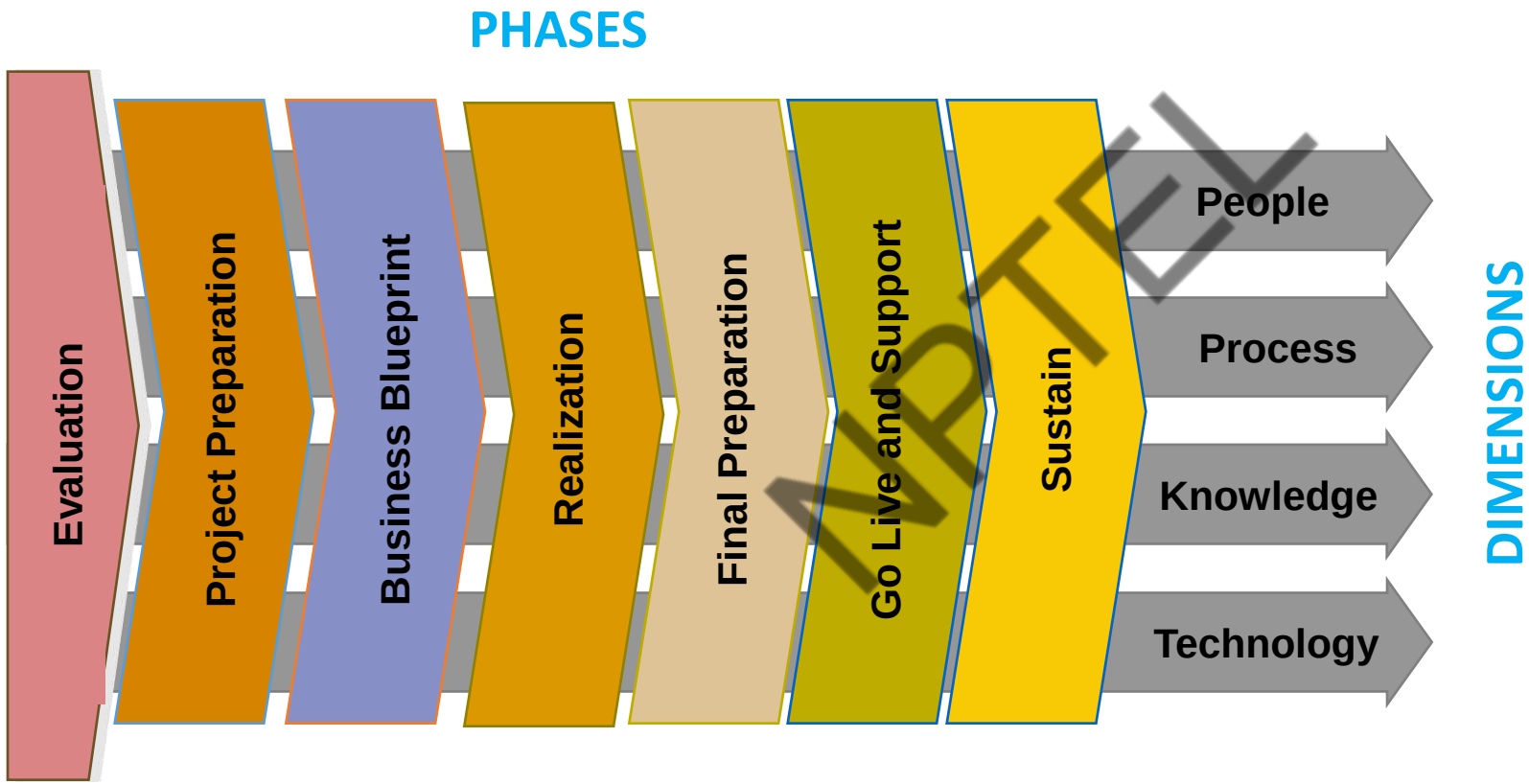
NPTEL

Define and manage the scope of the ERP Project

1. Process Scope
2. Functionality Scope
3. Organizational Scope
4. Technical Enhancement Scope
5. Technical Infrastructure Scope
6. Interface Scope
7. Data Conversion Scope
8. Reports and Forms Scope

NPTEL

ERP Implementation Method



NPTEL

Blueprint (Design) Phase – Best Practices

- ✓ It is critical to have overall end to end picture of your future state business processes. To-Be Scenario of the Organization.
- ✓ Business requirements should be as detailed as possible such that the business SMEs and ERP consultants can understand your exact business needs
- ✓ Only experts in ERP product knowledge should finalize the fit-gap analysis as Custom development is one of the major Cost and Time factor for the completion of the project.
- ✓ Custom development should be kept to the minimum to reduce cost and have high system performance.

NPTEL

Deployment & Go Live – Best Practices

- ✓ System Preparedness – Completeness of all System Integration Testing
- ✓ Completion of all End – User training
- ✓ System performance Testing (stress testing) to ensure better performance after go-live
- ✓ Checklist to verify internal and external constituent readiness :
  - Support Help desk in place
  - All IT equipment fully tested
  - Service Level Agreements in place
  - All stakeholders duly trained and made aware

NPTEL

Continuous Engagement with Stakeholders and Project Leaders (Do's and Don'ts) ……

- **Work closely with Implementer's senior executive (partner / delivery leader) to discuss avenues of improvements in project execution,** faster project delivery, resource challenges, solution optimization and scoping issues.

- **Review risks associated with project delivery**, budget overrun, solution quality and Q-gates with vendor executives and propose mitigation plan.

- **Oversee and advice the periodic (daily or weekly) project leadership meeting**. Ensure accurate project progress is reported and discussed. Advice on options to mitigate any known project risks or issues.

NPTEL

Continuous Engagement with Stakeholders and Project Leaders (Do's and Don'ts)

- **Manage disagreements or conflicts between business teams, internal leaderships and systems integrator**. Serve as an independent unbiased advisor to resolve these conflicts and keep the entire project team motivated throughout.

- **Evaluate and qualify project progress report for preparing periodic "steering committee presentation".** Provide expert opinion and advice to the Steering Committee when needed.

- **Ensure that all sign-off and deliverable approval procedures are followed throughout the project organization**.

NPTEL

ERP Challenges…(1of 3)

Project Implementation Challenges

- ✓ **Scope Changes**
- ✓ **Huge Budget**
- ✓ **Long Timeline**
- ✓ **Process redesign challenges**
- ✓ **Unrealistic expectation**
- ✓ **Complex Interfaces**

NPTEL

ERP Challenges…(2 of 3)

Support / Maintenance Challenges

- ✓ **Getting right people**
- ✓ **Managing multiple vendors**
- ✓ **Managing regular upgrades**
- ✓ **Technology obsolescence**
- ✓ **Large application portfolio**
- ✓ **Managing transition**
- ✓ **Realizing benefits**

NPTEL

ERP Challenges…(3 of 3)

Managing People

- ✓ **Managing Change**
- ✓ **Managing large project team**
- ✓ **Managing employee retention and relocation**
- ✓ **Top management support**

NPTEL

Reasons for ERP project risk

Organizational Politics

Multiple Projects

Inexperienced Core Team

Computer Literacy of Employees



Aggressive Time Frame

Frequent change in scope

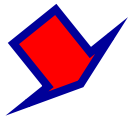
Connectivity in remote location

Lack of Top Mgmt Support

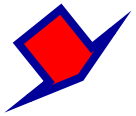
NPTEL

Five Step Project Risk Management Approach

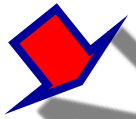
Find **potential risk elements** of the project



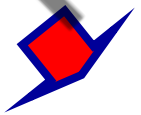
Analyze **Impact** of each of these elements



Assess the **Probability** of occurrence of these risk elements



Prioritize the risk



Develop a **Risk Mitigation Plan** for top risk elements

NPTEL

Some Critical Success factors ……

- **Ownership & Support from top management (Project Sponsor & Steering Committee)**
- **Commitment to the Product (e.g.SAP)**
- **COMMITMENT TO CHANGE**
- **Quick decision making and avoid lengthy discussions on trivial issues.**
- **Avoid frequent changes to requirements**
- **Strict adherence to the project plan**

NPTEL

ERP – Recent Trends

- ✓ ERP vendors are moving into Enterprise Applications space (CRM,SRM,PLM,SCM etc.)
- ✓ SME clients are everybody's focus
- ✓ ERP vendors are going vertical i.e. building more industry specific extensions
- ✓ New ERP deployment models are emerging in the form of SAAS, Cloud Computing etc.
- ✓ Newer types of ERP projects are emerging – Migration, Consolidation, Harmonization etc.

NPTEL

REFERENCES

- **Enterprise Resource Planning : Rajesh Ray**
- **Enterprise Resource Planning Systems : O'Leary**

NPTEL



NPTEL

THANK YOU !



NPTEL ONLINE CERTIFICATION COURSES

NPTEL

Management Information Systems

Prof. Surojit Mookherjee
VGSoM, IIT KHARAGPUR

Week 03: ERP

Lecture 05 : Class Discussion and Conclusion

Conclusion

- ERP software can be an expensive and time-consuming IT investment
- Implementing an ERP software is full of risks from a project management perspective.
- ERP applications facilitate the standardization and acceleration of business processes and contribute to the visibility necessary to run business well and grow revenue and profits.
- ERP applications form the information backbone of a business.

- ***ERP Implementation is NOT an IT program but a Business Program***

NPTEL

Class Discussion -

1. When , Manufacturing happens , how do Purchase , Sales and Finance get impacted ?
2. CTO (Chief Technical Officer) > CIO (Chief Information Officer) > CDO (Chief Digital Officer) : What has made these transitions / change in title ?
3. When a running ERP system fails , how large would be the loss for the organization ?

NPTEL

REFERENCES

- **Enterprise Resource Planning : Rajesh Ray**
- **Enterprise Resource Planning Systems : O'Leary**

NPTEL



NPTEL

THANK YOU !



NPTEL ONLINE CERTIFICATION COURSES

NPTEL

Management Information Systems

Prof. Surojit Mookherjee
VGSoM, IIT KHARAGPUR

Week 04: Customer Relationship Management

Lecture 01 :

Customer Relationship Management ……

- ✓ Customer relationship management (CRM) is a strategy for managing a company's interactions with it's customers and sales prospects.
- ✓ Often CRM is equated with automating business processes related to sales, marketing and customer service – CRM is not just about technology
- ✓ It involves strategy and redesign of business processes with the overall goal to find, attract, and win new customers, retaining those customers and finally getting feedback from them to better design company's product and customer facing processes.
- ✓ CRM can also reduce the costs of marketing, sales and customer service.

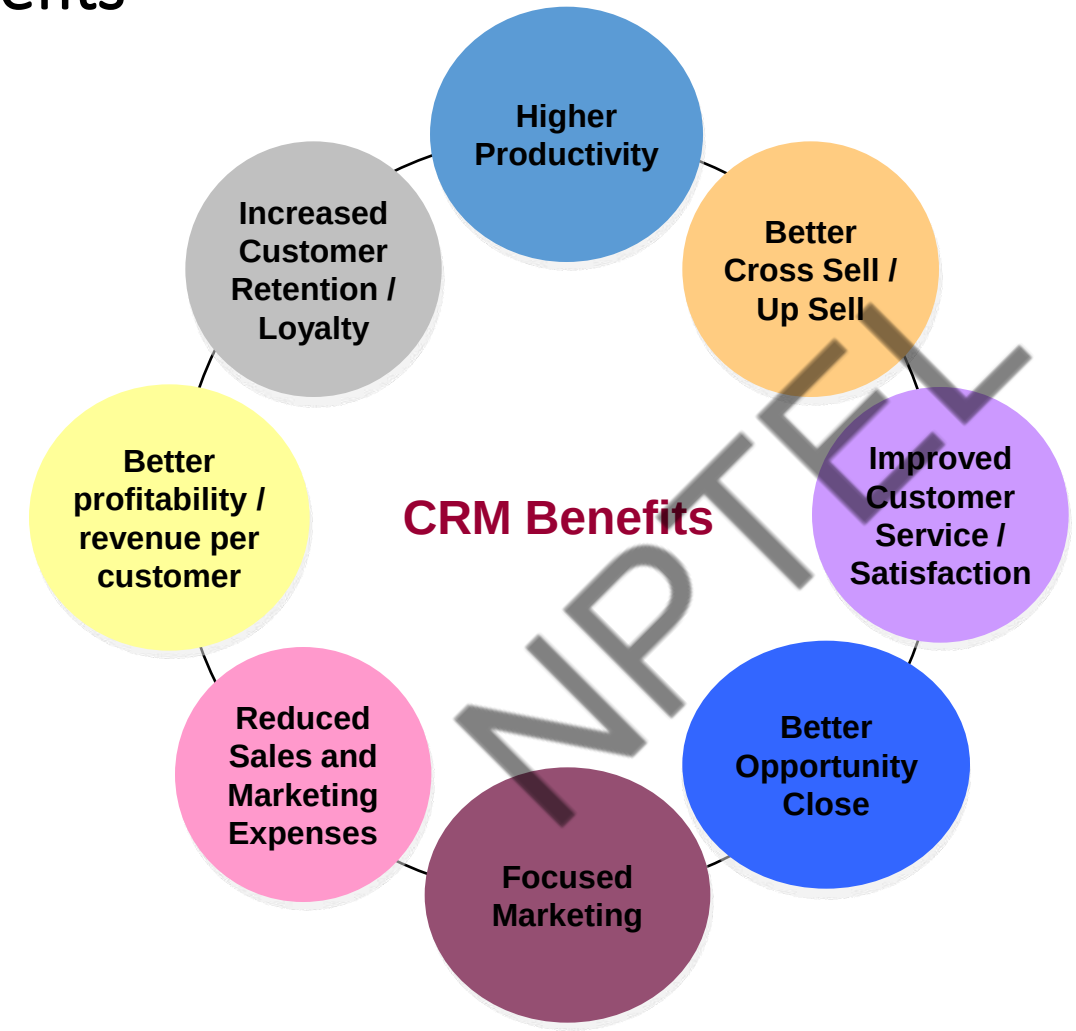
NPTEL

Customer Relationship Management

- ✓ CRM uses people, processes, and technology to gain insight into the behavior of customers
- ✓ CRM can bring together information from all data sources within an organization (and from outside the organization if needed) to give one holistic view of customer in real time.
- ✓ CRM allows customer facing employees from sales, customer support, and marketing to make informed decisions on everything from cross selling and upselling opportunities to target marketing strategies to competitive positioning tactics.

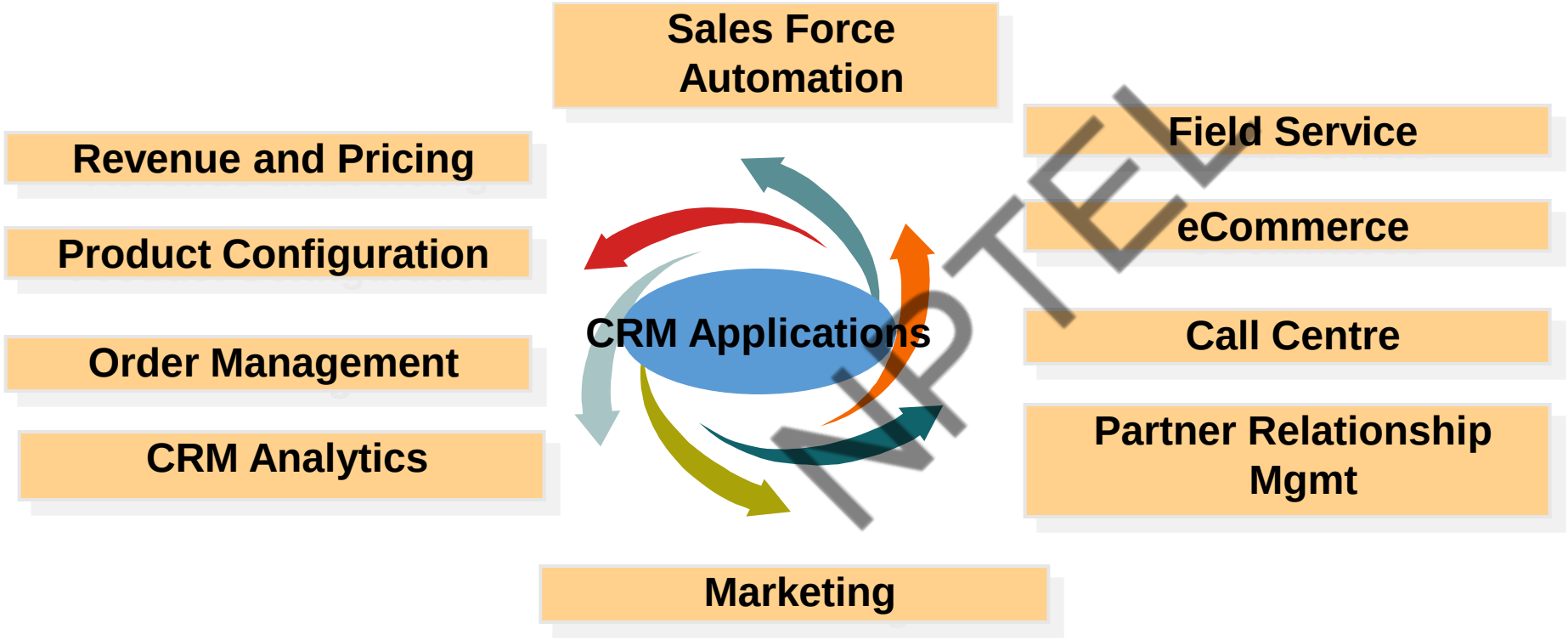
NPTEL

CRM Benefits



NPTEL

CRM Application Areas



NPTEL

Sales Force Automation (SFA)….

Typical areas of sales force automation applications are:

- ✓ Contact management
- ✓ Lead management / Opportunity management
- ✓ Sales forecasting and pipeline
- ✓ Sales performance management
- ✓ Territory management
- ✓ Quote generation

NPTEL

Sales Force Automation (SFA)

SFA applications help sales force in their daily jobs and aimed at increasing their productivity.

With these applications sales force need not remember or store contact information of thousands of company's existing customers and new prospects.

SFA applications help them to manage new leads and sales opportunities.

These applications helps sales force in managing sales in their territory and tracking sales performance, based on which they are provided incentive / compensation.

NPTEL

Field Service Application

As companies need to manage a large number of mobile technicians distributed across different customer sites, Mobile solutions are increasingly becoming an important component of field service applications.

Typical capabilities of field service applications are:

- ✓ Filed service management (Utility services like Power distribution , civic amenities)
- ✓ Scheduling workforce
- ✓ Managing spares / service parts
- ✓ Mobile Solutions capability

NPTEL

Field Service Application

These applications typically helps in different after sales service functions i.e. helps managing installation, service, or repairs of systems or equipment

These solutions also support scheduling workforce i.e. which worker should go to which customer site and attend which type of equipment / problem based on worker's availability and skill.

Some of these after sales service activities need some kind of spares, managing reverse logistics and spare parts management and these are under the domain of these applications.

NPTEL

E Commerce Application …….

Typical capabilities of E Commerce applications are:

- ✓ eMarketing
- ✓ Campaigns
- ✓ Personalization
- ✓ Online ordering
- ✓ Online product configuration
- ✓ Online auctions
- ✓ Online exchange and return
- ✓ Online Billing and Payment

NPTEL

E Commerce Application

These applications help companies to buy, sell, and do other transactions over the Web.

Companies use these applications for variety of purpose like product search / browse, buying through shopping cart, product promotion, personalization, cross-sell and up-sell, product configurations, registries, multichannel ordering, customer self service etc.

Today e-commerce is becoming even more powerful with Web 2.0. It has even spread to small towns and rural areas.

Easy availability of Smart phones and Data services have played a great riole in spreading the popularity of e-commerce

NPTEL

Call Centre / Contact Centre Application ….

Typical capabilities of call centre applications are:

- ✓ Predefined escalation route
- ✓ Previous records of the caller
- ✓ Support in different time zones in different languages
- ✓ Managing call assignments
- ✓ Knowledge Base
- ✓ Self service capabilities
- ✓ Workflow

NPTEL

Call Centre / Contact Centre Application

These solutions provide customer support by answering different types of customer queries / calls and routing the calls to the most competent person who can handle it.

Call centre uses a variety of technologies like: local area networks (LAN), automatic call distributors (ACD), computer technology integration (CTI), Web integration, interactive voice response, voice logging and messaging, IVR/speech portals etc. Call centers today are capable of supporting complex customer interactions, such as scheduling service requests, providing technical support, and handling financial transactions.

NPTEL

Partner Relationship Mgmt (PRM) Application…..

Typical capabilities of Partner Relationship Management applications are:

- Partner addition
- Distributing leads
- Partner profiling
- Managing different partner programs
- Partner opportunity tracking
- Special partner discounts and approvals
- Partner service
- Managing channel inventory / partner pipeline:

NPTEL

Marketing Mgmt Application…..

Typical capabilities of Marketing Management applications are:

- Campaign management
- Customer interaction management
- Marketing planning
- Marketing resource and asset management
- Opportunity / Lead management
- Loyalty marketing

NPTEL

Marketing Mgmt Application

These applications are used by companies to manage their end-to-end process from gathering and analyzing customer data across Web sites and other channels, to planning, budgeting, to executing customer communications and measuring results / effectiveness.

It helps the enterprise identify and target its best customers and generate qualified leads for the sales team.

Marketing automation also encompasses capabilities for managing customer loyalty, lists, collateral, and internal marketing resources.

These tools support an integrated approach to marketing strategy, development, delivery, and measurement across the marketing mix.

NPTEL

Revenue and Pricing Mgmt Application

Typical capabilities of Revenue and Pricing Management applications are:

- Trade promotion management
- Promotions
- Pricing and promotion optimization

NPTEL

These applications help companies to optimize and manage prices throughout the product life cycle, including: initial pricing, promotional, and markdown or clearance pricing.

Product Configuration Application

These tools are popular for complex products and services.

These applications help customers to configure a product as per their desired specification on the web (e.g Dell Desktops)

Based on this configuration, company's sales staff can quote and generate a price proposal quickly.

These applications can automate sales and product configuration, proposal configuration, cost estimation and pricing and can compress the entire lead-to-order process.

NPTEL

CRM Analytics

There can be different types of CRM analytics like:

- ✓ **Sales analytics:** This allow companies monitor and understand customer actions and preferences, through dashboards that graphically display KPIs.
- ✓ **Marketing analytics** Marketing applications generally come with predictive analytics to improve customer segmentation and targeting, and tools to measure the effectiveness (ROI) of a marketing / sales promotion campaign.
- ✓ **Customer service analytics** are increasing in popularity as companies demand greater visibility into the performance of call centers and other support channels, in order to correct problems before they affect customer satisfaction levels.

NPTEL

Leading CRM Products

- ✓ SAP CRM
- ✓ Oracle Siebel
- ✓ Salesforce.com
- ✓ Microsoft
- ✓ Amdocs

NPTEL

REFERENCES

- **Enterprise Resource Planning, Rajesh Ray, Tata McGraw-Hill – Chapter 28**
- **Management Information Systems: Managing the Digital Firm - Kenneth C. Laudon & Jane P. Laudon**

NPTEL



NPTEL

THANK YOU !



NPTEL ONLINE CERTIFICATION COURSES

NPTEL

Management Information Systems

Prof. Surojit Mookherjee
VGSoM, IIT KHARAGPUR

Week 04: Supply Chain Management

Lecture 02 : SCM – Part 1

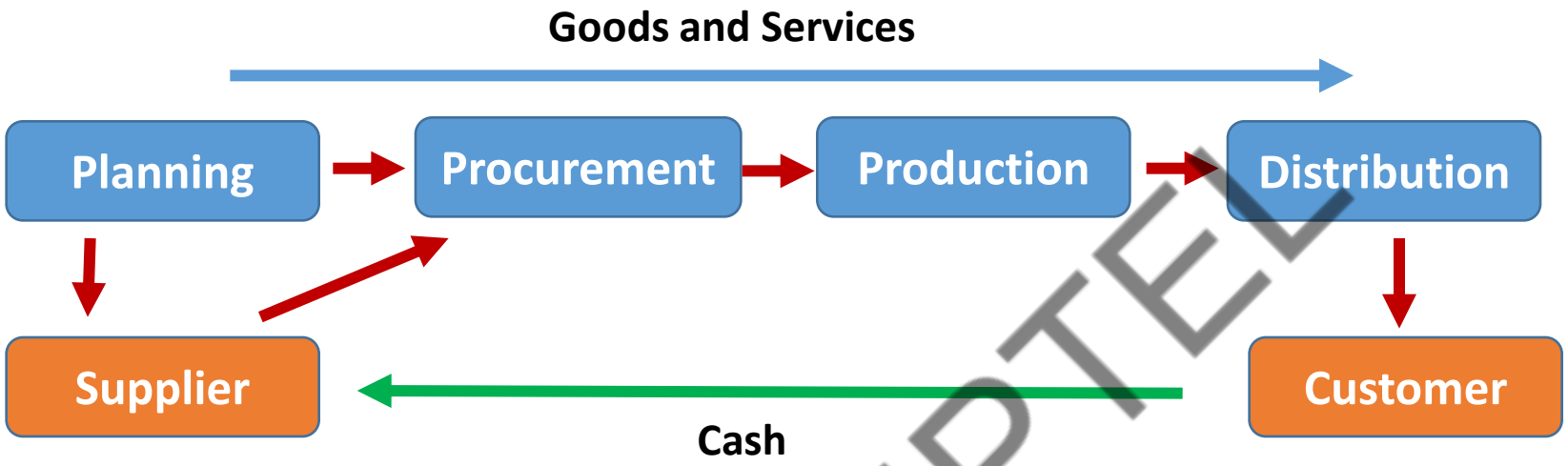
What is Supply Chain ?

Supply Chain Management is a complete process of changing the raw materials into goods and delivering them to the end-user.

- It is a process of managing the flow of goods/ services starting from the supply of raw materials to the manufacturer until the delivery of the final good to the consumer. It involves material flow and information flow.
- SCM involves the supply and storage of raw materials, production of goods, transporting and delivering it to the end-user.
- Supply Chain is a network of individuals, resources, organizations, processes, technologies connected together to produce a product/service.

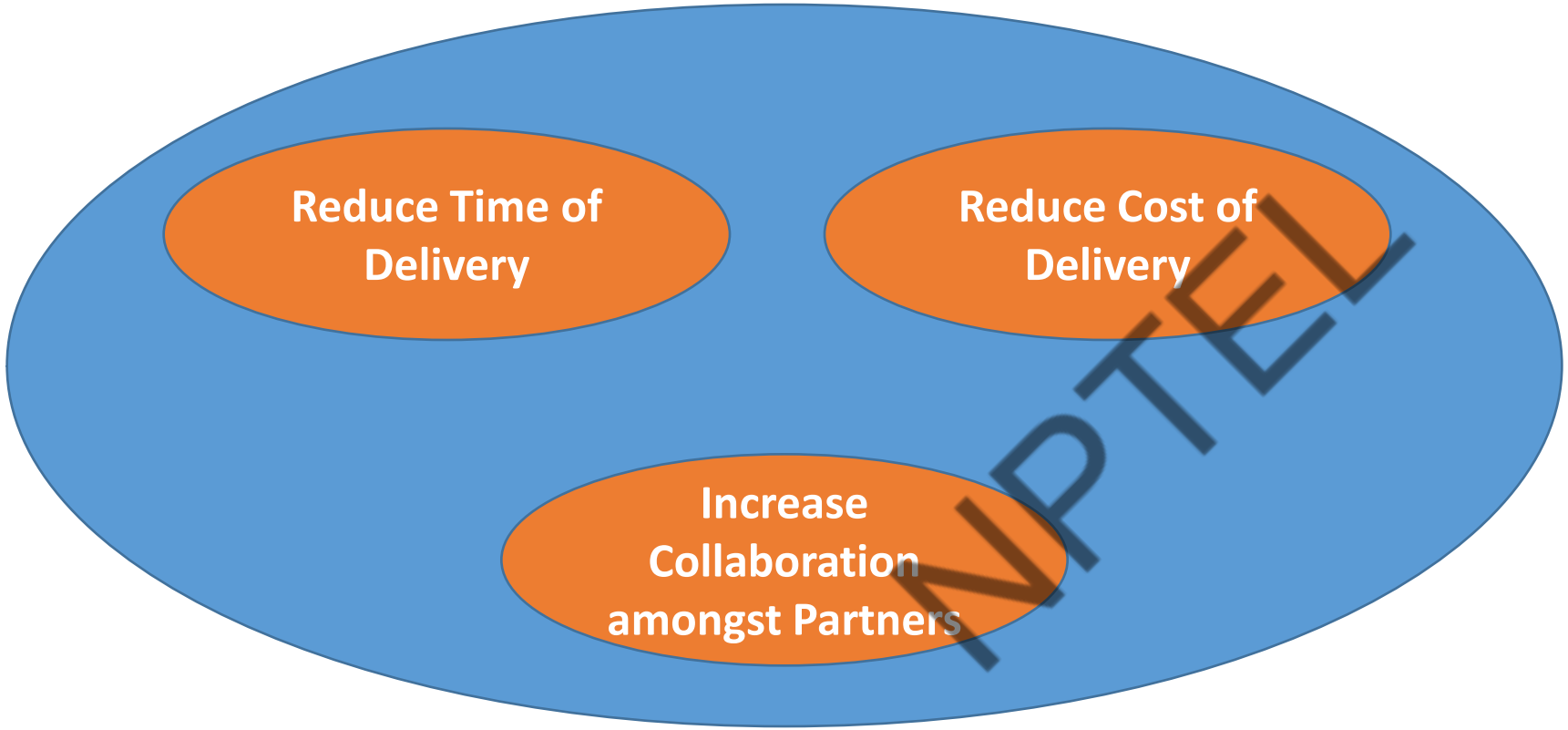
NPTEL

The Supply Chain Management Process



NPTEL

3 Key Features of SCM …. What it tries to achieve ?



NPTEL

What is Supply Chain Management

– **Manage a firm's relationships with Suppliers / Vendors and Customers**

– **Share information about :**

- ❖ Orders,
- ❖ Production,
- ❖ Inventory levels,
- ❖ Delivery of products and services

– **Goal:**

- **Right amount of products to destination with least amount of time and lowest cost**

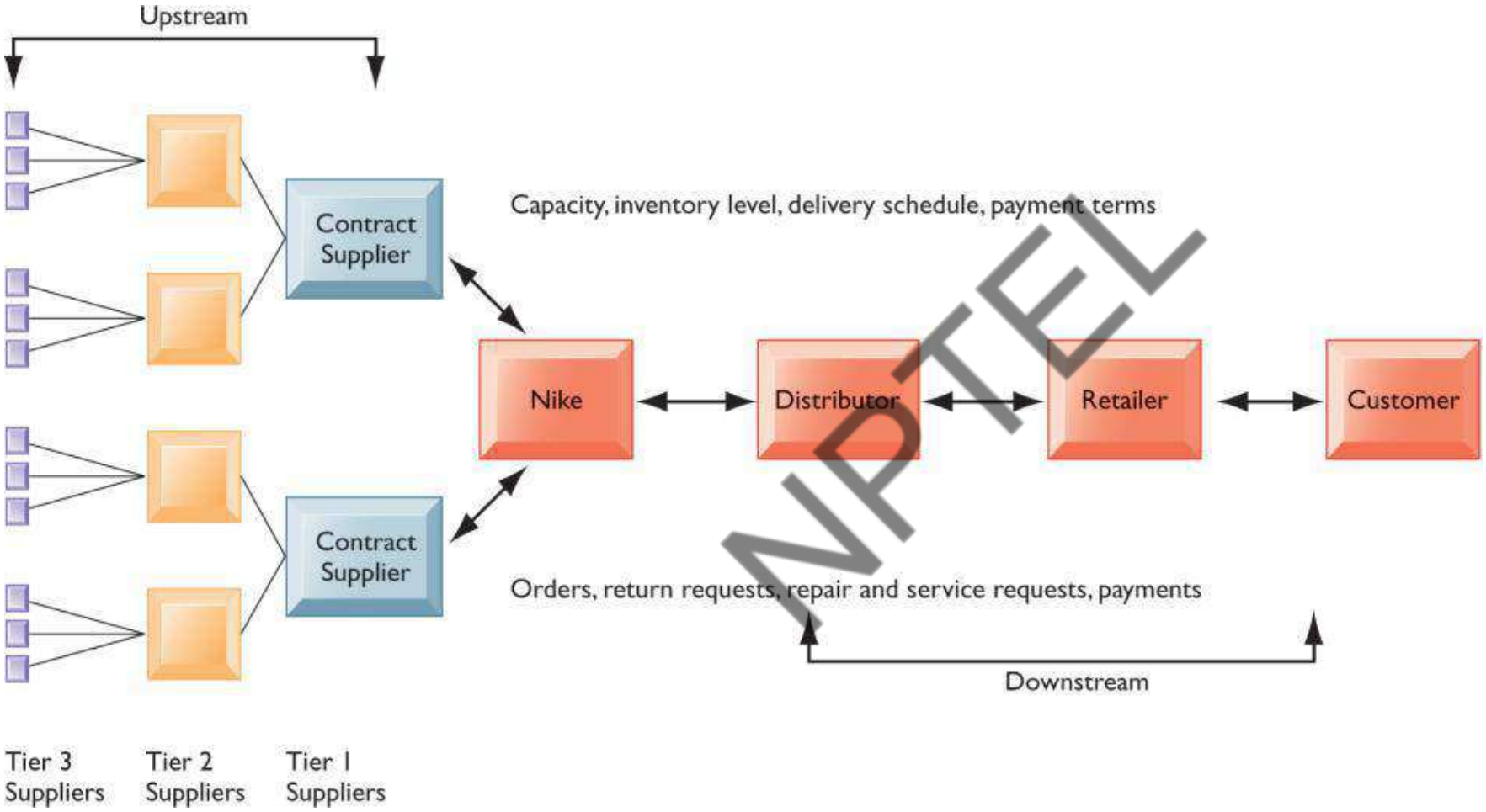
NPTEL

Supply Chain is …..

- **Network** of organizations and processes for:
  - Procuring materials, transforming them into products, and distributing the products
- **Upstream** supply chain
  - Firm's suppliers, suppliers' suppliers, processes for managing relationships with them
- **Downstream** supply chain
  - Organizations and processes responsible for delivering products to customers
- **Internal** supply chain
  - Inside the factory between stores , shop-floor and warehouse.

NPTEL

Nike's Supply Chain



NPTEL

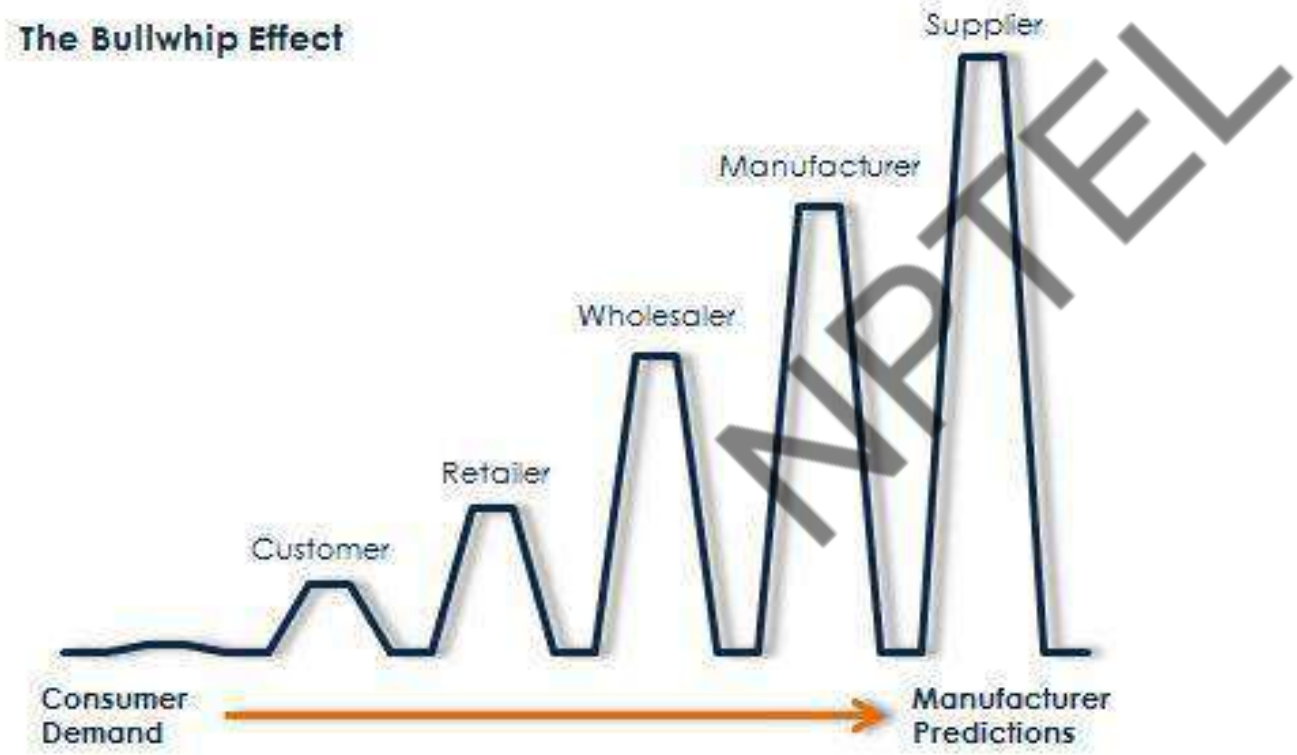
Supply Chain Management - Features

- Inefficiencies cut into a company's operating costs
  - Can waste up to 25 percent of operating expenses
- Just-in-time strategy
  - Components arrive as they are needed
  - Finished goods shipped after leaving assembly line
- Safety stock: Buffer for lack of flexibility in supply chain
- Bullwhip effect
  - Information about product demand gets distorted as it passes from one entity to next across supply chain

NPTEL

THE BULL-WHIP EFFECT

The bullwhip effect is the uncertainty caused from distorted information flowing up and down the supply chain.



NPTEL

Causes of the Bull-Whip effect

- Demand forecast updating
- Order batching
- Price fluctuation
- Rationing and shortage gaming
- Un-forecasted sales promotions
- Customers turning back sales orders

NPTEL

Impacts of the Bull-Whip effect

- Excess inventories
- Unnecessary costs
- Tense supplier relationships
- Stock outs
- Increased wastages
- Changes in buying patterns

NPTEL

Methods of coping with the Bull-Whip effect

- Improve communication along the supply chain
- Improve sources of forecast data
- Share Information
- Establish a demand-driven supply chain which reacts to actual customer orders – Pull instead of Push system
- Break order batches into smaller lots
- Stabilize prices
- Eliminate gaming in shortage situations

NPTEL

Supply Chain Management Systems

- Supply chain planning systems
  - Model To-Be supply chain based on Industry best practices
  - Enable demand planning through ERP
  - Optimize sourcing, manufacturing plans using Pull system
  - Establish inventory levels (e.g. Re-Order point planning)
  - Identify transportation modes
- Supply chain execution systems
  - Manage flow of products through distribution centers and warehouses

NPTEL

Global Supply Chains and the Internet

- Global supply chain issues
  - Greater geographical distances, time differences (e.g. most apparel sourced from South East Asian countries)
  - Participants from different countries
    - Different performance standards
    - Different legal requirements
- Internet helps manage global complexities
  - Warehouse management
  - Transportation management
  - Logistics
  - Outsourcing

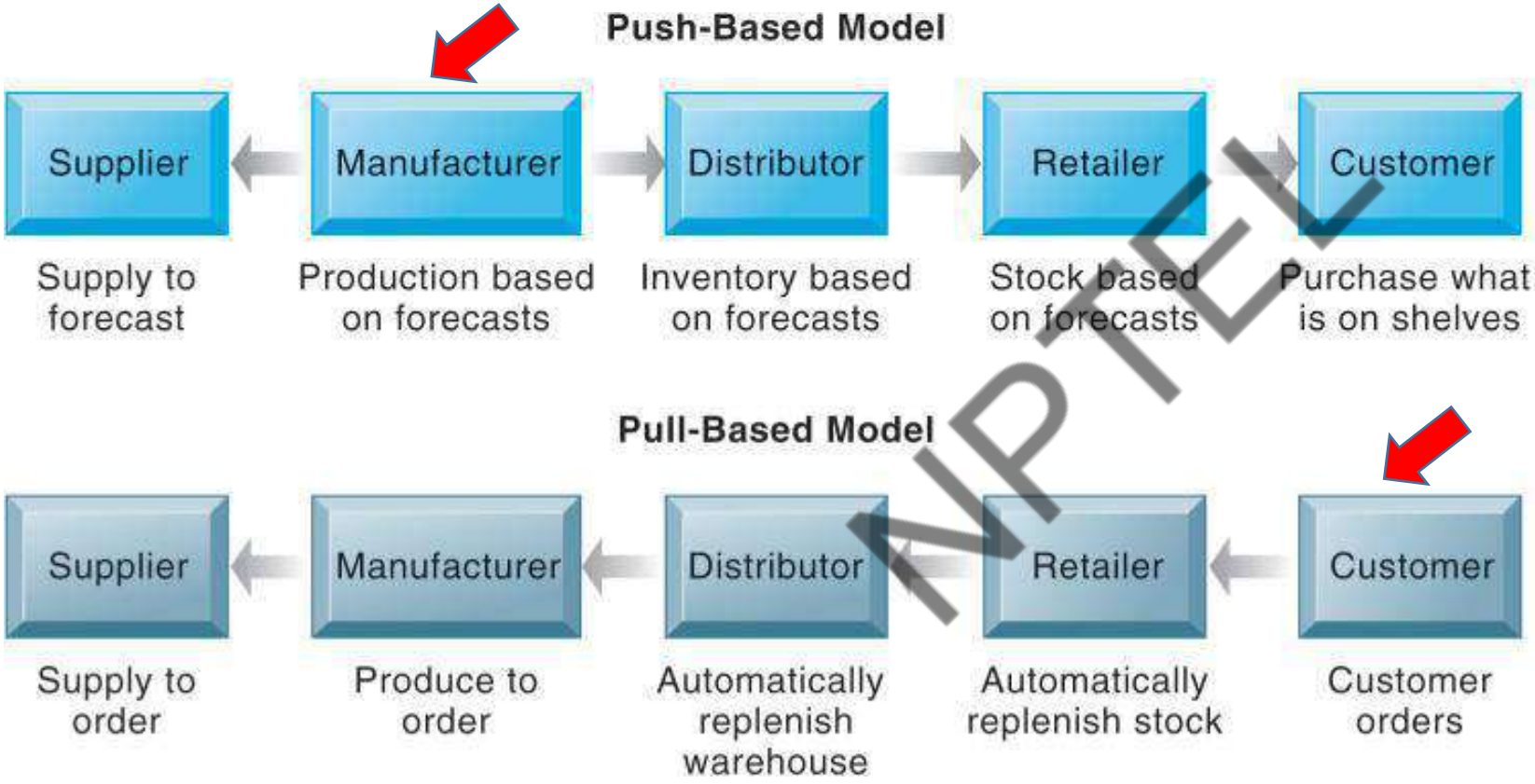
NPTEL

Demand-Driven Supply Chains: From Push to Pull Manufacturing and Efficient Customer Response

- Push-based model (build-to-stock)
  - Earlier SCM systems
  - Schedules based on best guesses of demand
- Pull-based model (demand-driven)
  - Web-based
  - Customer orders trigger events in supply chain (Dell Computers)
- Internet enables move from sequential supply chains to concurrent supply chains
  - Complex networks of suppliers can adjust immediately

NPTEL

Push- Versus Pull-Based Supply Chain Models



Car Manufacturers – Is it Push or Pull based ?

NPTEL

REFERENCES

- **Enterprise Resource Planning, Rajesh Ray, Tata McGraw-Hill – Chapter 25**
- **Management Information Systems: Managing the Digital Firm - Kenneth C. Laudon & Jane P. Laudon**

NPTEL



NPTEL

THANK YOU !



NPTEL ONLINE CERTIFICATION COURSES

NPTEL

Management Information Systems

Prof. Surojit Mookherjee
VGSoM, IIT KHARAGPUR

Week 04: Supply Chain Management

Lecture 03 : SCM – Part 2

JUST IN TIME AND LEAN SYSTEMS

NPTEL

Just-In-Time (JIT)

- **JIT** can be defined as an integrated set of activities designed to achieve high- volume production using minimal inventories (raw materials, work in process, and finished goods)
- JIT involves the elimination of waste in production effort
- JIT involves the timing of production resources (i.e., parts arrive at the next workstation "just in time")

NPTEL

Creating the Lean Supply Chain

- What is Lean System ?
  - "A philosophy that seeks to shorten the time between the customer order and the shipment to customer by **eliminating waste**"
- Three primary elements of Lean :
  - Just-in-time Purchasing
  - Just-in-time Transportation
  - Just-in-time Production

"Just-in-time" is a management philosophy, not technique, and the philosophy is simple - ***inventory is defined to be waste***

NPTEL

The Toyota Production System

- **Based on two philosophies:**
  - **Elimination of waste**
  - **Respect for people**

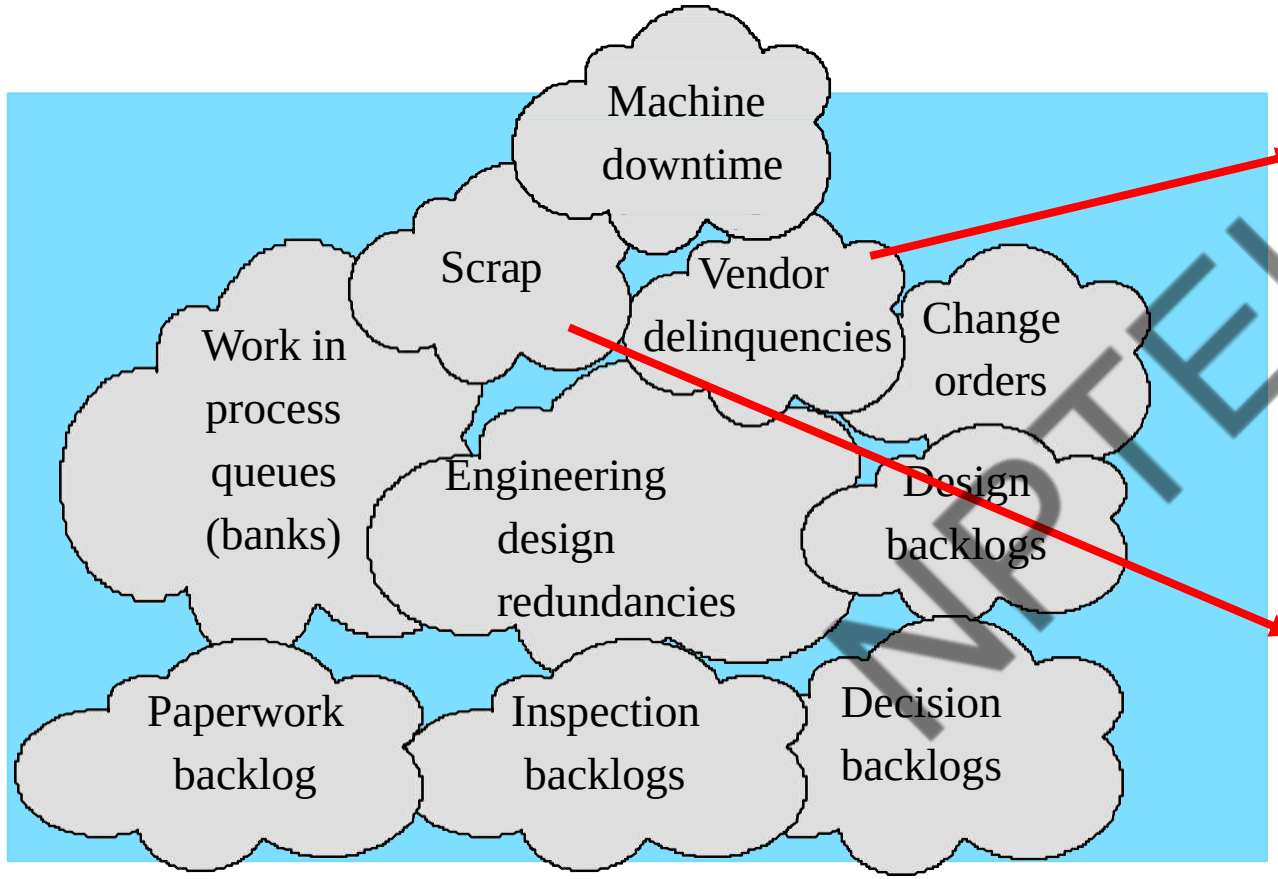
NPTEL

Waste in Operations

1. Waste from overproduction
2. Waste of waiting time
3. Transportation waste
4. Inventory waste
5. Processing waste
6. Waste of motion
7. Waste from product defects

NPTEL

Minimizing Waste: Inventory Hides Problems



Example: By identifying defective items from a vendor early in the production process the downstream work is saved

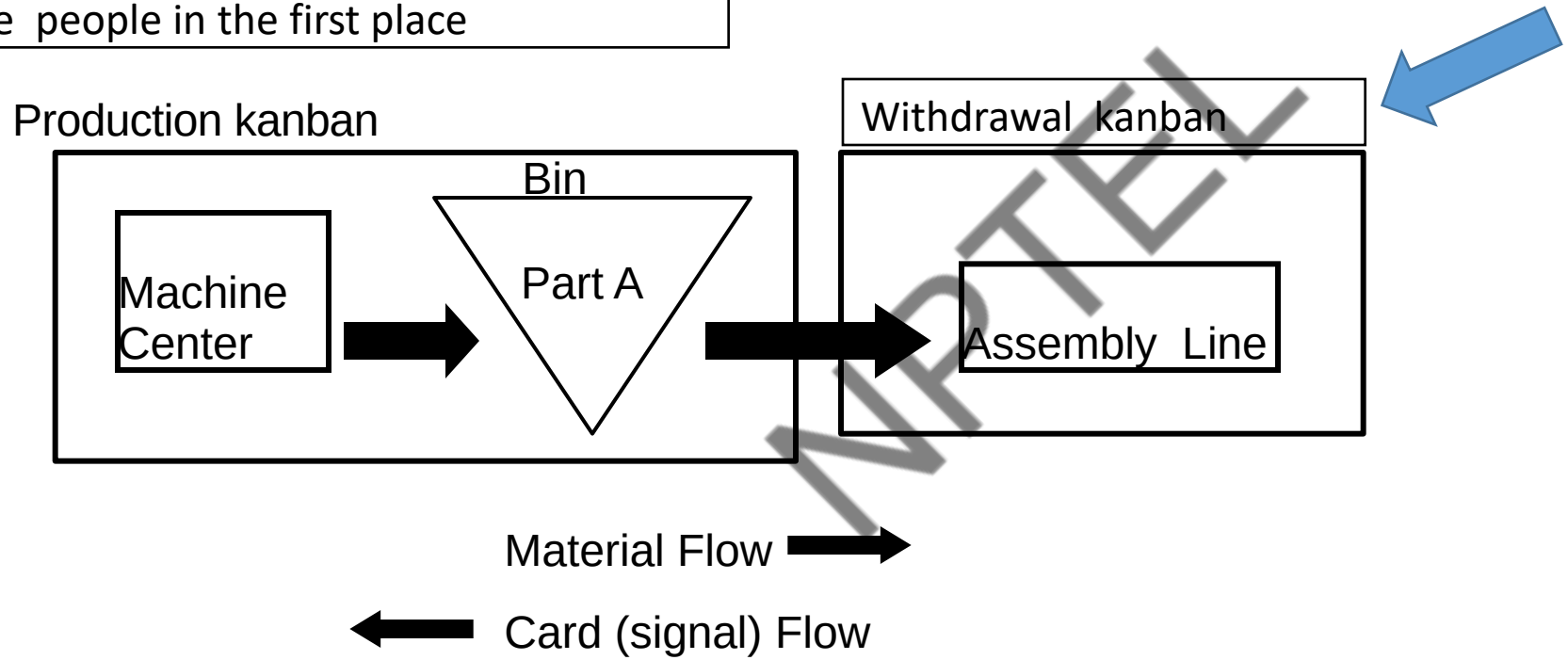
Example: By identifying defective work by employees upstream, the downstream work is saved.

NPTEL

Minimizing Waste: Kanban Production Control Systems

Once the Production kanban is received, the Machine Center produces a unit to replace the one taken by the Assembly Line people in the first place

The process begins by the Assembly Line people pulling Part A from Storage



NPTEL

Determining the Number of Kanban's Needed

- Setting up a kanban system requires determining the number of kanban cards (or containers) needed
- Each container represents the minimum production lot size
- An accurate estimate of the lead time required to produce a container is key to determining how many Kanban's are required

NPTEL

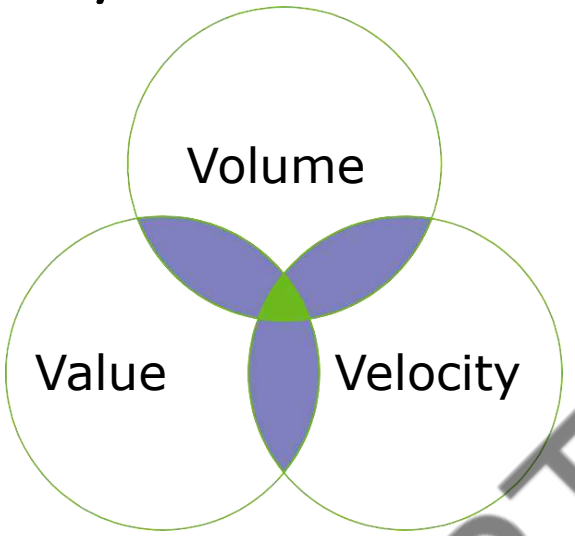
Managing Inventory Investment

Volume pertains to the amount of physical inventory a firm owns at any given time across the supply chain

Key Question: How much and what types of inventory do we own?

Key Measures: Total units, total pounds

Activities Affecting Volume: Improved forecasting techniques; supplier-provided consignment inventory



Value pertains to the unit cost and total value of inventory;

Key Measures: Total dollars; period-by-period unit value changes; ratio of sales to working capital?

Activities Affecting Volume: Product simplification/standardization

Velocity pertains to how quickly raw material/WIP become finished goods that accepted and paid by customer;

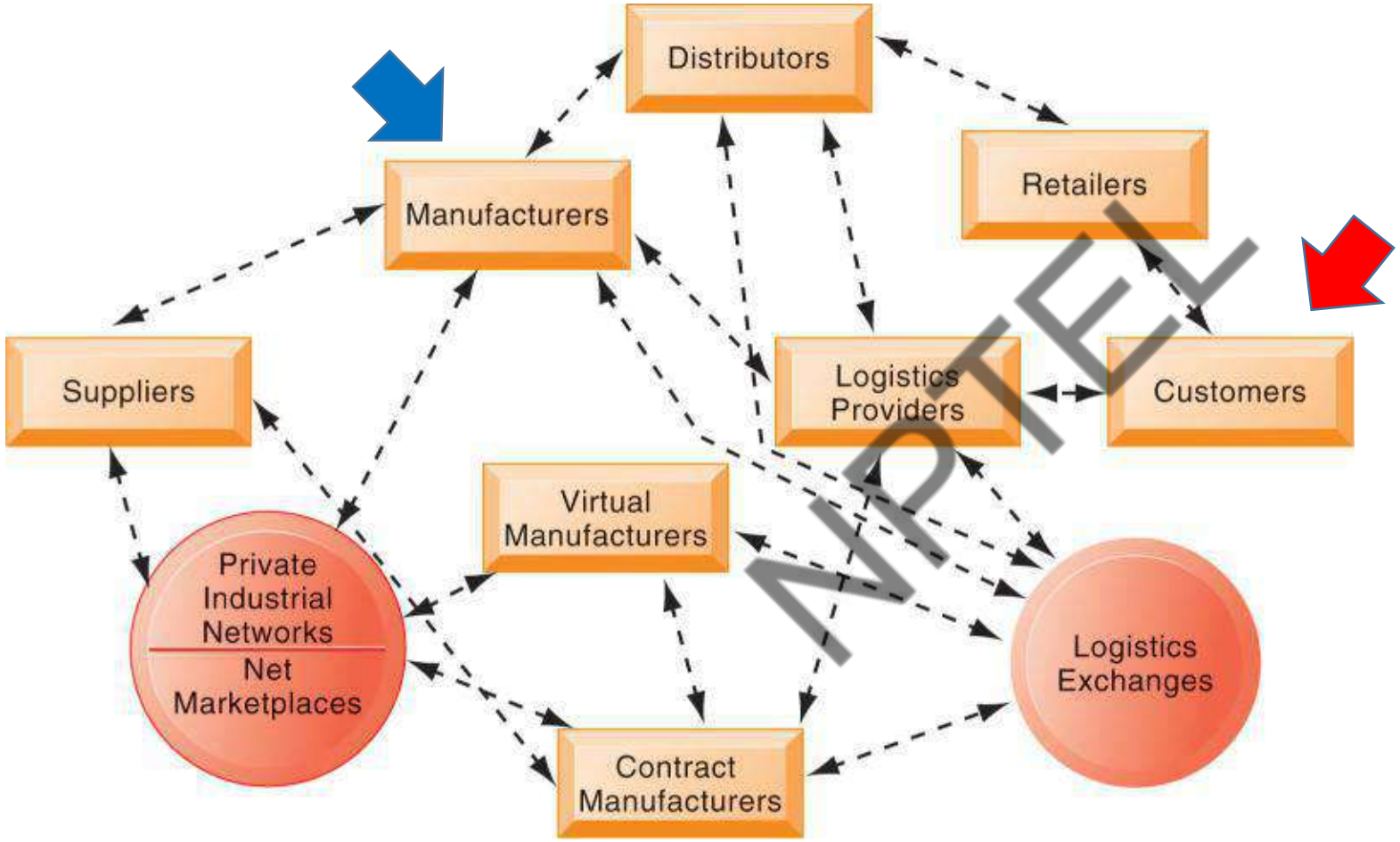
Key Question: How fast do me move inventory to the customer?

Key Measures: Inventory turns

Activities Affecting Volume: Made-to-order production, Lean supply chain practices;

NPTEL

The Emerging Internet-Driven Supply Chain



NPTEL

Business Value of Supply Chain Management Systems

- Match supply to demand
- Reduce inventory levels
- Improve delivery service
- Speed product time to market
- Use assets more effectively
  (Total supply chain costs can be 75 percent of operating costs)
- Increase sales

NPTEL

SCM – Challenges and New Technologies

NPTEL

Challenges in SCM today

TRACEABILITY

QUALITY OF FOOD SUPPLY

REGULATORY COMPLIANCE

COUNTERFEIT

COST

NPTEL

Technologies and Innovations in SCM



Big Data







Drones





Cloud Computing



Robotics



REFERENCES

- **Enterprise Resource Planning, Rajesh Ray, Tata McGraw-Hill – Chapter 25**
- **Management Information Systems: Managing the Digital Firm - Kenneth C. Laudon & Jane P. Laudon**

NPTEL



NPTEL

THANK YOU !



NPTEL ONLINE CERTIFICATION COURSES

NPTEL

Management Information Systems

Prof. Surojit Mookherjee
VGSoM, IIT KHARAGPUR

Week 04: Supplier Relationship Management

Lecture 04:

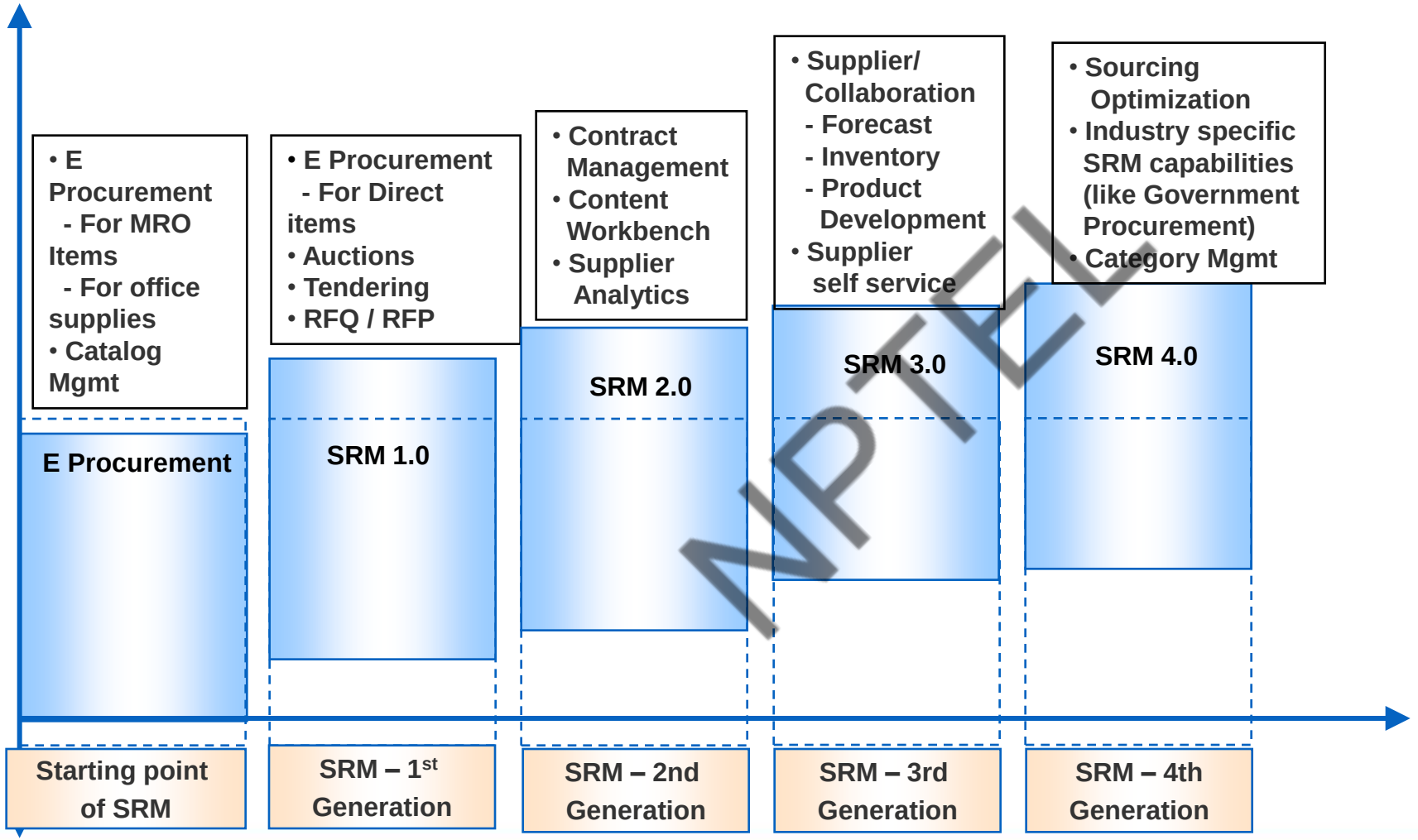
Supplier Relationship Management

Gartner defines SRM as: “The practices needed to establish the business rules and understanding needed for interacting with suppliers of products and services of varied criticality to the profitability of the enterprise

Meta defines SRM as “SRM is a system of applications that automates specific planning, scheduling, shipment, and payment processes between a manufacturer and its critical direct materials suppliers. SRM systems track both data and process- interchange flows, as well as provide the management and analytical tools to intelligently track the performance of direct suppliers based on benchmark, historical, and/or contractual conditions.”

NPTEL

Supplier Relationship Management – Application Evolution



NPTEL

SRM Vendors serve different purpose for different types of buyers



NPTEL

SRM – Five Important Areas

Five Important SRM areas are:

1. Electronic Procurement
2. Catalogue Management
3. Content Management
4. Contract Management
5. Supplier Collaboration

NPTEL

1. E Procurement

Common technology options available for E Procurement are:

A) Electronic Shopping Cart

Shopping cart is a common e procurement technology used by employees for procurement of goods and services. Employees can create shopping cart by searching products in electronic catalogs, start the shopping process, check the procurement status at any time, confirm goods and services once that is delivered and can then enter invoice for their purchase order.

B) Procurement Card

Procurement card can be used as a payment method in E Procurement systems. P cards are credit cards offered by companies like American Express and Visa.

C) Auction

This is a suitable process for selecting the most suitable offer in case an organization deals with a large number of requests for quotations (RFQs).

NPTEL

A) Shopping Cart Functions

Typical Shopping Cart functions are:

- ✓ Creating Shopping Cart (Cart have advanced search capabilities like Catalogs, Favorites etc. for searching right products)
- ✓ Shopping Cart approval
- ✓ Purchase Order and Goods Receipt
- ✓ Invoice Payment
- ✓ Tax Calculation
- ✓ Availability check
- ✓ Identifying sources of supply / possible vendor for the item
- ✓ Approval preview

NPTEL

B) Procurement Card Is a Win-Win for both Buyer and Supplier

Benefits to buyer:

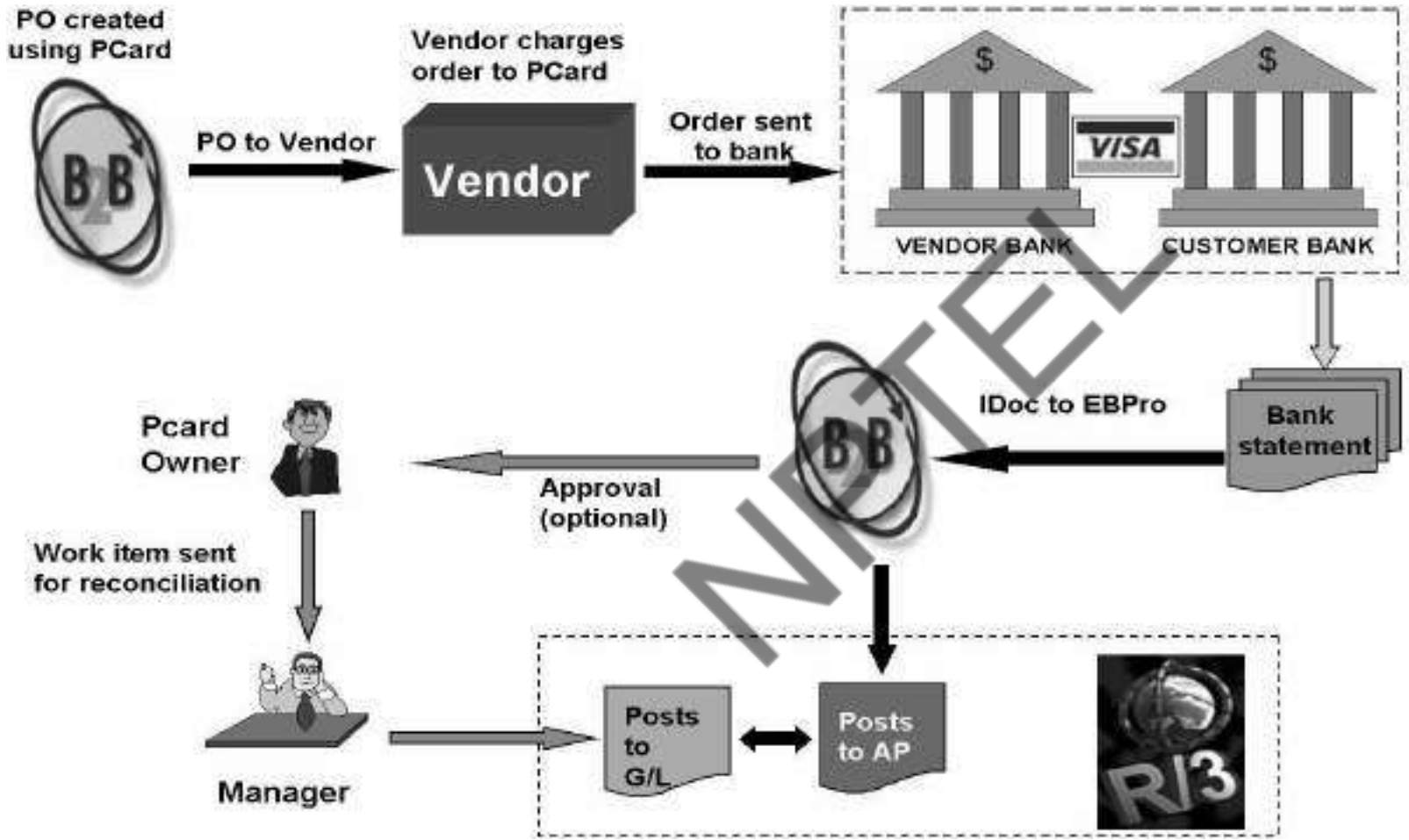
- ✓ Reduce cost of purchasing and payment by cutting out unnecessary administration of multiple ordering and payments to monthly one statement and payment.
- ✓ Replacing manual processes with electronic ordering, payment and reconciliation. This enables instant information sharing and processing.
- ✓ Reconciling transactions are easier as all transaction details are available on net – this help in quick reconciliation and resolve discrepancies

Benefits to supplier:

- ✓ Greater security for suppliers as here bank guarantees payment
- ✓ Low payment transaction costs as payment happens electronically

NPTEL

Procedure of Procurement Card with SAP…..example



NPTEL

Source: www.sap.com

C) Bids and Auctions

Types of Bids

- ✓ **Public Bids:** Here bid invitations are made accessible to potential bidders via marketplaces on the web. The bidder can reach company's web page via a hyperlink from the portal, log on to E Procurement system, and enter their bid.
- ✓ **Restricted Bids:** In this case bid invitations are made accessible to known bidders via e-mail.

Bids can also be classified as

- ✓ **Online Static Sealed Bid**: The bidders are allowed to bid only once with their price within a window of stipulated time
- ✓ **Online Dynamic Sealed Bid**: Here bidders are allowed to improve their price bids and till the hidden reserve price level is touched

NPTEL

2.Catalog Management ….

Catalog helps E Procurement in variety of ways:

- ✓ Catalogs list all products and detailed product information including links to supplier data. Catalogs help employees to easily find the products that they require and for this they support strong search engines that can locate product based on description, product attributes etc.
- ✓ Catalogs can provide price information.
- ✓ Catalogs can also check availability from a supplier

NPTEL

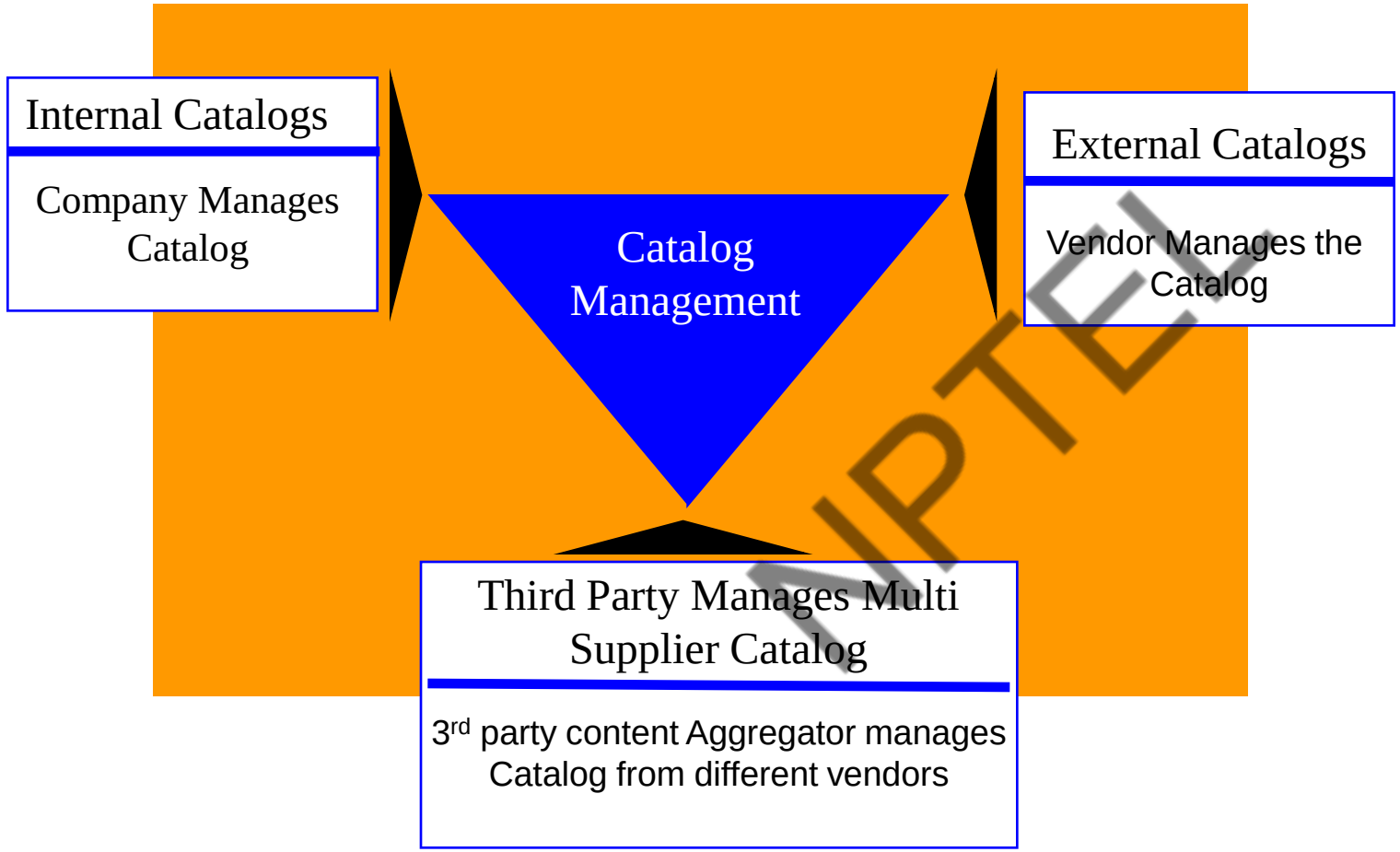
2.Catalog Management

Three Types Of Catalogs

- ✓ **External catalogs**: Catalog is managed, updated and hosted by the vendor at his own site. This is good for the company as vendor in this case is taking all the responsibility.
- ✓ **Internal catalogs**: Catalog is managed, updated and hosted by the company at his own site. The catalog can have products of various vendors. Advantage of this is look and feel of the catalog can be controlled and is uniform for all products making it easy for employees. The obvious disadvantage is the time and cost needed for company to maintain such catalogs.
- ✓ **A third party manages the multi supplier catalog on behalf of the company**: This multi supplier catalog is managed by a third party content aggregator on behalf of the company. Advantage of this is the consistent look and feel for the catalogs from multiple suppliers. The disadvantage is the recurring expenditure that needs to be paid to 3rd party.

NPTEL

Catalog Management – Options



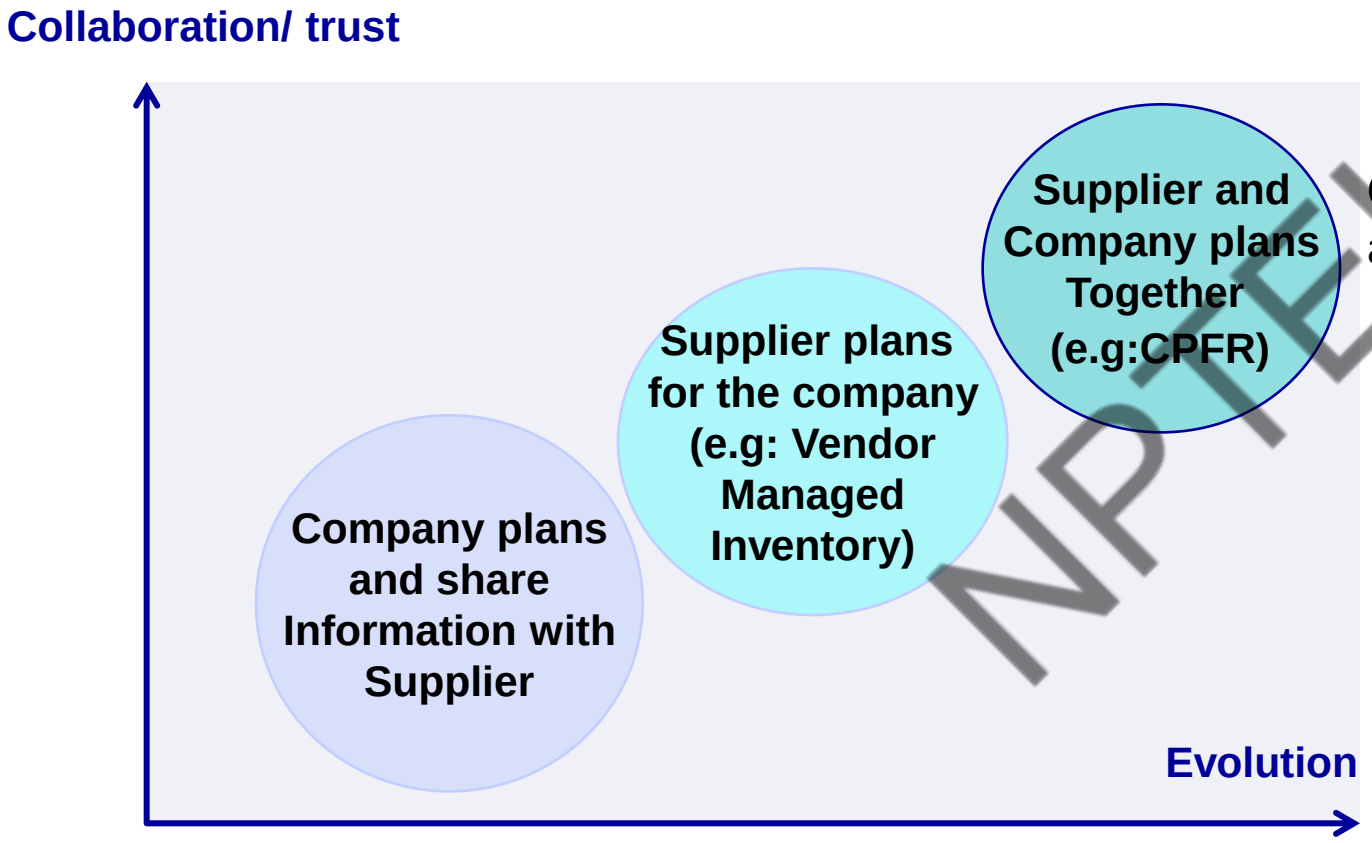
NPTEL

Areas Of Supplier Collaboration

- ✓ Demand / Forecast collaboration
- ✓ Inventory collaboration
- ✓ New product development collaboration
- ✓ Payment collaboration
- ✓ Logistics collaboration

NPTEL

Forms of Supplier Collaboration



Collaborative Planning, Forecasting and Replenishment (**CPFR**),

NPTEL

Technologies Of Supplier Collaboration

- ✓ **Electronic Data Interchange (EDI):** The oldest message based collaboration technology for transmitting things like Purchase order, Advanced shipping notification (ASN), Invoice etc.
- ✓ **Portal:** The company's portal give access to it's suppliers about delivery schedule, stock information, quality of receipts, rejection in case there is any, payment information etc. Supplier self service (SUS) portals are part of Supplier Relationship management solution of few ERP vendors like SAP.
- ✓ **Inventory collaboration**: Most of the ERP and SRM solutions provide features like Vendor managed inventory (VMI) for supporting inventory collaboration need.
- ✓ **Collaborative Forecasting**: ERP solutions provide advanced collaborative forecasting features in its Advanced Demand Planning solution.
- ✓ **Payment Collaboration - Evaluated receipts (ERS):** Most of the ERP solutions support ERS functionality where GR is settled automatically based on information received from Supplier and payment is released.

NPTEL

SRM 5.0 - Activities

- Purchasing Planning
- Category Management
- Supplier Qualification
- Supplier Negotiation
- Contract Management
- Requisitioning
- Order Management
- Receiving
- Financial Settlement
- Supplier Enablement

NPTEL

REFERENCES

- **Enterprise Resource Planning, Rajesh Ray, Tata McGraw-Hill – Chapter 23**
- **Management Information Systems: Managing the Digital Firm - Kenneth C. Laudon & Jane P. Laudon**

NPTEL



NPTEL

THANK YOU !



NPTEL ONLINE CERTIFICATION COURSES

NPTEL

Management Information Systems

Prof. Surojit Mookherjee
VGSoM, IIT KHARAGPUR

Week 04: Product Lifecycle Maintenace

Lecture 05 :

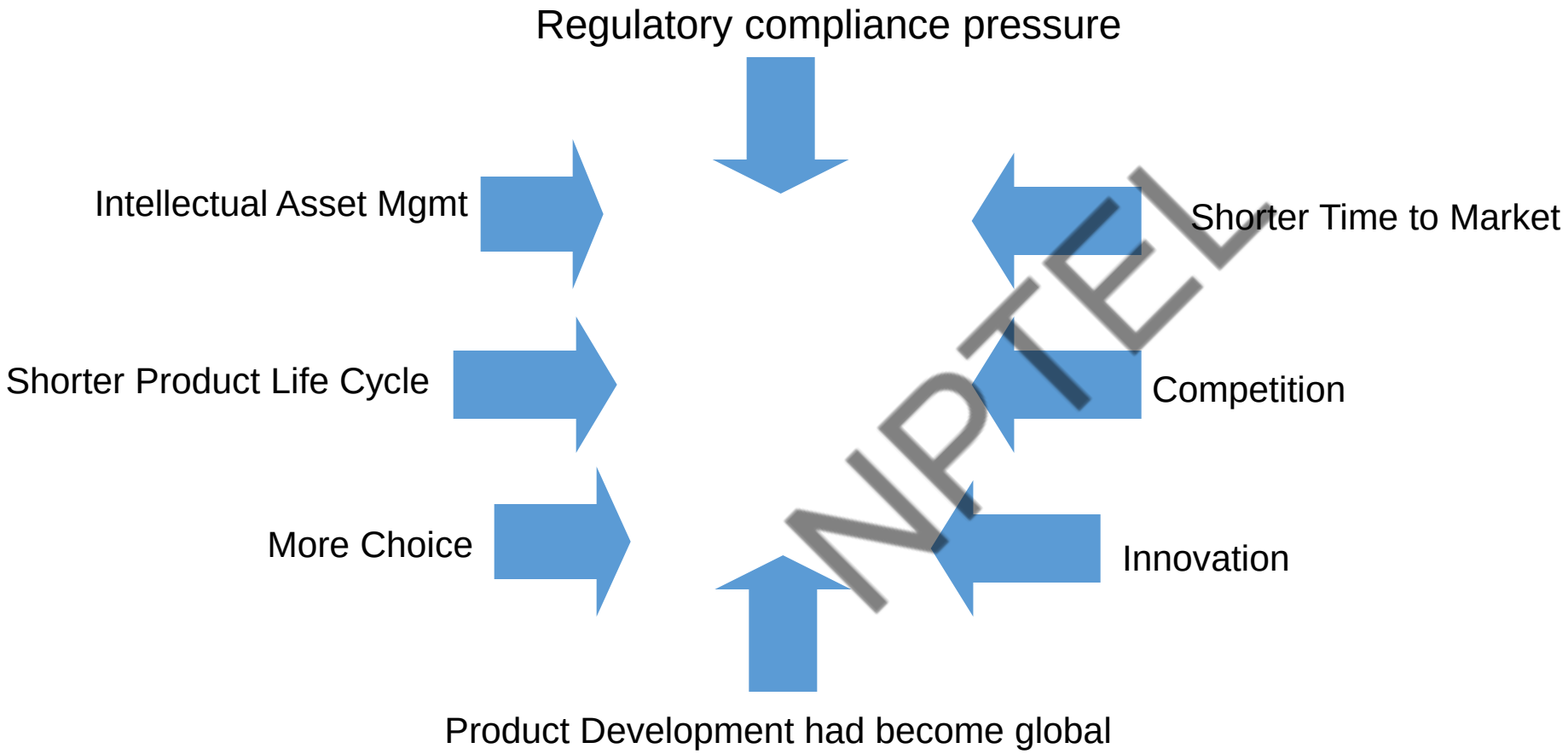
Product Lifecycle Management

Product lifecycle management (PLM) is a new set of enterprise application that manages all data and information about a product from its initial conception to retirement.

Product lifecycle includes design, manufacturing, ensuring quality of the product through out its life cycle, ensuring product safety during its usage and finally service till the product retires from service.

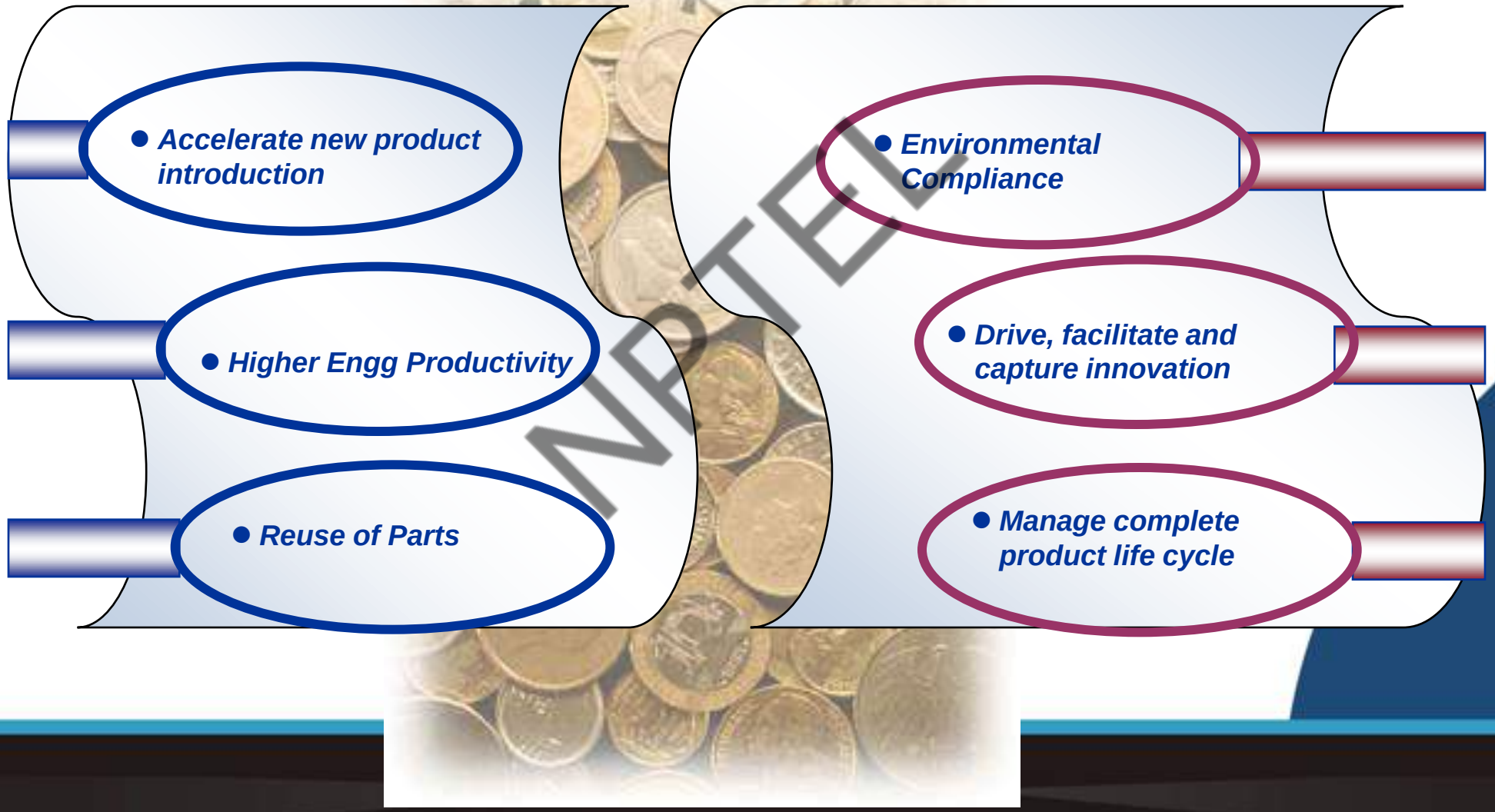
NPTEL

Business Drivers for PLM



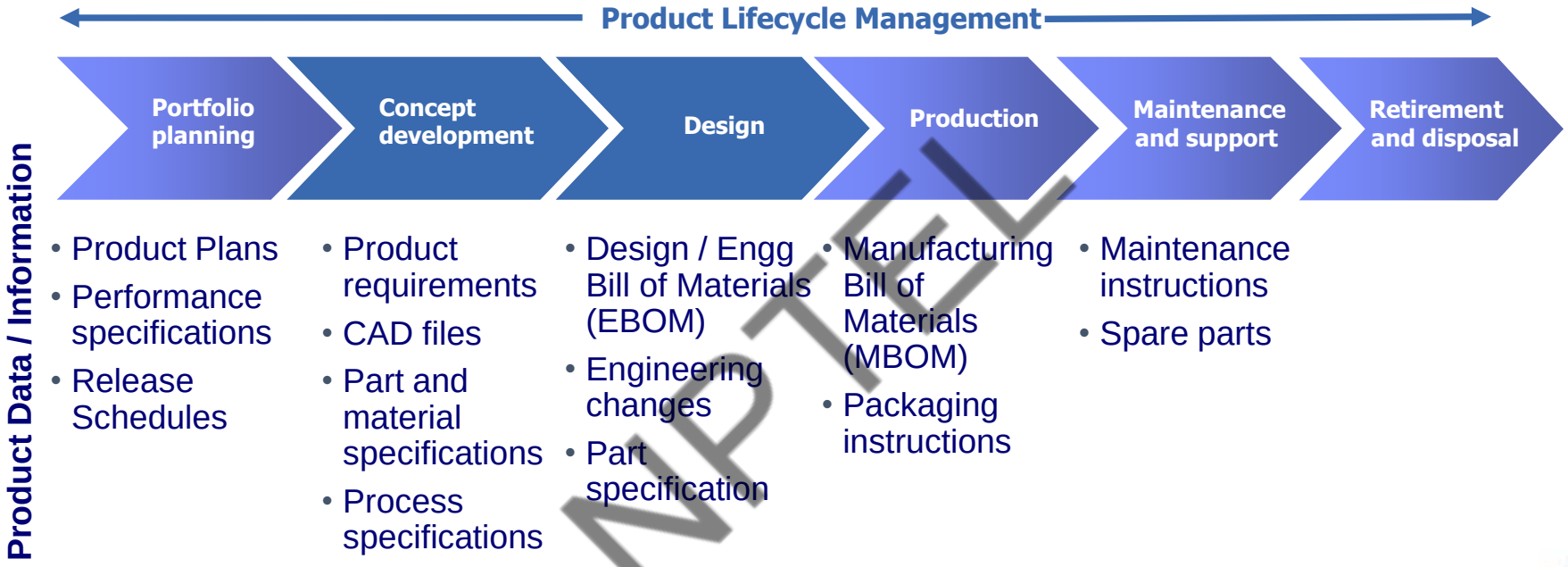
NPTEL

Value Drivers Of PLM



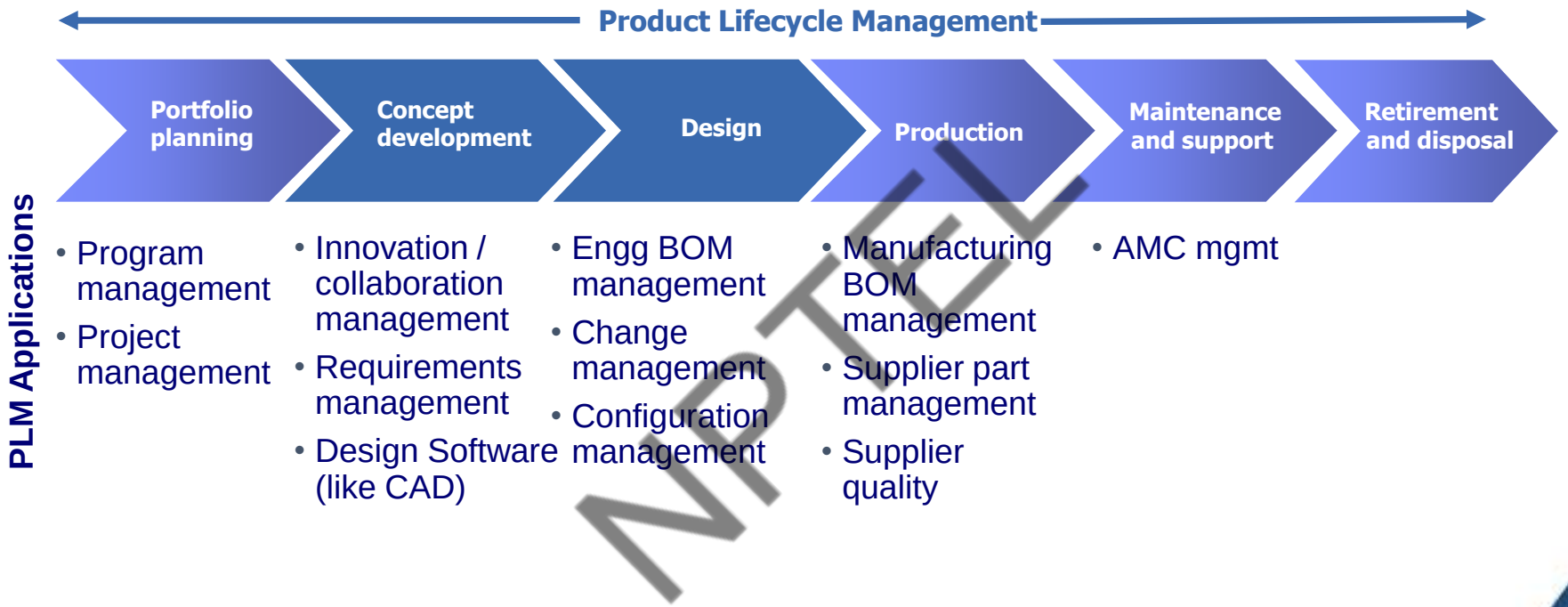
NPTEL

PLM: Data management



NPTEL

PLM need different applications to support needs of diff phases



NPTEL

Product Lifecycle Management Functionalities

Portfolio / Project

Portfolio Mgmt

Program Mgmt

Project Mgmt

Integration

- CAD
- ERP

Requirement Mgmt

Product Safety

Handling hazardous / dangerous goods

Environmental compliance

Product Data Mgmt (PDM)

Storing Product data in different formats

Managing BOM / Configuration / Structure

Visualization

Collaboration

Workflow

Change Mgmt

NPTEL

PLM Functionality – Document Management and CAD Integration

Document Management

Historically, PLM evolved from Document Management systems (and CAD Data Management systems). The document object can have attributes, revisions, versions, lifecycle states, approvals, controlled access, check-in, check- out, view, print, stamp, mark-up, and so on. Document management capabilities can be used for any document format including PDFs, office formats, JPEG, CAD and so on.

CAD Management / Integration

PLM solutions offer out of box integration with CAD software i.e. users can accesses PLM functions through the CAD application interface. Some of the functions supported include

- ✓ Common parts libraries
- ✓ Check-in and check-out
- ✓ Revision and version control
- ✓ Change and Release Management
- ✓ BOM Management

NPTEL

PLM Functionality – Search and Change Mgmt

Search

One of the primary goals of PLM is to allow for reuse of information and that's why PLM solutions provide sophisticated search capabilities. PLM solutions can offer different types of search capabilities like:

- ✓ Basic searching
- ✓ Parametric searching
- ✓ Search along relationships
- ✓ Full-text search

Change Management

Change Management is another key function of PLM products. Change Management is the process of creating a change request (can be a functionality change), for getting it approved, doing the change, testing it and finally transmitting the change to final design / BOM.

NPTEL

Chapter 31: PLM Functionality – Manufacturer Part Number and BOM Mgmt

Manufacturer Part Number (MPN)

- ✓ This is mainly for managing purchased parts.
- ✓ A manufacturer part number does not appear in company's Bill of Material.
- ✓ This linking is sometime required to quickly order a supplier a particular part and to avoid collisions when two manufacturers coincidentally use the same number for two different parts.
- ✓ MPN functionality helps in this.

Bill Of Material Management

- ✓ BOM Management capability includes building BOM structures, adding or removing parts, comparing BOMs, and managing BOM changes and transformation through their lifecycle.
- ✓ PLM systems help in transformation of a CADBOM to EBOM (Engineering BOM) to MBOM (Manufacturing BOM).
- ✓ PLM systems can manage complex BOMs through their lifecycle and configuration changes.
- ✓ PLM systems also have the ability to manage configurable options within a BOM.
- ✓ PLM systems also support BOM Cost Management through different BOM comparison.

NPTEL

Chapter 31: PLM Functionality - Product Visualization, Requirement Mgmt

Product Visualization

- ✓ PLM solutions provides a dynamic view of the product's 3D CAD model
- ✓ While designing, the designer can manipulate his inputs and based on that can simultaneously see it's effect.
- ✓ This helps in designing in an interactive mode where he can play with lot of parameters and can always see the effect immediately.

Requirements Management

- ✓ A new product always starts with a set of requirements. Functional requirements describe product behavior. Non-functional requirements describe how well this behavior should perform.
- ✓ A PLM system can manage requirements like any other object type, linking many-to-many to each other and to other requirements. So, a requirement can have sub-requirements that go into more detail. Requirements can lead to products feature.
- ✓ PLM systems manage all requirements through revision control and change management.

NPTEL

Chapter 31: PLM Functionality - Program and Product Management

- ✓ A program is basically a collection of projects so there can be a program at a product level or a product family level that has multiple projects underneath. The program is used for grouping and reporting purposes.
- ✓ A project contains tasks in a work breakdown structure, with associated resources, durations, deliverables, and workflows.
- ✓ PLM project management supports integration with third-party project management tools, such as MS-Project. Sometime Program Management is also referred as Portfolio Management.
- ✓ To support Program and Project Management capabilities, a PLM system needs to add additional object types like: Program, Project, Task, Work breakdown structure, Resource, Calendars (based on location, user, and so on).

NPTEL

Chapter 31: PLM EHS Functionality – Hazardous Material Tracking

- ✓ PLM EHS Software manages chemical inventory systems and track chemical storage (e.g., inventory), use, and transportation throughout the supply network and end-of-life disposal.
- ✓ This solution ensures that the company meets regulatory requirements relating to product safety and to the management and labeling of dangerous goods – while at the same time protecting employees and the environment.
- ✓ PLM EHS supports dangerous goods management by enabling companies to store and manage regulatory data, classify dangerous goods, perform checks during transport, and compile and update dangerous goods documentation. This data can be automatically transferred and used for packing lists.
- ✓ For any transportation channels (road, rail, sea, or air), PLM EHS ensure that shipments comply with regulations before leaving the premises, and manage dangerous goods storage, transportation, movement, delivery, compliance and reporting.

NPTEL

Chapter 31: PLM EHS Functionality – Material safety data sheet (MSDS)

- ✓ PLM EHS solution lets the company collect and process product compliance data automatically, enables to document product content and regulatory or sector-specific substance lists and integrates compliance checks and analyses.
- ✓ A specification management database allows to centrally managing substance information (including product composition).
- ✓ A global label management function supports management of many different types of labels that can be printed directly from various business processes (for example, from outbound delivery).
- ✓ A Material Safety Data Sheet (MSDS) documents proper procedures for handling of hazardous materials for workers. PLM EHS software support the generation and storage of MSDS information, such as physical data, toxicity levels, health effects/risks, remedy requirements for spills or accidents, and storage guidelines. PLM systems make this information readily available to workers and downstream consumers that are exposed to these materials.

NPTEL

Chapter 31: PLM EHS Functionality – Environmental Compliance

Environmental compliance allows an organization to capture the materials that are used in their products and track compliance against internal and external standards.

Environmental compliance capabilities maintain several types of information like:

- ✓ What is inside a part that could be a substance of concern for environmental regulatory purposes? The view must include both on purchased parts and also about where the final product is sold.
- ✓ What are the compliance standards (for example, WEEE, RoHS, REACH) that are in effect that the Part may need to comply with?
- ✓ Whether or the Part actually meets appropriate compliance standards or not

PLM products support compliance process in variety of ways like:

- ✓ Making sure that the latest standards are known
- ✓ Managing the process and correspondence with Part suppliers on their material content for each Part
- ✓ Performing BOM analysis to check compliance at the Product level
- ✓ Tracking the countries where products are sold
- ✓ Communicating the compliance status to suppliers, customers, and regulatory bodies

NPTEL

Chapter 31: PLM Vendors

Some of the leading vendors in PLM space are listed below:

- ✓ CAD (Computer Aided Design) / CAM (Computer Aided Manufacturing) / CAE (Computer Aided Engineering) / Document Management vendors moving into PLM space:
- ✓ PTC
- ✓ UGS
- ✓ Dassault Systémes / IBM (IBM recently acquired Dassault Systems and another PLM vendor MatrixOne)
- ✓ Agile Software

Leading ERP vendors building PLM solution

- ✓ SAP
- ✓ Oracle
- ✓ IFS

NPTEL

REFERENCES

- **Enterprise Resource Planning, Rajesh Ray, Tata McGraw-Hill – Chapter 31**
- **Management Information Systems: Managing the Digital Firm - Kenneth C. Laudon & Jane P. Laudon**

NPTEL



NPTEL

THANK YOU !



NPTEL ONLINE CERTIFICATION COURSES

NPTEL

Management Information Systems

Prof. Kunal Kanti Ghosh
VGSoM, IIT KHARAGPUR

Week 5: Module 1

Information Systems for Manufacturing Management

CONCEPTS COVERED

- Information Systems for Manufacturing Management

- Associated Business Processes and Subsystems for Manufacturing Management

NPTEL

Information Systems for Manufacturing Management

- The manufacturing/production function is responsible for actually producing the firm's goods and services.

- Manufacturing activities deal with:
  - planning, development, and maintenance of production facilities,
  - establishment of production goals,
  - acquisition and storage of production materials, and
  - scheduling of equipment, facilities, material and labour required to furnish finished products

NPTEL

Information Systems for Manufacturing Management

- **Overview of a simplified manufacturing process:**
  - **The manufacturing process mainly follows two different strategies:**
    - **Make-to-stock and**
    - **Make-to-order**
      - ✓ **Make-to-stock occurs when the company produces goods to create or increase an inventory, that is finished goods that are stored in the warehouse and are available for sales (push process)**
      - ✓ **Make-to-order occurs when production is generated by a specific customer order (pull process)**

NPTEL

Information Systems for Manufacturing Management

- **Overview of a simplified production process:**

- **Manufacturing companies that produce their own goods manage their interdepartmental production process across the Production and Warehouse departments.**

NPTEL

Information Systems for Manufacturing Management

- **Overview of a simplified production process:**

- **The production process involves the following steps:**
  - **The Warehouse department issues a planned order when the company needs to produce a finished product, either because the warehouse has insufficient inventory or because the customer placed a specific order for goods that are not currently in stock.**

NPTEL

Information Systems for Manufacturing Management

❖ **Overview of a simplified production process:**

➢ **Once the planned order reaches Production, the production controller authorizes the order and issues a production order, which is a written authorization to start the production of a certain amount of specific product.**

NPTEL

Information Systems for Manufacturing Management

❖ **Overview of a simplified production process:**

- **To assemble a finished product, Production department requires a number of materials (or parts)**

- **To acquire these materials, Production department generates a material withdrawal slip, which lists all of the needed parts, and forwards it to the Warehouse**

NPTEL

Information Systems for Manufacturing Management

❖ **Overview of a simplified production process:**

- **If the parts are available in the Warehouse, then the Warehouse delivers them to Production. If the parts are not available, then the company must purchase them via the procurement process.**

- **After Production has created the products, it updates the production order specifying that, as planned, a specific number of units of product can now be shipped to the Warehouse.**

NPTEL

Information Systems for Manufacturing Management

- As soon as the warehouse receives the finished goods, it issues a goods receipt document that certifies how many units of a product it received are available for sales.

NPTEL

Information Systems for Manufacturing Management

- The goal of production or manufacturing management is to optimally deploy human resources, machines, and materials to maximize production of goods and services by the organization

- In order to attain this goal, various types of information will be required.

NPTEL

Information Systems for Manufacturing Management

❖ **Strategic Information for:**

- **Location of new production facilities**
- **Design of new products (use of CAD)**
- **Yearly and monthly production quotas and alternate schedules**
- **Policies on machine replacement, augmentation, and modernization**

NPTEL

Information Systems for Manufacturing Management

❖ **Strategic Information for:**

- **Introduction of new production technologies, acquisition, and mergers**
- **Identification of best product mix**
- **Policy on quality assurance**

NPTEL

Information Systems for Manufacturing Management

❖ **Tactical Information for:**

- **Identification and control of areas of high cost such as WIP inventories**
- **Identification of critical bottlenecks in production**
- **Identification of alternate production schedules based on tools, machines, material, and personnel availability**

NPTEL

Information Systems for Manufacturing Management

- **Tactical Information for:**

  - **Performance measures of machines (breakdown histories and cost of repair and failure) to decide on replacement**
  - **Quality control measures**

NPTEL

Information Systems for Manufacturing Management

❖ **Operational Information for:**

- **Monitoring up to date production information by examining assemblies, detecting likely shortages and generating early warning**
- **Scheduling better production dynamically**
- **Preventive maintenance schedules of machines**

NPTEL

Information Systems for Manufacturing Management

❖ **Operational Information for:**

- **Monitoring tools, machines, and human resources availability**
- **Ensuring quality at each step in production**
- **Machine control (i.e., controlling the action of machines and equipment)**

NPTEL

Information Systems for Manufacturing Management

❖ **Production Planning:**

➢ **This sub-system maintains data on all production-related activities such as:**

✓ **raw materials,**

✓ **sales forecasts and plans,**

✓ **production schedules, and**

✓ **bill of material for different products**

NPTEL

Information Systems for Manufacturing Management

❖ **Production Planning:**

➢ **This sub-system also maintains data on:**

- **orders for repetitive manufacturing,**
- **costs of production data, and so on.**

➢ **It is a high-level module for managing the production function including activities such as routing, scheduling, and loading.**

NPTEL

Information Systems for Manufacturing Management

❖ **Production Planning:**

- **Production planning has evolved from Materials Requirement Planning (MRP).**

- **Independent demand items (e.g., the number of identical personal computers a computer manufacturer will sell).**

NPTEL

Information Systems for Manufacturing Management

❖ **Production Planning:**

- **In manufacturing operations, the demand for some items is interdependent.**
- **For example, a company that makes three types of chairs, all of which use the same screws and bolts.**

NPTEL

Information Systems for Manufacturing Management

❖ **Production Planning:**

- **In this case, the demand for screws and bolts depends on the total demand for all three types of chairs and their shipment schedules.**

- **The planning process that integrates production, purchasing and inventory management of interdependent items is called materials requirement planning (MRP).**

NPTEL

Information Systems for Manufacturing Management

❖ **Production Planning:**

➢ **The main functions of MRP are,**

❑ **to monitor stocks and**

❑ **to determine which material you need in what quantity at what time, and**

❑ **to create the corresponding order proposals from this subsystem**

✓ **Production orders**

✓ **Purchase orders**

NPTEL

Information Systems for Manufacturing Management

❖ **Production Planning:**

- **In MRP, the subsystem compares available stock or the scheduled receipts from purchasing or production with planned requirements in the net requirements calculation.**

- **Net Requirement = Planned Requirement – (Available stock at warehouse + Scheduled receipts from purchasing or production)**

- **Planned Requirement is calculated through explosion of bill of materials**

NPTEL

Information Systems for Manufacturing Management

❖ **Production Planning:**

- **Bill of Material (BOM) is a complete, formally structured list which make up the components of a product or assembly.**
- **The list contains the material identification number of each item, together with the quantity , unit of measure, and indicator for purchase or production.**
- **We can explode a BOM to show the total quantity of each item required to meet the planned requirement of the finished product.**

NPTEL

REFERENCES

- Laudon, K. C., and Laudon, J. P. (2016), Management Information Systems: Managing the Digital Firm
- Turban, E., Volonio, L., Wood, R. Gregory.(2016), Information Technology for Management
- Rainer, K., Prince, B., Watson, H.,(2015), Management Information Systems
- De, Rahul.,(2019), MIS: Managing Information Systems in Business, Government and Society
- Bagchi, Nirmalya., (2010), Management Information Systems

NPTEL



NPTEL

THANK YOU !



NPTEL ONLINE CERTIFICATION COURSES

NPTEL

Management Information Systems

Prof. Kunal Kanti Ghosh
VGSoM, IIT KHARAGPUR

Week 5: Module 2

Information Systems for Manufacturing Management (Continued)

CONCEPTS COVERED

- Information Systems for Manufacturing Management
- Associated Business Processes and Subsystems for Manufacturing Management

NPTEL

Information Systems for Manufacturing Management

❖ **Capacity Planning:**

- **Capacity refers to the ability to produce output per time period.**
- **Capacities can be allocated to work centres or resources.**
- **Capacity requirements specify how much capacity is required for individual orders at a specific time.**
- **Capacity Requirement Planning is a tool for determining available capacity and capacity requirements, and for carrying out capacity levelling.**

NPTEL

Information Systems for Manufacturing Management

❖ **Production Control:**

➢ **This function helps maintain quality of the products by keeping a close eye on activities such as ordering, follow-ups, and inventory control**

❖ **Product Design:**

➢ **This function deals with all the activities that give shape to the product**

NPTEL

Information Systems for Manufacturing Management

❖**Computer Aided Design (CAD):**

➢**This subsystem automates**

✓ **the creation and**

✓ **revision of designs,**

❑**using computers and sophisticated graphics software.**

NPTEL

Information Systems for Manufacturing Management

❖ **Computer Aided Design (CAD):**

➢ **Using a more traditional physical design methodology,**

❑ **each design modification requires**

✓ **a mould to be made and**

✓ **a prototype to be tested physically.**

NPTEL

Information Systems for Manufacturing Management

❖ **Computer Aided Design (CAD):**

➢ **This process is repeated many times**

✓ **which is very expensive and**

✓ **time consuming**

NPTEL

Information Systems for Manufacturing Management

❖ **Computer Aided Design (CAD):**

➢ **Using a CAD workstation,**

✓ **the designer need only make a physical prototype toward the end of the design process**

✓ **because the design can be**

❑ **easily tested and**

❑ **changed on the computer**

NPTEL

Information Systems for Manufacturing Management

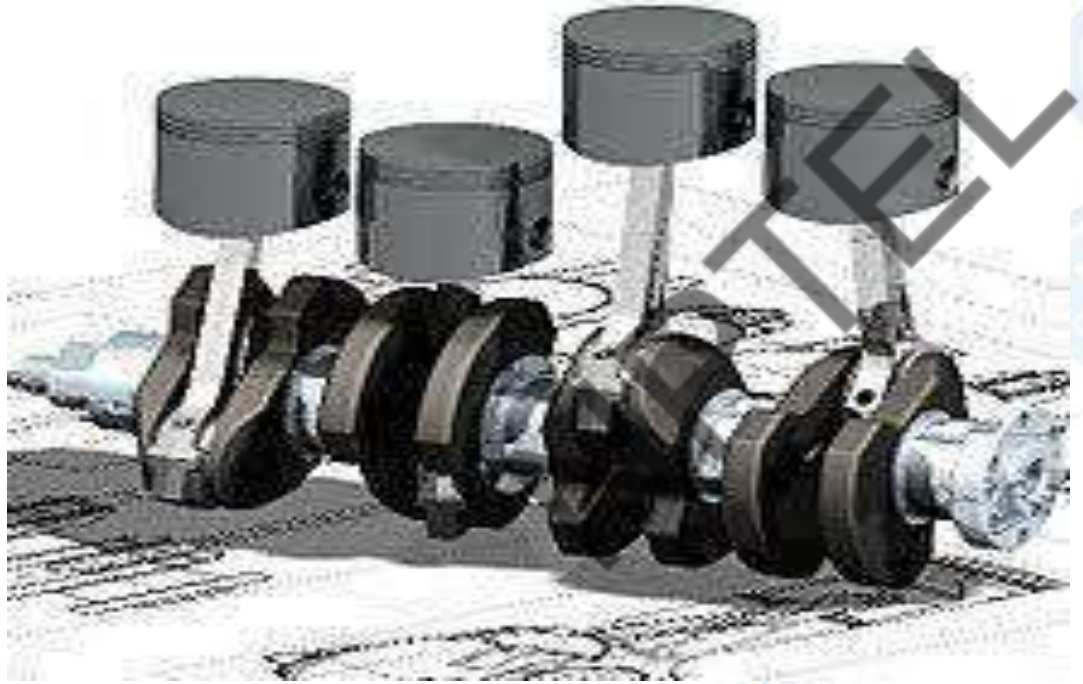
❖ **Computer Aided Design (CAD):**

- **For example, Ford Motor Company used a computer simulation**
  - ✓ **to create an engine cylinder that came up with the most efficient design possible**

- **CAD systems have also provided many benefits to companies like**
  - ✓ **Tata Motors as well as**
  - ✓ **Jaguar Land Rover**

NPTEL

Information Systems for Manufacturing Management



Design of Engine Components Using CAD System

Information Systems for Manufacturing Management

❖ **Computer Aided Design (CAD):**

➢ **The ability of CAD software to provide**

❑ **design specifications for the tooling and manufacturing processes**

✓ **saves a great deal of time and money**

✓ **while producing a manufacturing process with far fewer problems.**

NPTEL

Information Systems for Manufacturing Management

❖ **Computer Aided Design (CAD):**

- ➢ **CAD systems are able to supply data for 3-D printing also known as additive manufacturing**
- ✓ **Additive manufacturing uses machines to make solid objects, layer by layer, from specifications to a digital file**
- ➢ **3-D printing is currently being used for**
- ✓ **producing prototypes, and**
- ✓ **customized manufacturing work**

NPTEL

Information Systems for Manufacturing Management



Additive Manufacturing System

NPTEL

Information Systems for Manufacturing Management

❖ **Virtual Reality Systems:**

➢ **These subsystems have**

✓ **visualization,**

✓ **rendering, and**

✓ **simulation capabilities**

❑ **that go far beyond those of conventional CAD systems**

NPTEL

Information Systems for Manufacturing Management

❖ **Virtual Reality Systems:**

➢ **These systems use**

✓ **interactive graphics software to**

✓ **create computer generated simulations**

❑ **that are so close to reality that users almost believe they are participating in a real-world situation**

NPTEL

Information Systems for Manufacturing Management

❖ **Virtual Reality Systems:**

➢ **In many virtual reality systems, the user dons**

✓ **special clothing,**

✓ **headgear, and**

✓ **equipment depending on the application**

NPTEL

Information Systems for Manufacturing Management

❖ **Virtual Reality Systems:**

➢ **These clothing contains**

✓ **sensors that record the user's movements and**

✓ **immediately transmit the information back to the computer**

NPTEL

Information Systems for Manufacturing Management

❖ **Virtual Reality Systems:**

- For example, to walk through a virtual reality simulation of a house,
  - ✓ you would need special clothing that monitors the movement of your feet, hands, and head
  - ✓ you will also need
    - ❑ goggles containing video screens, and
    - ❑ sometimes audio attachments, and
    - ❑ feeling gloves
- so that you can be immersed in the computer feedback

NPTEL

Virtual Reality System



NPTEL

Information Systems for Manufacturing Management

- Thus the manufacturing management information system helps a firm accomplish all those objectives as mentioned before.
- In a highly dynamic business environment, with complex production processes, volumes of data are massive.
- This system stores and manages this data and guides the manager in achieving effective control of production activities.
- For example, the inventory control system, which is a part of the manufacturing management information system, makes it easy to manage inventory automatically.

NPTEL

REFERENCES

- Laudon, K. C., and Laudon, J. P. (2016), Management Information Systems: Managing the Digital Firm
- Turban, E., Volonio, L., Wood, R. Gregory.(2016), Information Technology for Management
- Rainer, K., Prince, B., Watson, H.,(2015), Management Information Systems
- De, Rahul.,(2019), MIS: Managing Information Systems in Business, Government and Society
- Bagchi, Nirmalya., (2010), Management Information Systems

NPTEL



NPTEL

THANK YOU !



NPTEL ONLINE CERTIFICATION COURSES

NPTEL

Management Information Systems

Prof. Kunal Kanti Ghosh
VGSoM, IIT KHARAGPUR

Week 5: Module 3

Information Systems for Materials Management

CONCEPTS COVERED

- Information Systems for Materials Management (MM)
- Associated Business Processes and Subsystems for MM

NPTEL

Information Systems for Materials Management

- Materials management is one of the important functions in any organization wherein the materials alone may account for as much as 50% to 60% of the cost of the final product

- Proper attention and tight control is bestowed on this function to ensure proper utilization of available resources

NPTEL

Information Systems for Materials Management

- The magnitude of this problem increases when
  - the operations are of a large scale,
  - spread over multiple locations, and
  - the company manufactures wide variety of products

NPTEL

Information Systems for Materials Management

- The main consideration that shapes management decisions in materials management is
  - to ensure maintenance of minimum stock all the time, subject to
    - ✓ availability of the right material,
    - ✓ at the right place, and
    - ✓ at the right time

NPTEL

Information Systems for Materials Management

- The inventory to be maintained depends on,
  - Demand
  - Lead time
  - Quality of material
  - Supplier reliability
  - Alternative material / sources

NPTEL

Information Systems for Materials Management

- And the most important factor on which the inventory to be maintained depends is,
- prompt and up-to-date information on
  - shipments,
  - receipts,
  - rejections,
  - stocks, and
  - consumption for enabling better control

NPTEL

Information Systems for Materials Management

- Thus, an efficient MIS almost always helps the managers,

- to reduce inventories and
- ensure availability of
  - ✓ the right material,
  - ✓ at the right place, and
  - ✓ at the right time

NPTEL

Information Systems for Materials Management

- **The primary objectives of MIS for "materials management" are,**

- **To maintain up-to-date stocks of all materials**

- **To immediately respond to any changes in the production plan, due to**

- **To respond to changes in the demand pattern, or actual production, by generating delivery schedules and purchase orders for suppliers**

NPTEL

Information Systems for Materials Management

- The other related objectives of MIS for “materials management” are,

- To report critical or excess inventory items
- To initiate follow-up action on the suppliers
- To ensure availability of the right material at the right time

NPTEL

Information Systems for Materials Management

❖ The other objectives of MIS for “materials management” are,

➢ To assist materials managers in proper assessment of the performance of the suppliers in terms of

✓ quality,

✓ reliability, and

✓ price

NPTEL

Information Systems for Materials Management

- The other related objectives of MIS for “materials management” are,

- To accurately report on the current inventories and forecast the same for the future periods

- To enable the top management in review and formulation of purchasing policies

- To provide demand forecasting for material planning

NPTEL

Information Systems for Materials Management

- Major goal is to have readily available materials needed by the organization and to keep optimal levels of stock of items

- To attain this goal, the information necessary at various levels of management may be classified as follows:

NPTEL

Strategic Information Requirement for Materials Management

❖ **Strategic Information**

- ❑ **Developing vendors for critical items**
- ❑ **Determining optimal levels of inventory**
- ❑ **Determining proportion of materials to be ordered from different vendors**
- ❑ **Reducing varieties of inventory**

NPTEL

Tactical Information Requirement for Materials Management

❖Tactical Information

❑Developing vendor performance measures

❑Determining optimal reorder levels

❑Determining issues of items to shops versus standard needs

NPTEL

Tactical Information Requirement for Materials Management

❖Tactical Information

❑Controlling high value of inventory

❑Determining impact of

✓material cost,

✓procurement with design changes, and

✓new project introduction

NPTEL

Operational Information Requirement for Materials Management

❖ **Operational Information**

- ❑ **List of excess and deficient items received**
- ❑ **List of items rejected**
- ❑ **List of critical items received**

NPTEL

Operational Information Requirement for Materials Management

❖ **Operational Information**

- ❑ **List of items in transit and under inspection**
- ❑ **Value of inventory-on-hand**
- ❑ **Goods received, rejected, and issued**

NPTEL

Operational Information Requirement for Materials Management

- Provide proper feedback on
  - the impact of current decisions, and
  - to help in formulating new guidelines consistent with the overall interests of the organization

- Such systems also highlight areas wherein there is a possibility of the guidelines not being fully implemented

NPTEL

MIS for Tactical and Operational Requirement for Materials Management

❖ Ordering subsystem enables materials managers to,

➢ Calculate optimum reorder quantity, taking into account

❑ manufacturing and supply lead times,

❑ such that the inventory will be maintained at the safety stock level

NPTEL

MIS for Tactical and Operational Requirement for Materials Management

❖ **Ordering subsystem enables materials managers to**

➢ **Prepare delivery schedules for each supplier of each item, taking into account**

✓ **the minimum lot size,**

✓ **transit time, and**

✓ **inter-shipment times**

NPTEL

MIS for Tactical and Operational Requirement for Materials Management

❖Ordering subsystem enables materials managers to

➢Maintain firm minimum schedules on the suppliers

❑for some period,

❑as per the organization policy,

❑so that the suppliers get sufficient lead time

❑to plan their production to meet the revised requirements

NPTEL

MIS for Tactical and Operational Requirement for Materials Management

❖Ordering subsystem enables materials managers to

➢ensure that all delivery schedules,

❑at least up to the firm period,

❑are covered by valid purchase orders

❑with the latest terms and conditions

NPTEL

Operational Information Requirement for Materials Management

- ❖System Deliverables include,
- ✓Monthly inventory statements
- ✓Inventory forecast and budget
- ✓Critical items report

NPTEL

Operational Information Requirement for Materials Management

- ❖System Deliverables include,
- ✓Effect of price changes of raw materials and components in the product
- ✓Theoretical versus Actual issues report
- ✓Ideal inventory report
- ✓Non-moving / Obsolete items report
- ✓Purchase Variance report

NPTEL

REFERENCES

- Laudon, K.C., and Laudon, J.P., (2016), Management Information Systems: Managing the Digital Firm
- Turban, E., Volonio, L., Wood, R. Gregory., (2016), Information Technology for Management
- Rainer, K., Prince, B., Watson, H., (2015), Management Information Systems
- De, Rahul.,(2019), MIS: Managing Information Systems in Business, Government and Society
- Bagchi, Nirmalya., (2010), Management Information Systems

NPTEL



NPTEL

THANK YOU !



NPTEL ONLINE CERTIFICATION COURSES

NPTEL

Management Information Systems

Prof. Kunal Kanti Ghosh
VGSoM, IIT KHARAGPUR

Week 5: Module 4

Information Systems for Quality Management

CONCEPTS COVERED

- Information Systems for Quality Management (QM)
- Associated Business Processes and Subsystems for QM

NPTEL

Information Systems for Quality Management

- Quality Management Systems integrate manufacturing, quality, and intelligence capabilities with the goal of continuously improving both product and process qualities

- These systems provide information about the quality of incoming material and parts as well as the quality of in-process, semi-finished and finished products

- These systems focus on preventing defects and as a result, minimize the number and impact of defects and quality recalls

NPTEL

Information Systems for Quality Management

- These systems record the results of all inspections and then compare the actual results with the expected results (established metrics)

- Quality control data may be collected by sensors or radio frequency identification (RFID) systems and interpreted in real time or they can be stored in a database for future analysis

- These systems also generate periodic reports on the percentage of defects or percentage of rework needed to keep managers informed of performance among departments

NPTEL

Information Systems for Quality Management

- All people participating in the value adding processes can connect to these systems and benefit from high quality, real-time information about all elements of the supply chain

NPTEL

Information Systems for Quality Management

❖ **Perspectives of Quality**

➢ **In a manufacturing-oriented company, quality has many facets including:**

❑ **Engineered and manufactured product quality**

❑ **Quality of supplied parts**

❑ **Control and documentation of all quality-related processes**

❑ **Quality inspections and tests**

NPTEL

Information Systems for Quality Management

- **Perspectives of Quality**

- **Quality inspections have until now been the cornerstones of quality management**

- **Now however, world-class manufacturing (WCM) focuses on engineering and management processes – the emphasis has clearly changed to engineered quality**

NPTEL

Information Systems for Quality Management

❖ **Perspectives of Quality**

- **Quality management system provides functionality for quality control throughout the supply chain**
- **Here you find a rich set of capabilities to plan and perform quality inspections of manufactured parts as well as of purchased parts or raw materials**
- **Organizations can define inspection operations for in-process inspections or they can use inspection orders to control purchased parts**

NPTEL

Information Systems for Quality Management

- **Perspectives of Quality**

- **Organizations may carry out inspections continually or periodically using statistical process control (SPC)**

- **All inspection data and quality information are available for online display**

NPTEL

Information Systems for Quality management

- Quality Management System uses the following basic data to control the tasks of quality management:
  - Material Master (containing the basic data for a material)
  - Inspection Characteristics
  - Inspection Methods
  - Sampling Procedures
  - Inspection Catalogs

NPTEL

Information Systems for Quality Management

- Inspection characteristics in a plan containing a description of what is to be inspected, an inspection method, a sampling procedure, and a dynamic modification rule (for sample determination based on the quality history)

- Inspection characteristics can be linked to a material specification, which the system can use as an inspection specification

NPTEL

Information Systems for Quality Management

- Inspection plans contain inspection operations, inspection characteristics and test equipment
- Inspection plans can be created using master inspection characteristics.
- Inspection plans can be assigned to the materials to be inspected

NPTEL

Information Systems for Quality Management

❖ **Quality Inspection at Goods Receipt:**

- **Based on the settings in the material master, the system creates an inspection lot at goods receipt and this specifies how a material is to be inspected**
- **An inspection lot is created automatically or manually**
- **After the inspection, results are recorded in the system and based on this usage decision (i.e., whether to accept the material or not) is taken**
- **For a certified vendor, there is no need of inspection lot during goods receipt**

NPTEL

Information Systems for Quality Management

❖ **Vendor Evaluation:**

- ❑ Based on the usage decision of the material (i.e., whether the material is accepted or rejected), a vendor's quality score can be calculated
- ❑ This quality score along with other criteria like vendor's price performance, delivery performance and service performance makes a vendor's overall score
- ❑ Weightage is given to each criteria
- ❑ Criteria to be used may differ by organizations
- ❑ However, quality, price, delivery and service are the most common criteria used across industry

NPTEL

Information Systems for Quality Management

❖ **Blocking of Invoices/Vendors based on Quality Inspection:**

- **A vendor's invoice can be blocked because of a quality inspection as and when an inspection lot is rejected**

- **If there is continuous rejection from the same vendor, the vendor itself can be blocked**

NPTEL

Information Systems for Quality Management

- QM Standard Reports

- QM systems can provide lot of standard reports on defects, inspection lots, inspection results, and so on either from the view point of vendors or from the perspectives of material

NPTEL

Information Systems for Quality Management

- QM in Production

- Following quality management processes are supported during production:
  - Inspection during production
  - Statistical process control
  - Test equipment calibration

NPTEL

Information Systems for Quality Management

❖Statistical Process Control (SPC)

- During production, the system uses statistical process control (SPC) techniques to monitor and control the production process.
- This process involves the use of control charts, for example, to display the mean value and standard deviation of measured values.
- Control charts are graphical tools that are used to determine whether or not a process is under statistical control and to detect external impacts on the process

NPTEL

Information Systems for Quality Management

- **Statistical Process Control (SPC)**

  - **If the upper or lower limits of a control chart are exceeded, the process is out of control and the concerned process owner must take appropriate corrective action**

NPTEL

Information Systems for Quality Management

- Quality Management in Sales and Service

- Following quality management processes are supported by the system:
  - Delivering Material as per Customer's Quality Specification
  - Quality Certificate
  - Batch Tracking
  - Batch Determination
  - Quality Inspection for Delivery

NPTEL

Information Systems for Quality Management

- ❖ **Delivering material as per customer's quality specification**
  - ❑ **In "make-to-order" scenario, the characteristics required by the customer are specified in the sales order and these characteristics need to be inspected during production**

- ❖ **Quality Certificate**
  - ❑ **When goods are shipped, a certificate can be produced by the system and can be included to document that the goods comply with the customer specification**

NPTEL

Information Systems for Quality Management

❖ **Quality Inspection for Delivery**

- **The system provides support to perform quality inspection before goods are issued for delivery to customer**

- **The quality inspection for a delivery is necessary to check the quality of a material or product before it leaves the premises of the manufacturer or vendor to ensure that the goods are in perfect condition before they are sent to the customer**

NPTEL

REFERENCES

- Laudon, K.C., and Laudon, J.P., (2016), Management Information Systems: Managing the Digital Firm
- Turban, E., Volonio, L., Wood, and R. Gregory., (2016), Information Technology for Management
- Rainer, K., Prince, B., and Watson, H., (2015), Management Information Systems
- De, Rahul., (2019), MIS: Managing Information Systems in Business, Government and Society
- Bagchi, Nirmalya., (2010), Management Information Systems

NPTEL



NPTEL

THANK YOU !



NPTEL ONLINE CERTIFICATION COURSES

NPTEL

Management Information Systems

Prof. Kunal Kanti Ghosh
VGSoM, IIT KHARAGPUR

Week 5: Module 5

Information Systems for Marketing

CONCEPTS COVERED

- Information Systems for Marketing (MKIS)
- Business Processes and Subsystems for Marketing

NPTEL

Information Systems for Marketing (MKIS)

- The role of this type of information system is to help the managers identify key customers

- It can be also used to generate reports on sales based on
  - ✓ time,
  - ✓ region, and so on.

NPTEL

Information Systems for Marketing (MKIS)

- If a customer is identified as a potential source of revenue, their loyalty can be appreciated in the form of;
  - ✓ personalized after-sales service,
  - ✓ discounts,
  - ✓ new offerings, or
  - ✓ prioritizing their orders

NPTEL

Information Systems for Marketing (MKIS)

- Customer satisfaction is key to the success of business today, and so it becomes important to identify the target market accurately

- Most businesses with similar offerings try to differentiate themselves through the experience they create for customers

NPTEL

Information Systems for Marketing (MKIS)

- Marketing information systems help with market segmentation
  - so that the appropriate marketing activities can be targeted at the relevant segment

- Very few firms today produce products for the mass market

NPTEL

Information Systems for Marketing (MKIS)

- Marketing information system processes data such as;
  - ✓ sales figures,
  - ✓ past trends, and
  - ✓ government policies to get the useful information that can be utilized by managers for taking decisions

NPTEL

Information Systems for Marketing (MKIS)

- When you buy an item from a retail store, your billing data is fed into the system

- Managers can,
  - ✓ take a look at the consolidated statements of such data. and
  - ✓ draw conclusions about the products that are in demand

NPTEL

Information Systems for Marketing (MKIS)

- If a discount scheme was offered, managers can understand the impact that the scheme had on sales

- Similarly, the performance of sales team in the store can also be evaluated based on,
  - the data from the marketing information system

NPTEL

Information Systems for Marketing (MKIS)

- **Marketing is concerned with:**
  - **Identification of the customer group for an organization's products or services,**
  - **Determination of what these customer group need,**
  - **Planning and enabling development of the identified products/services to meet those needs, and**
  - **Advertising and promoting these products and services**

NPTEL

Information Systems for Marketing (MKIS)

- A ”marketing information system” is defined as,
  - a "system in which marketing data is
    - ✓ formally gathered,
    - ✓ stored,
    - ✓ analyzed, and
    - ✓ distributed to marketing managers in accordance with their informational needs on a regular basis“ (Jobber, 2007)

NPTEL

Information Systems for Marketing (MKIS)

- Marketing information is gathered continuously from sources inside and outside an organization or a store

NPTEL

Information Systems for Marketing (MKIS)

- An overall “marketing information system” can be defined as,

- ✓“a set structure of procedures and methods for the regular and planned collection, analysis and presentation of information for use in making marketing decisions" (Kotler et al., 2006)

NPTEL

Information Systems for Marketing (MKIS)

- MKIS helps to monitor the degree of the marketing success
- It helps the managers to ensure that the objectives of the operations function is also achieved

NPTEL

Information Systems for Marketing (MKIS)

❖ **Relevance of MKIS:**

➢ **When companies diversify into new markets, both**

✓ **the companies and**

✓ **customer's point of view**

- **are needed to be handled by the marketing managers**

❑ **Therefore, there would be greater need for marketing information**

NPTEL

Information Systems for Marketing (MKIS)

❖ **Relevance of MKIS:**

- **Marketing managers should be aware of**
  - ✓ **the drivers behind consumers' preference for a particular brand**

- **They should know about the points**
  - ✓ **that distinguish their brands from that of the rivals**

- **This awareness is possible only with the help of a well-designed effective MKIS**

NPTEL

Information Systems for Marketing (MKIS)

❖ **Relevance of MKIS:**

➢ **Marketing managers should also be aware of**

✓ **the response of the consumers towards different strategic implementations, and**

✓ **technological developments adopted by the competitors with respect to non-price grounds of competition**

NPTEL

Information Systems for Marketing (MKIS)

❖ **Relevance of MKIS:**

➢ **An effective MKIS is designed to,**

❑ **gather,**

❑ **integrate,**

❑ **process, and**

❑ **distribute**

✓ **such information comprehensively from all sources, including that from marketing research**

NPTEL

Information Systems for Marketing (MKIS)

- Marketing information system consists of several subsystems

- Each of these subsystems,
  - perform a specific task, and
  - supplies specific information to the marketing managers

NPTEL

Information Systems for Marketing (MKIS)

- Some of the subsystems,
  - ✓ use advanced analytical models to provide
  - ❑ information about the present environment, as well as
  - ❑ the likely future scenarios

NPTEL

Information Systems for Marketing (MKIS)

❖ **Subsystems of MKIS:**

- **Marketing Intelligence Subsystem**
- **Market Research Subsystem**
- **Promotion and Advertising Subsystem**

NPTEL

Information Systems for Marketing (MKIS)

❖ **Marketing Intelligence Subsystem**

- **It is the main source used by managers for**
  - ✓ **obtaining daily information of the external environment,**
  - ✓ **hence, assists marketing managers to react to the changing scenario in a rapid manner**

NPTEL

Information Systems for Marketing (MKIS)

❖ **Marketing Intelligence Subsystem**

- **It filters data and stores critical information related to**
  - ✓ **competitor's transaction data, and**
  - ✓ **market-related intelligence data**

- **This subsystem works as a key information provider for other subsystems in MKIS**

NPTEL

Information Systems for Marketing (MKIS)

- ❖ **Marketing Intelligence Subsystem**

- ➢ **Sources of Marketing Intelligence**
  - ✓ **Marketing Managers**
  - ✓ **Sales Forces**

NPTEL

Information Systems for Marketing (MKIS)

- ❖ **Marketing Intelligence Subsystem**

- ➢ **Sources of Marketing Intelligence**
  - ✓ **Middlemen**
  - ✓ **Specialists**
  - ✓ **Marketing Information Section**

NPTEL

REFERENCES

- Laudon, K.C., and Laudon, J.P., (2016), Management Information Systems: Managing the Digital Firm
- Turban, E., Volonio, L., and Wood, R.G., (2016), Information Technology for Management
- Rainer, K., Prince, B., and Watson, H., (2015), Management Information Systems
- De, Rahul., (2019), MIS: Managing Information Systems in Business, Government and Society
- Bagchi, N., (2010), Management Information Systems

NPTEL



NPTEL

THANK YOU !



NPTEL ONLINE CERTIFICATION COURSES

NPTEL

Management Information Systems

Prof. Kunal Kanti Ghosh
VGSoM, IIT KHARAGPUR

Week 6: Module 1

Information Systems for Marketing (Continued)

CONCEPTS COVERED

- Information Systems for Marketing (MKIS)
- Business Processes and Subsystems for Marketing

NPTEL

Information Systems for Marketing (MKIS)

❖ **Marketing Research Subsystem**

➢ **The subsystem gathers data on**

✓ **consumers,**

✓ **markets and**

✓ **competitors from the open market**

- **by conducting surveys**

NPTEL

Information Systems for Marketing (MKIS)

❖ **Marketing Research Subsystem**

- **It is used to collect primary and secondary data, and displays the results in the forms of reports**
- **It collects and analyzes key data about the,**
  - ✓ **marketing process of the organization, as well as**
  - ✓ **the scenario of the market and competition**

NPTEL

Information Systems for Marketing (MKIS)

❖ Marketing Research Subsystem

➢ Areas of Marketing Research:

✓ Product Research

✓ Advertising Research

✓ Consumer Research

NPTEL

Information Systems for Marketing (MKIS)

❖ **Marketing Research Subsystem**

➢ **Areas of Marketing Research:**

✓ **Sales Research**

✓ **Corporate Research**

✓ **Competitor Research**

NPTEL

Information Systems for Marketing (MKIS)

❖ Marketing Research Subsystem

➢ Marketing Research Process:

✓ Define the problem and research objectives

✓ Develop the research plan

✓ Collect the information

✓ Analyse the information

✓ Present the findings

NPTEL

Information Systems for Marketing (MKIS)

❖ **Promotion and Advertising Subsystem**

➢ **Marketing managers need to determine**

❑ **the optimal ways to distribute their products and services through a combination of**

✓ **electronic,**

✓ **mobile, and**

✓ **physical channels**

NPTEL

Information Systems for Marketing (MKIS)

- **Promotion and Advertising Subsystem**

- **For example, integrating an advertising campaign with other online and offline advertising initiatives may generally provide the best overall results**

- **Self-service kiosks may cut labour costs for retailers and thereby, reduce the customers' check-out times**

NPTEL

Information Systems for Marketing (MKIS)

❖ **Promotion and Advertising Subsystem**

➢ **The promotion and advertising subsystem is often the most elaborated in the marketing information system, since it supports both**

✓ **personal selling, and**

✓ **advertising**

NPTEL

Information Systems for Marketing (MKIS)

❖ **Promotion and Advertising Subsystem**

- **Media selection packages assist in selecting a mix of avenues to persuade the potential purchaser, including**
  - **direct mail,**
  - **television,**
  - **print media, and**
  - **the electronic media such as the Internet and the web in particular**
- **The effectiveness of the selected media mix is monitored and its composition is continually adjusted**

NPTEL

Information Systems for Marketing (MKIS)

❖ **Promotion and Advertising Subsystem**

➢ **This subsystem,**

- ❑ **gives information about the success of each medium of communication and,**
- ❑ **helps the marketing department to formulate communication strategies**
- ✓ **based on the information of return on investment on each medium of communication**

NPTEL

Information Systems for Marketing (MKIS)

❖ **Promotion and Advertising Subsystem**

➢ **This subsystem gives detailed review reports on**

❑ **each rupee spent on,**

✓ **advertisement,**

✓ **promotion or**

✓ **publicity, and**

❑ **advertising techniques and analysis of their impact**

NPTEL

Information Systems for Marketing (MKIS)

- **Pricing Subsystem**

- **The subsystem uses advanced analytics to help managers take decisions on pricing of products/services**

- **Pricing decisions find support from this subsystem and access to databases that contain industry prices**

NPTEL

Information Systems for Marketing (MKIS)

❖ **Pricing Subsystem**

- **This subsystem provides an opportunity to**
  - **finely segment customer groups, and**
  - **charge different prices depending on,**
    - ✓ **the combination of products and services provided, as well as**
    - ✓ **the circumstances of the sales transaction**
- **Report on correlation of prices and sales**

NPTEL

Information Systems for Marketing (MKIS)

- **Internal Reports Subsystem**

- **It records various data from different departments of a company which is regarded as a major source of information**

NPTEL

Information Systems for Marketing (MKIS)

❖ **Internal Reports Subsystem**

➢ **Marketing managers depend on data from internal reports about,**

- ✓ **orders,**
- ✓ **sales,**
- ✓ **prices,**
- ✓ **costs,**
- ✓ **inventory levels,**
- ✓ **receivables,**
- ✓ **payables, and so on**

NPTEL

Information Systems for Marketing (MKIS)

- **Internal Reports Subsystem**

- **The primary process associated with internal reports subsystem is the order-to-cash cycle**

- **Sales representatives, dealers, and retailers are responsible for processing the orders**

- **The sales department prepares invoices and sends copies of the same to various departments while out-of-stock items are back-ordered**

NPTEL

Information Systems for Marketing (MKIS)

- **Internal Reports Subsystem**

  - **Shipped items are accompanied by shipping and billing documents that are sent to various departments**

  - **Companies need to perform these steps quickly and accurately because customers favor firms that can promise timely delivery**

NPTEL

Information Systems for Marketing (MKIS)

❖ **Internal Reports Subsystem**

➢ **Today many companies organize information in databases – customer databases, product databases, salesperson databases, and so forth – and then combine data from different databases**

❑ **The customer database contains every customer's name, address, past transactions, demographics, and psychographics (activities, interests, and opinions)**

NPTEL

Information Systems for Marketing (MKIS)

- **Internal Reports Subsystem**

- **Sales data presented in a graphic format, provide regular sales trend information and highlight whether certain customer types need to be targeted or focused**

NPTEL

Information Systems for Marketing (MKIS)

❖ **Internal Reports Subsystem**

❑ **Price information by product line and comparison with competitors help managers to monitor market trends;**

✓ **Analyzed by customer type, the subsystem helps to check price trends in customer groups**

✓ **Customer profitability analysis help identify high-profit and high-growth customers and target marketing efforts in order to retain and develop these accounts**

NPTEL

Information Systems for Marketing (MKIS)

❖ **Internal Reports Subsystem**

➢ **Stock level data and trends in key accounts or distributors, focus on whether different outlets need support**

NPTEL

Information Systems for Marketing (MKIS)

- ❖ Internal Reports Subsystem
  - ➢ Market support information
    - ❑ coordinate the effects of marketing promotions through,
      - ✓ advertising,
      - ✓ direct marketing,
      - ✓ trade incentives,
      - ✓ consumer competitions and so on
    - ❑ and helps to determine whether decisions are being made effectively

NPTEL

Information Systems for Marketing (MKIS)

❖ **Internal Reports Subsystem**

➢ **Competitive information and reviewing competitors' promotions and communications**

✓ **to see if the company is doing it better or worse than competitors, can improve market targets**

NPTEL

REFERENCES

- Laudon, K.C., and Laudon, J.P., (2016), Management Information Systems: Managing the Digital Firm
- Turban, E., Volonio, L., and Wood, G., (2016), Information Technology for Management
- Rainer, K., Prince, B., and Watson, H., (2015), Management Information Systems
- De, Rahul., (2019), MIS: Managing Information Systems in Business, Government and Society
- Bagchi, Nirmalya., (2010), Management Information Systems

NPTEL



NPTEL

THANK YOU !



NPTEL ONLINE CERTIFICATION COURSES

NPTEL

Management Information Systems

Prof. Kunal Kanti Ghosh
VGSoM, IIT KHARAGPUR

Week 6: Module 2:
Information System for Sales & Distribution

CONCEPTS COVERED

- Information System for Sales and Distribution (SDIS)
- Subsystems for Sales and Distribution

NPTEL

Information System for Sales and Distribution (SDIS)

- Sales and Distribution Information Systems (SDIS) provide support for,
  - ✓ sales and distribution function of,
  - ➢ any organization's products and services

NPTEL

Information System for Sales and Distribution (SDIS)

- Sales function is concerned with,
  - ✓ contacting the customers,
  - ✓ taking orders,
  - ✓ selling the products/services,
  - ✓ Following-up on sales, and
  - ✓ providing customer support

NPTEL

Information System for Sales and Distribution (SDIS)

- **Sales function is also responsible for,**
  - ✓ **ensuring that the accounts receivables due to the organization is minimized**

NPTEL

Information System for Sales and Distribution (SDIS)

- **Subsystems of SDIS:**
  - **Sales Forecasting**
  - **Sales Force Automation**
  - **Sales Order Processing**
  - **Sales Analysis and Reporting**

NPTEL

Information System for Sales and Distribution (SDIS)

❖ **Master Data related to SDIS:**

➢ **In order to provide operational support to sales and distribution function, the information system requires the following important master data:**

- ❑ **Customer Master**
- ❑ **Material Master**
- ❑ **Condition Master**
- ❑ **Route Master, and so on**

NPTEL

Information System for Sales and Distribution (SDIS)

❖**Sales Forecasting Subsystem:**

- **Based on the planned marketing mix and outstanding orders,**
  - **sales are forecast, and**
  - **a full marketing plan is developed**

- ***Sales forecasting*** **is an area where any quantitative methods employed must be tempered with human insight and experience**

NPTEL

Information System for Sales and Distribution (SDIS)

❖**Sales Forecasting Subsystem:**

➢**The actual sales will depend to a large degree on the dynamics of the environment**

➢**Qualitative techniques are generally used for,**

❑***Environmental Forecasting***

✓**An attempt to predict the social, economic, legal, and technological environment in which the company will try to realize its plans**

NPTEL

Information System for Sales and Distribution (SDIS)

❖Sales Forecasting Subsystem:

➢Sales forecasting uses numerous techniques, which include:

❑Group Decision-Making Techniques
✓that are used to elicit broad expert opinion

❑Scenario Analysis
✓in which each scenario in this process is a plausible future environment

❑ Time-Series Analysis
✓for extrapolation of trends and cycles

NPTEL

Information System for Sales and Distribution (SDIS)

❖**Sales Force Automation:**

➢**Involves equipping sales people with portable computers tied into the corporate information systems**

❑**This gives the sales people instantaneous access to information and frees them from the reporting paperwork**

NPTEL

Information System for Sales and Distribution (SDIS)

❖ **Sales Force Automation:**

➢ **This increases selling time and the level of performance**

➢ **Access to corporate databases is sometimes,**

❑ **accompanied by access to corporate expertise, either**

✓ **by being able to contact the experts, or**

✓ **by using expert systems that help specify the product meeting customer requirements**

NPTEL

Information System for Sales and Distribution (SDIS)

❖ **Sales Order Processing Subsystem:**

➢ **This subsystem provides support at every stage of the sales order processing cycle in the context of the following activities:**

❑ **Enquiry by customer (Verbal/Phone/Mail/Order Form)**

❑ **Inform the customer about availability of material**

NPTEL

Information System for Sales and Distribution (SDIS)

❖ **Sales Order Processing Subsystem:**

- **This subsystem provides support at every stage of the sales order processing cycle in the context of the following activities:**
- **Informing the customer about the date by when the material can be delivered if the stock is not readily available**
- **Sending quotations (price of the item, date of delivery, and other commercial terms) against enquiry**

NPTEL

Information System for Sales and Distribution (SDIS)

❖ **Sales Order Processing Cycle:**

➢ **Order confirmation after doing the credit check of customer**

- **Check for completeness and accuracy of order entry**
- **Update the sales person's incentive**

NPTEL

Information System for Sales and Distribution (SDIS)

❖ **Sales Order Processing Cycle:**

- **Inventory Sourcing (to define from where the material need to be sourced, externally from vendor or internally need to be produced)**

- **Locate the nearest warehouse from where order can be served and transmission of order information to the warehouse (or generation of challan)**

NPTEL

Information System for Sales and Distribution (SDIS)

❖ **Sales Order Processing Cycle:**

- **Order is picked and packed at the warehouse**
- **Arrange transportation for shipment**
- **Billing and Invoicing**
- **Dispatch of Goods**

NPTEL

Information System for Sales and Distribution (SDIS)

❖ **Sales Order Processing Cycle:**

- **Receipt of goods by customers and signing proof of delivery**
- **Collection of payment**
- **Communicate order status to customers on a regular basis**

NPTEL

Information System for Sales & Distribution (SDIS)

- **Analysis and Reporting Subsystem:**
  - **Analysis of Sales by Regions, Customer Groups, Sales Persons, and so on**
  - **Sales Target versus Achievement**
  - **Performance of the Sales Outlets**
  - **Effect of Model Changes on Sales**

NPTEL

Information System for Sales & Distribution (SDIS)

- **Analysis and Reporting Subsystem:**
  - **Seasonal Variations in Sales**
  - **Market Share and Trends**
  - **Costs of Sales Campaigns versus Benefits Achieved**

NPTEL

Information System for Sales & Distribution (SDIS)

- **Analysis and Reporting Subsystem:**
  - **Timing of Special Sales Campaigns**
  - **Report on Deployment of Sales Force, and**
  - **Consequent Achievement of Targets**

NPTEL

REFERENCES

- Laudon, K.C., and Laudon, J.P., (2016), Management Information Systems: Managing the Digital Firm
- Turban, E., Volonio, L., and Wood, G., (2016), Information Technology for Management
- Rainer, K., Prince, B., and Watson, H., (2015), Management Information Systems
- De, Rahul., (2019), MIS: Managing Information Systems in Business, Government and Society
- Bagchi, N., (2010), Management Information Systems

NPTEL



NPTEL

THANK YOU !



NPTEL ONLINE CERTIFICATION COURSES

NPTEL

Management Information Systems

Prof. Kunal Kanti Ghosh
VGSoM, IIT KHARAGPUR

Week 6: Module 3

Information Systems for Finance and Accounting

CONCEPTS COVERED

- Information Systems for Finance and Accounting
- Business Processes and Subsystems for Finance and Accounting

NPTEL

Information Systems for Finance and Accounting

- The finance function is responsible for
  - ✓ managing the organization's financial assets, such as cash, stocks, bonds, and
  - ✓ other investments
  - ▪ in order to maximize the return on those financial assets

- The finance function is also in charge of managing the capitalization of the firm

NPTEL

Information Systems for Finance and Accounting

- The finance function is also in charge of managing the capitalization of the firm

- In order to determine whether the firm is getting the best return on its investments, the finance function must obtain a considerable amount of information from sources external to the firm

NPTEL

Information Systems for Finance and Accounting

- The accounting function is responsible for maintaining and managing an organization's financial records –
  - ✓ receipts,
  - ✓ depreciation,
  - ✓ payroll, and so on

NPTEL

Information Systems for Finance and Accounting

- Strategic Information for:
  - Methods of Financing
  - Planning for Long-term Profits
  - Pricing Policies
  - Tax Planning
  - Design of the Firm's Portfolio of Investments

NPTEL

Information Systems for Finance and Accounting

- Tactical Information for:
  - Preparation of Short-term Budgets
  - Variation between Budget And Expenses
  - Monitoring Outstanding Payments / Receipts
  - Monitoring Credit and Payment Status
  - Monitoring Cost Increase and Pricing
  - Knowing the Impact of Taxation on Pricing

NPTEL

Information Systems for Finance and Accounting

❖Operational Information for:

- Periodic Financial Report
- Disseminating Budget Status to all Functional Managers
- Tax Returns
- Share Transfer
- Preparing Profit and Loss Account
- Tracking Payments and Receipts
- Maintaining Payroll and Provident Fund (PF) Accounts

NPTEL

Information Systems for Finance and Accounting

- The financial management information system (FMIS) finds extensive application in managing the financial matters of an organization

- An FMIS allows managers to evaluate the sources of funds that would be best suited for the firm, those that bear the least risk, and those that reaps the most gain/profit

NPTEL

Information Systems for Finance and Accounting

- The FMIS can also be used to decide how much money should be raised from equity and how much from debt
- Such a system has wide application in banks
- It helps to manage all the important daily transactional data from withdrawals to demand draft creation
- It also processes these data and converts them into useful information, and helps to store the information securely

NPTEL

Information Systems for Finance and Accounting

❖General Ledger (GL) Accounting:

- ➢It is a place to record all the business transactions occurring across the organization
- ➢Acts as a repository of all accounting transactions
- ❑All functions, namely AR, AP, and FA affect the general ledger in terms of passing accounting transactions
- ❑Similarly, other functions like inventory and production also pass accounting entries to the general ledger

NPTEL

Information Systems for Finance and Accounting

- ❖ General Ledger (GL) Accounting:
  - ➢ Provides reliable and relevant financial statements timely:
    - ❑ Financial statements like,
      - ✓ Profit and Loss Account
      - ✓ Balance Sheet
      - ✓ Financial Ratios
      - ✓ Budget versus Actual
      - ✓ Cash Flows
      - ✓ Current month/quarter/year financial analysis with respect to corresponding prior period, and so on

NPTEL

Information Systems for Finance and Accounting

❖ **General Ledger (GL) Accounting:**

- **The GL subsystem can produce the financial statements at any point of time for the decision-makers**
- **To consolidate financial statements of multiple units at different locations**
- **Records all the transactions of various units (organizations may have several units or entities) at different locations**
- **Generate periodic financial statements at the corporate level**

NPTEL

Information Systems for Finance and Accounting

❖General Ledger (GL) Accounting:

➢To perform other control functions:

❑GL subsystem also accepts standalone inputs such as adjustment transactions from one legal entity to another

❑Revaluation of foreign currency at the end of each quarter is also taken care of

NPTEL

Information Systems for Finance and Accounting

- **Accounts Payables (AP) and Accounts Receivables (AR):**

  - **It is a place where the data pertaining to vendor payments is recorded, along with cash and money received from the customers**

NPTEL

Information Systems for Finance and Accounting

❖ **Accounts Receivables (AR):**

- **To ensure proper and prompt billing of the goods sent to customer**

- **To accurately prompt the outstanding dues, preferably in accordance with the age (total number of days by which it is outstanding from the due date)**

NPTEL

Information Systems for Finance and Accounting

❖ **Accounts Receivables (AR):**

➢ **To identify the potential risk situation with respect to issues of collection**

❑ **If the collection of a particular invoice is not realized within certain days (due to a default from sales function or the customer creates an issue like unavailability of funds), then in such a case, the subsystem,**

✓ **marks this as bad debt and**

✓ **logically the customer can be put in high risk zone for future reference**

NPTEL

Information Systems for Finance and Accounting

❖Accounts Receivables (AR):

➢To manage the customer's credit notes

❑Credit note is passed on to the customer's account when

✓Wrong invoice is made, or

✓Customer has returned the goods

NPTEL

Information Systems for Finance and Accounting

❖ **Accounts Receivables (AR):**

➢ **To assist in the organization's collection efforts**

❑ **Collection is one of the most important activities for any organization and AR function plays an important role in sending reminders to the customers for payments**

NPTEL

Information Systems for Finance and Accounting

- **Accounts Receivables (AR):**

  - **To provide detailed customer history and sales analysis data for computing the credit limit**

  - **To assist in the cash management function**

  - **The subsystem helps AR function to coordinate with cash management team to reconcile the bank statements**

NPTEL

Information Systems for Finance and Accounting

❖ **Accounts Payable (AP):**

➢ **To accurately record the liabilities**

❑ **The subsystem records the liability of goods or raw materials or services purchased from suppliers to know the expected outflow**

❑ **Liability is booked on the basis of receipt of actual bills from the suppliers**

NPTEL

Information Systems for Finance and Accounting

❖ **Accounts Payable (AP):**

➢ **To disburse the payments and maintain the complete records**

❑ **Once the payment is disbursed, the subsystem maintains a complete trail of all payments**

❑ **All relevant documents are maintained including purchase orders, bills from suppliers, payment vouchers, cheques, bank references, and so on**

IPTEL

Information Systems for Finance and Accounting

❖ **Accounts Payable (AP):**

➢ **To coordinate with cash management team**

❑ **The subsystem provides all relevant information in bank reconciliation**

❑ **The subsystem helps in preparing cash flow statements taking into account all the payments**

NPTEL

Information Systems for Finance and Accounting

❖ **Accounts Payable (AP):**

➢ **To take rejection of materials into account**

❑ **Based on rejection of raw material from suppliers, a proper adjustment document (debit note) is sent to the suppliers' account to reduce the liability, thereby decreasing the amount of payment**

NPTEL

Information Systems for Finance and Accounting

- **Accounts Payable (AP):**
  - **deals with all payment-related issues covering**
    - **payments to the suppliers,**
    - **staff payments,**
    - **travel expenses,**
    - **incidental expenses for office work, and**
    - **payments to consultants and other professionals**

NPTEL

Information Systems for Finance and Accounting

❖**Asset Accounting:**

➢**The subsystem provides**

✓***support for procurement of capital equipment***

✓***support for monitoring movements of fixed assets***

✓***support for the sale of fixed assets***

✓***support for maintaining fixed asset register which is a statutory requirement***

✓***support for calculating depreciation***

NPTEL

Information Systems for Finance and Accounting

- **Financial and Accounting Information Systems also provides**
  - ✓ ***Support for managing the cash and bank-related processes***
  - ✓ ***Support for monitoring the organization's regulatory requirements***
  - ✓ ***Overhead cost control***
  - ✓ ***Activity Based Costing***
  - ✓ ***Profitability Analysis***
  - ✓ ***Business Planning and Budgeting***

NPTEL

REFERENCES

- Laudon, K.C., and Laudon, J.P., (2016), Management Information Systems: Managing the Digital Firm
- Turban, E., Volonio, L., and Wood, G., (2016), Information Technology for Management
- Rainer, K., Prince, B., and Watson, H., (2015), Management Information Systems
- De, Rahul., (2019), MIS: Managing Information Systems in Business, Government and Society
- Bagchi, N., (2010), Management Information Systems

NPTEL



NPTEL

THANK YOU !



NPTEL ONLINE CERTIFICATION COURSES

NPTEL

Management Information Systems

Prof. Kunal Kanti Ghosh
VGSoM, IIT KHARAGPUR

Week 6: Module 4

Information Systems for Human Resource Management

CONCEPTS COVERED

- Information Systems for Human Resources Management (HRM)
- Associated Business Processes and Subsystems for HRM

NPTEL

Information Systems for Human Resource Management

- Organizations need to,
  - find,
  - recruit,
  - motivate, and
  - train employees
  - ✓ to achieve their objectives and succeed in their workplace

NPTEL

Information Systems for Human Resource Management

- Human resources function is responsible for,
  - ✓ attracting,
  - ✓ developing, and
  - ✓ maintaining the firm's workforce

NPTEL

Information Systems for Human Resource Management

- In order to retain high performers, it is necessary to know,
  - how people feel about their
    - ✓ work environments,
    - ✓ their compensations, and
    - ✓ growth prospects

- In addition, maintenance of workplace health and safety is a prime requirement

NPTEL

Information Systems for Human Resource Management

- Human Resources Information Systems (HRIS) support activities such as,
- ✓ Identifying potential employees,
- ✓ Maintaining complete records on existing employees, and
- ✓ Creating programs to develop employees' talents and skills

NPTEL

Information Systems for Human Resource Management

- Strategic HRIS identifies the employees' requirements such as,
  - ✓ skills,
  - ✓ education level,
  - ✓ type of position,
  - ✓ number of position, and
  - ✓ cost
- ❑ for meeting the firm's long-term business plan

NPTEL

Information Systems for Human Resource Management

- The other dimensions of HRIS include,
  - ✓employment policies,
  - ✓procedures,
  - ✓communications, and
  - ✓compliance requirements

NPTEL

Information Systems for Human Resource Management

- Effective HR compliance programs are a necessity for all of the organizations in today's competitive and legal environment

- HR professionals need to monitor workplace and employment practices,
  - to ensure compliance with numerous laws and regulations

NPTEL

Information Systems for Human Resource Management

- HRIS focuses on,
  - ✓ legal and compliance responsibilities,
  - ✓ employee development,
  - ✓ talent management,
  - ✓ hiring, and
  - ✓ succession planning

NPTEL

Information Systems for Human Resource Management

- Major goal is to make best use of the available human resources in the organization

- To attain this goal, the information necessary at various levels of management may be classified as follows:

NPTEL

Information Systems for Human Resource Management

- ❖ Strategic Information
- ❑ Long-range Human Resource Requirements at Different Levels
- ❑ Policies on Wages And Incentives such as Stock Options and Bonus
- ❑ Policies on Human Resources Development and Training
- ❑ Policies on Personnel Welfare And Facilities
- ❑ Policies on Outsourcing Non-core Functions

NPTEL

Information Systems for Human Resource Management

- ❖ Tactical Information
- ❑ Performance Appraisal
- ❑ Demographic Make-up of Personnel and Its Impact on Retirement
- ❑ Production Incentives and Its Relation to Productivity

NPTEL

Information Systems for Human Resource Management

- ❖ Tactical Information
- ❑ Morale of Personnel
- ❑ Absentee Reduction
- ❑ Policies on Leave and Overtime
- ❑ Policies on Personnel Deployment

NPTEL

Information Systems for Human Resource Management

- ❖ Operational Information
- ❑ Routine Assessment
- ❑ Skills Inventory
- ❑ Loans / Advances and Recoveries
- ❑ Attendance Record
- ❑ Overtime Assignment

NPTEL

Information Systems for Human Resource Management

- Recruitment is a process of
  - ✓finding potential employees with the skills and talent needed by the organization,
  - ✓testing them, and
  - ✓deciding which ones to hire

NPTEL

Information Systems for Human Resource Management

- **Recruitment**
  - **Most organizations receive lot of applications for a job but might still have difficulty in finding the right person**
  - **With millions of resumes available online (for e.g., LinkedIn), companies are trying**
- **to find appropriate candidates on the "web", usually with the help of specialized search engines**

NPTEL

Information Systems for Human Resource Management

- **Recruitment**

  - **Companies also advertise thousands of jobs on the Web**

  - **Online recruiting can reach more candidates, which may bring in better applicants at a lower cost compared to traditional recruiting methods**

NPTEL

Information Systems for Human Resource Management

- **Recruitment**

  - **HRIS is connected to online job portals and through these job portals,**
  - **HRIS downloads and shortlists candidates (-via- screening and testing) for likely suitable positions in the organization**

  - **The subsystem also helps in work force planning and in managing bench**

NPTEL

Information Systems for Human Resource Management

- **Recruitment**

- **Once recruited, the employees become part of the corporate HR talent pool, which needs to be maintained and developed**

- **Several activities are supported by IT through the following subsystems:**

NPTEL

Information Systems for Human Resource Management

- **Performance Evaluation**

- **This subsystem maintains important measures of performance for each employee so that the performance of the employee can be rated**

NPTEL

Information Systems for Human Resource Management

- **Performance Evaluation**

- **Corporate managers can then analyse employees' performances with the help of intelligent systems,**
- **which provide systematic interpretation of performance over time**

NPTEL

Information Systems for Human Resource Management

- **Performance Evaluation**

- **The higher performers can then be rewarded and the lower level performers can be taken care of by appropriate training and development, or transfers and layoffs in some of the cases**

- **This type of information is very important for taking decisions on career planning of employees**

NPTEL

REFERENCES

- Laudon, K.C., and Laudon, J.P., (2016), Management Information Systems: Managing the Digital Firm
- Turban, E., Volonio, L., and Wood, G., (2016), Information Technology for Management
- Rainer, K., Prince, B., and Watson, H., (2015), Management Information Systems
- De, Rahul., (2019), MIS: Managing Information Systems in Business, Government and Society
- Bagchi, N., (2010), Management Information Systems

NPTEL



NPTEL

THANK YOU !



NPTEL ONLINE CERTIFICATION COURSES

NPTEL

Management Information Systems

Prof. Kunal Kanti Ghosh
VGSoM, IIT KHARAGPUR

Week 6: Module 5

Information Systems for Human Resource Management (Continued)

CONCEPTS COVERED

- Information Systems for Human Resources Management (HRM)
- Associated Business Processes and Subsystems for HRM

NPTEL

Training and Development Subsystem

- **Training and Development**

- **Employee training and development are important activities of the HR department**

- **Major issues include,**
  - ✓ **planning of classes, and**
  - ✓ **tailoring specific training programs to meet the needs of the organization and the employees**

NPTEL

Training and Development Subsystem

- **Training and Development**

- **HR departments may build a career development plan for each of the employees**

NPTEL

Training and Development Subsystem

- ❖ **Training and Development**

- ➢ **HRIS may support the,**
  - ✓ **planning,**
  - ✓ **monitoring, and**
  - ✓ **control of training and development related activities**
    - ▪ **by using workflow applications**

NPTEL

Training and Development Subsystem

❖Training and Development

➢This subsystem gives detailed information about,

✓Training Needs of Employees

✓Details of Trainers

✓Training Modules and Content

✓Training Feedback

✓Measure of Training Effectiveness

NPTEL

Training and Development Subsystem

❖ **Training and Development**

➢ **Now-a-days, innovative developments are taking place in the areas of,**

- **Intelligent Computer-aided Instruction, and**

❑ **Application of Multimedia Support for Instructional Activities**

NPTEL

Training and Development Subsystem

- **Training and Development**

- **For example, companies conduct much of their corporate trainings over their intranet or –via- the Web**

- **Use of Learning Management System**

NPTEL

Payroll Subsystem

- ❖ **Payroll Administration**

- ➢ **This subsystem provides support for payroll preparation**

- ➢ **The process is typically automated meaning that**
  - ✓ **pay checks are printed, and**
  - ✓ **money is transferred electronically into the employees' bank accounts**

NPTEL

Payroll Subsystem

- **Payroll Administration**

- **Several components need to be taken into consideration while computing the salary pay-out of employees**

NPTEL

Payroll Subsystem

- Payroll Administration

- These components might be,
  - ✓ tax deductions,
  - ✓ provident fund contributions,
  - ✓ leave and attendance data,
  - ✓ previous tax reports,
  - ✓ incentives and so on

NPTEL

Payroll Subsystem

- **Payroll Administration**

- **This module helps to manage the payment process by,**
  - **gathering data on all those components as mentioned earlier**

NPTEL

Payroll Subsystem

❖ **Payroll Administration**

- **This subsystem draws data from other modules such as,**
  - ✓ **the performance appraisal module,**
  - ✓ **attendance module, and so on,**
  - ▪ **which are integrated to generate the net payable amount**

- **This module is also linked to a financial management information system for the clearance of funds**

NPTEL

Payroll Subsystem

- **Payroll Administration**

- **This module is also linked to a financial management information system (FMIS) for the clearance of funds**

NPTEL

Payroll Subsystem

- ❖ Time and Attendance

- ➢ This module gathers data related to time and work

- ➢ It logs the number of hours each employee works every day, thereby, keeping tab on the efficiency levels of employees

NPTEL

Payroll Subsystem

❖Time and Attendance

➢This information can be used,

- to analyse the efficiency at the workplace on a given work day, or
- to calculate the average time required to perform a task

NPTEL

Payroll Subsystem

- **Benefits Administration**

- A major component of compensation is benefits
- Some of the benefits offered to the employees such as insurance or medical claims vary with the employees' job category

- This module can link this information to work out which benefit can be availed by which employee in a particular grade

NPTEL

Payroll Subsystem

- **Benefits Administration**

- **Employees' contributions to their organizations are rewarded by,**
  - ✓ **salaries/wages,**
  - ✓ **bonuses, and**
  - ✓ **other benefits (these benefits include those for health as well as contributions for pensions)**

NPTEL

Payroll Subsystem

- **Benefits Administration**

- **Providing flexibility in selecting benefits is viewed as a competitive advantage in large organizations – this can be supported by this subsystem**

- **Some companies have automated benefits enrolments**

NPTEL

Payroll Subsystem

- ❖ **Benefits Administration**

- ➢ **Employees can self-register for specific benefits using the,**
  - ✓ **corporate portal, or**
  - ✓ **voice technology**
- ❖ **Employees self-select desired benefits from a menu**

NPTEL

Payroll Subsystem

- **Benefits Administration**

- **This subsystem specifies the value of each benefit and the available benefits balance of each employee**

- **Some companies use intelligent agents to assist the employees and to monitor their actions**

NPTEL

Payroll Subsystem

- **Employee Self-Service Module**

- **This module allows employees to access data on their salary, deductions, leave, and attendance**

- **It reduces the need for employees to continually approach the HR department to avail such data**

NPTEL

Employee Relationship Subsystem

- **Employee Relationship Management**

- **Companies are developing employee relationship management applications with a view to manage their employees in a better manner**

- **A typical such application is a call-centre for the employees' problems**

NPTEL

REFERENCES

- Laudon, K.C., and Laudon, J.P., (2016), Management Information Systems: Managing the Digital Firm
- Turban, E., Volonio, L., and Wood, G., (2016), Information Technology for Management
- Rainer, K., Prince, B., and Watson, H., (2015), Management Information Systems
- De, Rahul., (2019), MIS: Managing Information Systems in Business, Government and Society
- Bagchi, N., (2010), Management Information Systems

NPTEL



NPTEL

THANK YOU !



NPTEL ONLINE CERTIFICATION COURSES

NPTEL

Management Information Systems

Prof. Kunal Kanti Ghosh
VGSoM, IIT KHARAGPUR

Week 7: Module 1

Information Systems (IS) Strategy

CONCEPTS COVERED

- **Information Systems (IS) Strategy**

NPTEL

Information Systems (IS) Strategy

- A strategy is a coordinated set of actions to fulfill objectives, purposes, and goals
- The essence of a business strategy is setting limits on what the business will seek to accomplish
- A business strategy is a plan articulating where a business seeks to go and how it expects to go there
- It is the means by which a business communicates its goals

NPTEL

Information Systems (IS) Strategy

- Strategy starts with a mission.
- A mission is a clear and compelling statement that states the purpose of existence of an organization
- It basically describes what the firm is all about.
- A mission statement sums up what is unique about the firm.

NPTEL

Information Systems (IS) Strategy

❖Mission statements about computer companies:

Company	Mission Statement
Amazon	We seek to be Earth's most customer-centric company for three primary customer sets: consumer customers, seller customers and developer customers
L L Bean	Sell good merchandise at a reasonable profit, treat your customers like human beings and they will always come back for more

NPTEL

Information Systems (IS) Strategy

- Management constructs this strategic plan in response to market forces, customer demands, and organizational capabilities.
- Some markets, such as those faced by manufacturers of laptop computers, and issuers of credit cards, are characterized by many competitors and a high level of competition such that product differentiation becomes increasingly difficult.

NPTEL

Information Systems (IS) Strategy

- Other markets, such as those for airlines and automobiles, are similarly characterized by high competition but product differentiation is better established
- Customer demands comprise the wants and needs of the individuals and companies who purchase the products and services available in the marketplace.
- Organizational capabilities include the skills and experience that give the corporation a currency that adds value in the marketplace

NPTEL

Information Systems (IS) Strategy

❖ **Generic Strategies Framework**

- **Michael Porter describes how businesses can build a sustainable competitive advantage**
- **"fundamental basis of above-average performance in the long run is sustainable competitive advantage."**
- **He identified three primary strategies for achieving competitive advantage:**
  - **Cost leadership – lowest-cost producer**
  - **Differentiation – product is unique**
  - **Focus – limited scope**

NPTEL

Information Systems (IS) Strategy

❖ **New Market Entrants:**

- **New entrants have a distinct advantage.**
- **They can consult the historical records on the existing players, analyze their weaknesses, and plug the mistakes committed by existing players.**
- **They are able to enter the market with innovative & improved products or services**

NPTEL

Information Systems (IS) Strategy

❖ **Porter's Competitive Forces Model:**

➢ **Traditional Players:**

- **There are certain companies that have been in the market for a long duration.**
- **These conventional market players pose stiff competition to other organizations.**
- **In the soft drink market, Coca-Cola and Pepsi can be considered the traditional competitors.**

NPTEL

Information Systems (IS) Strategy

❖ **New Market Entrants:**

➢ **Kinetic figured out that there was a need for a gearless scooter that could be started easily and effortlessly, and this is how it captured the market**

NPTEL

Information Systems (IS) Strategy

❖ **Substitute Products and Services:**

- **Constant innovation and upgradation can lead to the manufacturing of substitute products and new improved ways of rendering services.**
- **These substitute products and services are the greatest threat to an established organization**
- **For example, Airtel (provider of cell phone services) capturing a major portion of the market share of BSNL (landline provider)**

NPTEL

Information Systems (IS) Strategy

❖ **Customers:**

- **Organizations cannot take the loyalty of the customer for granted.**
- **The customer is always looking for better options.**
- **Organizations need to pay attention to retaining their customers by being sensitive to their changing demands**

NPTEL

Information Systems (IS) Strategy

❖**Suppliers:**

- **Organizations depend on their suppliers to provide them with right quantity & quality of products at the right price, and at the right time.**
- **Any delay in this chain can affect the delivery schedule of products or services**
- **In the Indian market, any sudden increase in the cost of material adversely impacts the organization.**

NPTEL

Information Systems (IS) Strategy

❖ **Suppliers:**

- **For example, to manufacture a car, a firm is dependent on suppliers for window glasses, tyres, electrical wires, lights and locks, and so on.**
- **A failure to procure any of the pieces, however minute, due to price or time constraints can throw off the entire production process.**

NPTEL

Information Systems (IS) Strategy

❖ **Strategies Formulated by Information Systems:**

➢ **Low-cost Leadership:**

- **Organizations need to reduce their operational costs so as to deliver the product or service at the most optimum price.**
- **Information systems play an important role in managing the operational cost of an organization.**

NPTEL

Information Systems (IS) Strategy

❖ **Strategies Formulated by Information Systems:**

➢ **Product Differentiation:**

- **Companies constantly strive to provide their customers with products or services that are different from those available in the market.**
- **This creates a positive attitude towards the brand in the eyes of the customer**

NPTEL

Information Systems (IS) Strategy

❖ **Strategies Formulated by Information Systems:**

➢ **Focus on Niche Markets:**

- **Information systems help organizations record the buying behavior and demands of the customers.**
- **This information enables companies to serve customers in an efficient manner.**

NPTEL

Information Systems (IS) Strategy

❖ **Strategies Formulated by Information Systems:**

➢ **Focus on Niche Markets:**

- **For example, organizations often offer loyalty cards to customers, so that when the customer make purchases, that information gets recorded in the central database of the organization and also on the magnetic chip on the loyalty card. Organizations give points for each purchase that can be accumulated and redeemed at a later date.**

NPTEL

Information Systems (IS) Strategy

❖ **Strategies Formulated by Information Systems:**

➢ **Strengthen Customer and Supplier Intimacy:**

- **Companies have found it to their advantage to encourage relations between customers and suppliers.**
- **e-Commerce companies like Flipkart allow customers to rate the suppliers.**
- **Since the companies keep track of all purchases made by the customer, and also of all the items viewed by the customers on their web-site, they can make suggestions to customers based on that data.**

NPTEL

Information Systems (IS) Strategy

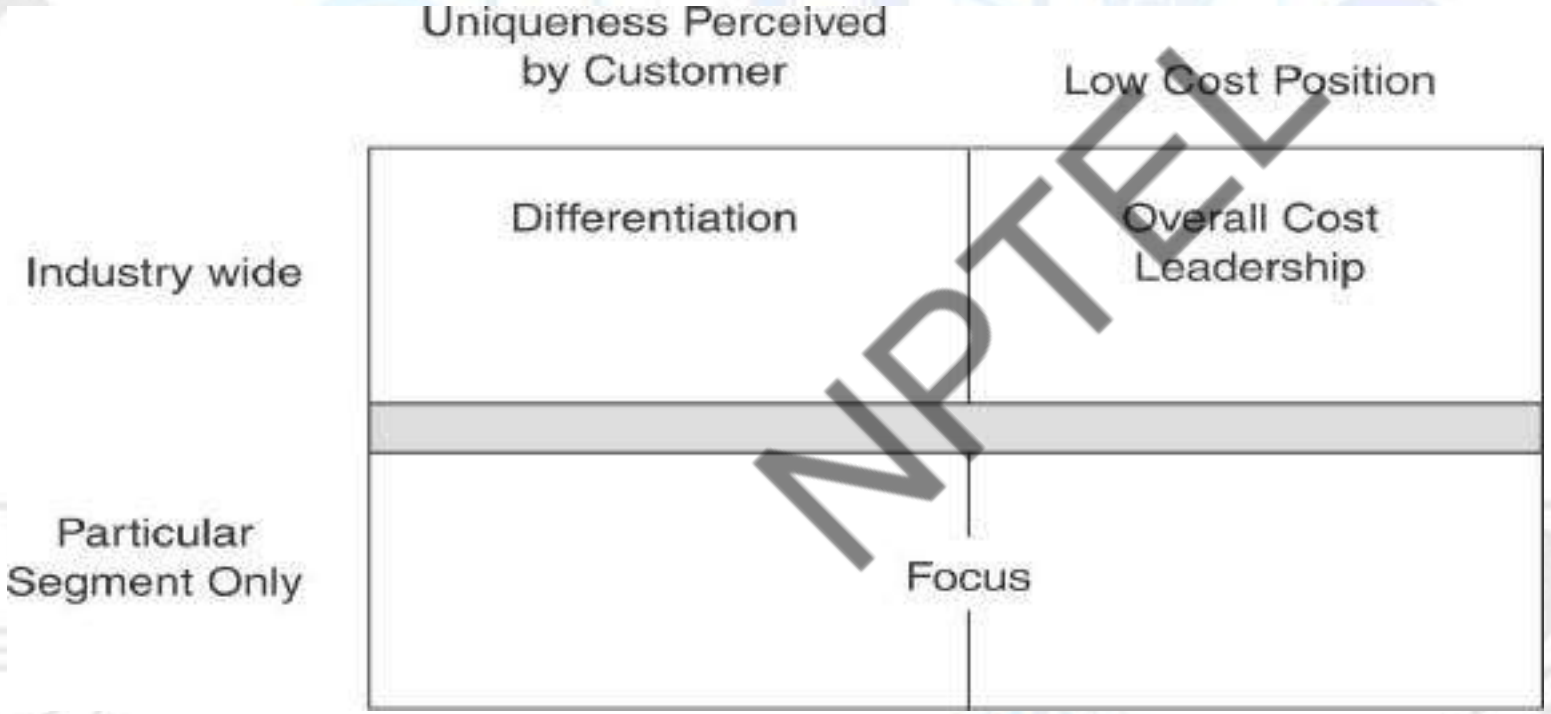
❖ **Generic Strategies Framework**

- **Michael Porter describes how businesses can build a sustainable competitive advantage**
- **"fundamental basis of above-average performance in the long run is sustainable competitive advantage."**
- **He identified three primary strategies for achieving competitive advantage:**
  - **Cost leadership – lowest-cost producer**
  - **Differentiation – product is unique**
  - **Focus – limited scope**

NPTEL

Information Systems (IS) Strategy

❖**Three Strategies for Competitive Advantage:**



NPTEL

Information Systems (IS) Strategy

❖ **Porter's Competitive Advantage:**

- **A company's overall business strategy drives all other strategies**
- **Porter defined these competitive advantages to represent various business strategies found in the marketplace**
- **Cost leadership results when the organization aims to be the lowest-cost producer in the marketplace**
- **Through differentiation, the organization qualifies its product or service in a way that allows it to appear unique in the marketplace**

NPTEL

Information Systems (IS) Strategy

❖ **Porter's Competitive Advantage:**

➢ **Focus allows an organization to limit its scope to a narrower segment of the market and tailor its offerings to that group of customers**

- ✓ **This strategy has two variants:**
  - **Cost focus**
  - **Differentiation focus**

NPTEL

Information Systems (IS) Strategy

❖Summary of strategic approaches and IT applications:

Strategic Approach	Key Idea	Application to Information Systems
Porter's generic strategies	Firms achieve competitive advantage through cost leadership, differentiation, or focus	Understanding which strategy is chosen by a firm is critical to choosing IS to complement the strategy

NPTEL

Information Systems (IS) Strategy

❖Summary of strategic approaches and IT applications:

Strategic Approach	Key Idea	Application to Information Systems
Dynamic environment strategies	Speed, agility, and aggressive moves and countermoves by a firm create competitive advantage	IS are critical to achieving the speed needed for moves and countermoves. IS are in a constant state of flux or development

NPTEL

Information Systems Strategy

- Information Systems (IS) strategy is the plan an organization uses to provide information services.
- IS allows a company to implement its business strategy.
- In IS strategy, the deployment plan for the required technology to support the business needs is documented.

NPTEL

Information Systems Strategy

- Business strategy is a function of:
  - Competition (What does the customer want and what does the competition do?)
  - Positioning (In what way does the firm want to compete?) and
  - Capabilities (What can the firm do?)
- IS strategy process help determine a company's capabilities

NPTEL

Information Systems Strategy

- Understanding business strategy means answering the following questions:
  - What is the business goal or objective?
  - What is the plan for achieving it?
  - What is the role of IS in this plan?
  - Who are the crucial competitors, and partners, and what is required of a successful player in this marketplace?
  - What are the industry forces in this marketplace?

NPTEL

Information Systems Strategy

- ❖Organizational strategy includes the organization's design as well as the choices it makes to define, set up, coordinate and control its work processes
- ❖Organizational strategy is a plan that answers the questions:
  - How will the company organize to achieve its goals and implement its business strategy?
- ❖Models for organizational design:
  - Business diamond
  - Managerial levers

NPTEL

Information Systems Strategy

❖Organizational strategy framework:

Framework	Key Idea	Usefulness in IS Discussions
Business Diamond	Four key components to an organization's design: people, structure, tasks & information control	Using IS in an organization will affect each of these components. This framework can be used to identify where these impacts are likely to occur

NPTEL

Information Systems Strategy

❖Organizational strategy framework:

Framework	Key Idea	Usefulness in IS Discussions
Managerial Levers	Organizational variables , control variables, and cultural variables are the levers managers can use to affect change in their organization	This model gives specific areas where IS can be used to manage and change the organization

NPTEL

Information Systems Strategy

- Understanding organizational strategy means answering the following questions:
  - What are the important structures & reporting relationships within the organization?
  - Who holds the decision rights to critical decisions?
  - What are the important people-based networks (social and informational) and how the managers can use them to get work done better

NPTEL

Information Systems Strategy

- Understanding organizational strategy means answering the following questions:
  - What are the characteristics, experiences, and skill levels of the people within the organization?
  - What are the key business processes?
  - What control systems (management and measurement systems) are in place?
  - What are the culture, values , and beliefs of the organization?

NPTEL

Information Systems Strategy

❖ **Infrastructure Components:**

- **Hardware: desktop units and servers**
- **Software: programs used to do business, to manage the computer itself, and to communicate between systems**
- **Network: physical means by which information is exchanged among hardware components, such as through a modem and dial-up network (in which case the service is provided by a vendor), or through a private digital network (in which case the service is provided by an internal unit)**
- **Data: bits & bytes of information (what data are in the system & where the rest are stored)**

NPTEL

Information Systems Strategy

- A basic framework to understand the decisions related to IS that an organization must make is shown in the next three slides.
- The purpose of this framework (IS Strategy Matrix) is to give the managers a high level view of the relation between the four IS infrastructure components & the other resource considerations that are the keys to IS Strategy

Infrastructure Components	What	Who	Where
Hardware	List of physical components of the system	System users and managers	Physical location of components

NPTEL

Information Systems Strategy

❖IS Strategy Matrix

Infrastructure Components	What	Who	Where
Software	List of programs, applications, and utilities	System users and managers	What hardware it resides on and physical location of hardware

NPTEL

Information Systems Strategy

❖IS Strategy Matrix

Infrastructure Components	What	Who	Where
Networking	Diagram of how hardware and software components are connected	System users and managers, company that provides the service	Where the nodes are located, and where the wires and other transport media are located

NPTEL

Information Systems Strategy

❖IS Strategy Matrix

Infrastructure Components	What	Who	Where
Data	Bits of information stored in the system	Owners of data, & data administrators	Where the information resides

NPTEL

Information Systems Strategy

❖ **Strategic Relationships:**

- **Organizational strategy and information strategy must complement each other.**
- **They must be designed so that they support, rather than hinder, each other.**
- **If a decision is made to change one corner of the IS strategy triangle, it is necessary to evaluate the other two corners to ensure that balance is preserved.**

NPTEL

Information Systems Strategy

❖ **Strategic Relationships:**

- **Changing business strategy without thinking through the effects on the organizational & IS strategies will cause the business to struggle until balance is restored.**
- **Likewise, changing IS or the organization alone will cause an imbalance**

NPTEL

REFERENCES

- Laudon, K. C., and Laudon, J. P. (2016), Management Information Systems: Managing the Digital Firm
- Turban, E., Volonio, L., Wood, R. Gregory.(2016), Information Technology for Management
- Pearlson, K.E., and Saunders, C.S. (2013), Strategic Management of Information Systems

NPTEL



NPTEL

THANK YOU !



NPTEL ONLINE CERTIFICATION COURSES

NPTEL

Management Information Systems

Prof. Kunal Kanti Ghosh
VGSoM, IIT KHARAGPUR

Week 7: Module 2
Aligning IT With Business Objectives

CONCEPTS COVERED

- **Aligning IT With Business Objectives**

NPTEL

Aligning IT With Business Objectives

- The more successfully a firm can align information technology with its business goals, the more profitable it will be
- Only one-quarter of firms achieve alignment of IT with the business
- About half of a business firm's profits can be explained by alignment of IT with business

NPTEL

Aligning IT With Business Objectives

- ❖Most businesses get it wrong.
- ❖Information technology takes on a life of its own and does not serve management and shareholder interests very well.
- ❖Instead of taking active role in IT, business people ignore it
  - Claim not to understand It
  - Tolerate failure

NPTEL

Aligning IT With Business Objectives

- Successful firms and managers understand
  - ✓ what IT can do and
  - ✓ how IT works,
  - ✓ take an active role in shaping its use, and
  - ✓ measure its impact on revenues and profits

NPTEL

Aligning IT With Business Objectives

❖Strategic Systems Analysis

➢To align IT with the business and

- use information systems effectively for competitive advantage,
- managers need to perform a strategic system analysis

NPTEL

Aligning IT With Business Objectives

❖Strategic Systems Analysis

➢To identify the types of systems that provide a strategic advantage to their firms,

- managers should ask the following questions:

NPTEL

Aligning IT With Business Objectives

- ❖Strategic Systems Analysis
- ✓What is the structure of the industry in which the firm is located?
- ✓What are some of the competitive forces at work in the industry?
- ✓Are there new entrants to the industry?
- ✓What is the relative power of suppliers, customers, and substitute products and services over prices?

NPTEL

Aligning IT With Business Objectives

- ❖ Strategic Systems Analysis
- ✓ What is the basis of competition
- ➢ quality, price, or brand?
- ✓ What are the direction and nature of change within the industry?
- ✓ From where are the momentum and change coming?

NPTEL

Aligning IT With Business Objectives

- **Strategic Systems Analysis**
  - ✓ **How is the industry currently using information technology?**
  - ✓ **Is the organization behind or ahead of the industry in its application of information systems?**

NPTEL

Aligning IT With Business Objectives

❖Strategic Systems Analysis

- What are the business, firm, and industry value chains for this particular firm?

✓How is the company creating value for the customer – through lower prices and transaction costs or higher quality?

NPTEL

Aligning IT With Business Objectives

- **Strategic Systems Analysis**
  - ✓ **Are there any places in the value chain where the business could create more value for the customer & additional profit for the company?**

NPTEL

Aligning IT With Business Objectives

- **Strategic Systems Analysis**
  - ✓ **Does the firm understand and manage its business processes using the best practices available?**
  - ✓ **Is it taking maximum advantage of SCM, CRM, and Enterprise Systems?**

NPTEL

Aligning IT With Business Objectives

- ❖ Strategic Systems Analysis
  - ✓ Does the firm leverage its core competencies?
  - ✓ Is the industry supply chain and customer base changing in ways that benefit or harm the firm?
  - ✓ Can the firm benefit from strategic partnerships?

NPTEL

Aligning IT With Business Objectives

- **Strategic Systems Analysis**
  - ✓ **Where in the value chain will information systems provide the greatest value to the firm?**

NPTEL

Aligning IT With Business Objectives

❖Strategic Systems Analysis

- Have we aligned IT with our business strategy and goals?

✓Have we correctly articulated our business strategy and goals?

NPTEL

Aligning IT With Business Objectives

- **Strategic Systems Analysis**
  - ✓ **Is IT improving the right business processes and activities to promote this strategy?**
  - ✓ **Are we using the right metrics to measure progress toward these goals?**

NPTEL

REFERENCES

- Laudon, K. C., and Laudon, J. P. (2016), Management Information Systems: Managing the Digital Firm
- Turban, E., Volonio, L., Wood, R. Gregory.(2016), Information Technology for Management
- Rainer, K., Prince, B., Watson, H.,(2015), Management Information Systems
- De, Rahul.,(2019), MIS: Managing Information Systems in Business, Government and Society
- Bagchi, Nirmalya., (2010), Management Information Systems

NPTEL



NPTEL

THANK YOU !



NPTEL ONLINE CERTIFICATION COURSES

NPTEL

Management Information Systems

Prof. Kunal Kanti Ghosh
VGSoM, IIT KHARAGPUR

Week 7: Module 3
Balanced Score Card

CONCEPTS COVERED

- **Basic Concepts of Balanced Score Card (BSC)**

NPTEL

Introduction to Balanced Score Card (BSC))

- As companies around the world transform themselves for competition that is based on information, their ability to exploit intangible assets has become more decisive than their ability to invest in and manage physical assets.

NPTEL

Introduction to Balanced Score Card (BSC))

- A balanced scorecard focuses on measurable outcomes on four dimensions of a business's performance: Financial, business process, customer, and learning and growth. Each dimension uses key performance indicators (KPIs) to understand how well an organization is performing on any of the dimensions at any time.

NPTEL

Introduction to Balanced Score Card (BSC))

- The framework of a balanced scorecard requires managers to focus on more than just financial performance. They must focus on things they are able to influence at the present time like customer satisfaction, business process efficiency, or employee training. The KPIs are developed by senior executives and are automatically provided to users through executive support systems.

NPTEL

Introduction to Balanced Score Card (BSC))

- Today's managers recognize the impact that measures have on performance.
- But they rarely think of measurement as an essential part of their strategy.

NPTEL

Introduction to Balanced Score Card (BSC))

- For example, executives may introduce new strategies and innovative operating processes intended to achieve breakthrough performance, then continue to use the same short-term financial indicators they have used for decades (e.g., ROI, Sales growth, Operating income and so on).

NPTEL

Introduction to Balanced Score Card (BSC))

- Hence they fail not only to introduce new measures to monitor new goals and processes but also to question whether or not their old measures are relevant to the new initiatives.

NPTEL

Introduction to Balanced Score Card (BSC))

- BSC enables companies to track financial results while simultaneously monitoring progress in building the capabilities and acquire the intangible assets they need for future growth.

NPTEL

Introduction to Balanced Score Card (BSC))

- BSC addresses a serious deficiency in traditional management systems : their inability to link a company's long-term strategy with its short term actions.
- BSC is a comprehensive framework that translates a company's strategic objectives into a coherent set of performance measures.

NPTEL

Introduction to Balanced Score Card (BSC))

- Balanced Score Card is a framework for operationalizing a firm's strategic plan by focusing on measurable outcomes on four dimensions of a firm's performance:
  - Financial
  - Customer
  - Business process
  - Learning and Growth

NPTEL

Introduction to Balanced Score Card (BSC))

- Managers need to select a limited number of critical indicators within each of the four perspectives.
- Thereby BSC helps managers to focus on strategic vision.

NPTEL

Introduction to Balanced Score Card (BSC))

- While traditional financial measures report on what happened last period without indicating how managers can improve performance in the next, BSC functions as the cornerstone of a company's current and future success.

NPTEL

Introduction to Balanced Score Card (BSC))

- Unlike conventional metrics , BSC provides balance between external measures like operating income and internal measures like new product development.

NPTEL

Introduction to Balanced Score Card (BSC))

- This balanced set of measures both reveals the trade-offs that managers have already made among performance measures and there is no need to make any further trade-offs among key success factors

NPTEL

Introduction to Balanced Score Card (BSC))

- Many companies that attempt to implement local improvement programs (e.g., process reengineering, total quality, employee empowerment) lack a sense of integration.

NPTEL

Introduction to Balanced Score Card (BSC))

- BSC is used as the language, the benchmark against which all new projects and businesses are evaluated.

NPTEL

Introduction to Balanced Score Card (BSC))

- Performance on each dimension is measured using Key Performance Indicators (KPIs), which are the measures proposed by senior management for understanding how well the firm is performing along any given dimension.

NPTEL

Introduction to Balanced Score Card (BSC))

- For example, one key indicator of how well an online retail firm is meeting its customer performance objectives is the average length of time required to deliver a package to consumer.

NPTEL

Introduction to Balanced Score Card (BSC))

- For a bank, a key indicator of business process performance is the length of time required to perform a basic function like creating a new customer account

NPTEL

Introduction to Balanced Score Card (BSC))

- The balanced scorecard framework is thought to be “balanced” because it causes managers to focus on more than just financial performance

NPTEL

Introduction to Balanced Score Card (BSC))

- In this view, financial performance is past history – the result of past actions – and
- ✓ managers should focus on the things they are able to influence today, such as
  - business process efficiency,
  - customer satisfaction, and
  - employee training

NPTEL

Introduction to Balanced Score Card (BSC))

- Once a scorecard is developed, the next step is
- ✓automating a flow of information to executives and managers for
- ✓each of the key performance indicators.

NPTEL

Introduction to Balanced Score Card (BSC))

- Clearly communicating the Vision & Strategy of the Company.
- Selecting a balanced set of measures.
- Aligning action plans to the Objectives.
- Linking organizational objectives to individual goals

NPTEL

Introduction to Balanced Score Card (BSC))

- **Translating the Vision : By relying on measurement, BSC forces the managers to come to agreement on the metrics they will operationalize to realize their vision.**

NPTEL

Introduction to Balanced Score Card (BSC))

- **Communicating & Linking : When a scorecard is disseminated up and down the organizational chart, strategy becomes a tool available to everyone.**

NPTEL

Introduction to Balanced Score Card (BSC))

- **Business Planning: BSC forces companies to integrate strategic planning and budgeting , thereby ensuring that financial budgets do indeed support strategic goals.**

NPTEL

Introduction to Balanced Score Card (BSC))

- **Feedback and Learning: Based on a mechanism for strategic feedback and review, BSC fosters a kind of learning – the ability to reflect on inferences & adjust theories about cause and effect relationships.**

NPTEL

REFERENCES

- Laudon, K. C., and Laudon, J. P. (2016), Management Information Systems: Managing the Digital Firm
- Turban, E., Volonio, L., Wood, R. Gregory.(2016), Information Technology for Management
- Rainer, K., Prince, B., Watson, H.,(2015), Management Information Systems
- De, Rahul.,(2019), MIS: Managing Information Systems in Business, Government and Society
- Bagchi, Nirmalya., (2010), Management Information Systems

NPTEL



NPTEL

THANK YOU !



NPTEL ONLINE CERTIFICATION COURSES

NPTEL

Management Information Systems

Prof. Kunal Kanti Ghosh
VGSoM, IIT KHARAGPUR

Week 7: Module 4

Data Centers, Virtualization and Cloud Computing

CONCEPTS COVERED

- **Data Centers, Virtualization and Cloud Computing**

NPTEL

Introduction to Virtualization and Cloud Computing

❖ **Data Centers, Virtualization and Cloud Computing**

- **On-premises data centers, virtualization, and cloud computing are types of IT infrastructures or computing systems**
- **Previously, companies owned their servers, storage, and network components to support their business applications, and these computing resources used to reside in their own premises**
- **Today, there exists several choices, each with their strengths, weaknesses and cost considerations**

NPTEL

Introduction to Data Centers, Virtualization and Cloud Computing

❖ **Data Centers:**

- A data center consists of a large number of network servers used for the storage, processing, management, distribution, and archiving of data, systems, web traffic, and enterprise applications
- It also refers to the building or facility that houses the servers and equipment

NPTEL

Introduction to Data Centers, Virtualization and Cloud Computing

❖ **Data Centers:**

- A data center is mainly used in organizations that run many different types of applications and have complex workloads
- However, a data center has limited capacity
- Once it is built, its capacity cannot be changed without installing additional equipment

NPTEL

Introduction to Data Centers, Virtualization and Cloud Computing

❖ **Data Centers:**

- **Data center failures disrupt all operations regardless of who owns the data center**
- **The outages in data centers indicate the risks of maintaining the complex and sophisticated technology needed to maintain digital services used by millions of people**

NPTEL

Introduction to Data Centers, Virtualization and Cloud Computing

❖ **Virtualization:**

- **Physical corporate data centers are rapidly being replaced by virtual infrastructure, called virtualization.**
- **A virtual infrastructure originally referred to one where software replaced hardware in a way that a "virtual machine" was accessible to provide computing power**

NPTEL

Introduction to Data Centers, Virtualization and Cloud Computing

❖ **Virtualization:**

- **Typically, computing capabilities, storage, and networking are provided by a third party or group of vendors, usually over the Internet or through a private network.**
- **Facilities available through virtual architectures are servers, storage, backup, network, and disaster recovery**

NPTEL

Introduction to Data Centers, Virtualization and Cloud Computing

❖ **Virtualization:**

- In a virtualized desktop, the user's device locally accesses desktop software on a remote server.
- Virtualization is a way to design architecture because it enables resources to be shared and allocated as needed by the user, and makes maintenance of the systems and equipment easier since resources are centralized

NPTEL

Introduction to Data Centers, Virtualization and Cloud Computing

❖ **Virtualization:**

- **Increases the flexibility of IT assets,**
- **Allows companies to consolidate IT infrastructure,**
- **Reduces maintenance and administration costs and**
- **Prepares for strategic initiatives**

NPTEL

Introduction to Data Centers, Virtualization and Cloud Computing

❖ **Virtualization:**

- **Cost-cutting aspect of virtualization is for tactical reasons**
- **Flexible sourcing and cloud computing are the most important strategic reasons for virtualization**

NPTEL

Introduction to Data Centers, Virtualization and Cloud Computing

❖**Virtualization:**

❖**The characteristics and benefits of virtualization are as follows:**

➢***Memory-intensive*:**

- **Need a huge amount of RAM (or primary memory) because of their massive processing requirements.**

➢***Energy-efficient*:**

- **Minimizes energy consumed for running and cooling servers in the data centers (95% reduction in energy use per server)**

NPTEL

Introduction to Data Centers, Virtualization and Cloud Computing

❖**Virtualization:**

❖**The characteristics and benefits of virtualization are as follows:**

➢***Scalability and load balancing*:**

- **Provides load balancing to handle the demand for requests to the site.**
- **The infrastructure automatically distributes the load across a cluster of physical servers to ensure the maximum performance of all running virtual machines.**

NPTEL

Introduction to Data Centers, Virtualization and Cloud Computing

❖ **Cloud Computing:**

- **Cloud computing is another term used to describe an architecture based on services provided by a third party over the internet and private network.**
- **Companies offering cloud computing make an entire data center's worth of servers, networking devices, systems management, security, storage, and other infrastructure available to their clients.**

NPTEL

Introduction to Data Centers, Virtualization and Cloud Computing

❖ **Cloud Computing:**

- **Cloud systems are scalable.**
- **Clients can buy the exact amount of storage, computing power, security, or other IT functions that they need, when they need it, and pay for what they use.**
- **Cloud systems can be adjusted to meet changes in business needs**

NPTEL

Introduction to Data Centers, Virtualization and Cloud Computing

❖ **Cloud Computing:**

- **At an extreme, the cloud's capacity is unlimited depending on the vendor's offerings and service level agreements (SLAs).**
- **However, a drawback of the cloud system is control because it is managed by a third party.**
- **Companies do not have as much control as they do with a data center.**

NPTEL

Introduction to Data Centers, Virtualization and Cloud Computing

❖ **Cloud Computing:**

➢ **Service Level Agreements (SLAs):**

- **A negotiated agreement between a company and service provider that can be a legally binding contract or an informal contract.**
- **The goal is not building the best SLA terms, but getting the terms that are most meaningful to the business.**

NPTEL

Introduction to Data Centers, Virtualization and Cloud Computing

❖ **Cloud Computing:**

➢ **Unless a client company uses a *private cloud* within its network, it shares computing and storage resources with other cloud users in the vendor's *public cloud.***

NPTEL

Introduction to Data Centers, Virtualization and Cloud Computing

❖ **Cloud Computing:**

- ***Public clouds*** **allow multiple clients to access the same virtualized services and utilize the same pool of servers across a public network.**
- **In contrast, *private clouds* are single-tenant environments with stronger security and control for regulated industries and critical data**

NPTEL

Introduction to Data Centers, Virtualization and Cloud Computing

❖ **Cloud Computing:**

- In effect, *private clouds* retain all the IT security and control provided by traditional data center infrastructure with the advantage of cloud computing
- Companies often use an arrangement of both on-premises data centers and cloud computing.

NPTEL

REFERENCES

- Laudon, K. C., and Laudon, J. P. (2016), Management Information Systems: Managing the Digital Firm
- Turban, E., Volonio, L., Wood, R. Gregory.(2016), Information Technology for Management
- Pearlson, K.E., and Saunders, C.S. (2013), Strategic Management of Information Systems

NPTEL



NPTEL

THANK YOU !



NPTEL ONLINE CERTIFICATION COURSES

NPTEL

Management Information Systems

Prof. Kunal Kanti Ghosh
VGSoM, IIT KHARAGPUR

Week 7: Module 5
Selection of Cloud Vendor

CONCEPTS COVERED

- **Selection of Cloud Vendor**

NPTEL

Introduction to Data Centers, Virtualization and Cloud Computing

❖ **Cloud Computing: XAAS: "As A SERVICE" Delivery Models**

➢ **Infrastructure as a Service (IaaS):**

- **Provides infrastructure through virtualized servers, networks, storage, and systems software to augment or replace the functions of an entire data center**
- **The customer may have full control of the actual server configuration**

NPTEL

Introduction to Data Centers, Virtualization and Cloud Computing

❖ **Cloud Computing: XAAS: "As A SERVICE" Delivery Models**

➢ **Infrastructure as a Service (IaaS):**

- **A way of delivering cloud computing infrastructure as an on-demand service.**
- **Rather than purchasing servers, software, data center space, or networks, companies instead buy all computing resources as a fully outsourced service**
- **Amazon Web Services is an example of IaaS provider**

NPTEL

Introduction to Data Centers, Virtualization and Cloud Computing

❖ **Cloud Computing: XAAS: "As A SERVICE" Delivery Models**

➢ **Platform as a Service (PaaS):**

- **Provides services using virtualized servers on which clients can run existing applications or develop new ones**
- **The cloud provider manages the hardware and underlying operating system**
- **Tools and services make coding and deployment faster and more efficient, like Google App Engine**

NPTEL

Introduction to Data Centers, Virtualization and Cloud Computing

❖ **Cloud Computing: XAAS: "As A SERVICE" Delivery Models**

➢ **Platform as a Service (PaaS):**

- **Provides a standard unified platform for app development, testing, and deployment**
- **Allows the creation of Web applications quickly and easily without the complexity of buying and maintaining the underlying infrastructure**

NPTEL

Introduction to Data Centers, Virtualization and Cloud Computing

❖ **Cloud Computing: XAAS: "As A SERVICE" Delivery Models**

➢ **Platform as a Service (PaaS):**

- **Without PaaS, the cost of developing some apps would be prohibitive**

NPTEL

Introduction to Data Centers, Virtualization and Cloud Computing

❖Cloud Computing: XAAS: "As A SERVICE" Delivery Models

➢Software as a Service (SaaS):

- Provides software application functionality through a web browser
- Sometimes known as an ASP, or Application Services Provider
- Apps are designed for end users
- Users can run applications via multiple devices on cloud infrastructure

NPTEL

Introduction to Data Centers, Virtualization and Cloud Computing

❖ **Cloud Computing: XAAS: "As A SERVICE" Delivery Models**

➢ **Software as a Service (SaaS):**

- **Other terms for SaaS are *on-demand computing* and *hosted services***
- **Instead of buying and installing expensive packaged enterprise applications , users can access software apps on demand over a network, with an internet browser *paying the license fee as required***

NPTEL

Introduction to Data Centers, Virtualization and Cloud Computing

❖Cloud Computing: XAAS: "As A SERVICE" Delivery Models

➢Software as a Service (SaaS):

- Salesforce.com is one of the most widely known SaaS providers.
- Another example is Google Docs.

✓Instead of installing MS-Word on ones own computer, and then loading Word to create a document, one can use a browser to log into Google Docs. Only the browser uses one's computer resources.

NPTEL

Introduction to Data Centers, Virtualization and Cloud Computing

❖Cloud Computing: XAAS: "As A SERVICE" Delivery Models

➢Data as a Service (DaaS):

- Data shared among clouds, systems, apps, regardless of the data source or storage location.
- DaaS makes it easier for data architects to select data from different pools, filter out sensitive data, and make the remaining data available on-demand
- Eliminates the risks and burdens of data management to a third-party cloud provider

NPTEL

Selection of Cloud Vendor

- Decision to select a cloud service provider requires greater diligence compared to any other IT decisions
- Assessment and selection of the right cloud services provider is an important strategic decision for business leaders since cloud computing is becoming an increasingly important part of the IT delivery model
- It is important to investigate each provider's offerings prior to subscribing

NPTEL

Selection of Cloud Vendor

- A vendor needs to address several service factors for being selected as the right cloud services provider
- The move to the cloud is a move to vendor-managed services wherein cloud service-level agreements is the most important element of a contract

NPTEL

Selection of Cloud Vendor

❖Service Level Agreements (SLAs):

- A negotiated agreement between a company and service provider that can be a legally binding contract or an informal contract.
- The goal is not building the best SLA terms, but getting the terms that are most meaningful to the business.
- Staffs experienced in managing outsourcing projects may have the required expertise for framing and managing SLAs with vendors.

NPTEL

Selection of Cloud Vendor

❖Service Level Agreements (SLAs):

- The Practical Guide to Cloud Service Level Agreements (published by the Cloud Standards Customer Council) brings together numerous customer experiences into a single guide for IT and business leaders who would like to consider cloud adoption.

NPTEL

Selection of Cloud Vendor

❖ **Service Level Agreements (SLAs):**

➢ **As per the Practical Guide to Cloud Service Level Agreements, an SLA serves:**

✓ **as a means of formally documenting the services, performance expectations, responsibilities and limits between cloud service providers and their users**

NPTEL

Selection of Cloud Vendor

❖ **Service Level Agreements (SLAs):**

➢ **A typical SLA describes levels of service using various attributes such as :**

- **Availability**
- **Serviceability**
- **Performance**
- **Operations**

NPTEL

Selection of Cloud Vendor

❖ **Service Level Agreements (SLAs):**

➢ **A typical SLA describes levels of service using various attributes such as :**

- **Billing and**
- **Penalties associated with violations of such attributes**

NPTEL

Selection of Cloud Vendor

❖Service Factors to Consider When Evaluating Cloud Vendors or Service Providers

Factors	Examples of Questions to Be Addressed
Delays	What are the estimated server delays & network delays
Workloads	What is the volume of data & processing that can be handled during a specific amount of time
Costs	What are the costs associated with workloads across multiple cloud computing platforms?

NPTEL

Selection of Cloud Vendor

❖Service Factors to Consider When Evaluating Cloud Vendors or Service Providers

Factors	Examples of Questions to Be Addressed
Security	How are data and networks secured against attacks? Are data encrypted & how strong is the encryption? What are network security practices?

NPTEL

Selection of Cloud Vendor

❖Service Factors to Consider When Evaluating Cloud Vendors or Service Providers

Factors	Examples of Questions to Be Addressed
Disaster recovery & business continuity	How is service outage defined? What level of redundancy is in place to minimize outages, including backup services in different geographical regions? If a natural disaster or outage occurs, how will cloud services be continued?

NPTEL

Selection of Cloud Vendor

❖Service Factors to Consider When Evaluating Cloud Vendors or Service Providers

Factors	Examples of Questions to Be Addressed
Technical expertise and understanding	Does the vendor have expertise in one's industry or business processes? Does the vendor understand what one needs to do Does the vendor have the technical expertise to fulfill those obligations?

NPTEL

Selection of Cloud Vendor

❖Service Factors to Consider When Evaluating Cloud Vendors or Service Providers

Factors	Examples of Questions to Be Addressed
Insurance in case of failure	Does the vendor provide cloud insurance to mitigate user losses in case of service failure or damage? This is a new and important concept

NPTEL

Selection of Cloud Vendor

❖Service Factors to Consider When Evaluating Cloud Vendors or Service Providers

Factors	Examples of Questions to Be Addressed
Third-party audit or an unbiased assessment of the ability to rely on the service provided by the vendor	Can the vendor show objective proof with an audit that it can live up to the promises it is making?

NPTEL

Selection of Cloud Vendor

- Cloud services can advance the core business of delivering superior services to optimize business performance
- Cloud can cut costs and add flexibility to the performance of critical business apps.
- Cloud can improve responsiveness to end-consumers, application developers, and business organizations.

NPTEL

Selection of Cloud Vendor

❖In order to achieve those benefits, there must be IT, legal, and senior management oversight because a company still must meet its

➢ legal obligations and responsibilities to

- employees,
- customers,
- business partners, and
- society.

NPTEL

REFERENCES

- Laudon, K. C., and Laudon, J. P. (2016), Management Information Systems: Managing the Digital Firm
- Turban, E., Volonio, L., Wood, R. Gregory.(2016), Information Technology for Management
- Pearlson, K.E., and Saunders, C.S. (2013), Strategic Management of Information Systems

NPTEL



NPTEL

THANK YOU !



NPTEL ONLINE CERTIFICATION COURSES

NPTEL

Management Information System

Saini Das

Vinod Gupta School of Management, IIT Kharagpur

Module 08: E-commerce : Digital Markets, Digital Goods

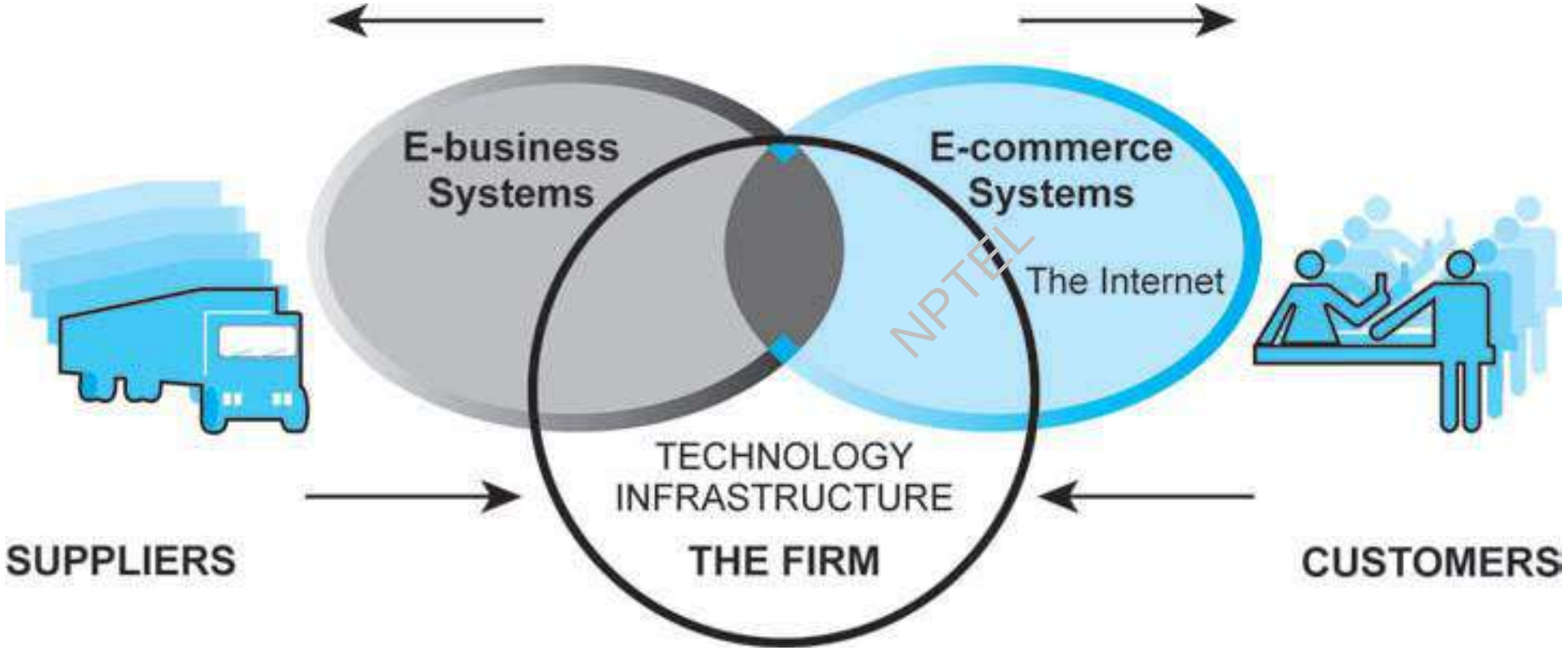
Introduction to E-commerce

E-commerce

Digitally enabled commercial transactions (involving exchange of value) between and among organizations and individuals.

NPTEL

E-commerce v/s E-business



NPTEL

Unique Features of E-commerce

- Ubiquity
- Global reach
- Richness
- Interactivity
- Information density (plentiful, cheap and accurate)
- Personalization/customization
- Network effect
- Hit and run pricing

NPTEL

Personalization

- Merchants can target their marketing message to specific individuals by adjusting it to a persons' clickstream behavior, names, interests and past purchases.



NPTEL

Customization

- Customization refers to changing the delivered product or service based on a user's preference or past purchases.



"I Built It, Therefore I Own It"

Types of E-commerce

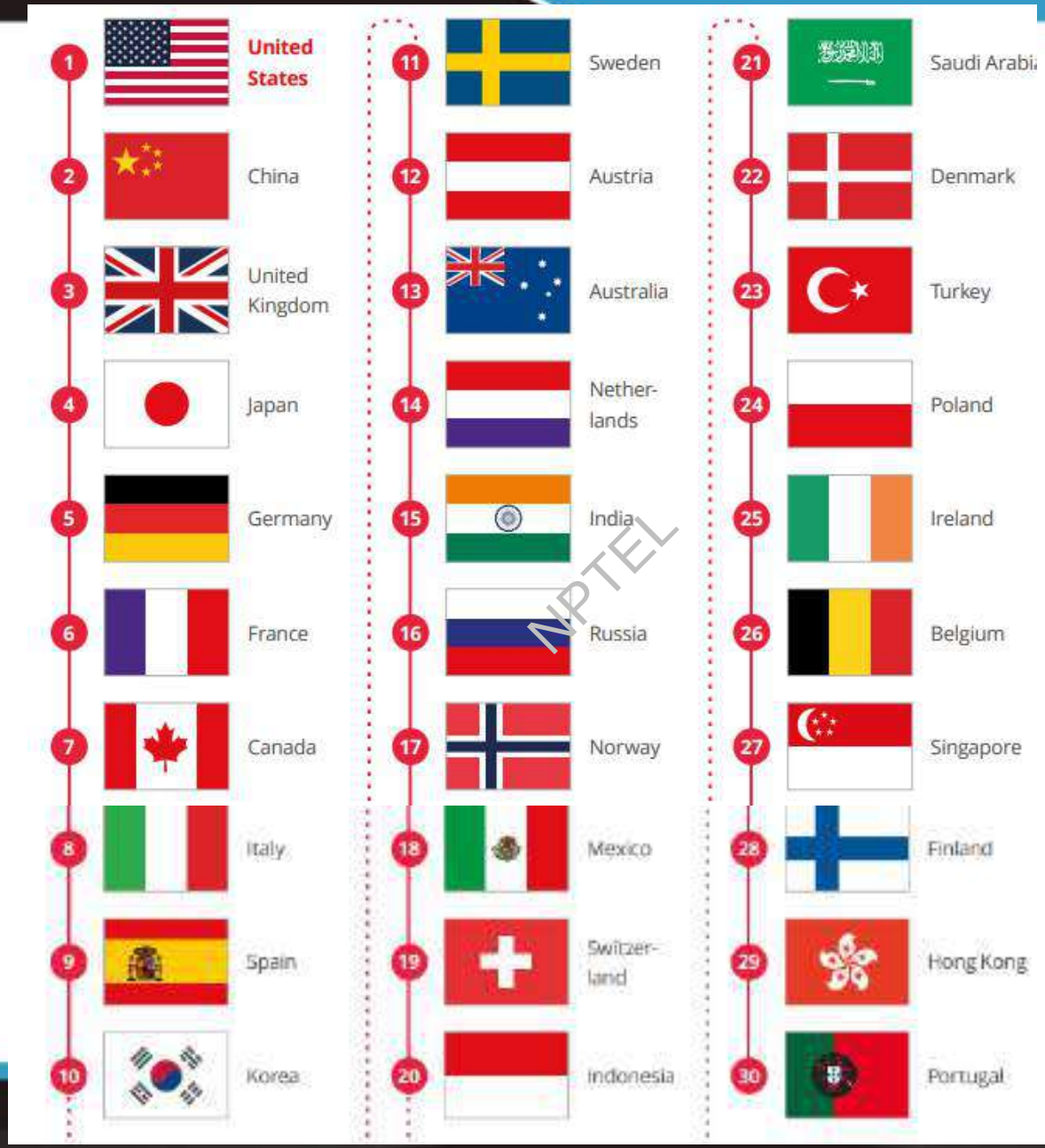
- **Business-to-consumer (B2C)**
  - Amazon.com
- **Business-to-Business (B2B)**
  - Indiamart.com
- **Consumer-to-consumer (C2C)**
  - e-Bay, Etsy
- **Mobile e-commerce**

NPTEL

Global E-commerce Market Ranking

- Logistics ranking
- Average revenue per shopper
- Total market revenue
- Percentage of population shopping online
- Number of online shoppers
- Percentage of cross-border shoppers
- Number of cross-border shoppers
- CAGR 2018-2022

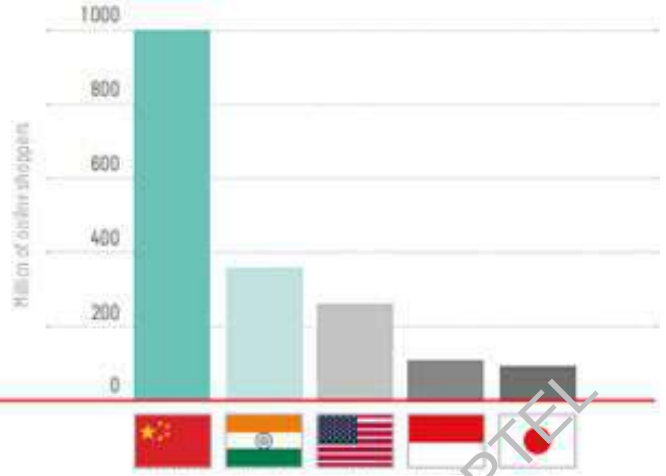
NPTEL


1 United States
2 China
3 United Kingdom
4 Japan
5 Germany
6 France
7 Canada
8 Italy
9 Spain
10 Korea
11 Sweden
12 Austria
13 Australia
14 Nether-lands
15 India
16 Russia
17 Norway
18 Mexico
19 Switzer-land
20 Indonesia
21 Saudi Arabia
22 Denmark
23 Turkey
24 Poland
25 Ireland
26 Belgium
27 Singapore
28 Finland
29 Hong Kong
30 Portugal
NPTEL

Global E-commerce Market Ranking (contd..)

Top 5 markets by total number of online shoppers

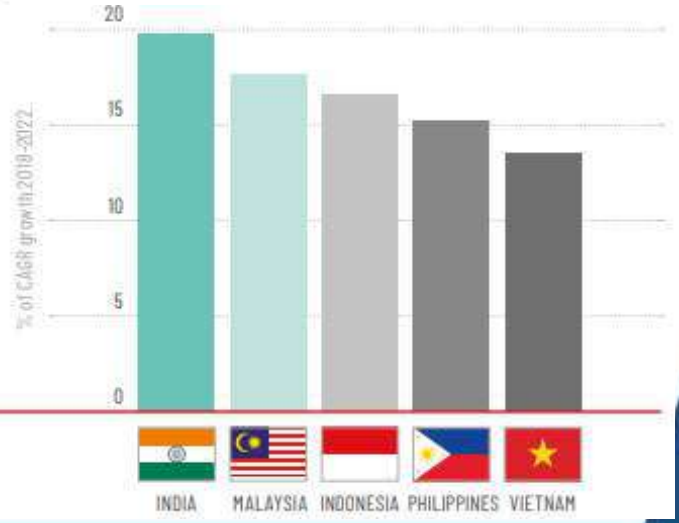
1. China	**1 billion**
2. India	360.1 million
3. USA	258.5 million
4. Indonesia	107 million
5. Japan	89.9 million



NPTEL

Top 5 countries by % of CAGR growth 2018-2022

1. India	**19.8%**
2. Malaysia	17.6%
3. Indonesia	16.6%
4. Philippines	15.2%
5. Vietnam	13.5%



E-commerce Challenges

NPTEL

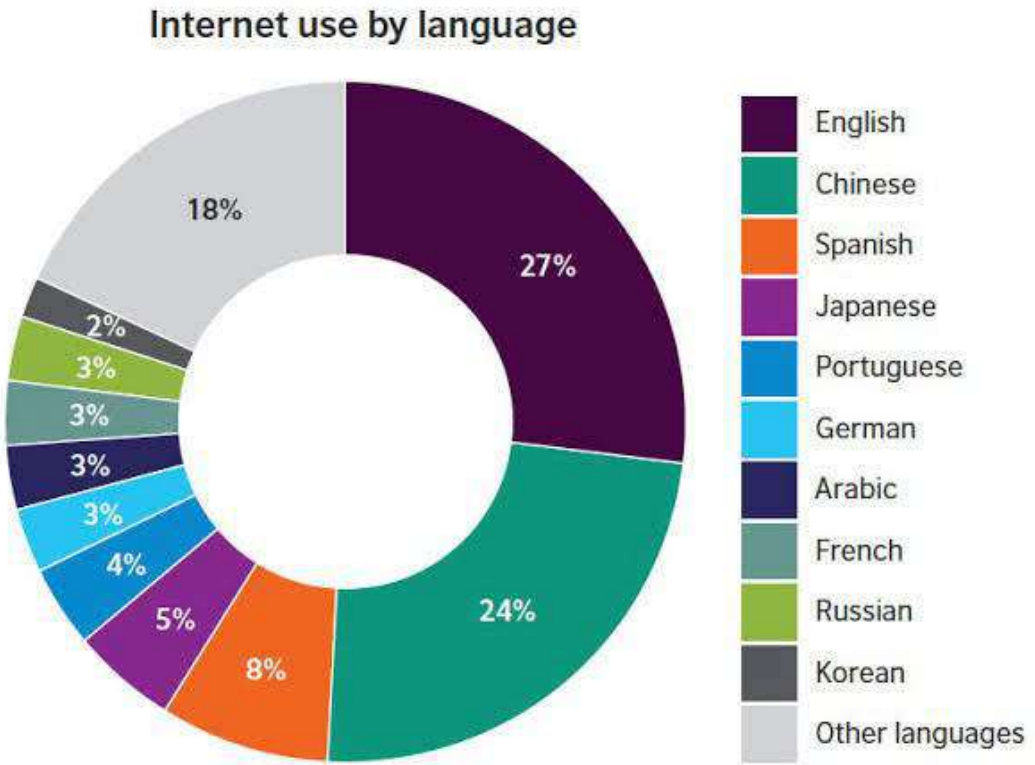
Trust Issues



"On the Internet, nobody knows you're a dog."

NPTEL

Language and Cultural Issues





NPTEL

DOLCE & GABBANA

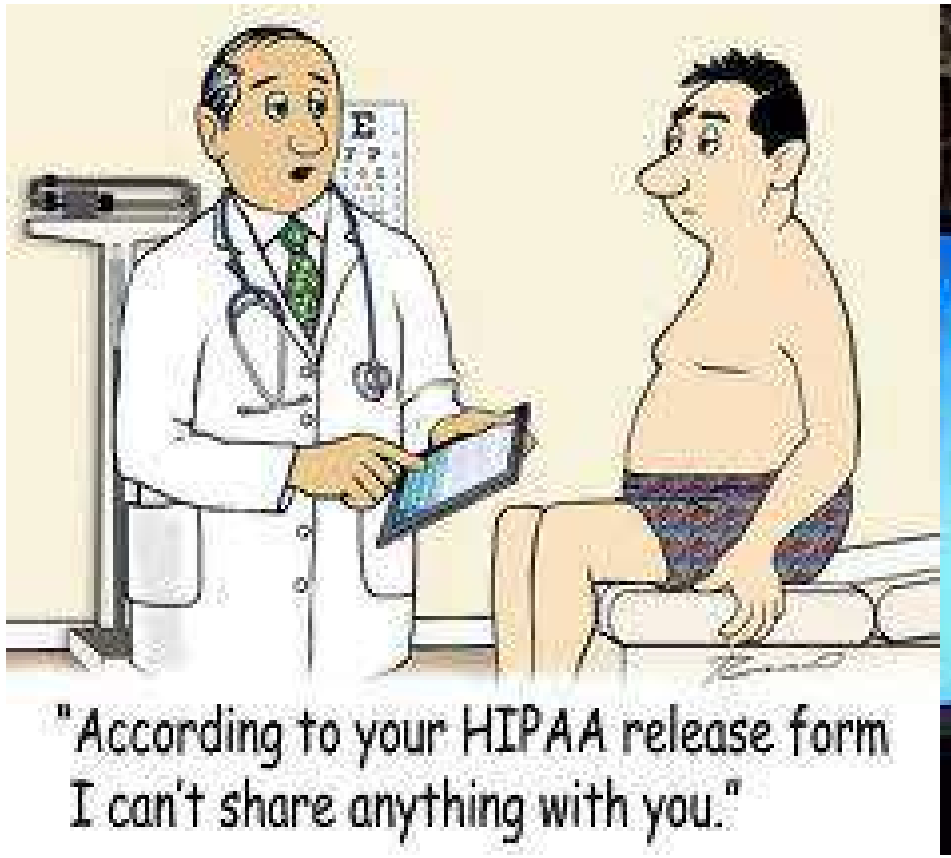
https://www.youtube.com/watch?v=TBFC5isQuYA

Security Issues





Legal and Ethical Issues





References

- K. Laudon and J. Laudon (2016). Management Information Systems Publisher: Pearson. Edition 14e.

- R. De. (2018). MIS Managing Information Systems in Business, Government and Society. Publisher: Wiley. Second Edition.

- Laudon, K. C., & Traver, C. G. (2019). E-commerce: Business, technology, society (15th ed.). Upper Saddle River, NJ: Pearson/Prentice Hall

NPTEL



NPTEL

THANK YOU !



NPTEL ONLINE CERTIFICATION COURSES

NPTEL

Management Information System

Saini Das

Vinod Gupta School of Management, IIT Kharagpur

Module 08: E-commerce : Digital Markets, Digital Goods
B2C Business Models

Business model

- Business model ascertains the **value proposition** of your business and how your **business makes money (revenue model).**

NPTEL

Key Elements of a Business Model

- Value Proposition: How a company's product or services fulfills customer needs?
- Revenue Model: How to earn money? (Sales/Advertising/Transaction fee)
- Market Opportunity: What marketspace you intend to serve and target audience?
- Competitive Environment: Who else occupies your intended marketspace?
- Competitive Advantage: What special advantages does your firm bring to the marketspace?
- Market Strategy: How to promote products and attract target audience?
- Organizational Development: How the company needs to organize?
- Management Team: What experiences and backgrounds are important for company's leaders?

NPTEL

Five Primary Revenue Models

REVENUE MODEL	EXAMPLES	REVENUE SOURCE
Advertising		Fees from advertisers in exchange for advertisements
Subscription	WSJ.com Consumerreports.org	Fees from subscribers in exchange for access to content or services
Transaction Fee	eBay	Fees (commissions) for enabling or executing a transaction
Sales	LLBean Gap JCPenny.com	Sales of goods, information, or services
Affiliate		Fees for business referrals

NPTEL

Affiliate Revenue Model



B2C Business Models

NPTEL

Business Model	Variations	Examples	Description	Revenue Model
Portal	Horizontal/ General	Yahoo AOL MSN	Offers an integrated package of content and services. Seeks to be a user's home base	Advertising
	Vertical/Specialized (Vortal)	Sailnet	Offers services and products to specialized domains	Effective for advertisers
	Search	Google	Search Services	Advertising
Virtual Merchant	E-tailers/ Pureplay/ Resellers	Flipkart(pre-2013) Netmeds Amazon	Online version of retail store	Sales of goods
	Brick and click	Walmart.com Shoppers Stop	Online distribution channel for a company that already has physical stores	-do-
	Click and brick	Lenskart.com Bluestone.com Pepperfry.com	Physical stores for an existing online company	-do-
	Manufacturer direct	Dell.com Sony.com	Manufacturer uses online channel to sell directly to customer	-do-

NPTEL

Business Model	Examples	Description	Revenue Model
Content Provider	HBR WSJ.com CNN.com	Information and entertainment providers such as newspapers, sports and special interest guidance	Advertising, Subscription fees (Freemium), Affiliate Referral fees
Transaction Broker	Makemytrip Hotels.com RedBus Yatra.com	Processors of online sales transactions such as stockbrokers and travel agents (OTAs).	Transaction fees
Market Creator/ Marketplace	Flipkart (post 2013) Housing.com	Web-based businesses that use Internet technology to create markets where buyers and sellers meet, search for products and complete transactions,	Transaction fees Listing fees
Service provider	VisaNow.com BharatMatrimony.com MOOCs RocketLawyer	Online Service sellers	Sales of services, Commission fees, Listing Fees
Community provider	Facebook Linkedin WebMD The Motley Fool	Social networking for people with common interests, hobbies.	Advertising, Subscription fees, Referral fees

NPTEL

References

- K. Laudon and J. Laudon (2016). Management Information Systems Publisher: Pearson. Edition 14e.

- R. De. (2018). MIS Managing Information Systems in Business, Government and Society. Publisher: Wiley. Second Edition.

- Laudon, K. C., & Traver, C. G. (2019). E-commerce: Business, technology, society (15th ed.). Upper Saddle River, NJ: Pearson/Prentice Hall

NPTEL



NPTEL

THANK YOU !



NPTEL ONLINE CERTIFICATION COURSES

NPTEL

Management Information System

Saini Das

Vinod Gupta School of Management, IIT Kharagpur

Module 08: E-commerce : Digital Markets, Digital Goods
B2B Business Models

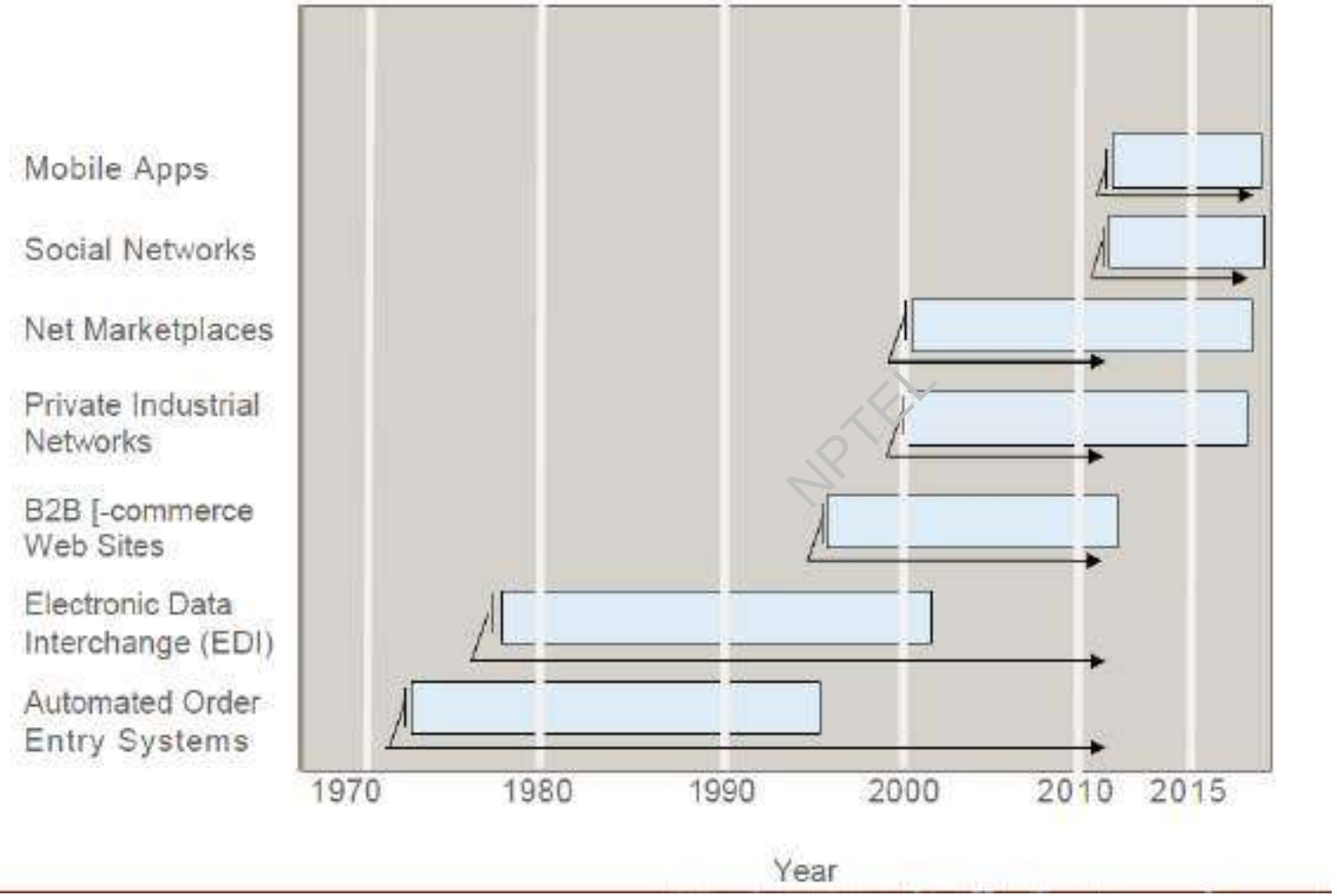
B2B E-commerce

- Inter-firm trade to exchange value across organizational boundaries enabled by the Internet and mobile apps.
- Business Processes involved:
  - Demand Management
  - Order fulfilment
  - Manufacturing management
  - Procurement
  - Product development
  - Returns
  - Logistics/transportation
  - Inventory management
  - CRM

NPTEL

- The global B2B e-Commerce market value of US$12.2 trillion in 2019 is over **6 times** that of the B2C market.

Evolution of technology platforms in B2B E-commerce



NPTEL

Automated Order Entry Systems

- Telephone modems used to send digital orders
- Some companies automated reordering by placing the modems in their procurement offices.
- They are seller-side solutions as they showed goods only from a single seller.

NPTEL

Electronic Data Interchange

- One of the earliest uses of IT for SCM.
- Involves the electronic exchange of business transaction (BT) documents between supply chain trading partners
- BT documents are converted to standardized EDI formats as specified by industry and international protocols.
- Owned by buyers, hence they are buyer-side solutions.
- Important industrial network technology to support communication among a small set of strategic partners.
- Not suitable for net marketplaces

NPTEL

Net Marketplace

- Digital platform that brings B2B buyers and sellers together

CHARACTERISTIC	MEANING
Bias	Sell-side vs. buy-side vs. neutral. Whose interests are advantaged: buyers, sellers, or no bias?
Ownership	Industry vs. third party. Who owns the marketplace?
Pricing mechanism	Fixed-price catalogs, auctions, bid/ask, and RFPs/RFQs.
Scope/Focus	Horizontal vs. vertical markets.
Value creation	What benefits do they offer customers or suppliers?
Access to market	In public markets, any firm can enter, but in private markets, entry is by invitation only.

Characteristics of Net Marketplaces

NPTEL

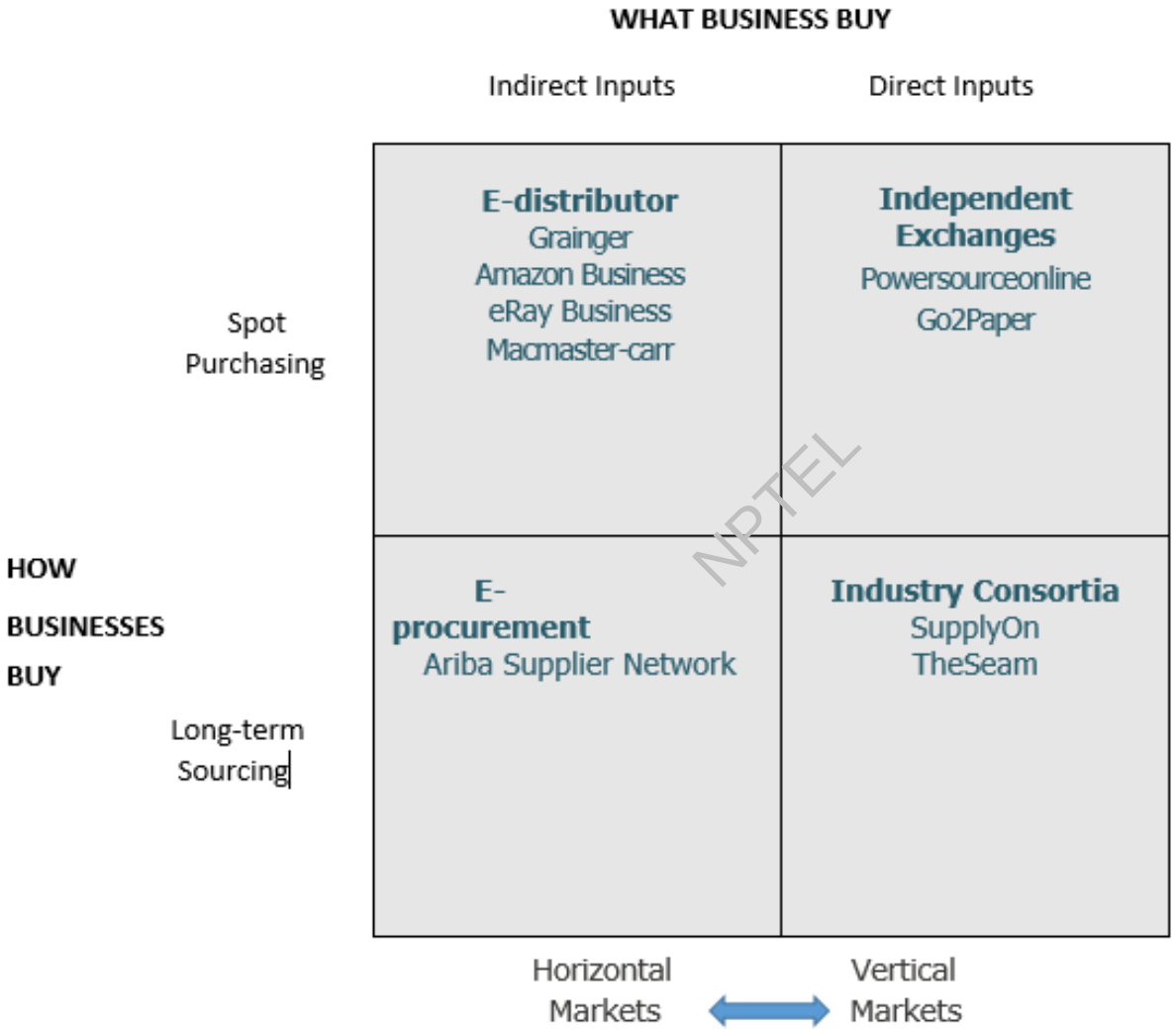
Classification of Net Marketplaces

- **What Businesses Buy:**
  - Direct Input
  - Indirect Input

NPTEL

- **How Businesses Buy:**
  - Spot purchasing (Episodic and anonymous purchases)
  - Long-term sourcing (Purchases that take place over many years according to a contract between a firm and its vendor)

Types of Net Marketplaces



NPTEL

Types of Net Marketplaces (contd..)

Business Model	Examples	Description	Revenue Model
E-Distributor (Indirect input; Spot purchasing)	Grainger.com; Amazon Business	Independently owned B2B equivalent of a B2C reseller; Horizontal platform; public Market; seller-biased.	Sales of goods
E-procurement (Indirect input; long-term sourcing)	SAP-Ariba; IBM Emptoris Sourcing;	Independently owned digital marketplace where sellers and buyers transact for products and services; Horizontal platform; Neutral; public market.	Transaction fees; listing fees
Exchange (Direct input; Spot purchasing)	MolBase IronPlanet PowerSource Online	Independently owned vertical digital marketplace for many suppliers and small number of large commercial buyers; public market; buyer-biased.	Transaction fees
Industry Consortium (Direct input; long-term sourcing)	E2Open	Industry-owned vertical digital market open to select suppliers for optimizing the industry supply-chain; buyer-biased; Private market.	Transaction fees; Payment from industry members

NPTEL

Examples

EXCHANGE	FOCUS
PowerSource	Computer parts exchange, new and used computer equipment
Online Converge	Electronic components
IronPlanet	Used heavy equipment
EquipNet	Used industrial equipment and online auctions
Intercontinental Exchange	International online marketplace for over 600 commodities

Examples of Independent Exchanges

INDUSTRY	NAME OF INDUSTRY CONSORTIA
Agribusiness	The Seam
Automotive	SupplyOn
Chemical	Elemica
Food	Dairy.com
Hospitality	Avendra

Examples of Industry Consortia

NPTEL

Examples of Industry Consortia


GRAINGER
[+] Tell us what you think of the NEW Grainger.com >
Catalog | Find A Branch | Cart Contains (0) Products
PRODUCTS
SERVICES
RESOURCES
REPAIR PARTS
WORLDWIDE
Enter keyword or item number
Search
CYBERSALE
From batteries to power tools, don't miss these limited-time offers of 10 to 50% off popular products!
Don't wait! Deals end 12/6/13
Start Saving Now
Bulk Order Pad
Product Categories
Abrasives
Adhesives, Sealants and Tape
Cleaning
Electrical
Electronics, Appliances, and Batteries
Fasteners
Fleet and Vehicle Maintenance
Furniture, Hospitality and Food Service
HVAC and Refrigeration
Hand Tools
Hardware
Hydraulics
Lab Supplies
Lubrication
Machining
Material Handling
Motors
Office Supplies
Outdoor Equipment
Paint, Equipment and Supplies
Plumbing
Production
Power Tools
Power Transmission
Pumps
Raw Materials
Reference and Learning Supplies
Safety
Security
Test Instruments
Welding
Brands
50% Off Duracell Batteries
Save big on select Coppertop Bulk Packs. Hurry! Ends 12/6/13.
SAVE NOW
CYBERSALE
Limited-time offers on some of our most popular products!
Check Out These Top Sellers
1 of 5
Prev | Next
Extension Cord, 50ft
Item # 3EB10
POWER FIRST
Price: $75.95
Add to Cart
Extension Cord, 100 Ft
Item # 1FD57
POWER FIRST
Price: $120.10
Add to Cart
Telescoping Ladder, 19 ft., IA, Aluminum
Item # 3EJ77
LITTLE GIANT
Price: $619.00
Add to Cart

NPTEL

Private Industrial Networks

Business Model	Examples	Description	Revenue Model
Single Firm	Walmart Procter & Gamble	Company-owned network to coordinate trans-organizational business processes/collaborative commerce with a limited set of partners; private market	Cost absorbed by network owner and recovered through production and distribution efficiencies

NPTEL

References

- K. Laudon and J. Laudon (2016). Management Information Systems Publisher: Pearson. Edition 14e.

- R. De. (2018). MIS Managing Information Systems in Business, Government and Society. Publisher: Wiley. Second Edition.

- Laudon, K. C., & Traver, C. G. (2019). E-commerce: Business, technology, society (15th ed.). Upper Saddle River, NJ: Pearson/Prentice Hall

NPTEL



NPTEL

THANK YOU !



NPTEL ONLINE CERTIFICATION COURSES

NPTEL

Management Information System

Saini Das

Vinod Gupta School of Management, IIT Kharagpur

Module 08: E-commerce : Digital Markets, Digital Goods
Unique e-commerce business models

Differences between B2B and B2C e-commerce

- B2C transactions are ad-hoc in nature, with one seller, selling to customers in a single transaction; B2B involves many buyers and sellers who establish long-term purchasing contracts.
- B2C transactions are of low value vis-à-vis B2B transactions.
- The delivery schedule and timing of goods are critical and require high degree of commitment in B2B, since they tie into organisational supply-chains.
- B2B do not have fixed prices and may involve auctions.

NPTEL

Auctions

- Websites (usually C2C) that allow sellers and buyers to meet over the Internet.
- Pricing is generally through bidding, where the highest bidder wins.
- Usually C2C companies engage in auctions, example: eBay.
- After the auction buyers and sellers rate each other.
- Auction sites are subject to strong network effects.



NPTEL

Crowdsourcing/Crowdfunding

- A disruptive business model!
- Advantages:
  - Market validation
  - Consumer Feedback
  - Goal attainment
- Disadvantages:
  - You pay a certain portion of your raise to the platform
  - Many projects fail due to difficulty in attaining a set goal
- Example: Kickstarter

https://www.youtube.com/watch?v=-38uPkyH9vI



NPTEL

Online shopping mall/store builder

- Offers the products of all registered sellers in one place
- Web store builder allows sellers to have their own custom online store
- All listings are "fixed price" and buyers can add products from several sellers into their shopping cart and buy them in one single order
- Logistics and distribution provided by the mall provider
- Example: eCrater, Shopgo, ezstore, Shopify, Rakuten

NPTEL

The Coupon Business

Deal-of-the-day website that features discounted gift certificates usable at local or national companies

- If a certain number of people sign up for the offer, then the deal becomes available to all
- Helps merchants build a loyal customer base
- Focuses on group unity
- Larger the minimum threshold, larger the discount

Revenue Model:

Portion of the amount paid by the customer

Example: Groupon; Nearbuy

NPTEL

Is the coupon business model sustainable?



- Long term goals of sellers may not succeed
- Servicing huge number of customers if a deal is very attractive
- Threat from merchant companies that have upper hand in any deal
- Threat from larger companies with huge user base

NPTEL

Online Staffing Platform/Freelance marketplace

Online marketplace in which independent professionals and their clients can establish and fulfill work arrangements.

Clients

- Allows businesses to post jobs, search for freelance professionals, and solicit proposals.
- Evaluate contracts and communicate

NPTEL

Freelancers

- Search jobs
- Post a profile displaying past jobs and feedback, a portfolio, and specific skill and educational-background information.
- Examples: Elance; oDesk (video)

Online Staffing Platform/Freelance marketplace (contd..)

- ***Online Staffing*** *— "A type of* ***Talent Exchange*** *consisting of an online platform where contingent workers, contractors, freelancers can offer their skills and services for limited projects or even on-going assignments and where organizations and individuals can post their requirements or put tasks/projects out to bid".*

NPTEL

- **Revenue Model –** Membership fees or marking-up the billings of the contractors/freelancers by 5-15%
- https://www.youtube.com/watch?v=xbjkIE_CQI8

References

- K. Laudon and J. Laudon (2016). Management Information Systems Publisher: Pearson. Edition 14e.

- R. De. (2018). MIS Managing Information Systems in Business, Government and Society. Publisher: Wiley. Second Edition.

- Laudon, K. C., & Traver, C. G. (2019). E-commerce: Business, technology, society (15th ed.). Upper Saddle River, NJ: Pearson/Prentice Hall

NPTEL



NPTEL

THANK YOU !



NPTEL ONLINE CERTIFICATION COURSES

NPTEL

Management Information System

Saini Das

Vinod Gupta School of Management, IIT Kharagpur

Module 08: E-commerce : Digital Markets, Digital Goods
Social Commerce & M-commerce

Social Commerce

- Social Commerce is based on the idea of digital **social graphs**.
- Digital social graph is a mapping of all significant online social relationships.
- In an inter-connected social network, the products and services you buy will influence the decisions of your friends, and their decisions will in turn influence you.
- Marketers should take advantage of the fact that people are enmeshed in social networks, share interests and values and communicate and influence one another.

NPTEL

Social Commerce



- Online communities are also ideal venues to employ viral marketing techniques.
- Viral marketing in an online community spreads at the speed of light and goes much faster geographically than a small network of friends.
- “The wisdom of the crowd” is often used by companies to source advice, funds, ideas from users on social media. Examples: Kickstarter, Milaap, etc.

NIELIT

Features of Social Commerce

- **Newsfeed:** A stream of notifications from friends and advertisers that social users find on their home pages.

- **Timelines:** A stream of photos and events in the past that create a personal history for users, one that can be shared with friends.

- **Social sign-on:** Web sites allow users to sign into their sites through their social network pages on Facebook or another social site. This allows Web sites to receive valuable social profile information from Facebook and use it in their own marketing efforts.

NPTEL

Features of Social Commerce (contd…)

- **Collaborative shopping:** Creating an environment where consumers can share their shopping experiences with one another by viewing products, chatting, or texting. Friends can chat online about brands, products, and services.

- **Network notification:** Creating an environment where consumers can share their approval (or disapproval) of products, services, or content, or share their geo-location, perhaps a restaurant or club, with friends.



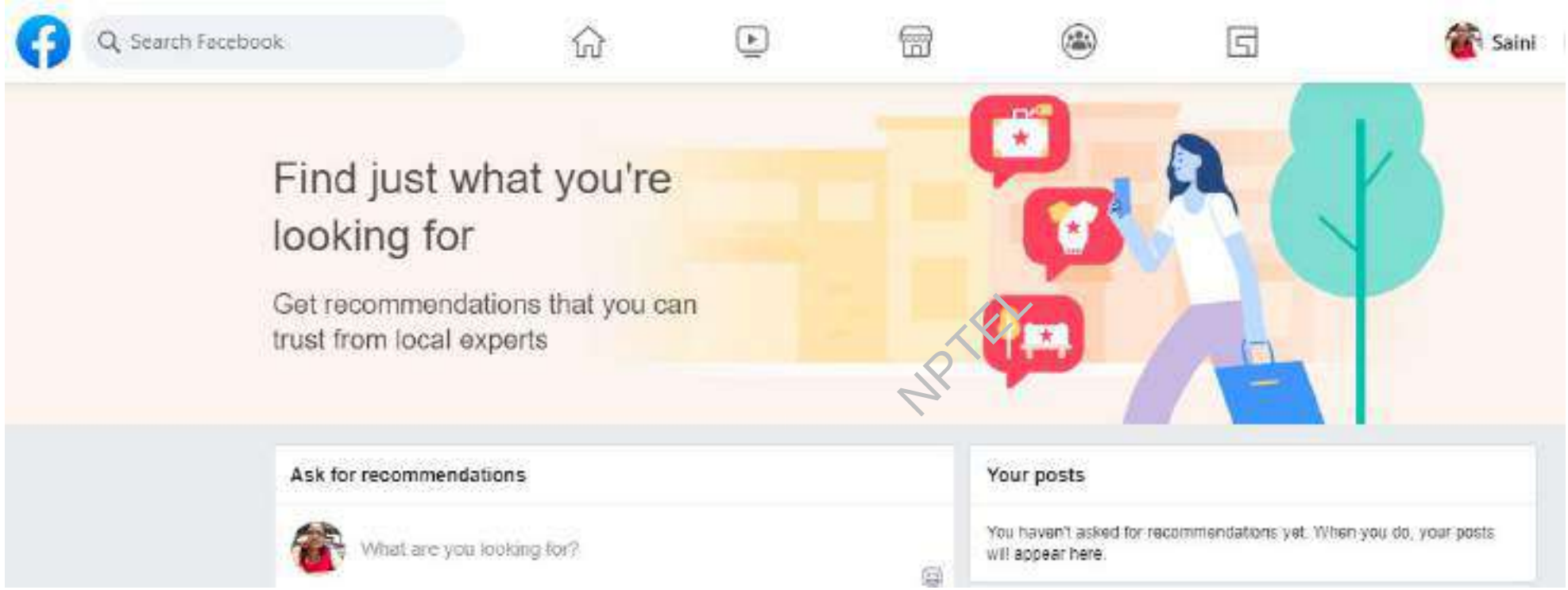
NPTEL

Features of Social Commerce (contd…)

- **Social search (recommendations):** Enabling an environment where consumers can ask their friends for advice on purchases of products, services, and content. While Google can help you find things, social search can help you evaluate the quality of things by listening to the evaluations of your friends, or their friends. For instance, Amazon's social recommender system can use your Facebook social profile to recommend products.

NPTEL

Facebook Recommendations



Google Social Search

Top Social Networks Search Results Dashboard

Aarogya Setu

Search

NPTEL

Facebook

Web Image

About 276,000 results (0.23 seconds)

Sort by: Relevance

Download the **Aarogya Setu** App to... - MyGov Corona Hub | Facebook
https://www.facebook.com/...**aarogya-setu**.../2880119372102233/



Download the **Aarogya Setu** App to strengthen our fight against COVID-19. The App is available in 11 languages. #IndiaFightsCorona मैं सुरक्षित। हम...

Twitter

Web Image

About 120,000 results (0.26 seconds)

Sort by: Relevance

Aarogya Setu (@SetuAarogya) | Twitter
https://twitter.com/**setuaarogya**

The latest Tweets from **Aarogya Setu** (@SetuAarogya). GoI's mobile application for contact tracing & dissemination of medical advisories to contain spread of ...

Instagram

Web Image

About 809 results (0.19 seconds)

Sort by: Relevance

TechoGhetto | Tech News on Instagram: "**Aarogya Setu** is a mobile ...
https://www.instagram.com/p/B_aG7Ljp3Sh/

25 Apr 2020 ... techoghetto **Aarogya Setu** is a mobile app developed by the ministry of electronics and IT India, to help citizens identify their risk... more.

Aarogya Setu

Mobile Commerce



- https://www.youtube.com/watch?v=Fdt5rRgimao (till1m59secs

- Mobile commerce is the fastest growing form of e-commerce.

- Mobile commerce applications have taken off for services that are time-critical, that appeal to people on the move, or that accomplish a task more efficiently than other methods.

- The main areas of growth in m-commerce are retail sales and sales of digital content such as, music, TV shows and movies.

NPTEL

Mobile Commerce (contd..)

- Mobile commerce leverages security and ease of payments on the pervasive mobiles, supporting innovative payment solutions through mobiles.

- Benefits include the ability to make payments without the use of a computer or access to Internet and without a credit card.

- Mobile payment services are similar to credit card transactions, in that the merchant must have an account with the service.

NPTEL

Location-based Services and Applications

- A **geosocial service** can tell you where your friends are meeting. Example: Foursquare https://www.youtube.com/watch?v=rq62BOcQde8

- A **geoadvertising service** is the practice of delivering ads based on geographic location of the targeted population. https://www.youtube.com/watch?v=98K-G8zdbVk

- A **geoinformation service** can tell you the price of a house you are looking for or about special exhibits at a museum you are passing by. Example: Wikitude.me

- The foundation for mobile commerce is the **Global Positioning System (GPS)** enabled map services available on smartphones.

NPTEL

The way forward….

- https://www.youtube.com/watch?v=hj14BD-Uyvs

NPTEL

References

- K. Laudon and J. Laudon (2016). Management Information Systems Publisher: Pearson. Edition 14e.

- R. De. (2018). MIS Managing Information Systems in Business, Government and Society. Publisher: Wiley. Second Edition.

- Laudon, K. C., & Traver, C. G. (2019). E-commerce: Business, technology, society (15th ed.). Upper Saddle River, NJ: Pearson/Prentice Hall

NPTEL



NPTEL

THANK YOU !

Management Information System

Saini Das
Vinod Gupta School of Management, IIT Kharagpur

NPTEL

Module 09: Emerging Technologies
Cloud Computing Part - I

Cloud Computing



"Cloud computing is cool technology,
but every time it rains I lose my data!"

buzzingup.com

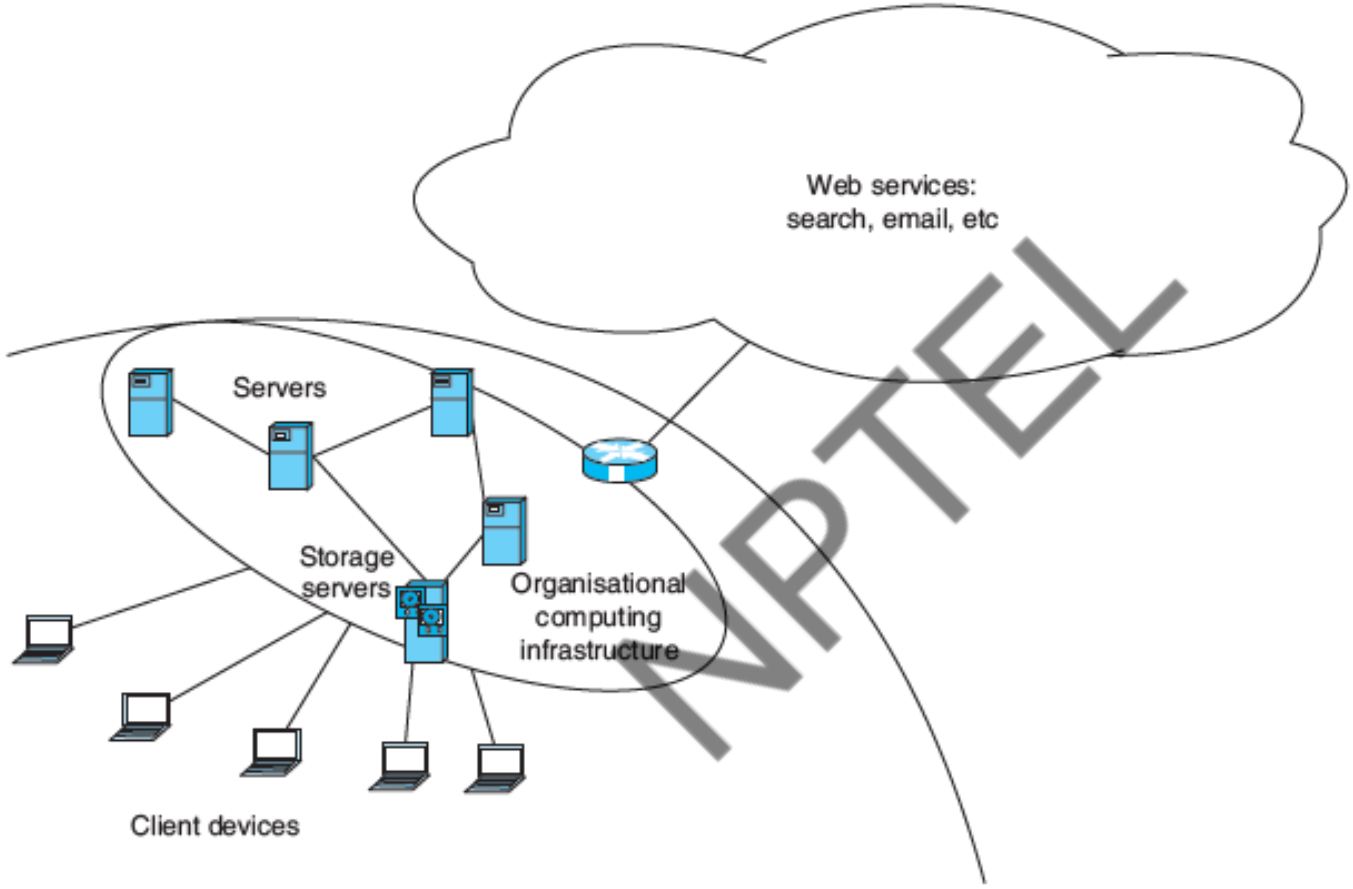
NPTEL

What is Cloud Computing?

- *"It is an **information technology service model** where computing services (both hardware and software) are delivered **on-demand** to customers over a network in a self-service fashion, independent of device and location. The resources required to provide the requisite quality of service levels are **shared, dynamically scalable, rapidly provisioned and released with minimal client interaction.** Users pay for the service as an **operating expense** without incurring any significant initial **capital expenditure**" (Marston et al., 2012)*

NPTEL

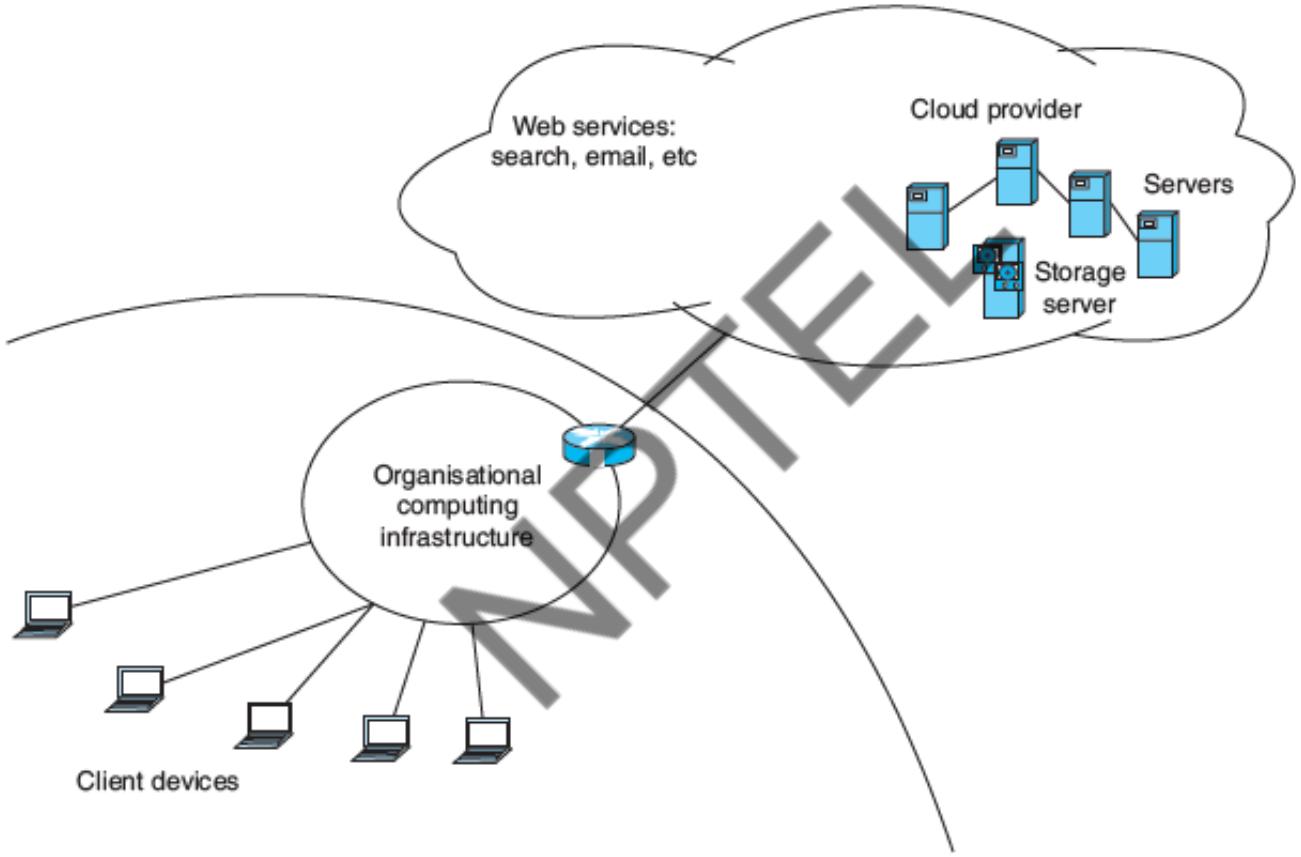
Traditional Computing Architecture



Hosted infrastructure. Here the organisation owns and maintains its own infrastructure of servers and draws on some web-based services from the internet.

NPTEL

Cloud Computing Architecture



Cloud infrastructure: Here an organisation draws on server and storage resources from the cloud and maintains a minimal infrastructure internally.

NPTEL

Benefits of Cloud Computing

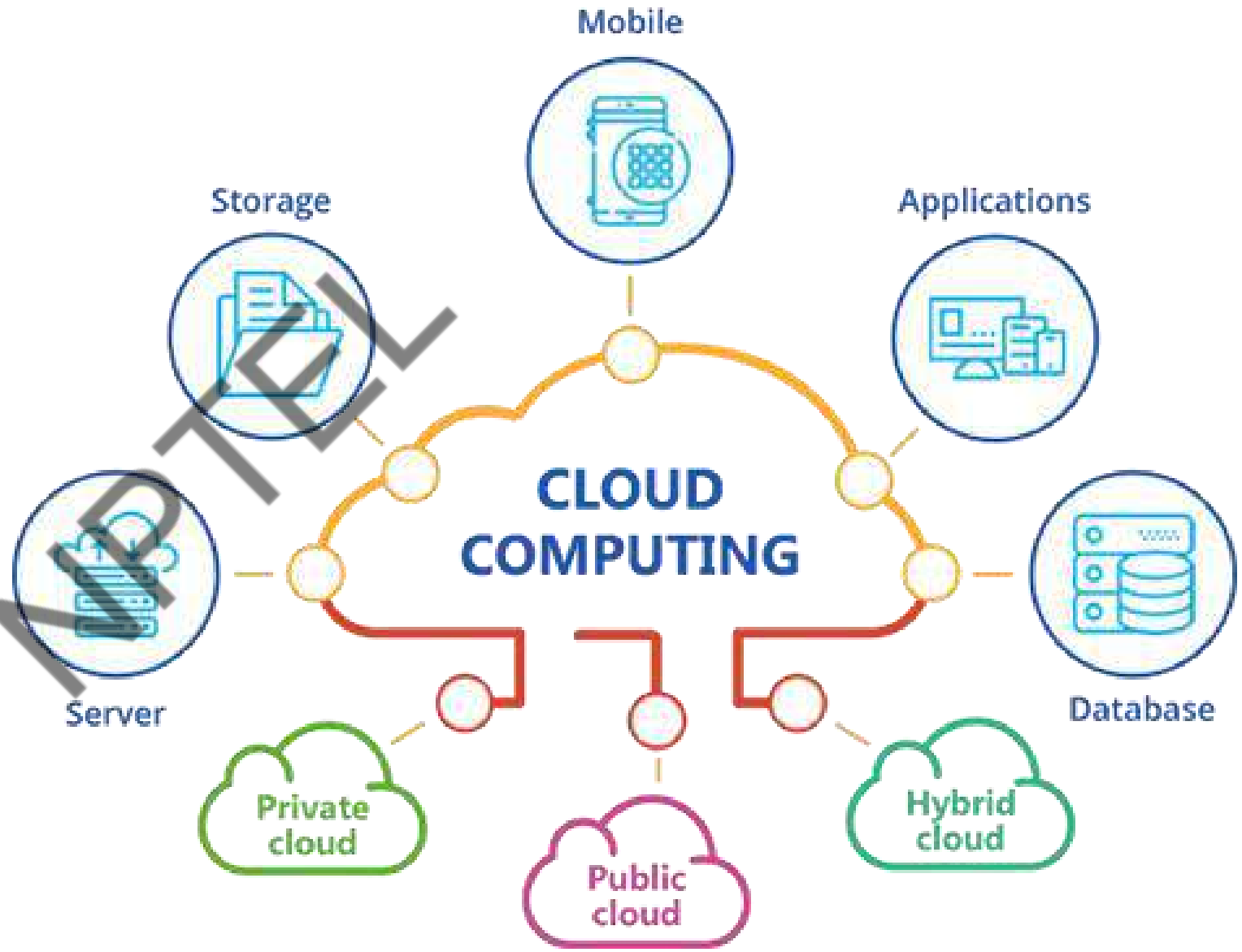
- Scalability
- Measured Use
  - Memory usage in unit time; CPU utilization in cycles per unit time; data usage in bits per unit time etc.
- Usage-based pricing
- Managed Services
- Service levels
  - Duration of time an infrastructure is available without interruption; speed of a service over a period of time etc.
- Ubiquitous Access
- Reduced Set-up Time
- Resource Pooling

NPTEL

Cloud Service Providers

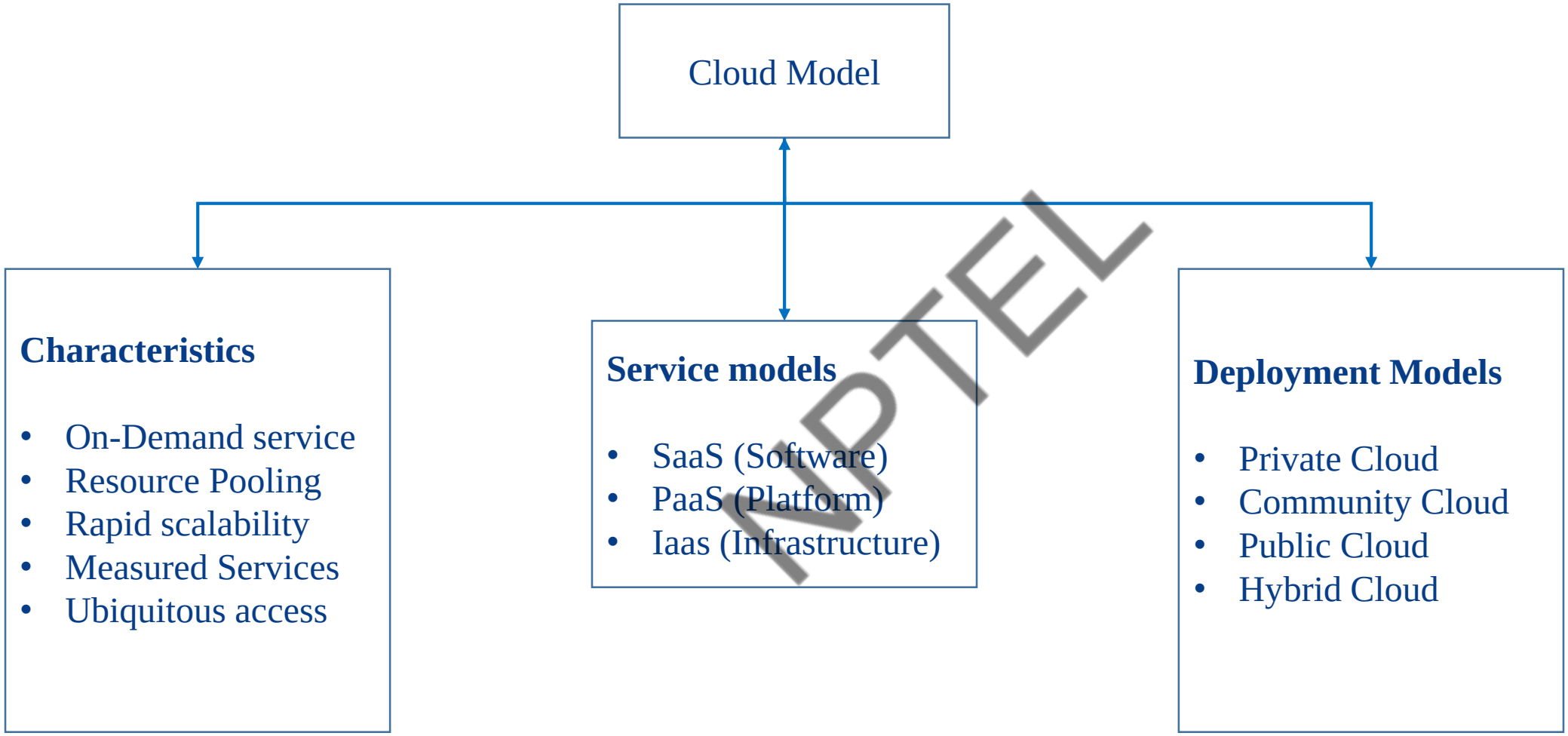
Top 12 Service Providers

- Amazon Web Services
- Kamatera
- DigtialOcean
- Rackspace
- MassiveGrid
- Alibaba Cloud
- LiquidWeb
- Microsoft Azure
- Google Cloud Platform
- VMware
- Salesforce
- Oracle Cloud



NPTEL

Cloud Model Hierarchy



NPTEL

Cloud Service Models

- **IaaS** : Single server or group of servers, CPU, memory, bunch of storage capacity, networking facilities, security solutions such as firewalls or bandwidth in the cloud. Eg: Amazon web services, Microsoft Azure etc.

- **PaaS** : Cloud-based platform that companies can use to develop their custom applications or write software that integrates with existing applications. The vendor hosts the application development tools, packages, plugins and libraries and makes it widely available. Vendors are Microsoft, Amazon, etc.

- **SaaS** : Application or suite of applications that resides in the cloud instead of on a user's hard drive or in a data center. Most mature.
- Eg: Salesforce.com's CRM software

NPTEL

IaaS provides data centre facilities; PaaS provides a software development facility; SaaS provides applications over the Internet.

Cloud Service Models Comparison

Feature	IaaS	PaaS	SaaS
What is provided	Hardware + System software + Networking software + System management tools	IaaS + Application development tools and libraries	Cloud-based Application
Management	Vendor manages infrastructure; everything else managed by client	Vendor manages infrastructure and development tools; client manages everything else	Vendor manages infrastructure and application; client manages user accounts only
Typical usage	Infrastructure for multiple servers and applications	Build and deploy new applications; maybe for new markets	Single application for focussed use

NPTEL

Challenges

- Security and privacy
- Availability and reliability
- Service quality
- Integration
- Regulations

NPTEL

Cloud Deployment models

- **Public** : Services are rendered over a network that is open for public use. Truly shared facility that allows multiple clients to use the resources provided.
- **Private** : Cloud infrastructure operated solely for a single organization, either managed internally or by a third-party, and hosted either internally or externally.
- **Hybrid** : Cloud computing service that is composed of some combination of private and public cloud services, sometimes from different service providers.
- **Community Cloud**: Shared computing infrastructure set up as a public, private or hybrid cloud used by a specified community of users such as hospitals or banks.

NPTEL

References

- K. Laudon and J. Laudon (2016). Management Information Systems Publisher: Pearson. Edition 14e.

- R. De. (2018). MIS Managing Information Systems in Business, Government and Society. Publisher: Wiley. Second Edition.

NPTEL

NPTEL

Management Information System

Saini Das
Vinod Gupta School of Management, IIT Kharagpur

NPTEL

Module 09: Emerging Technologies
Cloud Computing – Part II

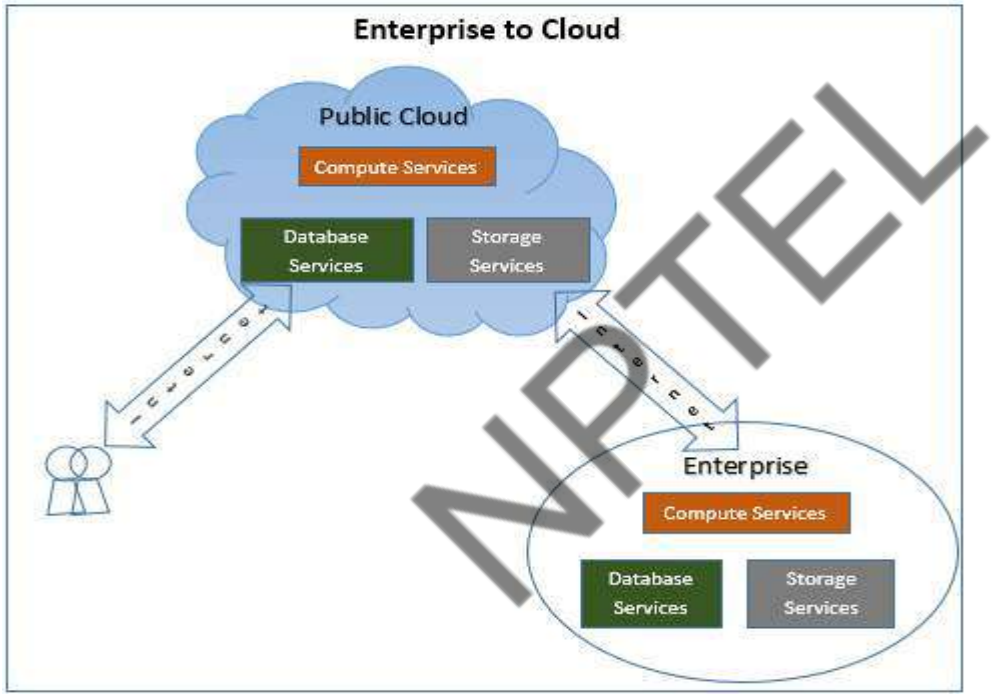
Cloud Deployment models

- **Public** : Services are rendered over a network that is open for public use. Truly shared facility that allows multiple clients to use the resources provided.
- **Private** : Cloud infrastructure operated solely for a single organization, either managed internally or by a third-party, and hosted either internally or externally.
- **Hybrid** : Cloud computing service that is composed of some combination of private and public cloud services, sometimes from different service providers.
- **Community Cloud**: Shared computing infrastructure set up as a public, private or hybrid cloud used by a specified community of users such as hospitals or banks.

NPTEL

Public Cloud

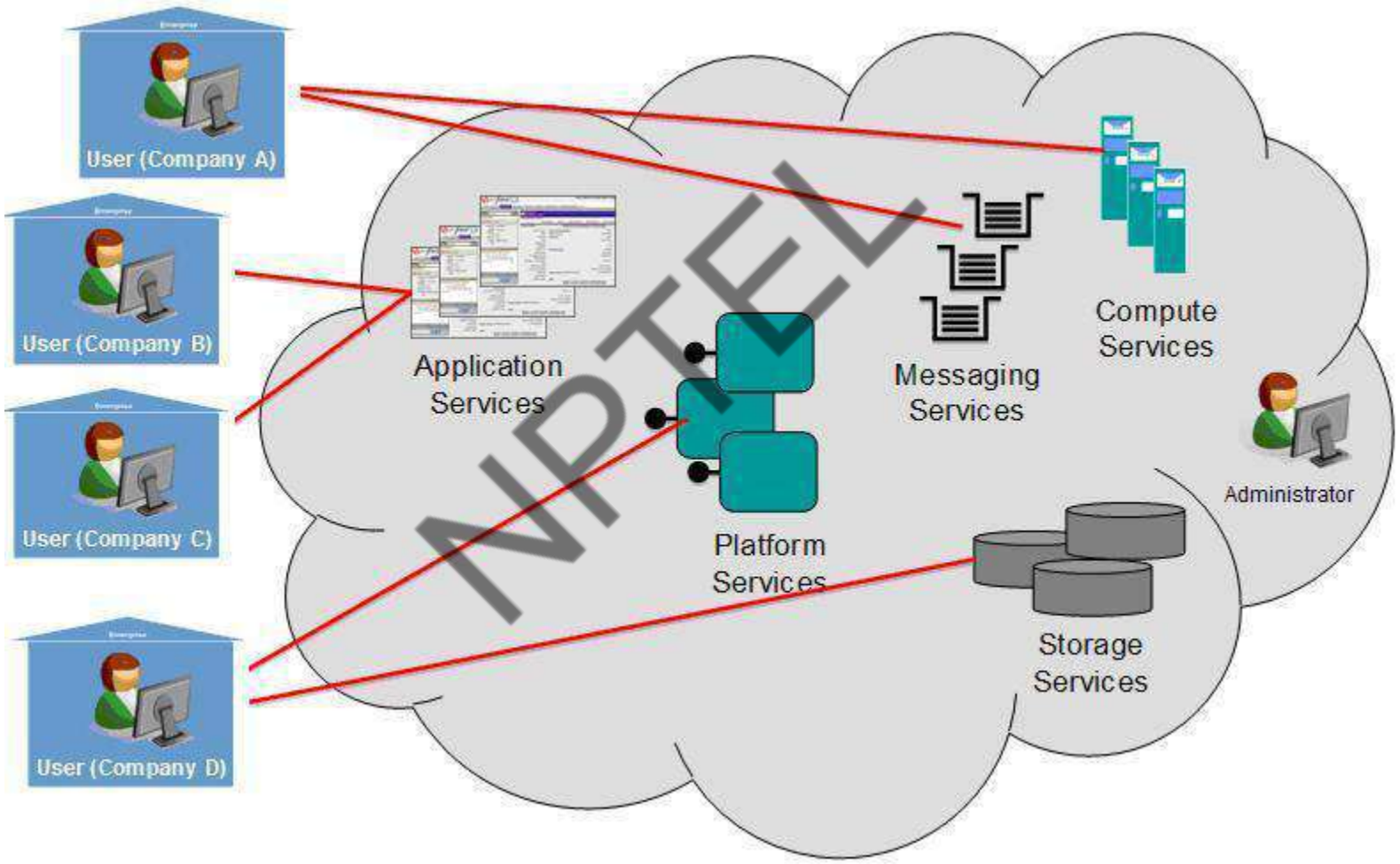
- Cloud infrastructure is provisioned for open use by the general public. It may be owned, managed, and operated by a business, academic, or government organization, or some combination of them. It exists on the premises of the cloud provider.



NPTEL

- In Public setting, the provider's computing and storage resources are potentially large; the communication links can be assumed to be implemented over the public internet; and the cloud serves a diverse pool of clients (and possibly attackers).

Public Cloud (contd..)



NPTEL

Advantages of Public Cloud

- Low up-front costs to migrate to the cloud
- Reduced set-up time
- Cheaper compared to private cloud; pay-per-use
- Scalable; unlimited resources available on demand

NPTEL

Disadvantages of Public Cloud

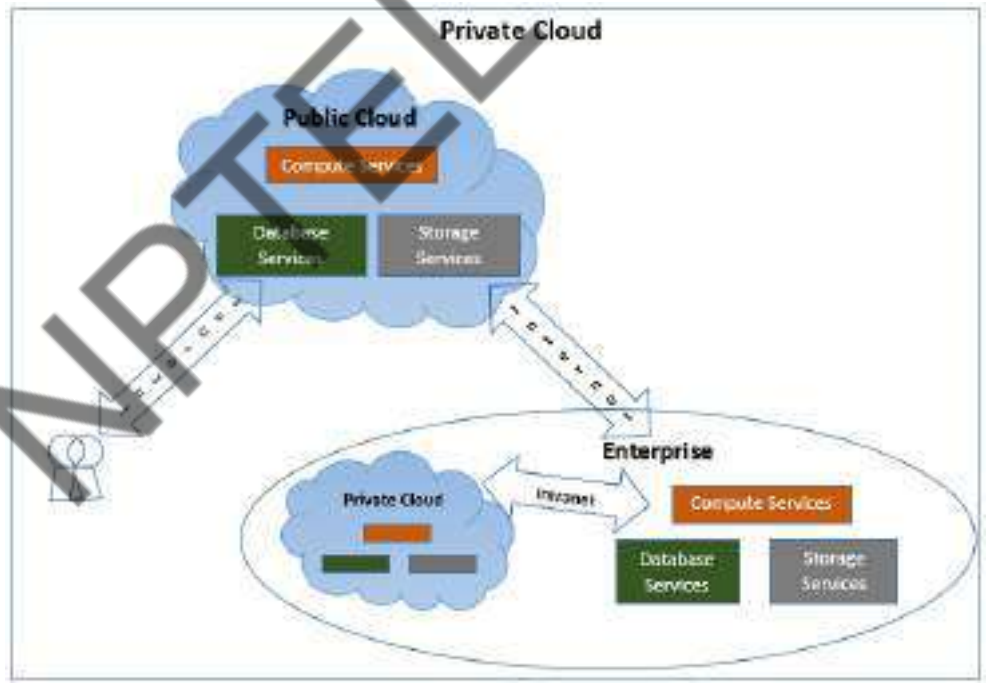
- Workload locations are hidden from clients:
  - A provider may migrate a subscriber's workload, whether processing or data, anywhere at any time depending on cost and regulatory restrictions unless the provider has offered (optional) location restriction polices.
- Risks from multi-tenancy:
  - A single machine may be shared with the workloads of competitors or adversaries.
  - Introduces both reliability and security risk.
- Restrictive default service level agreements :
  - The default service level agreements of public clouds specify limited promises that providers make to subscribers
- Limited visibility and control over data regarding security :
  - The details of provider’s system operation and software are usually considered proprietary information and are not divulged to subscribers.
  - Subscribers cannot verify if data has been completely deleted from a provider's systems.

NPTEL

Private Cloud

- The Cloud infrastructure is provisioned for exclusive use by a single organization comprising multiple consumers (e.g., business units). It may be owned, managed, and operated by the organization, a third party, or some combination of them, and it may exist on or off premises.

- Examples of Private Cloud:
  - Eucalyptus
  - Ubuntu Enterprise Cloud- UEC
  - Amazon VPC (Virtual Private Cloud)
  - VMware Cloud Infrastructure Suite
  - Microsoft ECI data center.



NPTEL

Private Cloud (contd..)

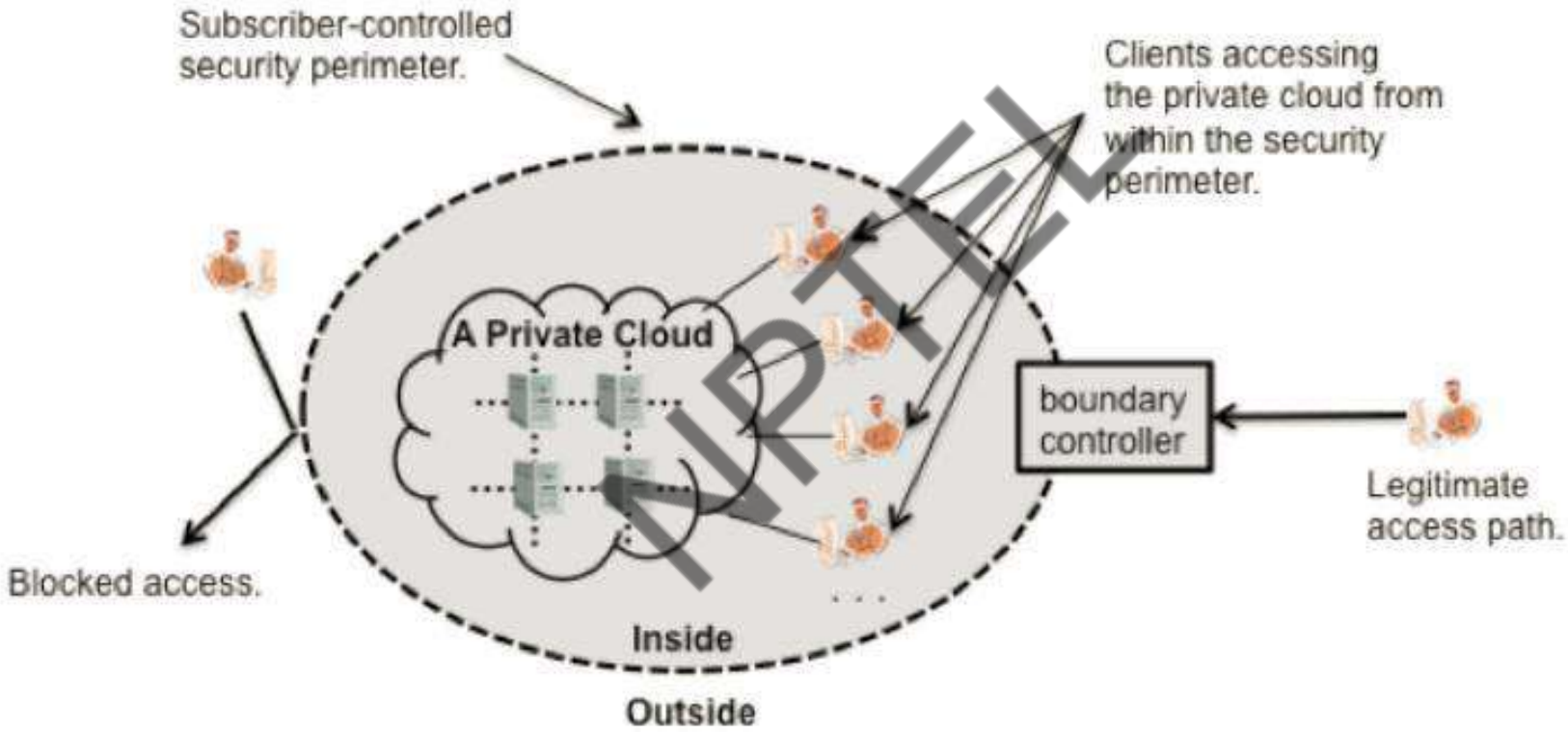
- Contrary to popular belief, private cloud may exist off premises and can be managed by a third party. Thus, two private cloud scenarios exist, as follows:

- On-site Private Cloud
  - Applies to private clouds implemented at a customer's premises.

- Outsourced Private Cloud
  - Applies to private clouds where the server side is outsourced to a hosting company.

NPTEL

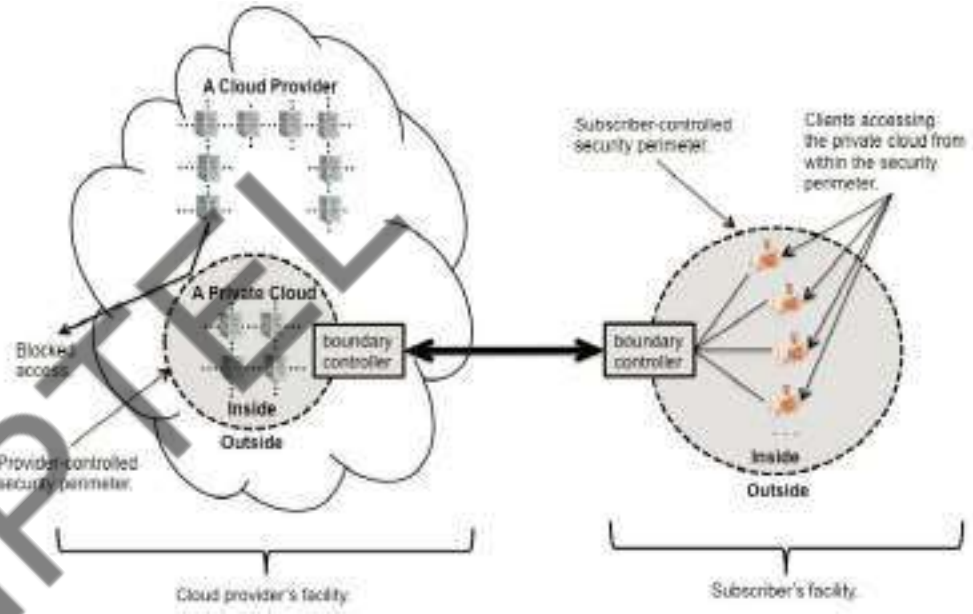
On-site Private Cloud



NPTEL

Outsourced Private Cloud

- Outsourced private cloud has two security perimeters, one implemented by a cloud subscriber (on the right) and one implemented by a provider.
- Two security perimeters are joined by a protected communication link.
- The security of data and processing depends on the strength and availability of both security perimeters and of protected communication link.



NPTEL

Advantages of Private Cloud

- Potentially strong security from external threats (both on-site and outsourced private)

NPTEL

Disadvantages of Private Cloud

- Subscribers require IT skills (on-site private)
- Significant-to-high up-front costs to migrate into the cloud (on-site-private).
- Network dependency (on-site-private).
- Modest-to-significant up-front costs to migrate into the cloud (outsourced-private).

NPTEL

Community Cloud

- Cloud infrastructure is provisioned for exclusive use by a specific community of consumers from organizations that have shared concerns (e.g., mission, security requirements, policy, and compliance considerations). It may be owned, managed, and operated by one or more of the organizations in the community, a third party, or some combination of them, and it may exist on or off premises.

Examples: Microsoft Government Community Cloud
Open Cirrus

On-site Community Cloud

- Community cloud is made of a set of participant organization. Each participant organization may provide cloud services, consumes cloud services, or both
- At least one organization must provide cloud services
- Each organization implements a security perimeter

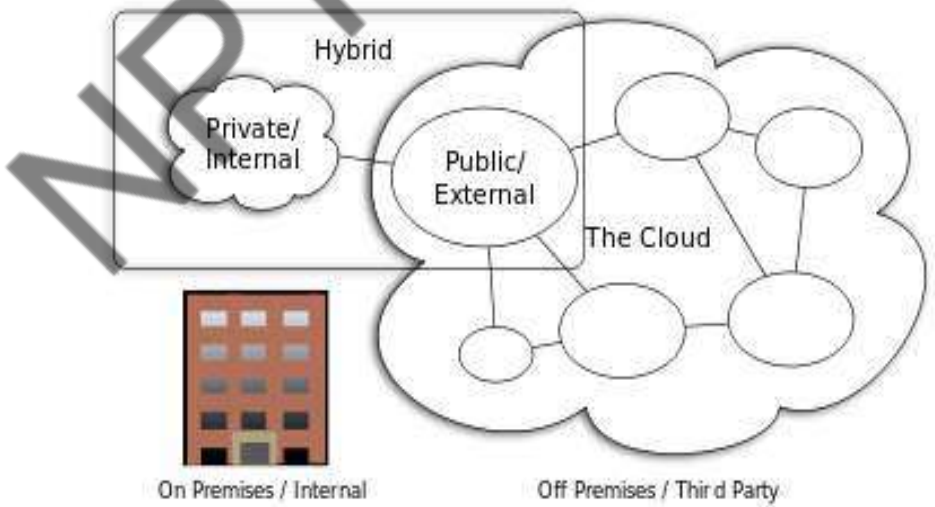
Outsourced Community Cloud

NPTEL

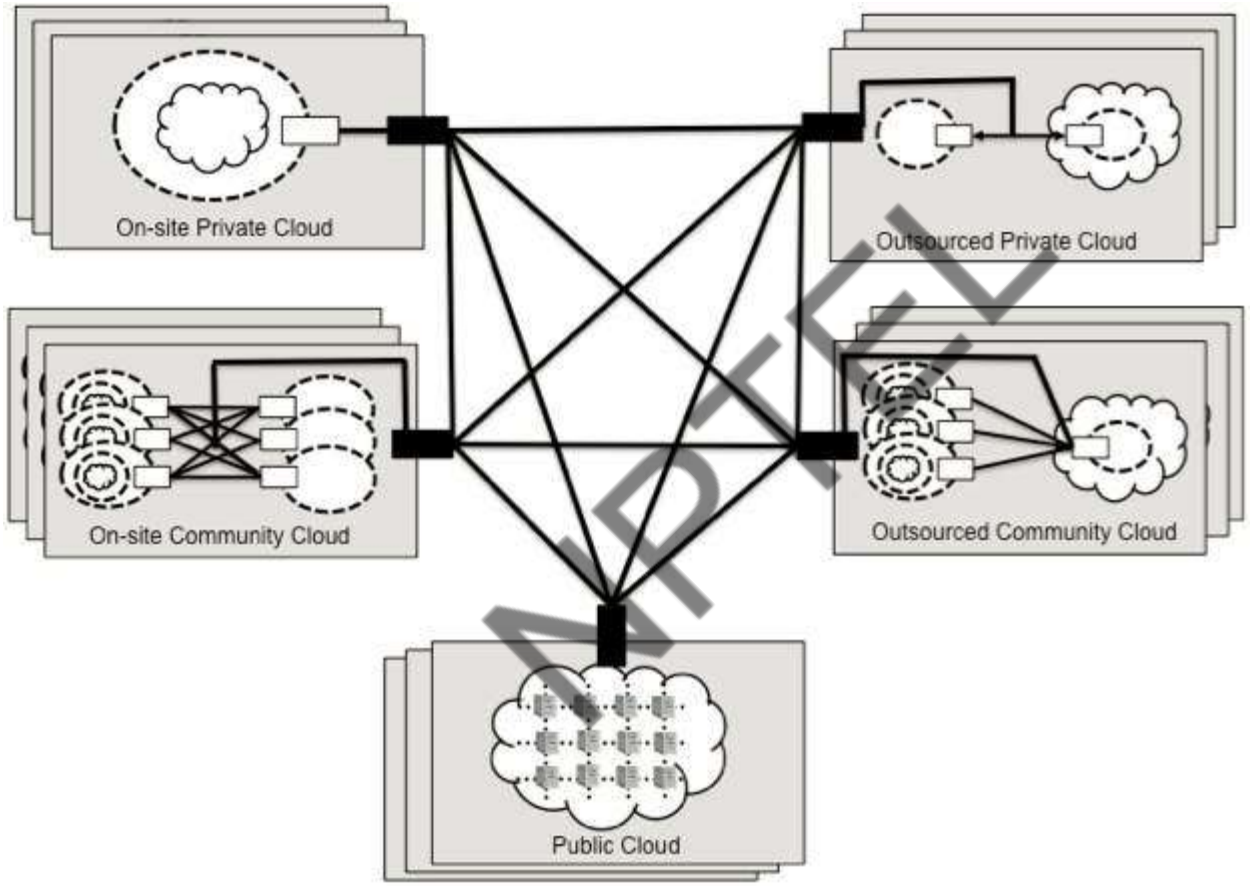
Hybrid Cloud

- A hybrid cloud is a composition of two or more clouds where each constituent cloud is one of the five variants. Hybrid clouds may change over time with constituent clouds joining and leaving.
- Examples:
  - Microsoft Azure (capable of hybrid cloud)
  - VMware vCloud

NPTEL



Hybrid Cloud (contd..)



NPTEL

Examples of hybrid cloud usage

- An e-commerce site relies on Salesforce in the public cloud to manage its customer relationship management (CRM) functions while also using a private cloud to test and build new analytics products based on that data.

- A parts manufacturer relies on a private cloud to collect and analyse billions of points of data coming in from IoT sensors but also needs to enable customers on the public cloud to see real-time order-status updates that depend on that sensor data.

- A major health care provider needs the ability to compartmentalize patient data in compliance with HIPAA through a private cloud while also enabling patients the ability to access some of their information through the provider's web app on a public cloud.

- A defence research organization performs mission-critical defence research using a private cloud but also uses a public cloud for enabling its day-to-day operational matters.

NPTEL

Advantages and Disadvantages of Hybrid Cloud

- Advantages:
  - IT manpower rationalization
  - Adequate amount of security
  - Flexibility
  - Cloud bursting is possible
  - Cost rationalization

- Disadvantages:
  - Complex to manage and handle
  - Volatile

NPTEL

Which Cloud Deployment Model to Choose?

- Spending power
- Security
- Reliability
- In-house IT expertise
- Possibility of cloud bursts
- Flexibility
- Urgency to deploy

NPTEL

References

- K. Laudon and J. Laudon (2016). Management Information Systems Publisher: Pearson. Edition 14e.

- R. De. (2018). MIS Managing Information Systems in Business, Government and Society. Publisher: Wiley. Second Edition.

- L. Badger, T. Grance, R. Patt-Corner & J. Voas. (2012). NIST Special Publication 800-146. Cloud Computing Synopsis and Recommendations available at https://nvlpubs.nist.gov/nistpubs/Legacy/SP/nistspecialpublication800-146.pdf.

NPTEL

NPTEL

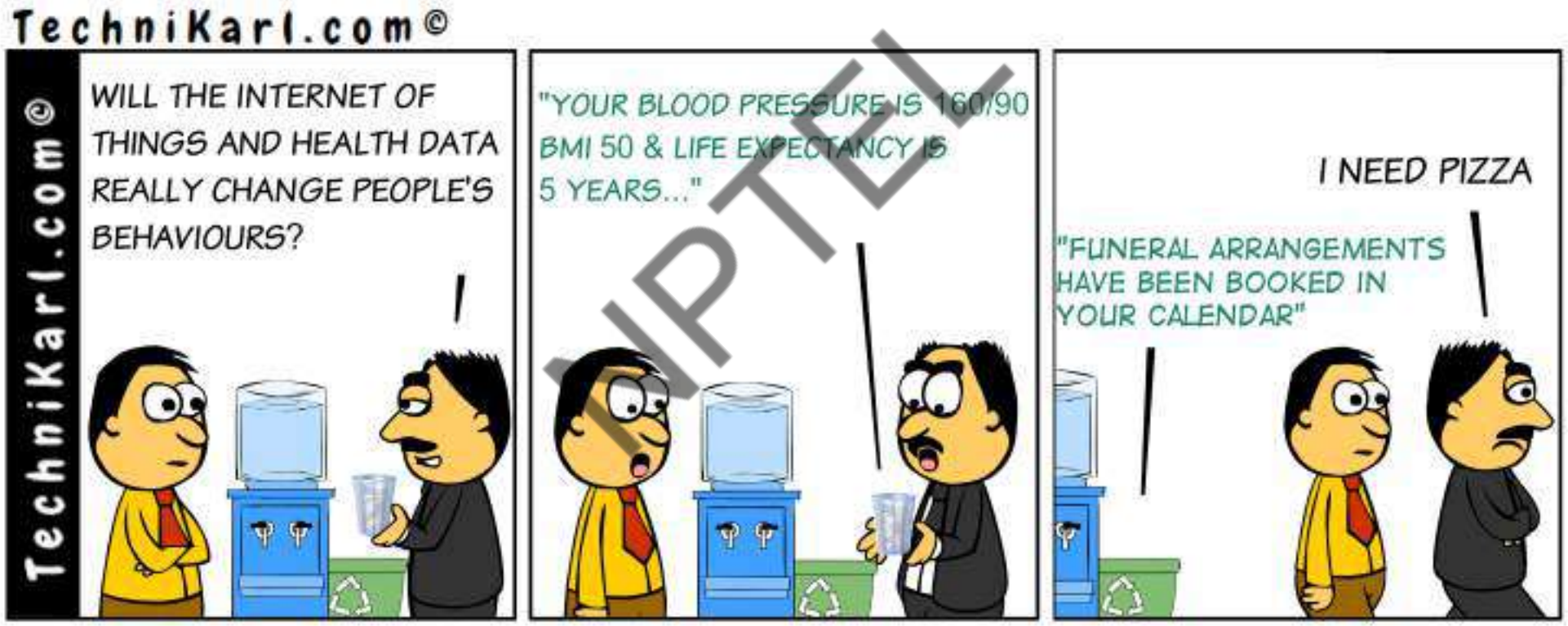
Management Information System

Saini Das
Vinod Gupta School of Management, IIT Kharagpur

NPTEL

Module 09: Emerging Technologies
Internet of Things Part - I

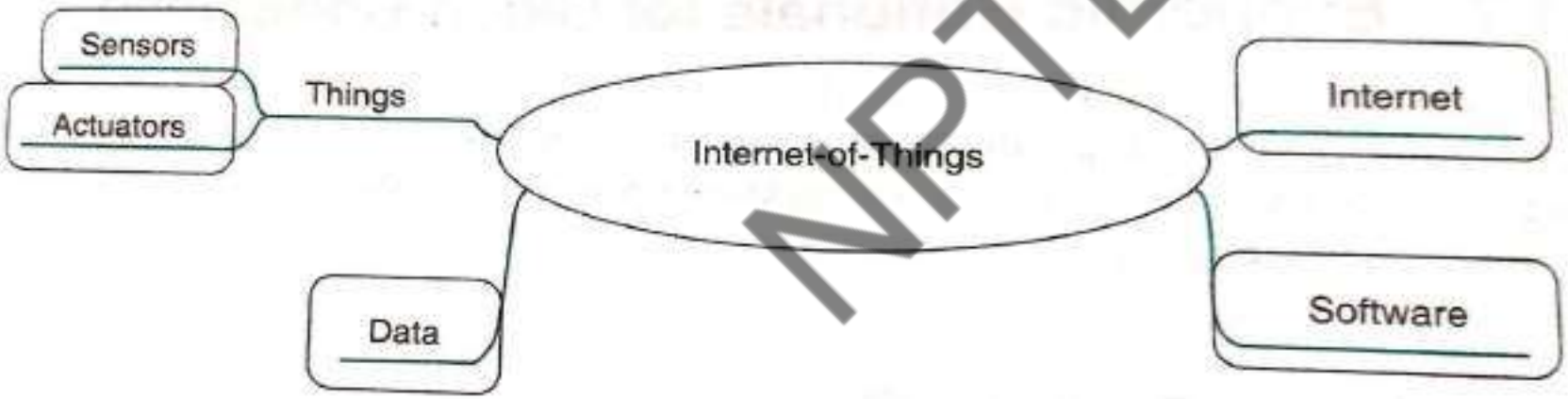
Internet of Things



NPTEL

What is IoT?

- IoT consists of a large number of devices that are connected via the Internet to sense and respond to conditions in their environments. These devices are typically **sensors** and **actuators** that exchange information with a controlling software for a defined purpose.



NPTEL

Sensors

- A sensor is a device that detects events or changes in its physical environment and provides an electronic output.
    - Simple sensor
    - Smart sensor





NPTEL

- Selection of sensor depends on attributes such as :
    - data filter capacity
    - power consumption
    - size
    - sensitivity and accuracy.

Types of Sensors

Type of Sensor	Usage
Temperature sensor	Detects changes in temperature or absolute temperature values. Wide applications in industrial machines for chemical and mechanical processes and in biological applications.
Pressure sensor	Measures the pressure of gases or liquids. Is used to measure the levels, flows and altitudes. Also used in touchscreen devices.
Flow sensor	Is used to record the rate of fluid and gas flow in pipes.
Level sensor	Is used to measure the level of liquids in containers.
Image sensor	Records images, including details of colour, intensity and brightness, for devices such as cameras.
Noise sensor	Monitors level and intensity of ambient noise. Is used to gauge noise pollution.
Air particle sensor	Monitors the level of air particles and gases such as carbon dioxide and other pollutants.
Proximity sensor	Detects the presence or absence of objects in the vicinity.
Moisture sensor	Measures the relative humidity of air. Is used inside buildings and factories.
Speed sensors	Measures the speed of objects, such as vehicles.

NPTEL

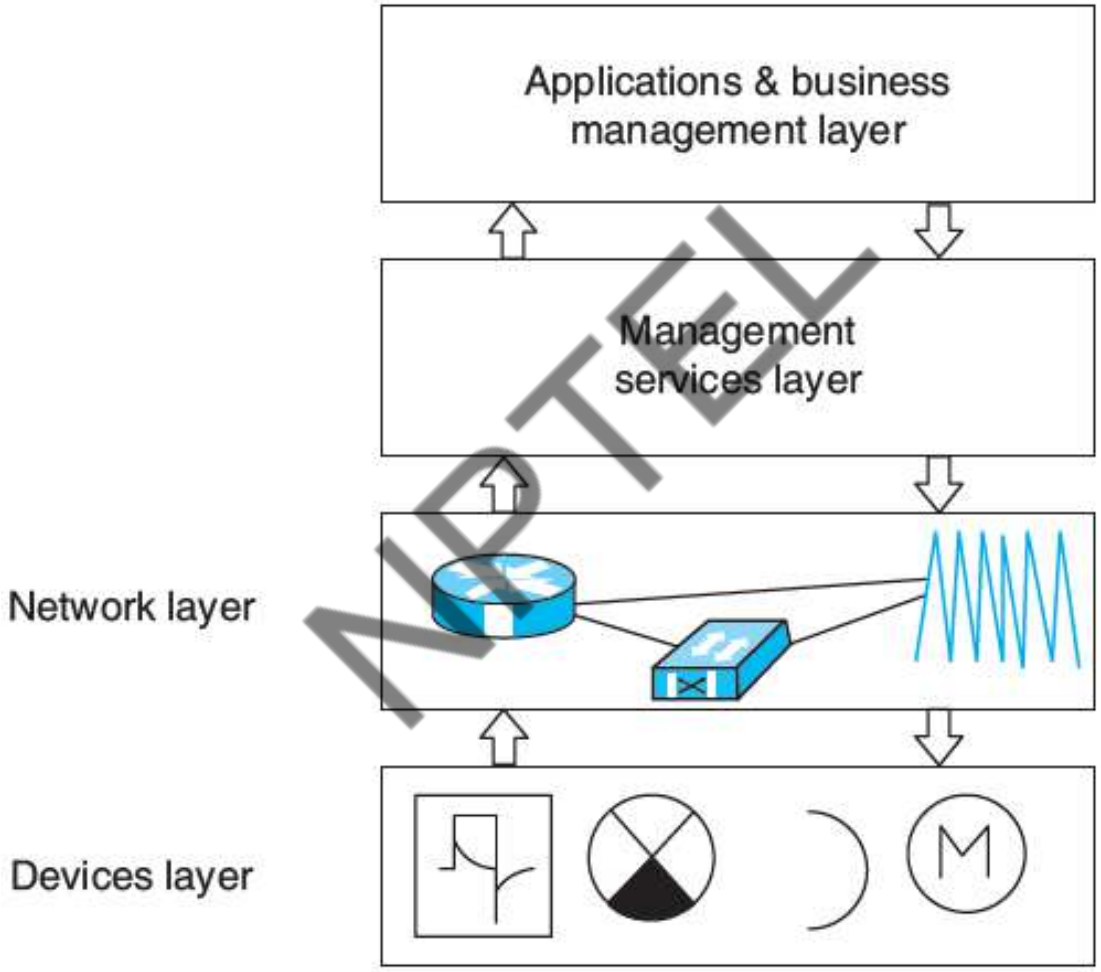
Actuators

- Actuators are tools or mechanisms that can change their own state and or the state of other devices and machines.
- They are often motors that respond to electrical or data signals and change the state of something they are connected to.

Actuator Type	Usage
Electrical	Motors that convert electrical signals to mechanical motion, e.g., change the position of a valve.
Mechanical	Devices that convert motion or energy into a different movement, e.g., movement of gears or levers.
Hydraulic and Pneumatic	Devices that respond to pressure in liquids and gases, e.g., convert pressure in liquids into pressure on container walls.

NPTEL

IoT Architecture



Four layers of the IoT architecture.

NPTEL

IoT in Smart Cities

- In the context of smart cities there may be sensors in the water infrastructure that sense and collect information on water pressure and water flows.
- This information would be relayed over Internet to a software that would identify the location of the information and process it to determine if any action needs to be taken.
- In case some action is required information will be sent over Internet to the actuators that can effect the changes in pressure and flow of water. For example: Actuator may be a motor that responds to an electronic signal to open, close or change the position of a valve.
- https://www.youtube.com/watch?v=uEsKZGOxNKw

NPTEL

IoT in Residential Water Supply

- Residential water supply is a problem in many parts of the city of Bangalore.
- Due to inadequate water supply households rely either on groundwater that is pumped up to their overhead tanks or purchase water commercially.
- Residents face problem of knowing exactly how much water they have in their storage tanks, on the basis of which they can start or stop motors or place order for water tankers.
- Gnarus Solutions has developed an loT system that addresses this problem.

NPTEL

IoT in Residential Water Supply (contd..)

- They placed sensor devices in water storage tanks in buildings that sensed the water level and sent out the data over a wifi gateway to a controlling software.
- The software processed the data and sent out updates about the water level to an app that could be accessed from the user's smart phone.
- The software also analyzed the data to produce reports on weekly and monthly water consumption patterns.
- The app has additional features that can be used to alert plumbers, tank cleaners or electricians if potential problems arise and users want something fixed or their tanks cleaned.

NPTEL

Healthcare with IoT

- Healthcare devices such as wrist bands and health bands measure body parameters such as heart rate, BP and oxygen levels.

- These parameters are relayed using Bluetooth wireless protocol, to smartphones that forward them to controlling applications.

- The Giraff project in Sweden uses a telepresence robot that is placed in the homes of patients who need monitoring and round-the-clock healthcare support. It promotes “digital aging” at home.



NPTEL

References

- K. Laudon and J. Laudon (2016). Management Information Systems Publisher: Pearson. Edition 14e.

- R. De. (2018). MIS Managing Information Systems in Business, Government and Society. Publisher: Wiley. Second Edition.

NPTEL

NPTEL

Management Information System

Saini Das
Vinod Gupta School of Management, IIT Kharagpur

NPTEL

Module 09: Emerging Technologies
Internet of Things Part - II

Classifying IOT devices by application

- **Wearables**

- Entertainment
- Fitness
- Smart watch
- Location and tracking

- **Building and Home Automation**

- Access control
- Light and temperature control
- Energy optimization
- Predictive maintenance
- Connected appliances

Health Care

- Remote monitoring
- Ambulance telemetry
- Drug tracking
- Hospital asset tracking
- Access control
- Predictive maintenance

Smart Manufacturing

- Flow optimization
- Real-time inventory
- Asset tracking
- Employee safety
- Predictive Maintenance

NPTEL

Classifying IOT devices by application (contd..)

- **Smart Cities**

- Residential e-meters
- Smart street lights
- Pipeline leak detection
- Traffic control
- Surveillance cameras
- Centralized and integrated system control

Automotive

- Infotainment
- Wire replacement
- Predictive Maintenance
- Telemetry

NPTEL

https://www.youtube.com/watch?v=NjYTzvAVozo

IoT Ecosystem

The Internet of Things **ecosystem** includes all the components and players that enable businesses, governments, and consumers to connect to their **IoT** devices. The **ecosystem** includes core, i.e., hardware, software, connectivity, standards, and application services, supporting services.

Roles include:

- Chip manufacturer,
- SIM provider,
- Module provider,
- Network operator,
- Network equipment provider
- M2M service provider
- Integrator
- Sensor and actuation service broker
- (App) Developer
- User
- Standard development org. (SDO)
- Regulatory body etc.

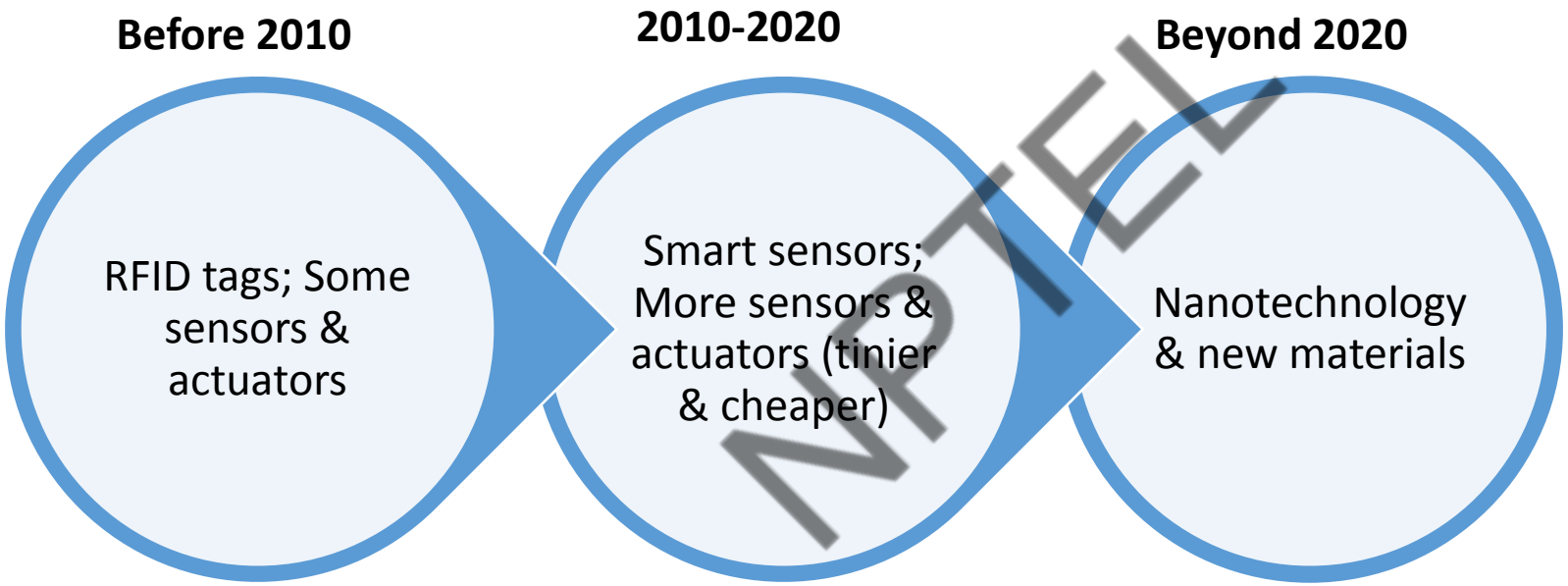
NPTEL

Key IoT Technologies

- Hardware
- Software & Algorithms
- Network
- Data processing
- Data storage

NPTEL

Hardware (Things)



NPTEL

IoT >>>>> IoNT

Software & Algorithms

- Objects rely on software to communicate effectively with each other and to deliver enhanced functionality and connectivity.
- The focus of software development is shifting to user-oriented, distributed intelligence and machine-to-machine and machine-to-human collaboration. The evolution of IoT software is as follows:

- IoT oriented Relational DBMS (pre 2010)

- Sensor middleware (2010-2020)

- Goal & User-oriented, easy-to-deploy IoT software (beyond 2020)

NPTEL

Network Technologies

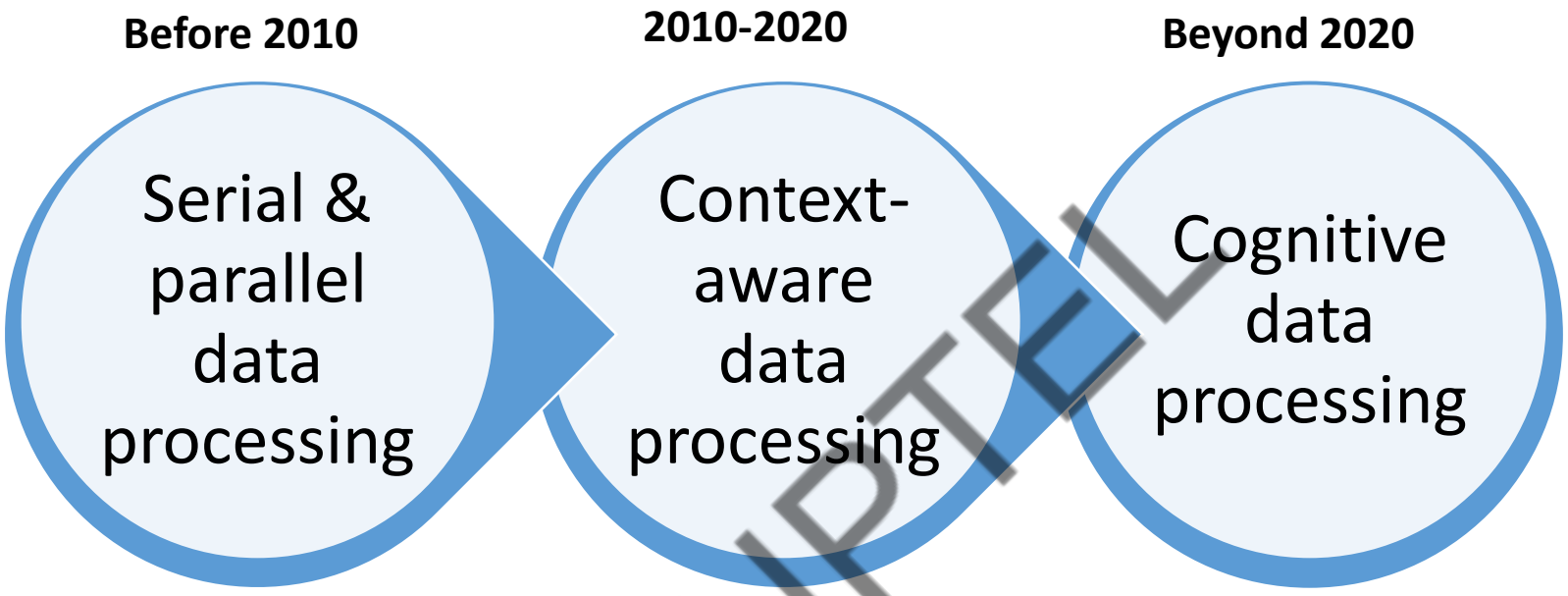
- Network is the backbone of the IoT environment.
- Over the years network technology is moving to unobtrusive wire-free communication that allows more flexible device-to-device application deployment. The evolution is as follows:

- Sensor Network (pre 2010)

- Autonomous self- organizing & self-repairing network (2010 onward).

NPTEL

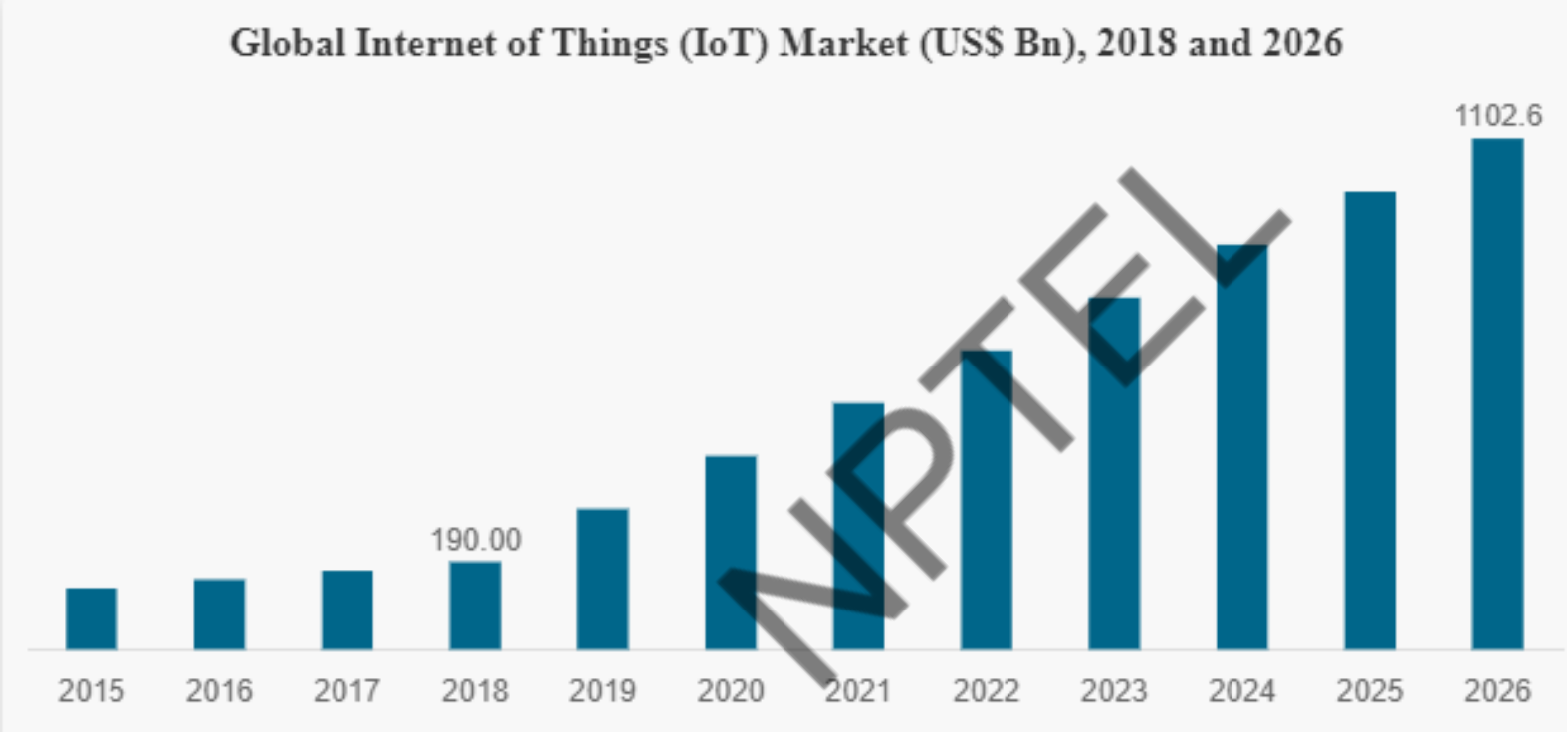
Data Processing



NPTEL

- **Context-aware data processing** enables sensors and devices to deliver relevant information to a user based on context-specific data such as users current location, climate and physical conditions (eg: within a home, dept. store, park or museum).
- **Cognitive data processing** optimizes data processing by using AI & machine learning to sense, predict, infer and learn tasks and environment based on user feedback.
- https://www.youtube.com/watch?v=LVlT4sX6uVs

IoT Markets



CAGR of 24.7% in the forecast period

Benefits

- Enhances data collection
- Real-time monitoring and decision-making
    - A CMO could view a dashboard that shows real-time traffic flow patterns within a retail outlet, and have control of in-store elements, such as display monitors, audio, lighting, and shelf stocking.
    - Smart Cities
- Greater access and control of Internet-connected devices
- Increasing efficiency and productivity
    - RFID tags on a variety of manufacturing materials to provide real-time information to assembly workers, supervisors, and managers.Connecting technologies
- Connected technologies
    - Homeowners could have a system where a variety of property-related elements—such as lighting, locks, televisions, and kitchen appliances—could be tied together into a ‘connected’ home.

NPTEL

References

- R. De. (2018). MIS Managing Information Systems in Business, Government and Society. Publisher: Wiley. Second Edition.

- I. Lee & K. Lee. (2015). The Internet of Things (IoT): Applications, investments, and challenges for enterprises. *Business Horizons.* 58, pp. 431-440.

- B. Weinberg, G. Milne, Y. Andonova, & F. Hajjat. (2015). Internet of Things: Convenience vs. privacy and secrecy. *Business Horizons.* 58, pp.615-624.

NPTEL

NPTEL

Management Information System

Saini Das
Vinod Gupta School of Management, IIT Kharagpur

NPTEL

Module 09: Emerging Technologies
Internet of Things Part - III

Benefits

- Enhances data collection
- Real-time monitoring and decision-making
    - A CMO views a dashboard that shows real-time traffic flow patterns within a retail outlet and controls in-store elements, such as audio, lighting, and shelf stocking.
    - Smart Cities
    - Healthcare
- Greater access and control of Internet-connected devices
- Increasing efficiency and productivity
    - RFID tags on a variety of manufacturing materials provide real-time information to assembly workers, supervisors, and managers.
- Connected technologies
    - Homeowners could have a system where a variety of elements—such as lighting, locks, televisions, and kitchen appliances—could be tied together into a ‘connected’ home.

NPTEL

Challenges in IoT

- Data Management: Storage and processing of zettabytes of data
- Data Ownership
- Data mining sophistication
  - In 2015 McKinsey Global Institute estimated that the United States needs 140,000 to 190,000 more workers with analytical skills and 1.5 million managers and analysts with analytical skills to make business decisions based on the analysis of big data.
- Privacy and Security
- Legal
- Chaos
- Internet Access

NPTEL

Privacy Challenges of IoT

"The right to be left alone and the right to be free of unreasonable personal intrusions, surveillance or interference from other individuals or organizations".








Tradeoff between privacy and QoS

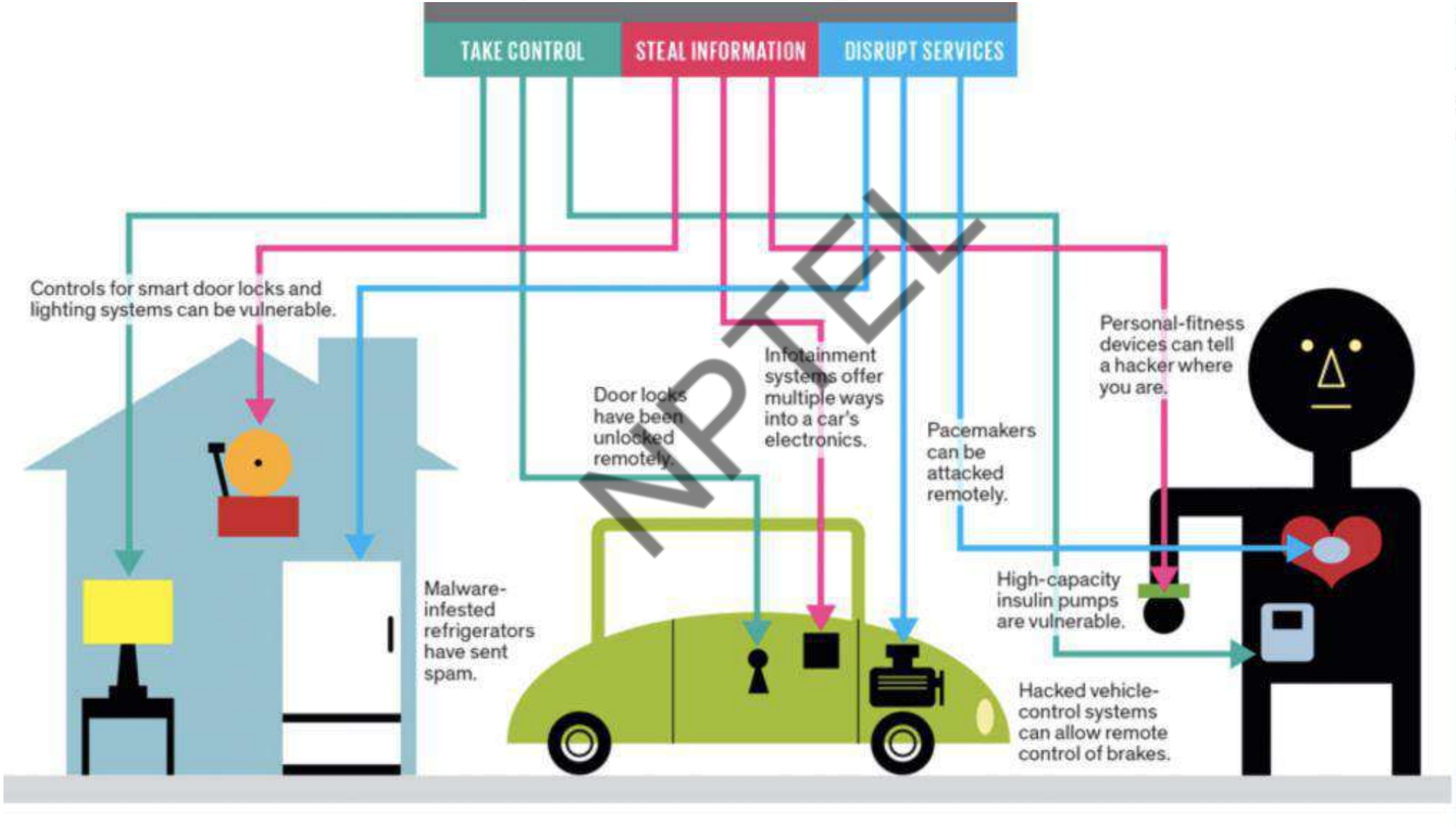
NPTEL

Privacy by Design

- Privacy by design is a process that calls for proactive consideration of privacy objectives and aims from the start, then continues throughout the design and delivery process of products and related actions.
- Seven foundational principles associated with a privacy by design process are:
  - Privacy should be proactive and preventative rather than reactive and remedial
  - Privacy is a default condition rather than an option to be selected
  - Privacy is embedded in the design of a product and is not an add-on
  - Privacy does not impede the full functionality of a product
  - Adequate security measures are applied to protect privacy throughout an entire system wherever sensitive data may travel and throughout the lifecycle of the data.
  - Privacy procedures are visible and transparent to support accountability and trust in delivering on privacy related promises and objectives.
  - Privacy design reflects respect for users' interests and empowers them to manage their data through actions pertaining to consent, accuracy, access, and compliance.

NPTEL

Security Challenges of IoT



NPTEL

Security Challenges of IoT (contd..)

- IoT devices have vulnerabilities due to lack of encryption, insecure web interfaces, inadequate software protection, and insufficient authorization.

- On average, each device contained 25 loopholes, or risks of compromising the home network (HP survey 2014)

- Security challenges may be resolved by training developers to incorporate security solutions (e.g., intrusion prevention systems, firewalls) into products and encouraging users to utilize IoT security features that are built into their devices.

NPTEL

Legal Challenges of IoT

- **Data Ownership and liability issues** : Multiple stakeholders, MGI and M2M communication generated in an IoT environment pose ownership and liability issues.

- **Intellectual Property rights**: When an original data is created by virtue of the interaction of various devices in an IoT environment, which may include, *inter alia*, a new process of arriving at desired results, who claims the IP Rights in such content/data/process?

NPTEL

Chaos challenge of IoT

- In a hyper-connected world, an error in one part of a system can cause disorder throughout.
- A single device may have an insignificant problem, but for the system as a whole, the chain reactions of other connected devices can become disastrous.
- If not designed carefully, multi-purpose devices and collaborative applications can turn our lives into chaos.
- To prevent chaos in the hyper-connected IoT world, businesses need to make effort to reduce the complexity of connected systems, enhance the security and standardization of applications, and guarantee the safety and privacy of users anytime, anywhere.

NPTEL

What is Big Data?

- In order to address the issues of storing and processing huge amounts of data generated by IoT systems Big Data technology comes into picture.
- Huge volume of data produced by both humans and machines, at a very high rate and with massive variety is referred to as Big Data.
- Key enablers for the growth of "Big Data" are:
  - Increase of storage capacities
  - Increase of processing power
  - Increased availability of data

  https://www.youtube.com/watch?v=eVSfJhssXUA



NPTEL

References

- R. De. (2018). MIS Managing Information Systems in Business, Government and Society. Publisher: Wiley. Second Edition.

- I. Lee & K. Lee. (2015). The Internet of Things (IoT): Applications, investments, and challenges for enterprises. *Business Horizons.* 58, pp. 431-440.

- B. Weinberg, G. Milne, Y. Andonova, & F. Hajjat. (2015). Internet of Things: Convenience vs. privacy and secrecy. *Business Horizons.* 58, pp.615-624.

NPTEL

NPTEL

Management Information System

Saini Das
Vinod Gupta School of Management, IIT Kharagpur

NPTEL

Module 09: Emerging Technologies

Big Data and other Emerging Technologies

What is Big Data?

- Huge volume of data produced by both humans and machines, at a very high rate and with massive variety is referred to as Big Data.

- Key enablers for the growth of “Big Data” are:
  - Increase of storage capacities
  - Increase of processing power
  - Availability of data



NPTEL

4Vs of Big Data

Volume:

- Around 40,000 exabytes of data are being produced annually.

- Sources of huge volumes of data:
  - Posts, images and videos shared on social media sites
  - Sensors attached to IoT devices
  - Online transactions for banking and e-commerce
  - Scientific research and experiments such as weather analysis
  - Astronomical observations.

NPTEL

4Vs of Big Data (contd..)

Velocity:

- The rate at which data is produced or accumulated is very high.

NPTEL

Source	Volume of Data
YouTube videos	300 hours of video uploaded every minute. This is about 1.6 terabytes of data per minute.
IoT devices	A jet engine produces about 300 gigabytes of data per minute.
Emails	There were 187 million emails generated per minute, worldwide.
Social media	There were more than 455,000 tweets generated per minute. 15 million text messages were sent every minute. Facebook logged 4 million posts per minute (in 2014).

4Vs of Big Data (contd..)

Variety:

- Big Data may be structured, semi-structured or unstructured.
- Unstructured data constitutes almost 80% of Big Data and consists of files that are independent and not relationally linked to other files.
- Examples:
  - Log files from servers
  - Collection of tweets
  - Collection of texts
  - Video or audio posts on social media sites
  - Logs of chat sites

NPTEL

4Vs of Big Data (contd..)

Variety (contd..):

- Semi-structured data has some inherent structure, either corresponding to a hierarchy or graph.
- Lacks relationship between the files
- Often maintained in XML or JSON languages that are readable both by humans and computer programs.
- Data stored in such formats is text-based and one of the major sources of such data is sensors.
- Structured data have some relationships between them

  and is highly organized.

NPTEL

4Vs of Big Data (contd..)

Veracity:

- Refers to the validity of data.
- Data that can used to process for valuable information is called signal; the rest is considered as noise.
- Big data often has noise that is generated at the source.
  - Incorrect names/hashtags; incorrect or misleading addresses
  - Incorrect data readings from sensors; missing data values, etc.
- High values for signal-to-noise ratio are always preferred for Big Data environments.

NPTEL

Technology behind Big Data

- Several Big Data database products are used in the industry such as, Hadoop and Mongo DB.

- These databases are able to process massive volumes of data across distributed databases.

- Their architecture is very different from that of traditional relational databases and is beyond the scope of this course.

NPTEL

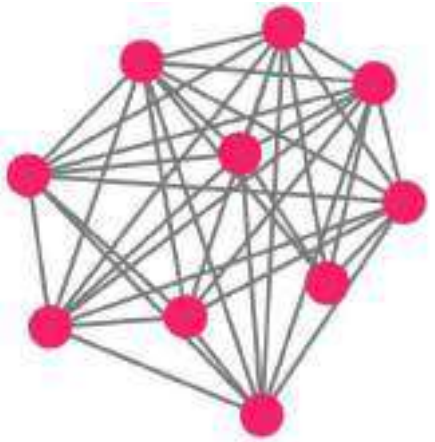
Applications of Big Data

- A major U.S airline used publicly available data about weather, flight schedules, and other factors with proprietary data the company itself collected, including feeds from a network of passive radar stations it had installed near airports to gather data about every plane in the local sky, to calculate the ETAs of it's aircrafts at the airports.

- Flipkart collects big data about :
  - customers page visits, logins, product browsing behaviour, bounce off pages, product purchases etc.
  - Logistics
  - Previous 'Big Billion Day' sales.

NPTEL

Blockchain Technology



- Blockchains are software artifacts that enable the creation of public ledgers that are a record of transactions maintained on widely distributed network of clients.
- Once a ledger entry is made it cannot be tampered with and it becomes a permanent entry.
- The entries in the ledger are visible to all collaborating parties.
- The blockchain data is maintained on distributed entries that are secure and cannot be hacked.
- Blockchains provide a proof of authenticity in that only verified and legitimate parties can make the ledger entries.

NPTEL

Applications of Blockchains

- Blockchains are used to manage supply chains. With a blockchain-enabled supply chain, a client can know exactly when a particular shipment has been made by a global supplier, the conditions under which they were stored and transferred and when they will arrive.

- Blockchains can be used by banks who can set up blockchain ledgers for lending money to customers and also recording payments through the system.

- Blockchains can be used for distribution of artistic works, like songs or books directly from songwriters and authors to buyers, without the intermediaries.

NPTEL

Virtual Reality



- **Virtual Reality (VR)** is a computer-generated simulation of an alternate world or reality. It is used in 3D movies and video games. Completely immersive virtual environment. It helps to create simulations similar to the real world and "immerse" the viewer using computers and sensory devices like headsets and gloves. VR is 75% virtual and 25% real.

- Examples:
- Ford Motor Company uses VR to design vehicles.
- At NYU Medical Centre, students wearing 3D-glasses are able to “dissect” a virtual cadaver projected on a screen.

NPTEL

Augmented Reality



- **Augmented Reality (AR)** is a perfect blend of the digital world and the physical elements to create an artificial environment. The system augments the real-world scene. AR is 25% virtual and 75% real.

- Example:
  - Sephora Virtual Artist.
  - Image guided surgery uses AR where images obtained from ultrasound, MRI or CT scans are superimposed on the patient in the operating room.
  - Ikea

NPTEL

References

- K. Laudon and J. Laudon (2016). Management Information Systems Publisher: Pearson. Edition 14e.

- R. De. (2018). MIS Managing Information Systems in Business, Government and Society. Publisher: Wiley. Second Edition.

NPTEL

NPTEL



NPTEL ONLINE CERTIFICATION COURSES

NPTEL

Management Information Systems

Prof. Kunal Kanti Ghosh
VGSoM, IIT KHARAGPUR

Week 10: Module 1
Knowledge Management Systems (KMS)

CONCEPTS COVERED

- Basic Concepts of Knowledge Management Systems
- Knowledge Management Value Chain
- Types of Knowledge Management Systems

NPTEL

Knowledge Management Systems

- Knowledge management refers to the set of business processes developed in an organization to create, store, transfer, and apply knowledge

- Knowledge management increases the ability of an organization to learn from its environment and to incorporate knowledge into its business processes

NPTEL

Knowledge Management Systems

- **Important Dimensions of Knowledge**

- **Data: Flow of captured events or transactions**

- **Information: Data organized into categories of understanding**

- **Knowledge: Concepts, experience, and insight that provide a framework for creating, evaluating, and using information. Can be tacit (undocumented) or explicit (documented)**

NPTEL

Knowledge Management Systems

- Important Dimensions of Knowledge

- Wisdom: The collective and individual experience of applying knowledge to the solution of problem; knowing when, where, and how to apply that knowledge
- Knowledge is a firm asset:
  - Intangible asset
  - Requires organizational resources
  - Value increases as more people share it

NPTEL

Knowledge Management Systems

❖ Important Dimensions of Knowledge

❑ Knowledge has different forms:

- Tacit or explicit
- Know-how, craft, and skill
- Knowing how to follow procedures; why things happen

NPTEL

Knowledge Management Systems

❖ **Important Dimensions of Knowledge**

❑ **Knowledge has a location:**

- **Cognitive event**
- **Social and individual bases of knowledge**
- **Sticky, situated, contextual**

NPTEL

Knowledge Management Systems

❖ **Important Dimensions of Knowledge**

❑ **Knowledge is situational:**

- **Conditional**
- **Contextual**

NPTEL

Knowledge Management Systems

❖ **Organizational Learning and Knowledge Management:**

❑ **Organizational learning: Adjusting business processes and patterns of decision making to reflect knowledge gained through information and experience gathered**

NPTEL

Knowledge Management Systems

❖ **Knowledge Management Value Chain**

- **Knowledge acquisition**
- **Knowledge storage**
- **Knowledge dissemination**
- **Knowledge application**
- **Building organizational and management capital: collaboration, communities of practice, and office environments**

NPTEL

Knowledge Management Systems

❖ **Knowledge Acquisition:**

➢ **Organizations acquire knowledge in various ways:**

- **Corporate repositories of documents, reports, presentations and best practices**
- **Unstructured documents like e-mails**
- **Online expert networks**
- **Knowledge discovery through identification of patterns in corporate data**
- **Knowledge workstations for discovering new knowledge**

NPTEL

Knowledge Management Systems

❖ **Knowledge Storage:**

➢ **Documents, patterns, and expert rules must be stored through various means so they can be retrieved and used by employees:**

- **Creation of a database**
- **Document management systems (for digitizing, indexing & tagging)**
- **Expert systems**

NPTEL

Knowledge Management Systems

- **Knowledge Dissemination:**
- **Various means through which organization knowledge is disseminated are:**
  - **Portals, e-mail, instant messaging, wikis, social business tools**
  - **Collaboration tools for sharing calendars, documents, data, and graphics**
  - **Search engines technology**

NPTEL

Knowledge Management Systems

- **Knowledge Application:**
- **In order to add business value and justify ROI, organizational knowledge must become a part of management decision making and incorporated into**
  - **Decision support systems**
  - **Enterprise applications for managing key internal business processes and relationships with customers and suppliers**

NPTEL

Knowledge Management Systems

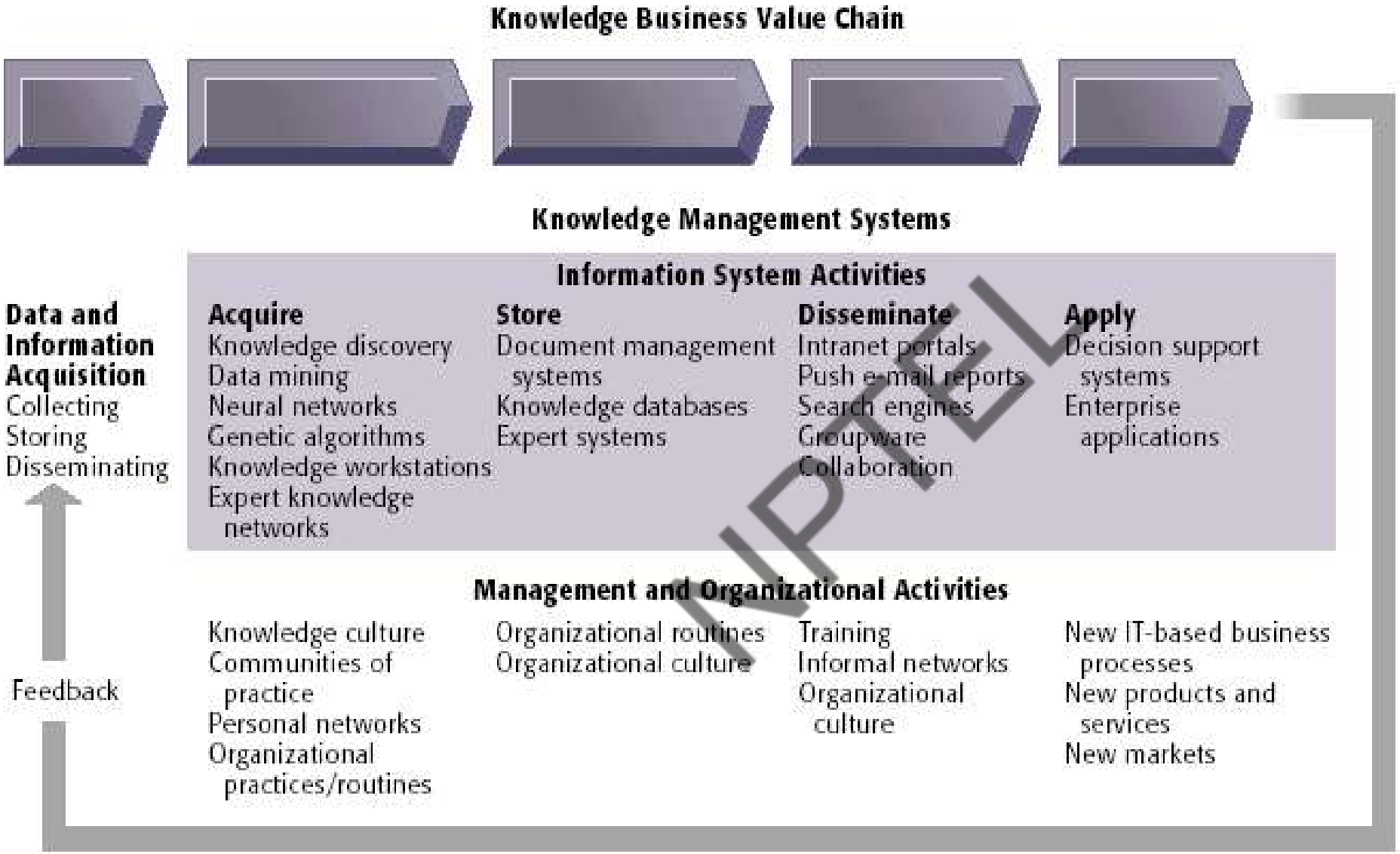
❖ **Building Organizational and Management Capital:**

➢ **Collaboration, and Office Environment**

➢ **New roles and responsibilities:**

- **Chief Knowledge Officer**
- **Knowledge Managers**
- **Communities of Practices (informal social networks of professionals and employees within and outside the firm who have similar work-related activities and interests)**

NPTEL



NPTEL
NPTEL Online Certification Courses
IIT Kharagpur

Knowledge Management Systems

- **Types of Knowledge Management Systems:**
  - **Enterprise-wide knowledge management systems**
  - **Knowledge work systems**
  - **Intelligent techniques**

NPTEL

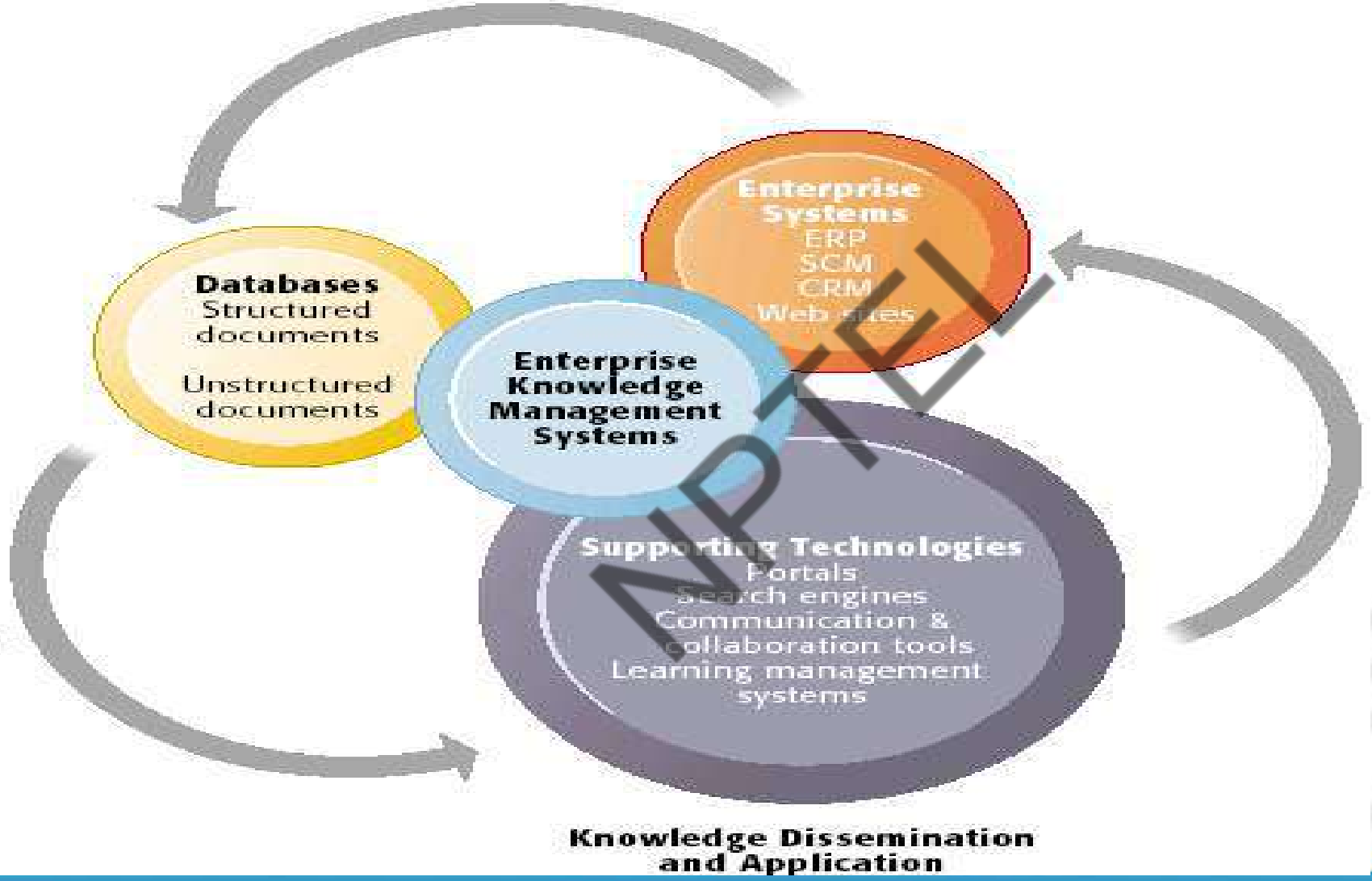
Knowledge Management Systems

- ❖ Enterprise-wide knowledge management systems
- ➢ General-purpose firm wide efforts to collect, store, distribute, and apply digital content and knowledge
- ➢ These systems include capabilities for
  - Searching for information,
  - Storing both structured and unstructured data, and
  - Locating employee expertise within the firm
  - Supporting technologies (e.g., portals, search engines, collaboration and social business tools, and learning management systems)

NPTEL

Enterprise-Wide Knowledge Management Systems Overview

Data and information generation



NPTEL

Knowledge Management Systems

- **Knowledge work systems:**
  - **Specialized systems built for engineers, scientists, and other knowledge workers charged with discovering and creating new knowledge for a company**

NPTEL

Knowledge Management Systems

❖Examples of knowledge work systems:

- CAD: Information system that automates the creation and revision of industrial and manufacturing designs using sophisticated graphics software
- Virtual Reality Systems: Interactive graphics software and hardware that create computer-generated simulations that emulate real-world activities or photorealistic simulations
- Investment Workstations: Powerful desktop computer for financial specialists, which is optimized to access and manipulate massive amounts of financial data

NPTEL

Knowledge Management Systems

❖Intelligent Techniques:

- Data Mining (focus on discovering knowledge)
- Expert Systems (distilling knowledge in the form of rules for a computer program)
- Neural Networks(focus on discovering knowledge)
- Fuzzy Logic
- Genetic Algorithms (for optimal/effective solutions)
- Intelligent Agents

NPTEL

Knowledge Management Systems

❖ **Intelligent Techniques:**

➢ **Expert system:**

- **An intelligent technique for capturing tacit knowledge in a very specific and limited domain of human expertise**

➢ **Fuzzy logic:**

- **Rule-based technology that can represent imprecise values or ranges of values by creating rules that use approximate or subjective values**
- **Used for problems that are difficult to represent by IF-THEN rules**

NPTEL

Knowledge Management Systems

❖Intelligent Techniques:

➢Neural Networks:

- Hardware or software that emulates the processing patterns of the biological brain to discover patterns and relationships in massive amounts of data

- Use large numbers of sensing and processing nodes that interact with each other

NPTEL

Knowledge Management Systems

❖**Intelligent Techniques:**

➢**Neural Networks:**

- **Uses rules it 'learns" from patterns in data to construct a hidden layer of logic that can be applied to model new data**

- **Applications are found in medicine, science, and business**

NPTEL

Knowledge Management Systems

❖ **Intelligent Techniques:**

➢ **Genetic Algorithms:**

- **Adaptive computation that examines very large number of solutions for a problem to find optimal solution**
- **Programmed to "evolve" by changing and reorganizing component parts using processes such as reproduction, mutation, and natural selection: worst solutions are discarded and better ones survive to produce even better solutions**

NPTEL

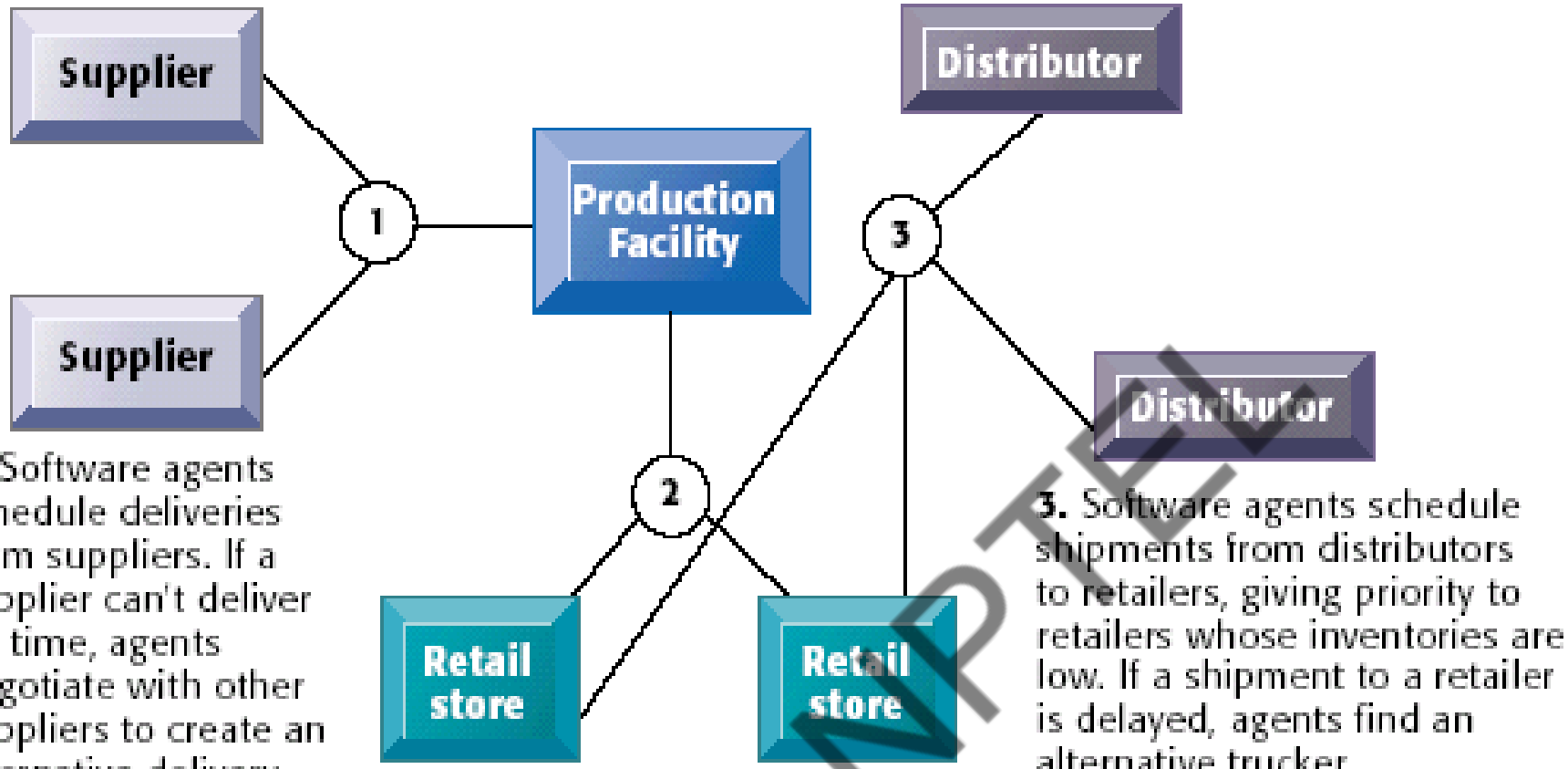
Knowledge Management Systems

❖ **Intelligent Techniques:**

➢ **Intelligent Agents:**

- **Software programs that work in the background without direct human intervention to carry out specific, repetitive, and predictable tasks for an individual user, business process, or software application**

NPTEL



1. Software agents schedule deliveries from suppliers. If a supplier can't deliver on time, agents negotiate with other suppliers to create an alternative delivery schedule.

2. Software agents collect real-time sales data on each P&G product from multiple retail stores. They relay the data to P&G production for replenishing orders and to sales and marketing for trend analysis.

3. Software agents schedule shipments from distributors to retailers, giving priority to retailers whose inventories are low. If a shipment to a retailer is delayed, agents find an alternative trucker.

NPTEL

Knowledge Management Systems

❖Management Challenges:

- Insufficient resources are available to structure and update the content in repositories.
- Poor quality and high variability of content quality results from insufficient validating mechanisms.
- Content in repositories lacks context, making documents difficult to understand.

NPTEL

Knowledge Management Systems

❖**Management Challenges:**

- **Individual employees are not rewarded for contributing content, and many fear sharing knowledge with others on the job.**

- **Search engines return too much information, reflecting lack of knowledge structure or taxonomy.**

NPTEL

Knowledge Management Systems

❖Steps in Development of Successful Knowledge Management Projects:

- Develop in stages
- Choose a high-value business process
- Choose the right audience
- Measure ROI during initial implementation
- Use the preliminary ROI to project enterprise-wide values

NPTEL

REFERENCES

- Laudon, K. C., and Laudon, J. P. (2016), Management Information Systems: Managing the Digital Firm
- Turban, E., Volonio, L., Wood, R. Gregory.(2016), Information Technology for Management
- Rainer, K., Prince, B., Watson, H.,(2015), Management Information Systems
- De, Rahul.,(2019), MIS: Managing Information Systems in Business, Government and Society
- Bagchi, Nirmalya., (2010), Management Information Systems

NPTEL



NPTEL

THANK YOU !



NPTEL ONLINE CERTIFICATION COURSES

NPTEL

Decision Support Systems

Prof. Kunal Kanti Ghosh
VGSoM, IIT KHARAGPUR

Week 10: Module 2

Decision Support Systems

CONCEPTS COVERED

- **Introduction to Decision Support Systems (DSS)**

NPTEL

Introduction to Decision Support Systems (DSS)

- Man-Computer interactive system
- Aid management in making decisions
- Combines Computer's ability to process databases and models with manager's ability to draw on his experience and exercise his judgement

NPTEL

Introduction to Decision Support Systems

- Scope
  - Role played by DSS in solving managerial decision problems
  - Basic components of a DSS
  - Basic steps involved in design of a DSS

NPTEL

Introduction to Decision Support Systems

- Types of Managerial Decision Problems
  - Structured
  - Unstructured
  - Semi-structured
- The category into which a decision problem falls depends on how well its objectives can be defined and whether there are well-established procedures for solving it

NPTEL

Introduction to Decision Support Systems

❖ **Types of Managerial Decision Problems**

- **Problems with well defined objectives and well established solution procedures are said to be Structured (e.g., Inventory Replenishment)**

- **Problems with vague objectives and with no well established solution techniques are Unstructured (e.g., New products to be manufactured)**

- **Semi-Structured problems combine the features of both**

NPTEL

Introduction to Decision Support Systems

❖ **Types of Managerial Decision Problems**

- **The structured problems can be completely delegated to the computer whereas the unstructured problems have to be solved by the manager alone**
- **The semi-structured problem which lie between the structured and unstructured problems can be solved neither wholly by a computer nor wholly by a manager;**
- **They require a man-computer team for their solution**
- **It is, therefore, in handling the semi-structured problems that the decision support systems can be effective**

NPTEL

Introduction to Decision Support Systems

- **Different Phases of a Decision-Making Process**
  - **Intelligence: Here the decision maker scans his environment, processes raw data and identifies the problem areas**
  - **Design: For the problems identified in the above phase, various alternatives are formulated and analysed in this phase**
  - **Choice: Here, a selection is made from the alternatives proposed in the above phase, and the chosen alternative is implemented**

NPTEL

Introduction to Decision Support Systems

- **Problem Solving Process in a DSS:**
  - **Iterative and interactive**
  - **The computer provides some output to the manager, the manager then interprets the output, gives new input, modifies the existing data or asks new questions**
  - **The computer's major tools are models, both simple and sophisticated, whereas the manager uses mostly his experience and judgment**

NPTEL

Introduction to Decision Support Systems

- **Problem Solving Process in a DSS:**
  - **The models used by the computer consist of relationships between the major elements of the system being studied.**
  - **Such relationships usually involve a certain degree of abstraction**
  - **The output of the models will be an approximation of the behaviour of the real system**
  - **The models will be valid only if certain assumptions about the real system are satisfied**

NPTEL

Introduction to Decision Support Systems

- **MIS vs DSS**
  - **DSS designer views effectiveness as more important than efficiency**
    - **Effectiveness involves identifying what should be done and ensuring that the chosen criterion is a relevant one**
    - **Efficiency involves minimization of time, cost or effort in performing a specified activity**
  - **In DSS, the computer is used not to replace the decision maker but to enhance his/her decision making abilities**

NPTEL

Introduction to Decision Support Systems

- **MIS vs DSS**
  - **In DSS, the manager plays a very active role, using his/her experience and judgment wherever necessary, and controls as well as directs the computer in its tasks**

NPTEL

Introduction to Decision Support Systems

- ❖ **Range of Capabilities of a DSS**
- ➢ **Not all DSS need to include models in order to be effective**
- ➢ **A DSS can provide simple data retrieval facilities and yet be useful in aiding decision making .**

NPTEL

Introduction to Decision Support Systems

❖ **Range of Capabilities of a DSS**

➢ **The range of facilities offered by DSS can be listed as:**

- **Retrieving information**
- **Providing mechanism for adhoc data analysis**
- **Providing pre-specified aggregation of data in the form of reports**
- **Estimating the consequences of proposed decisions**
- **Proposing decisions**

NPTEL

Introduction to Decision Support Systems

- ❖ Need for DSS Approach
- ➢ Problem is semi-structured
  - ➢ All the parameters of the problem are not clearly defined
  - ➢ The parameters are firmed up as one proceeds through the solution process
- ➢ Inputs are dynamic in nature
  - ➢ The input data for the problem are not fixed
  - ➢ These data change from time to time
  - ➢ For example, incorporation of new customer or change in demand from customer

NPTEL

Introduction to Decision Support Systems

- ❖ **Range of Capabilities of a DSS**
- ➢ **DSS rely more heavily on modelling than MIS**
- ➢ **Use mathematical or analytical models to perform 'what-if' or other kinds of analysis**
- ➢ **'What-if' analysis, works forward from known or assumed conditions**
- ➢ **It allows the manager to vary certain values or test results to predict outcomes if changes occur in those input values**

NPTEL

Introduction to Decision Support Systems

- **Range of Capabilities of a DSS**

- **What happens if we raise product prices by 10 percent or increase the advertising budget by Rs. 10 lakhs?**

- ***Sensitivity analysis*** **models ask** ***what-if*** **questions repeatedly to predict a range of outcomes when one or more input variables are changed multiple times**

NPTEL

Introduction to Decision Support Systems

- **Range of Capabilities of a DSS**
  - ***Backward Sensitivity analysis*** **helps decision makers with goal seeking.**
  - **For example, if the user wants to sell one million product units next year, how much must he or she should reduce the price of the product?**

NPTEL

Introduction to Decision Support Systems

- **Range of Capabilities of a DSS**
  - ***Multidimensional data analysis and OLAP*** **are some of the key business intelligence techniques that are deployed in DSS**
  - ***Spreadsheets*** **have similar feature for *multidimensional analysis* called *'pivot table'***
  - **Managers use *pivot table* to identify and understand patterns in business information that may be useful for semi structured decision making**

NPTEL

Introduction to Decision Support Systems

- **Range of Capabilities of a DSS**

- **In the past, much of this modelling was done with spreadsheets and small stand-alone databases.**

- **Today these capabilities are incorporated into large enterprise BI systems where they are able to analyse data from large corporate databases**

NPTEL

Introduction to Decision Support Systems

- There are several other problems for which decision support systems have been reported to have been successfully employed
- Some of these problems are
  - ✓ Decisions on merger of companies
  - ✓ Budget planning
  - ✓ Portfolio management in banks
  - ✓ Corporate planning and
  - ✓ Capacity planning in production

NPTEL

REFERENCES

- Laudon, K. C., and Laudon, J. P. (2016), Management Information Systems: Managing the Digital Firm
- Turban, E., Volonio, L., Wood, R. Gregory.(2016), Information Technology for Management
- Rainer, K., Prince, B., Watson, H.,(2015), Management Information Systems
- De, Rahul.,(2019), MIS: Managing Information Systems in Business, Government and Society
- Bagchi, Nirmalya., (2010), Management Information Systems

NPTEL



NPTEL

THANK YOU !



NPTEL ONLINE CERTIFICATION COURSES

NPTEL

Management Information Systems

Prof. Kunal Kanti Ghosh
VGSoM, IIT KHARAGPUR

Week 10: Module 3

Executive Support Systems

CONCEPTS COVERED

- **Introduction to Executive Support Systems (ESS)**

NPTEL

Introduction to Executive Support Systems (ESS)

- ESS is also referred to as Executive Information Systems (EIS)
- Senior managers need systems that focus on strategic issues and long-term trends, both in the firm and in the external environment
- They are concerned with questions such as:
  - What will be the employment levels in five years?
  - What are the long-term industry cost trends?
  - What products should the firm make in the next five years?

NPTEL

Introduction to Executive Support Systems (ESS)

- Combines many features of MIS and DSS
- Provide top executives with immediate and easy access to information
- Identify factors that are critical to accomplishing strategic objectives (critical success factors)
- So popular that it has been expanded not only to managers, but also to analysts, and other knowledge workers

NPTEL

Introduction to Executive Support Systems (ESS)

- Information is presented in forms tailored to the preferences of the executives using the system
- Customizable graphical user interfaces
- Exception reports
- Trend analysis
- Drill down capability

NPTEL

Introduction to Executive Support Systems (ESS)

- Executive Support Systems (ESS) based on Business Intelligence (BI) address such decision making needs of senior management
- They address non-routine decisions requiring judgment, evaluation, and insight because there is no agreed-on procedure for arriving at a solution
- These systems are intended to be used by the senior managers directly to provide support to non-programmed decisions in strategic environment

NPTEL

Introduction to Executive Support Systems (ESS)

❖ **Programmed and Non-programmed Decisions**

- **There are two types of decisions – programmed and non-programmed decisions**
- **Programmed decisions are basically automated processes, general routine work, where**
  - **These decisions have been taken several times**
  - **These decisions follow some guidelines or rules**

✓ **For example, selecting a reorder level for inventories, is a programmed decision**

NPTEL

Introduction to Executive Support Systems (ESS)

- **Non-programmed Decisions:**
  - **Occur in unusual situations**
  - **Decisions are new**
  - **There will not be any rules to follow**
  - **These decisions are made based on the available information**
  - **These decisions are based on the manager's discretion, instinct, perception and judgment**
  - ✓ **Investing in new technology is an example of a non-programmed decision**

NPTEL

Introduction to Executive Support Systems (ESS)

- Executive Support Systems (ESS) present graphs and data from many sources through an interface that is easy for senior managers to use

- Often the information is delivered to senior executives through a portal, which uses a Web interface to present integrated personalized business content

NPTEL

Introduction to Executive Support Systems (ESS)

- Executive Support Systems (ESS) are designed to incorporate data about external events, such as new tax laws or competitors, but they also draw summarized information from internal MIS and DSS
- ESS are designed to filter, compress, and track critical data, displaying the data of greatest importance to senior managers
- Increasingly, such systems include business intelligence analytics for analysing trends, forecasting, and 'drilling down' to data at greater levels of detail

NPTEL

Introduction to Executive Support Systems (ESS)

- For example, the CEOs of many multi-national companies in various sectors has Executive Support Systems that provide on their desktop a minute-to-minute view of:
  - their organization's financial performance as measured by
    - ✓ working capital,
    - ✓ accounts receivable,
    - ✓ accounts payable,
    - ✓ cash flow and
    - ✓ inventory

NPTEL

Introduction to Executive Support Systems (ESS)

- The information is presented in the form of digital dashboard, which displays on a single screen graphs and charts of key performance indicators for managing a company

- Digital dashboards are becoming an increasingly popular tool for management decision makers

NPTEL

Introduction to Executive Support Systems (ESS)

- Contemporary business intelligence and analytics technology have promoted data-driven management, where decision makers rely heavily on analytical tools and data at their fingertips to guide their work
- Data captured at the factory or sales floor level are immediately available for high-level or detailed views in executive dashboards and reports
- It is real-time management

NPTEL

Introduction to Executive Support Systems (ESS)

- ❖ Information base for ESS is often external, unstructured, and even uncertain.
- ➢ Exact scope and context of such information is often not known beforehand
- ➢ The required information is intelligence based-
  - Market intelligence
  - Investment intelligence
  - Technology intelligence

NPTEL

Introduction to Executive Support Systems (ESS)

- **Examples of Intelligent Information**
  - **Following are some examples of intelligent information, which is often the source of an ESS –**
    - **External databases**
    - **Technology reports like patent records and so on**
    - **Technical reports from consultants**
    - **Market reports**

NPTEL

Introduction to Executive Support Systems (ESS)

- **Examples of Intelligent Information**
  - **Following are some examples of intelligent information, which is often the source of an ESS –**
    - **Confidential information about competitors**
    - **Speculative information like market conditions**
    - **Government policies**
    - **Financial reports and information**

NPTEL

Introduction to Executive Support Systems (ESS)

- Features of Executive Information System
  - Drill Down Capabilities
  - Designed With Management CSFs in Mind
  - Personalized Analysis
  - Navigation of Information
  - Graphical and Textual Information
  - Status Access, Trend Analysis, and Exception Reporting

NPTEL

Introduction to Executive Support Systems (ESS)

- **Advantages of ESS**
  - **Easy for upper level executive to use**
  - **Ability to analyse trends**
  - **Augmentation of managers' leadership capabilities**
  - **Enhance personal thinking and decision-making**
  - **Contribution to strategic control flexibility**
  - **Enhance organizational competitiveness in the market place**

NPTEL

Introduction to Executive Support Systems (ESS)

- Advantages of ESS
  - Instruments of change
  - Increased executive time horizons
  - Better reporting system
  - Improved mental model of business executive
  - Help improve consensus building and communication

NPTEL

Introduction to Executive Support Systems (ESS)

- **Advantages of ESS**
  - **Improve office automation**
  - **Reduce time for finding information**
  - **Early identification of company performance**
  - **Detail examination of critical success factor**
  - **Better understanding**
  - **Time management**

NPTEL

Introduction to Executive Support Systems (ESS)

- Limitations of ESS
  - Functions are limited
  - Hard to quantify benefits
  - Executive may encounter information overload
  - System may become slow
  - Difficult to keep current data
  - May lead to less reliable and insecure data
  - Executive cost for small company

NPTEL

REFERENCES

- Laudon, K. C., and Laudon, J. P. (2016), Management Information Systems: Managing the Digital Firm
- Turban, E., Volonio, L., Wood, R. Gregory.(2016), Information Technology for Management
- Rainer, K., Prince, B., Watson, H.,(2015), Management Information Systems
- De, Rahul.,(2019), MIS: Managing Information Systems in Business, Government and Society
- Bagchi, Nirmalya., (2010), Management Information Systems

NPTEL



NPTEL

THANK YOU !



NPTEL ONLINE CERTIFICATION COURSES

NPTEL

Management Information Systems

Prof. Kunal Kanti Ghosh
VGSoM, IIT KHARAGPUR

Week 10: Module 4

Expert Systems

CONCEPTS COVERED

- **Introduction to Expert Systems**

NPTEL

Introduction to Expert Systems

- Expert System is a computer system that emulates the decision-making ability of an expert
- Based on Artificial Intelligence, these systems rely on intelligent techniques for capturing tacit knowledge in a very specific and limited domain of human expertise
- These systems capture the knowledge of skilled employees in the form of a set of rules in a software system that can be used by others in the organization

NPTEL

Introduction to Expert Systems

- A knowledge-based information system
- Contain knowledge about a specific, complex application area
- Acts as an expert consultant to end users

NPTEL

Introduction to Expert Systems

❖**Components of an Expert System**

➢**Knowledge Base**

- Facts about a specific subject area
- Heuristics that express the reasoning
- procedures of an expert (rules of thumb)

NPTEL

Introduction to Expert Systems

❖**Components of an Expert System**

➢**Software Resources**

- An **inference engine** processes the knowledge and recommends a course of action
- **User interface programs** communicate with the end user
- **Explanation programs** explain the reasoning process to the end user

NPTEL

Introduction to Expert Systems

❖ Knowledge Representation Methods

➢Case-Based

- Knowledge organized in the form of cases
- Cases are examples of past performance, occurrences, and experiences

➢Frame-Based

- Knowledge organized in a hierarchy or network of frames
- A frame is a collection of knowledge about an entity, consisting of a complex package of data values describing its attributes

NPTEL

Introduction to Expert Systems

❖ **Knowledge Representation Methods**

➢ **Object-Based**

- Knowledge represented as a network of objects
- An object is a data element that includes both data and the methods or processes that act on those data

➢ **Rule-Based**

- Knowledge represented in the form of rules and statements of fact
- Rules are statements that typically take the form of a premise and a conclusion (If, Then)

NPTEL

Introduction to Expert Systems

- They perform very limited tasks
- For example, diagnosing a malfunctioning machine or determining whether to grant credit for a loan
- Problems that cannot be solved by human experts in the same short period of time are far too difficult for an expert system

NPTEL

Introduction to Expert Systems

- However, by capturing human expertise in limited areas, expert systems can provide benefits, helping organizations make high-quality decisions with fewer people
- Today, expert systems are widely used in business in discrete, highly structured decision-making situations

NPTEL

Introduction to Expert Systems

❖How Expert Systems Work

- Human knowledge must be modelled or represented in a way that a computer can process
- Expert systems model human knowledge as a set of rules that collectively are called the knowledge base
- The rules are obtained by
  - carefully interviewing one or several "experts" who have a thorough command of the knowledge base for the system or
  - by documenting business rules found in manuals, books, or reports

NPTEL

Introduction to Expert Systems

❖How Expert Systems Work

- Expert systems have from 200 to many thousands of these rules, depending on the complexity of the problem
- These rules are much more interconnected and nested than in a traditional software program
- The strategy used to search through the knowledge base is called the inference engine
- Two strategies are commonly used forward chaining and backward chaining

NPTEL

Introduction to Expert Systems

❖How Expert Systems Work

- In forward chaining, the inference engine begins with the information entered by the user and searches the rule base to arrive at a conclusion
- The strategy is to carry out the action of the rule when a condition is true
- In the figure, beginning on the left, if the user enters a client's name with income greater than $100,000, the engine will fire all rules in sequence from left to right. (Reference Laudon, page 473)

NPTEL

Intelligent Techniques

Rules in an Expert System

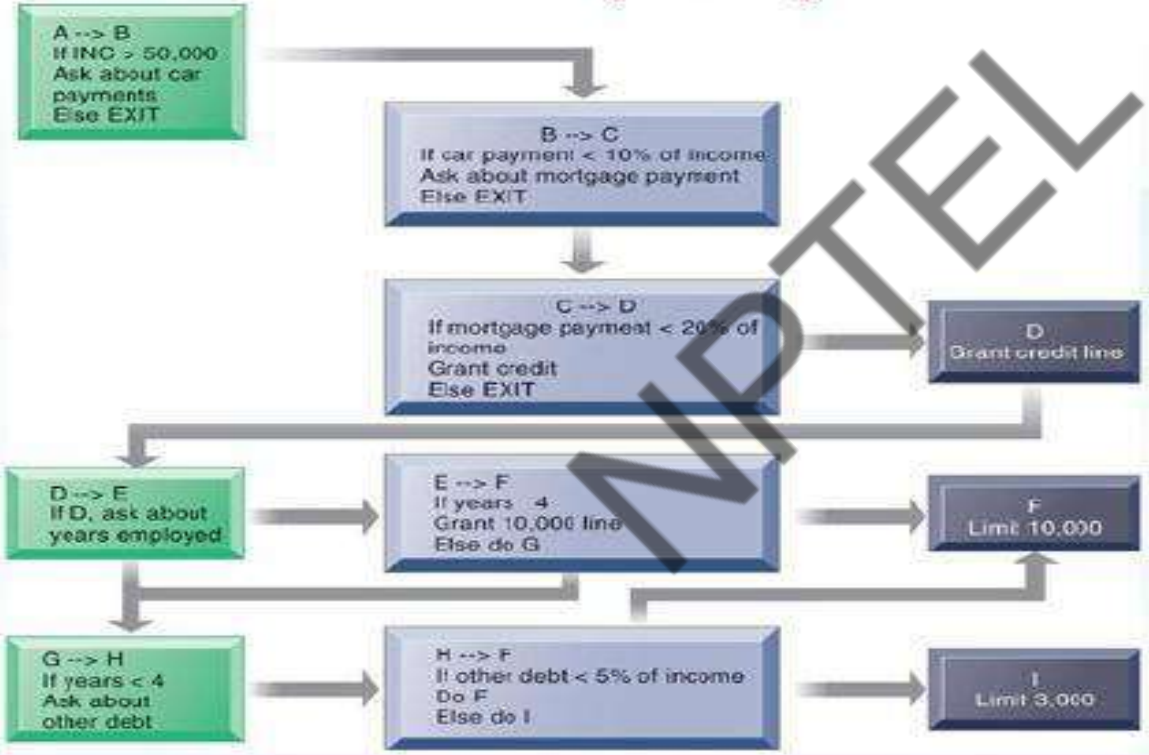


Figure 11-9

An expert system contains a number of rules to be followed. The rules are interconnected; the number of outcomes is known in advance and is limited; there are multiple paths to the same outcome; and the system can consider multiple rules at a single time. The rules illustrated are for simple credit-granting expert systems.

NPTEL

NPTEL Online Certification Courses
IIT Kharagpur

Introduction to Expert Systems

❖**How Expert Systems Work**

- If the user then enters information indicating that the same client owns real estate, another pass of the rule base will occur and more rules will fire
- Processing continues until no more rules can be fired

NPTEL

Inference Engines in Expert Systems



Source: Kenneth C. Laudon & Jane P. Laudon (2012), Management Information Systems: Managing the Digital Firm, Twelfth Edition, Pearson.

NPTEL Online Certification Courses
IIT Kharagpur

Introduction to Expert Systems

❖How Expert Systems Work

- In backward chaining, the strategy for searching the rule base starts with a hypothesis and proceeds by asking the user questions about selected facts until the hypothesis is either confirmed or disproved.
- In our example, in figure 11.6, ask the question “Should we add this person to the prospect database?”
- Begin on the right of the diagram and work toward the left
- You can see that the person should be added to the database if a sales representative is sent, term insurance is granted, or a financial adviser visits the client

NPTEL

Introduction to Expert Systems

- **Benefits of Expert Systems**

➢ Captures the expertise of an expert or group of experts in a computer-based information system

- Faster and more consistent than an expert
- Can contain knowledge of multiple experts
- Does not get tired or distracted
- Cannot be overworked or stressed
- Helps preserve and reproduce the knowledge of human experts

NPTEL

Introduction to Expert Systems

➢ **Business Benefits of Expert systems:**

- Reduced errors
- Reduced costs
- Reduced training time, and
- Higher levels of quality and service

NPTEL

Introduction to Expert Systems

❖Limitations of Expert Systems

- Although expert systems lack the robust and general intelligence of human beings, they can provide benefits to organizations if their limitations are well understood
- Only certain classes of problems can be solved using expert systems
- Virtually all successful expert systems deal with problems of classification in limited domains of knowledge where there are relatively few alternative outcomes and these possible outcomes are all known in advance

NPTEL

Introduction to Expert Systems

❖ **Limitations of Expert Systems**

- Expert systems are much less useful for dealing with unstructured problems typically encountered by managers
- Many expert systems require large, lengthy, and expensive development efforts

NPTEL

Introduction to Expert Systems

❖ **Limitations of Expert Systems**

- Hiring and training more experts may be less expensive than building an expert system
- Typically, the environment in which an expert system operates is continually changing so that the expert system must also continually change
- Some expert systems, especially large ones, are so complex that in a few years the maintenance costs equal the development costs.

NPTEL

REFERENCES

- Laudon, K. C., and Laudon, J. P. (2016), Management Information Systems: Managing the Digital Firm
- Turban, E., Volonio, L., Wood, R. Gregory.(2016), Information Technology for Management
- Rainer, K., Prince, B., Watson, H.,(2015), Management Information Systems
- De, Rahul.,(2019), MIS: Managing Information Systems in Business, Government and Society
- Bagchi, Nirmalya., (2010), Management Information Systems

NPTEL



NPTEL

THANK YOU !



NPTEL ONLINE CERTIFICATION COURSES

NPTEL

Management Information Systems

Prof. Kunal Kanti Ghosh
VGSoM, IIT KHARAGPUR

Week 10: Module 5
Learning Management Systems (LMS)

CONCEPTS COVERED

- **Introduction to Learning Management Systems (LMS)**

NPTEL

Introduction to Learning Management Systems (LMS)

- Companies need ways to keep track of and manage employee learning and to integrate it more fully into their knowledge management and other corporate systems

NPTEL

Introduction to Learning Management Systems (LMS)

- A learning management system (LMS) provides tools for the management, delivery, tracking, and assessment of various types of employee learning and training

- Contemporary LMS support multiple modes of learning, including CD-ROM, downloadable videos, Web-based classes, live instruction in classes or online, and group learning in online forums and chat sessions

NPTEL

Introduction to Learning Management Systems (LMS)

- The LMS consolidates mixed-media training, automates the selection and administration of courses, assembles and delivers learning content, and measures learning effectiveness
- Businesses run their own learning management systems, but they are also turning to publicly-available massive open online courses (MOOCS) to educate their employees.
- A MOOC is an online course made available via the Web to very large number of participants

NPTEL

Introduction to Learning Management Systems (LMS)

- A learning management system (LMS) is a software application for
  - the administration,
  - documentation,
  - tracking,
  - reporting, and
  - delivery of
    - educational courses,
    - training programs, or learning and development programs.

NPTEL

Introduction to Learning Management Systems (LMS)

- The learning management system concept emerged directly from e-Learning.
- Although the first LMS appeared in the higher education sector, the majority of the LMSs today focus on the corporate market.

NPTEL

Introduction to Learning Management Systems (LMS)

- A Learning Management System (or LMS) is a software package that enables the management and delivery of online content to learners.

- Most LMSs are web-based to facilitate "anytime, anyplace, any pace" access to learning content and administration (Coates, 2005).

NPTEL

Introduction to Learning Management Systems (LMS)

- LMS can also be defined as a software application or Web-based technology used to plan, implement, and assess a specific learning process (Dougiamas & Taylor, 2003).

NPTEL

Introduction to Learning Management Systems (LMS)

- Learning management systems were designed to identify training and learning gaps, utilizing analytical data and reporting.
- An LMS delivers and manages all types of content, including video, courses, and documents.

NPTEL

Introduction to Learning Management Systems (LMS)

- Through LMS, teachers may
  - ✓create and integrate course materials,
  - ✓articulate learning goals,
  - ✓align content and assessments,
  - ✓track studying progress, and
  - ✓create customized test for students.
- LMS allows the communication of learning objectives, and organize learning timelines.

NPTEL

Introduction to Learning Management Systems (LMS)

- LMS allows the communication of learning objectives, and organize learning timelines.
- LMS delivers learning content and tools straight to learners, and
- It can also reach student groups through special settings.
- Such systems have built in customizable features including assessment and tracking.

NPTEL

Introduction to Learning Management Systems (LMS)

- Thus, learners can see in real time their progress and instructors can monitor and communicate the effectiveness of learning.

- One of the most important features of LMS is trying to create a streamline communication between learners and instructors.

NPTEL

Introduction to Learning Management Systems (LMS)

- Such systems, besides
  - facilitating online learning,
  - tracking learning progress,
  - providing digital learning tools,
  - manage communication,
  - selling content,
- may also be used to provide different communication features.

NPTEL

Introduction to Learning Management Systems (LMS)

❖**There are six major advantages of LMS:**

- Interoperability,
- Accessibility,
- Reusability,
- Durability,
- Maintenance ability and
- Adaptability

NPTEL

Introduction to Learning Management Systems (LMS)

❖**Other advantages include:**

- Content in various formats: text, video, audio, and so on.
- One can access materials anytime, from everywhere, teachers can modify the content, and students can see the updated material.
- The evaluation of students is easier and fair, based on student attendance and online quizzes.
- Students and teachers can re-use the material every time they need.

NPTEL

Introduction to Learning Management Systems (LMS)

❖ **Disadvantages**

- Implementing LMS requires a well-built technology infrastructure.
- Teachers have to be willing to adapt their curricula from face to face lectures to online lectures.

NPTEL

Introduction to Learning Management Systems (LMS)

- LMS is a software that automates the administration of training events.
- LMS registers users, tracks courses in a catalogue, and records data from learners
- It also provides reports to management.

NPTEL

Introduction to Learning Management Systems (LMS)

- **Modular Object Oriented Dynamic Learning Environment (MOODLE) is a Learning Management System (LMS)**
- It is a free, Open Source Software package designed using sound pedagogical principles

- It helps educators to create effective quality courses and
- It helps students learn.

NPTEL

Introduction to Learning Management Systems (LMS)

❖ **MOODLE:**

- One advantage of the open source format is that it enables users to download a variety of learning modules that fit their needs.
- Educators can add email, discussion boards, live chats, online quizzes, and a variety of other modules that differentiate their system from commercially available content management systems.

NPTEL

Introduction to Learning Management Systems (LMS)

❖ **MOODLE:**

- Another advantage of open source systems is that users can add new features and enhancements that are available to all users at no additional cost

NPTEL

Introduction to Learning Management Systems (LMS)

❖ **MOODLE:**

- Information about everything that is added to Moodle is saved in a database.
- This information includes the name of all students and teachers, the individual quiz items, journal entries, activities that students have done, and so on.
- The front end of Moodle, the part that students see, is therefore a database driven web site, as opposed to the traditional static site that is same regardless of who looks at it.

NPTEL

Introduction to Learning Management Systems (LMS)

❖**MOODLE:**

- This comprehensive data tracking allows teachers to keep abreast of the activities of their students.
- They are able to see what their students have been doing via the "logs", as well as being able to see in detail how students have done in the various tasks made by the course teachers

NPTEL

Introduction to Learning Management Systems (LMS)

❖ **MOODLE:**

- As a ‘website’, Moodle is by definition a form of communication
- Teachers can use Moodle to keep in touch with their students by adding text, images and sound

NPTEL

Introduction to Learning Management Systems (LMS)

❖**MOODLE:**

- More specifically, there are modules within Moodle like
  - email,
  - chat, etc.
- that are specifically geared to facilitate communication between students and teachers and between students and students.

NPTEL

Introduction to Learning Management Systems (LMS)

❖**MOODLE:**

- What is more, Moodle manages all these modes of communication
- For example, forums (bulletin boards) can be set to automatically send out emails to subscribers informing them of new postings.

NPTEL

Introduction to Learning Management Systems (LMS)

❖**MOODLE:**

- Moodle is designed to help educators create online courses with opportunities for rich interaction.
- Its open source license and modular design means that many people can develop additional functionality

NPTEL

REFERENCES

- Laudon, K. C., and Laudon, J. P. (2016), Management Information Systems: Managing the Digital Firm
- Turban, E., Volonio, L., Wood, R. Gregory.(2016), Information Technology for Management
- Rainer, K., Prince, B., Watson, H.,(2015), Management Information Systems
- De, Rahul.,(2019), MIS: Managing Information Systems in Business, Government and Society
- Bagchi, Nirmalya., (2010), Management Information Systems

NPTEL



NPTEL

THANK YOU !



NPTEL ONLINE CERTIFICATION COURSES

NPTEL

Management Information System

Saini Das

Vinod Gupta School of Management, IIT Kharagpur

Module 11: Ethical, Social and Security issues in MIS

Ethical and Social Issues in MIS - I

Ethical and Social Issues in MIS

- Ethics refers to the principles of right and wrong that individuals use to make choices to guide their behaviour.
- Information Systems (IS) raise new ethical questions because they provide opportunities for intense social change, and thus threaten existing distributions of money, power, rights and obligations.
- Ethical issues in IS have been amplified by the rise of Internet that makes it easier to assemble, integrate and distribute information.
- Some of important ethical and social issues in IS are:
  - Information Privacy Issues
  - Intellectual Property Rights Infringement
  - Workplace Monitoring

NPTEL

Ethical Issue : Privacy



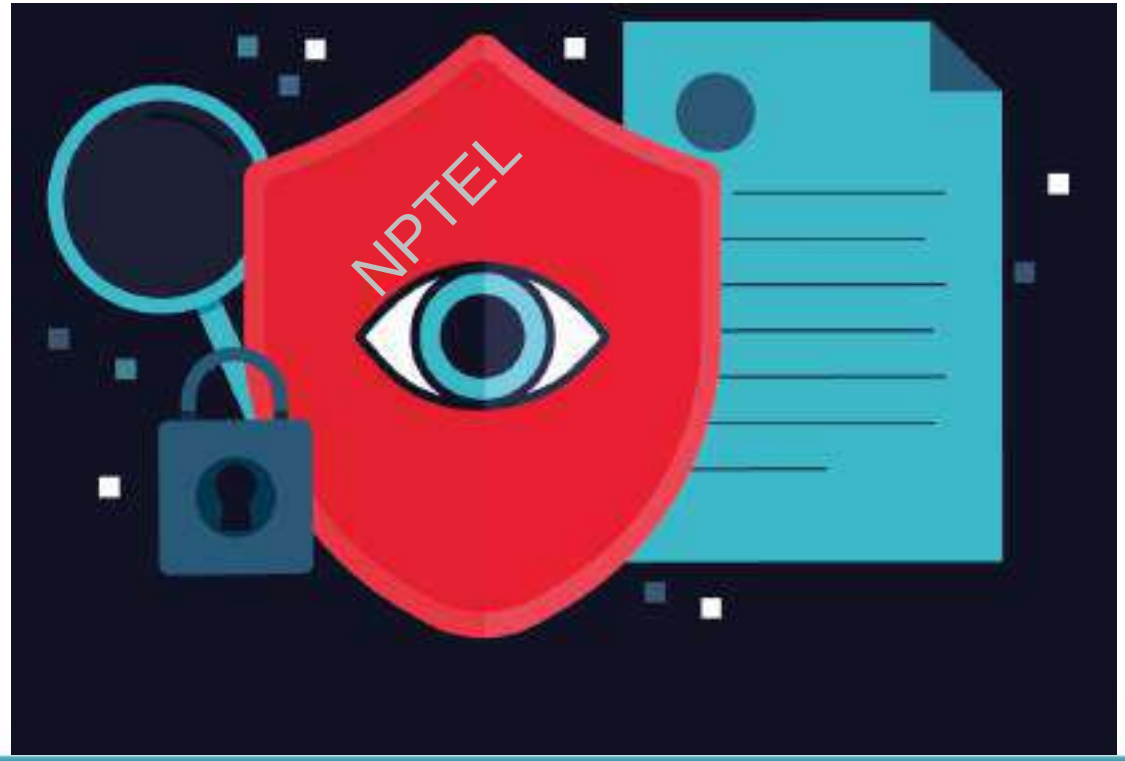
"The right to be left alone and the right to be free of unreasonable personal intrusions".

- The right of privacy is not absolute. Privacy must be balanced against needs of society.
- The publics' right to know is superior to individuals' right to privacy

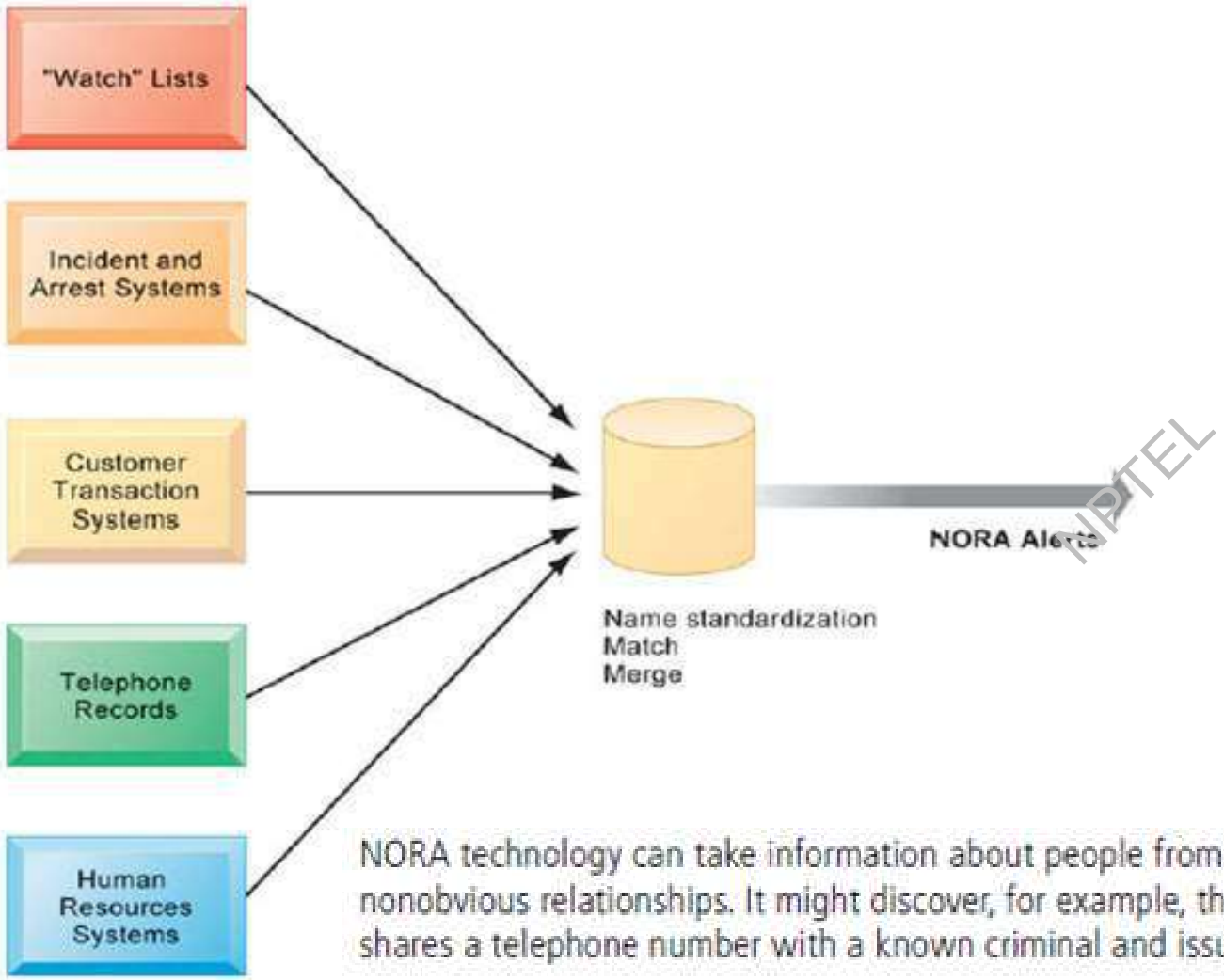
NPTEL

Key terminologies

- **Profiling:** The use of computers to combine data from multiple sources and create electronic dossiers of detailed information on individuals.



NORA



NORA technology can take information about people from disparate sources and find obscure, nonobvious relationships. It might discover, for example, that an applicant for a job at a casino shares a telephone number with a known criminal and issue an alert to the hiring manager.

NPTEL

Information Collected on the Internet

- **Personally identifiable information (PII)**

Name	Gender	Education
Address	Age	Preference data
Phone number	Occupation	Transaction data
E-mail address	Location	Clickstream data
Social security number	Location history	Device used for access
Bank accounts	Likes	Browser type
Credit card accounts	Photograph	

NPTEL

- **Anonymous information**
  - Age; occupation; income; zip code; ethnicity

Mechanisms to collect information about individuals on the Internet

- Web Site Registration
- Cookies
- Web Beacons
- Spyware

NPTEL

Cookies

- Cookies are small text files deposited on a computer hard drive when a user visits websites. Cookies identify the visitor's Web browser software and track visits to the Web site. When the visitor returns to a site that has stored a cookie, the Web site software will search the visitor's computer, find the cookie, and know what that person has done in the past. It may also update the cookie depending on the activity during the visit.

NPTEL

How cookies identify visitors?



NPTEL

1. The Web server reads the user's Web browser and determines the operating system, browser name, version number, Internet address, and other information.
2. The server transmits a tiny text file with user identification information called a cookie, which the user's browser receives and stores on the user's computer hard drive.
3. When the user returns to the Web site, the server requests the contents of any cookie it deposited previously in the user's computer.
4. The Web server reads the cookie, identifies the visitor, and calls up data on the user.

Web Beacons

- Web beacons, also called Web bugs, are tiny objects invisibly embedded in e-mail messages and Web pages that are designed to monitor the behavior of the user visiting a Web site or sending an e-mail.
- Web beacons are placed on popular websites by “third party” firms who pay the Web sites a fee for access to their audience. Typical popular Web sites contain 25–35 web beacons.

NPTEL

Spyware



- **Spyware** is a software that aims to gather information about a person or organization without their knowledge and that may send such information to another entity without the consumer's consent, or that asserts control over a computer without the consumer's knowledge.
- Spyware can collect almost any type of data, including personal information like internet surfing habits, user logins, and bank or credit account information.
- Spyware can also interfere with user control of a computer by installing additional software.
- Some spyware can change computer settings, which can result in slow Internet connection speeds, un-authorized changes in browser settings, or changes to software settings.

NPTEL

Technical Solutions

- Anti-spyware technologies
- Email encryption
- Do Not Track options
- Opt-out model of informed consent
- Opt-in model of informed consent
- Making email or surfing anonymous

NPTEL

References

- K. Laudon and J. Laudon (2016). Management Information Systems Publisher: Pearson. Edition 14e.

- R. De. (2018). MIS Managing Information Systems in Business, Government and Society. Publisher: Wiley. Second Edition.

NPTEL



NPTEL

THANK YOU !



NPTEL ONLINE CERTIFICATION COURSES

NPTEL

Management Information System

Saini Das

Vinod Gupta School of Management, IIT Kharagpur

Module 11: Ethical, Social and Security issues in MIS

Ethical and Social Issues in MIS – II

Legal Solutions to Privacy Issues

- Laws
  - HIPAA
  - FIP
  - GDPR

NPTEL

Health Insurance Portability and Accountability Act (HIPAA)



- HIPAA of 1996 is a federal law that required the creation of national standards to protect sensitive patient health information from being disclosed without the patient's consent or knowledge.

- The US Department of Health and Human Services (HHS) issued the HIPAA Privacy Rule to implement the requirements of HIPAA.

- The Privacy Rule ensures that the privacy of individuals' health information is properly protected while allowing the flow of health information needed to provide and promote high quality health care.

NPTEL

FTC's Fair Information Practices (FIP)

Notice/Awareness (core principle)	Sites must disclose their information practices before collecting data. Includes identification of collector, uses of data, other recipients of data, nature of collection (active/inactive), voluntary or required, consequences of refusal, and steps taken to protect confidentiality, integrity, and quality of the data.
Choice/Consent (core principle)	There must be a choice regime in place allowing consumers to choose how their information will be used for secondary purposes other than supporting the transaction, including internal use and transfer to third parties. Opt-in/opt-out clause must be available.
Access/Participation	Consumers should be able to review and contest the accuracy and completeness of data collected about them in a timely, inexpensive process.
Security	Data collectors must take reasonable steps to assure that consumer information is accurate and secure from unauthorized use.
Enforcement	There must be a mechanism to enforce the principles in place. This can involve self-regulation or legislation giving consumers legal remedies for violations.

NPTEL

General Data Protection Regulation (GDPR)

Purpose	• Harmonize data privacy laws across Europe • Reshape the way organizations across the region approach data privacy. • Protect and empower EU citizens' data privacy
Scope	• Applies to all firms and organizations worldwide that collect, process, or use personal information of EU citizens
Administration and enforcement	• Creates a new EU-wide Information Commissioners Office to enforce the regulation in the European Union. Each country also has its own Data Protection Agency
Individual rights	• Easier access to all personal data without charge within one month • Right to be forgotten (power to erase data) • Data portability: allow people to move their data to other providers • Give users more control over the use of their data by third parties and partners • Right to seek damages for abuse, including class action suits
Organizational requirements	• Data protection officer in all firms with more than 250 employees, reporting to senior management • Requires explicit consent before collecting data on people (positive opt-in) • Published rationale for data collection and how long it will be held • Requires firms to report breaches, hacks, and unauthorized disclosure within 72 hours • Third-party risk management. Firms are liable for data shared with partners and must maintain a list of all sharing firms • Requires firms to maintain a record of all EU personal data • Privacy by design of all new systems • Targeting limits: allows anonymized data for audience targeting, but targeting based on social media or other personal profiles remains a grey area • New schedule of fines: up to $20 million or 4% of global revenue • Privacy shield: agreements with non-EU countries to ensure any data processed outside the European Union meets EU GDPR standards

NPTEL

Ethical issue : Intellectual Property Rights Infringement

- Intellectual Property (IP) refer to creations of the mind, such as inventions, literary and artistic works, symbols, names, images, and designs, used in commerce.
- IT has made it difficult to protect intellectual property because computerized information can be easily copied or distributed on networks.
- IP is subject to a variety of protections under three different legal traditions:
  - Copyright
  - Trade Secrets
  - Patents



NPTEL

Copyright

- Copyright is a statutory grant that protects creators of intellectual property from having their work copied by others for any purpose during the life of the author plus an additional 70 years after the author's death.
    - Literary works
    - Musical works
    - Dramatic works
    - Artistic works
    - Sound recordings, films, broadcasts, cable programs

NPTEL

- Illegal file sharing services such as Kazaa, Napster, Torrent etc. help users share digital copyrighted software, music and video files.

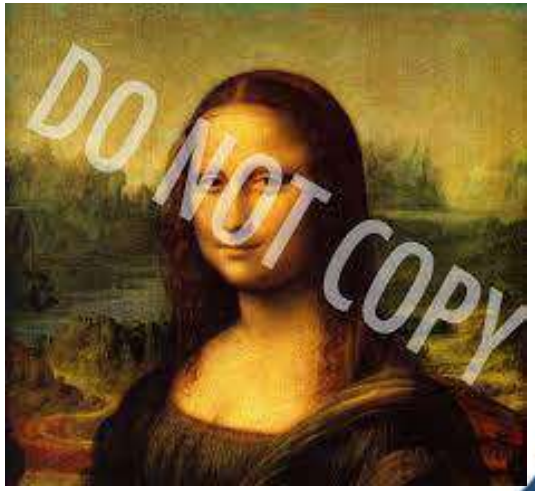
Copyright Protection Approaches

- Legitimate online digital content stores and services such as, iTunes, Pandora, Kindle, Netflix have resulted in decline of some forms of piracy.

- Digital watermarks are unique identifiers embedded in digital content that make it possible to identify ownership or pirated works

- Digital Millennium Copyright Act 1998 makes it illegal to circumvent technology-based protections of copyrighted materials.

- Software and Information Industry Association (SIIA) runs an anti-piracy hotline for individuals to report piracy activities.



NPTEL

Trade Secrets



- Any intellectual work product – a formula, device, pattern or compilation of data used for a business purpose can be classified as a trade secret provided it is not based on information available in the public domain.

NPTEL

- Trade secret law protects the actual ideas in a work product.

- The creator or owner must take care to bind employees and customers with nondisclosure agreements and to prevent the secret from falling into the public domain.

Patent



- A document that grants the holder exclusive rights to an invention for a fixed number of years (usually 20 years).

- Patent infringement occurs when another party makes, uses, or sells a patented item without the permission of the patent holder.

- The patent holder may choose to sue the infringing party to stop his or her activities, as well as to receive compensation for the unauthorized use.

NPTEL

Ethical Issue : Workplace Monitoring



- The need for monitoring has existed through the ages.
- Managements justify this need in two ways: to boost productivity – to ensure no 'free-loading' and to spot best performers and reward them; .
- Critique of monitoring – it is obtrusive, leads to lowered trust in the workforce, which lowers confidence.
- Pervasive use of information systems has increased the scale and precision of monitoring.

NPTEL

Workplace Monitoring: Use of Information System

- Use of digital video cameras
- Use of computer network logs
- Scanning and storage of outgoing and incoming emails
- Use of employee identity cards to monitor physical movements
- Access to all files on employees' personal computers and laptops

NPTEL

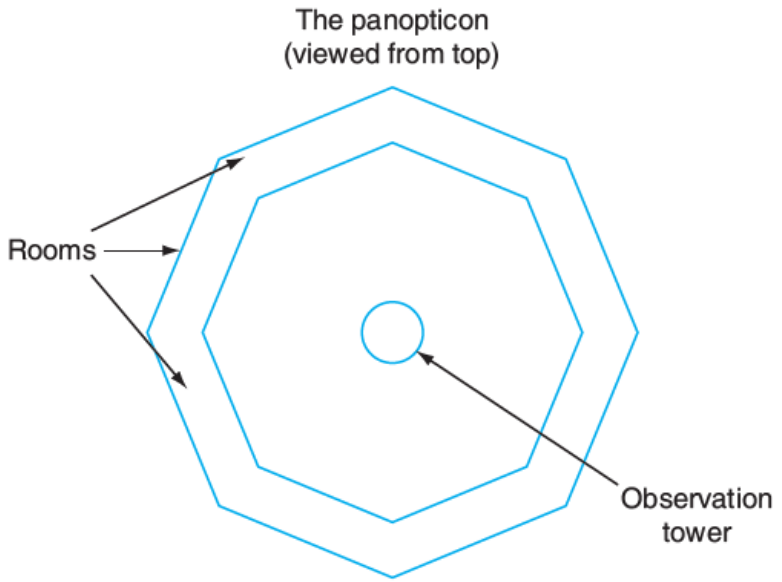
Information Panopticon

- Conceived by British economist-philosopher Jeremy Bentham in the 19th century, the Panopticon is an octagonal structure with rooms on each side, with windows opening towards the central watch-tower.

- This structure made disciplining possible, since it ensured the source of power was always present, though this could not be verified.

- When organisation implement rules of conduct and other monitoring policies, the Panopticon effect helps ensure conformity and discipline.

- IS have the Panopticon effect where employees are aware they are being monitored but are not sure.



NPTEL

References

- K. Laudon and J. Laudon (2016). Management Information Systems Publisher: Pearson. Edition 14e.

- R. De. (2018). MIS Managing Information Systems in Business, Government and Society. Publisher: Wiley. Second Edition.

NPTEL



NPTEL

THANK YOU !



NPTEL ONLINE CERTIFICATION COURSES

NPTEL

Management Information System

Saini Das

Vinod Gupta School of Management, IIT Kharagpur

Module 11: Ethical, social and security issues in MIS

Security issues in MIS

Apr 2019



May 2017



Jul 2015



Dec 2013



Apr 2011



?

NPTEL

Ashley Madison Data Breach



"Customer data is a liability, not an asset".

Sony PlayStation data breach



Apr 27, 2011

"Sony suffered a massive breach in its video game online network that led to the theft of names, addresses and possibly credit card data belonging to **77 million user accounts** in what is one of the largest-ever Internet security break-ins.

Sony learned that user information had been stolen from its PlayStation Network, prompting it to shut down the network immediately."

Economic Losses from Cybercrime

- Losses from piracy and theft of confidential business information
- Losses from theft of money and customer profile information
- Social cost of implementing laws and monitoring mechanisms for sites hosting illegal content.
- Damage to reputation of organizations and loss in stock valuations
- Increased cost of acquiring security technology
- Opportunity costs of lost business

NPTEL

The Cyber Black Market for Stolen Data

DATA	PRICE*
Individual U.S. card number with expiration date and CVV2 (the three-digit number printed on back of card) (referred to as a CVV)	$5–$8
Individual U.S. card number with full information, including full name, billing address, expiration date, CVV2, date of birth, mother's maiden name, etc. (referred to as a Fullz or Fullzinfo)	$30
Dump data for U.S. card (the term "dump" refers to raw data such as name, account number, expiration data, and CVV encoded on the magnetic strip on the back of the card)	$110–$120
Online payment service accounts	$20–$300
Bank account login credentials	$80–$700
Online account login credentials (Facebook, Twitter, eBay)	$10–$15
Medical information/health credentials	$10–$20
1,000 e-mail addresses	$1–$10
Scan of a passport	$1–$2

NPTEL

Motivation for online security breaches

- Monetary motivation
  - personal use
  - selling stolen information (underground black marketplace)
- Fun, thrill, challenge
  - Vandalize, deface, disrupt a website
- Malicious intention
- Identifying vulnerabilities apriori

Ethical Hacker/Cracker/Script kiddies

NPTEL

Pillars of Information Security

- **Confidentiality:** Is the confidential data accessible to anyone other than those authorized to view them?

- **Integrity :** Has the data been altered or manipulated without authorization?

- **Availability :** Can I access the information when required?

NPTEL

Security Threats in Organizations

NPTEL

Denial-of-Service Attack

- Purpose is to disrupt or deny normal computer processing
- DoS/DDoS attacks
  - Flooding a website with useless, continuous packets, page requests or pings in order to slow down or pull down its services.
  - A Distributed DoS attack uses numerous computers to inundate and overwhelm the network from numerous launch points.

NPTEL







Malware



- Malicious external software that pose a threat to the security of organizations:
  - **Viruses:** They infiltrate and spread in organizational networks, infecting PCs and destroying files and data. They spread when humans take an action such as downloading an email attachment, copying an infected file or using an infected device.
  - **Worms:** They are independent computer programs that copy themselves from one computer to another over a network. Unlike viruses worms do not rely on human action and can replicate on their own.
  - **Trojan Horses:** Infiltrates computers and secretly allows external software and people to invade the computer and use its resources.
  - **Ransomware:** They try to extort money from users by taking control of their computers and displaying annoying pop-up messages.
  - **Spyware:** Small program files that install themselves surreptitiously to monitor user web surfing activity and serve up advertisements.

NPTEL

Website Defacement/Cyber Vandalism

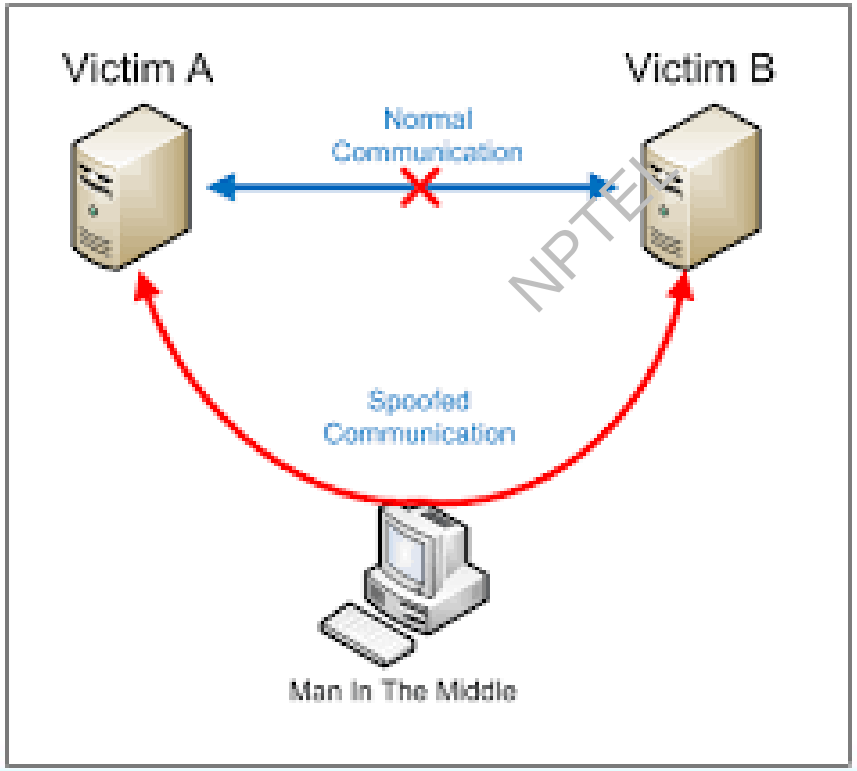
- Website defacement is an attack on a website that makes unauthorised changes to the visual appearance of a website or a web page. These are typically the work of defacers, who break into a web server and replace the hosted website with one of their own.



NPTEL

Man-in-the-Middle Attack

- An attack that intends to intercept/alter a message between a sender and a recipient.
- The attacker eavesdrops and intercepts all messages between two victims and injects new and modified messages to one or both of them.



NPTEL

References

- K. Laudon and J. Laudon (2016). Management Information Systems Publisher: Pearson. Edition 14e.

- R. De. (2018). MIS Managing Information Systems in Business, Government and Society. Publisher: Wiley. Second Edition.

NPTEL



NPTEL

THANK YOU !



NPTEL ONLINE CERTIFICATION COURSES

NPTEL

Management Information System

Saini Das

Vinod Gupta School of Management, IIT Kharagpur

Module 11: Ethical, Social and Security issues in MIS

Security issues in MIS - II

Identity Theft

- Phishing attack
- Spear Phishing attack
- Pharming attack

NPTEL

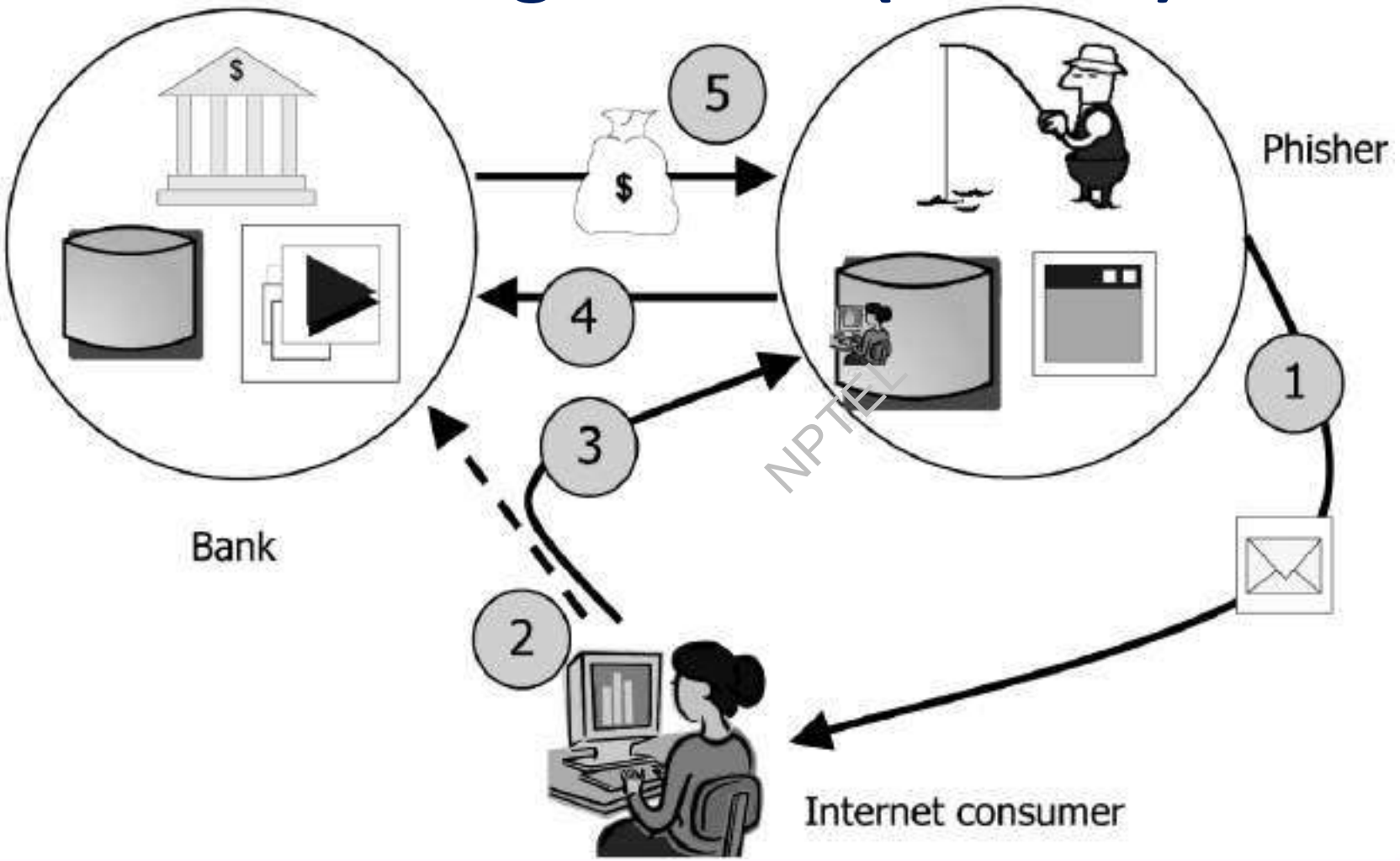
Phishing Attack



- Attacker sends e-mail messages to a large number of recipients
- Message states that an account has been compromised and the matter should be corrected
- Message includes a link which takes the user to a fake site that resembles the authentic site.
- User enters a login name and password and gets an error message.
- Perpetrator captures the user details.
- Once inside a victim's account, the perpetrator can access personal information

NPTEL

Phishing Attacks (contd..)



NPTEL



Dear valued customer of TrustedBank,

We have recieved notice that you have recently attempted to withdraw the following amount from your checking account while in another country: $135.25.

If this information is not correct, someone unknown may have access to your account. As a safety measure, please visit our website via the link below to verify your personal information:

http://www.trustedbank.com/general/custverifyinfo.asp

Once you have done this, our fraud department will work to resolve this discrepency. We are happy you have chosen us to do business with.

Thank you,
TrustedBank

NPTEL

Spear Phishing Attacks



From:
Date: 22 September 2010 16:06
To:
Subject: Salaries 2011 (confidential) - Fixed
Attach: new_salaries_2011.pdf (263 bytes)

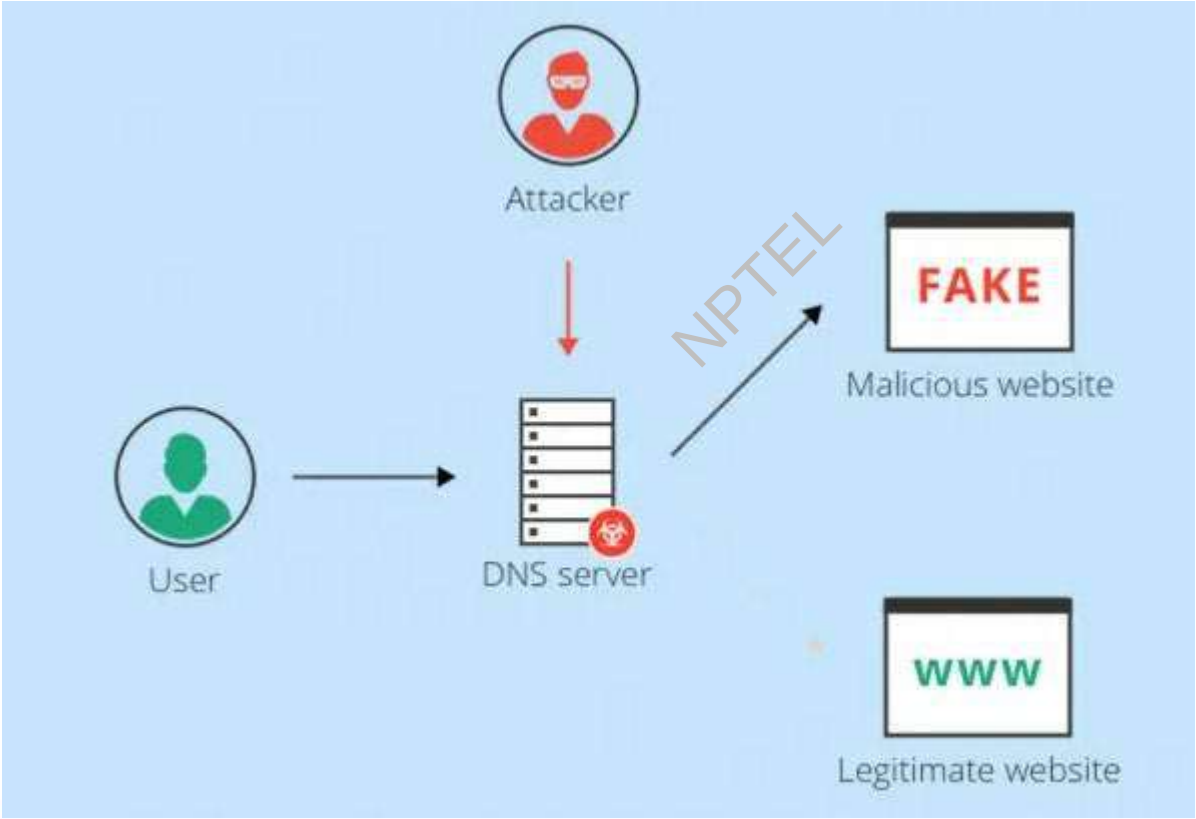
Hi, here you have the fixed version of the salaries. Sorry about the errors in the yesterdays email...
Please let me know if you have any comments.

--
Human Resources and
Distribution Services

NPTEL

Pharming Attack

- Pharming redirects users to a bogus Web page even when the user types the correct website address into his or her browser.



NPTEL

Solutions

NPTEL

Organizational Security Environment



NPTEL

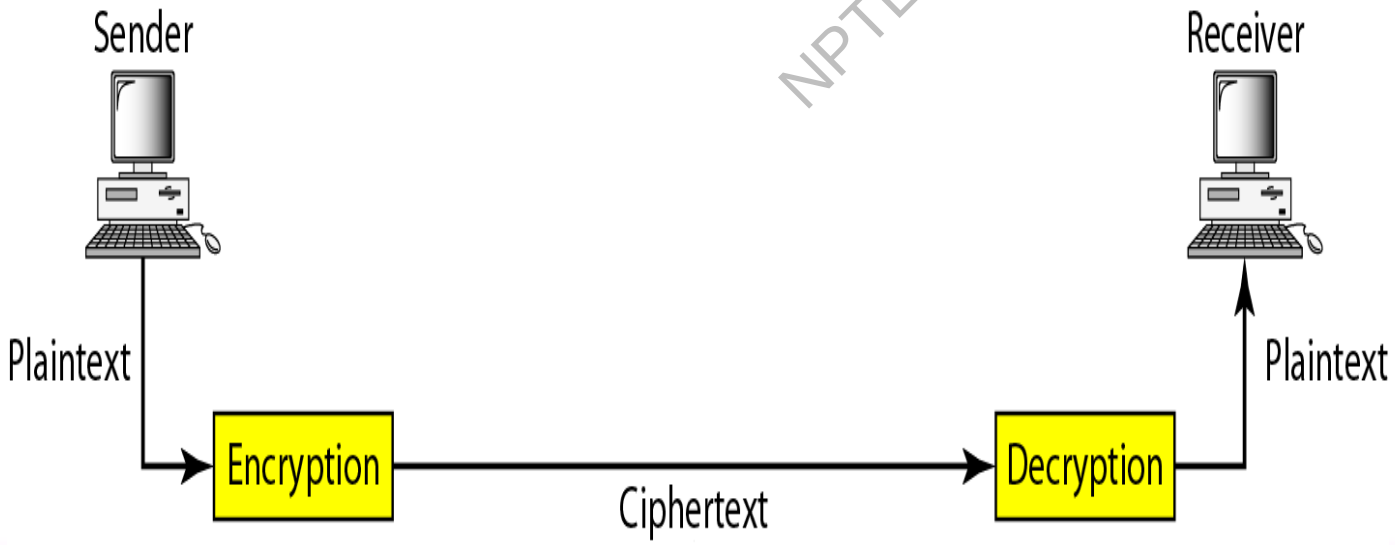
Technology Solutions

NPTEL

Encryption

- Using a mathematics based program and a secret key to produce a string of characters that is unintelligible.

- Cryptography
  - Science and art that studies encryption



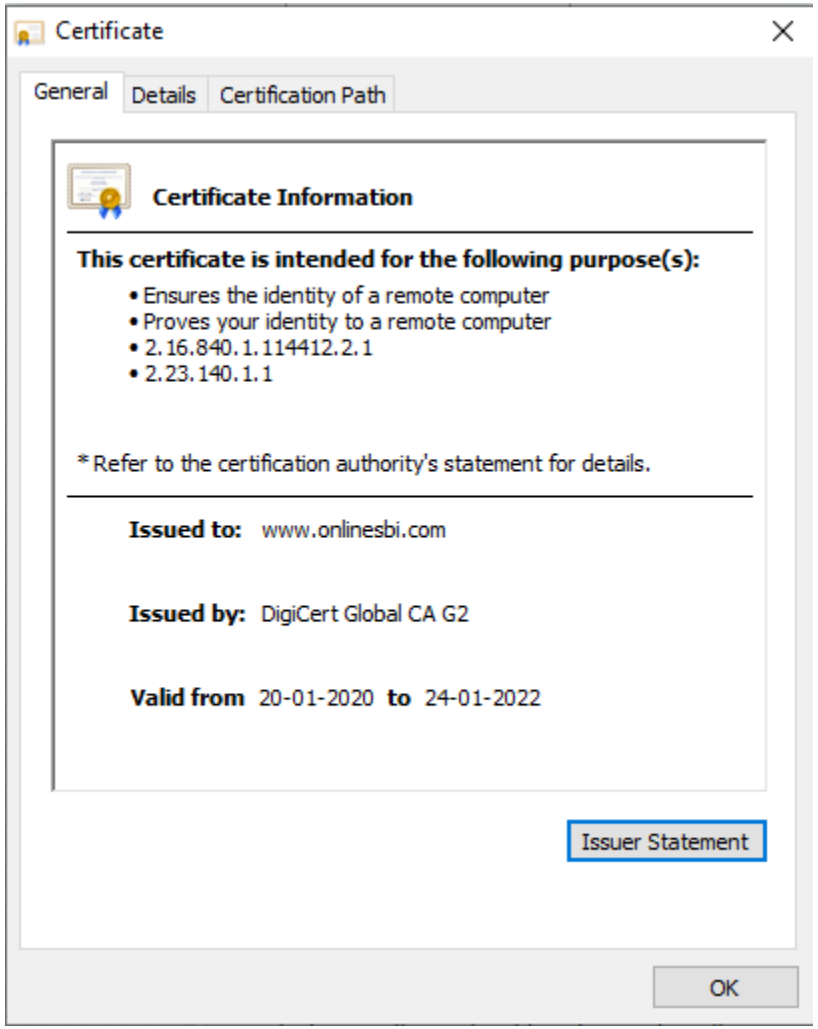
NPTEL

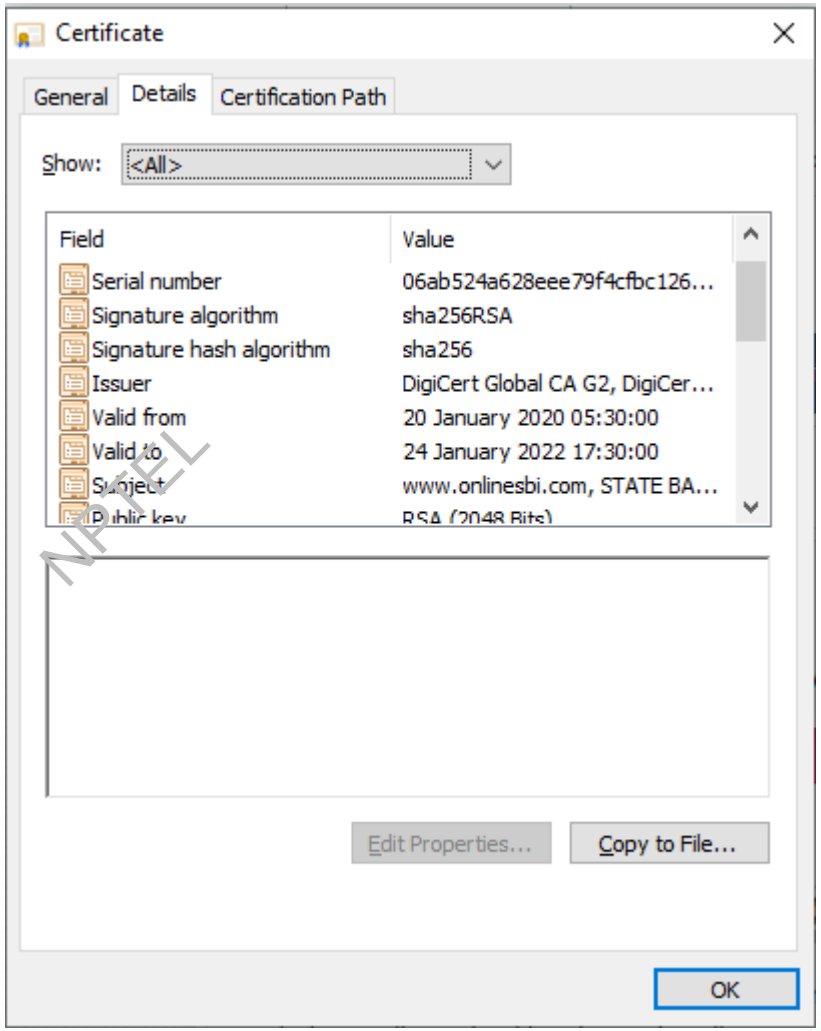
Digital Certificates

- A digital certificate is a program embedded in a Web page or email that verifies that the sender or Web site is who or what it claims to be
  - A certificate is signed code or message that provides proof that the holder is the person identified by the certificate.
  - Certification Authority(CA) issues digital certificates: Verisign, RapidSSL, GeoTrust
- Main elements of a digital certificate are:
  - Certificate owner's identifying information
  - Certificate owner's public key
  - Dates between which the certificate is valid
  - Number of the certificate
  - Digital signature of the certificate issuer

NPTEL

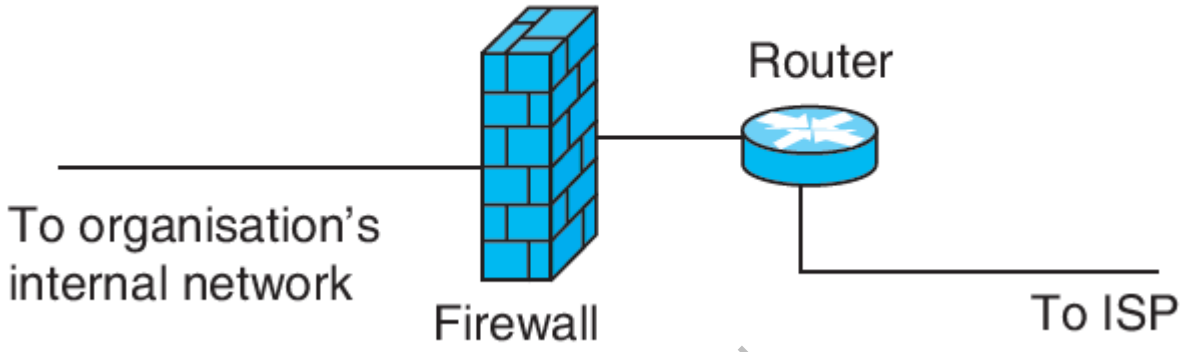
Digital Certificates (contd..)





NPTEL

Firewall



NPTEL

- Firewalls are filtering and protection devices - usually a combination of hardware and software
- Packet-level filtering based on rules
- Firewalls slow down traffic at the perimeter – to overcome this firewalls are built into hardware.

Virtual Private Network (VPN)

- A technology that enables clients or employees of an organisation, who are outside the network, to connect securely to the organisation on the public Internet.
- It creates a 'tunnel' relying on authentication and encryption.



NPTEL

Phishing Attack Countermeasures

- Most important step that companies can take today is to educate web site users
- Many companies contract consulting firms that specialize in anti-phishing filters
- Report Phishing emails

www.millersmiles.co.uk

NPTEL

References

- K. Laudon and J. Laudon (2016). Management Information Systems Publisher: Pearson. Edition 14e.

- R. De. (2018). MIS Managing Information Systems in Business, Government and Society. Publisher: Wiley. Second Edition.

NPTEL



NPTEL

THANK YOU !



NPTEL ONLINE CERTIFICATION COURSES

NPTEL

Management Information System

Saini Das

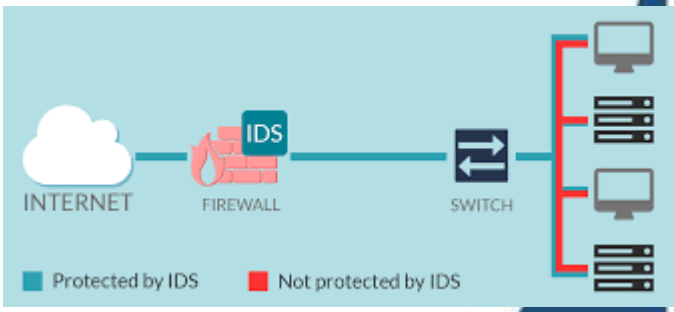
Vinod Gupta School of Management, IIT Kharagpur

Module 11: Ethical, Social and Security issues in MIS

Security issues in MIS - III

Intrusion Detection Systems

- Intrusion detection systems feature full-time monitoring tools placed at the most vulnerable points or “hot spots” of corporate networks to detect and deter intruders continually.
- The system generates an alarm if it finds a suspicious or anomalous event.
- Scanning software looks for patterns indicative of known methods of computer attacks and sends warnings of vandalism or system administration errors.
- Monitoring software examines events as they are happening to discover security attacks in progress.
- The intrusion detection tool can also be customized to shut down a particularly sensitive part of a network if it receives unauthorized traffic.



NPTEL

Antivirus and Antispyware Software

- Antivirus software is designed to check computer systems and drives for the presence of computer viruses.
- Often the software eliminates the virus from the infected area.
- However, most antivirus software is effective only against viruses already known when the software was written.
- To remain effective, the antivirus software must be continually updated.
- Leading antivirus software vendors, such as McAfee, Symantec, and Trend Micro, have enhanced their products to include protection against spyware.



NPTEL

Unified Threat Management Systems

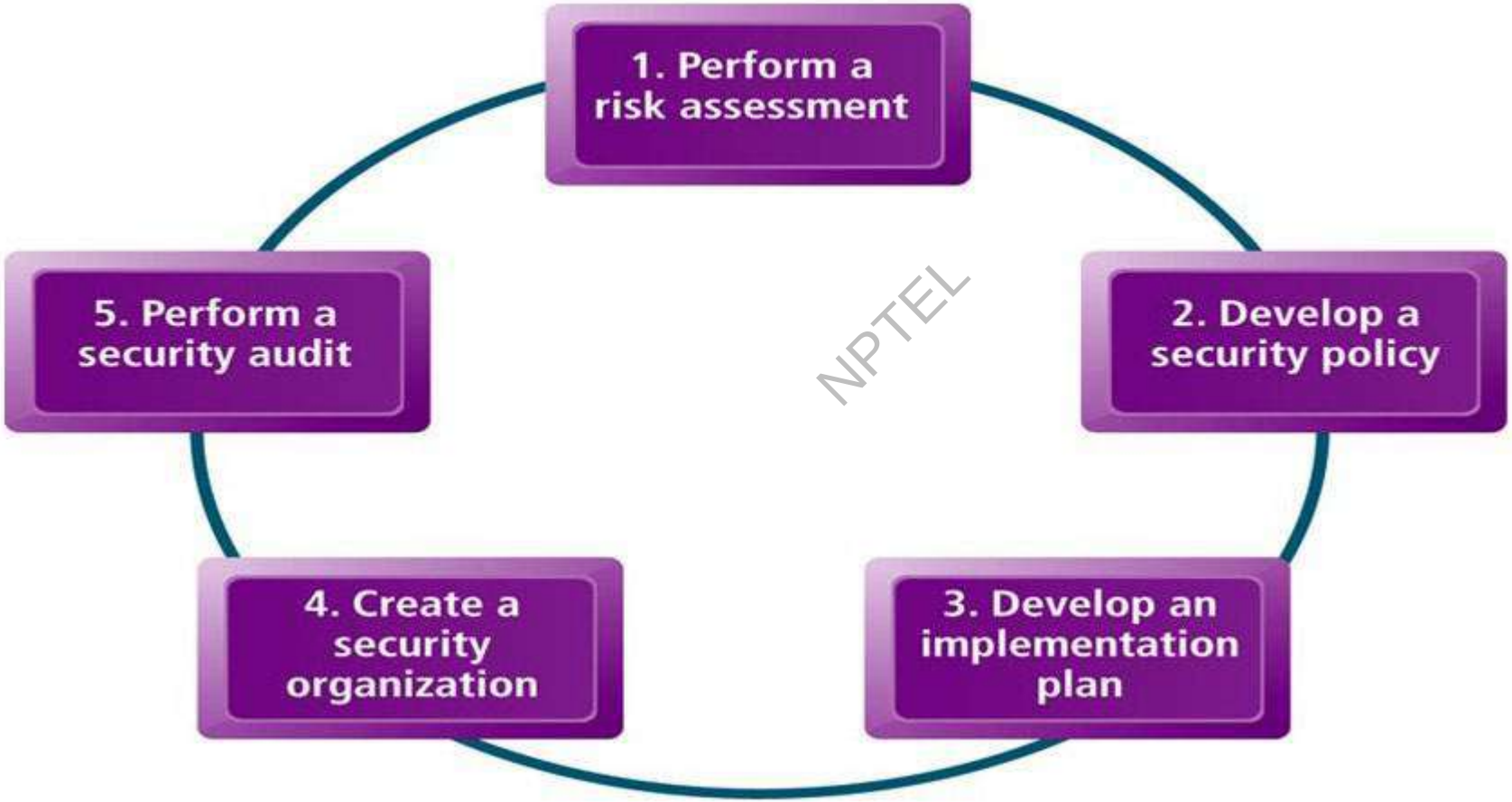
- A comprehensive appliance having various security tools, including firewalls, virtual private networks, intrusion detection systems, and anti-spam software.
- Helps businesses reduce costs and improve manageability.
- Although initially aimed at small and medium-sized businesses, UTM products are available for all sizes of networks
- Leading UTM vendors include Crossbeam, Fortinent, and Check Point.

NPTEL

Organizational Policies
&
Procedures

NPTEL

Organizational Security Plan



NPTEL

Risk Assessment

- Before a company commits resources to security it must know which assets require protection and the extent to which these assets are vulnerable.
- Risk assessment determines the level of risk to the firm if a specific activity or process is not properly controlled.
- IS specialists should try to determine the value of information assets, points of vulnerability, the likelihood of a problem, and the potential for damage.
- The expected annual loss for each exposure can be determined by multiplying the likelihood of a breach, and the potential for damage.
- Once the risks have been assessed, management will concentrate on the control points with the greatest vulnerability and potential for loss.

NPTEL

Security Policy



- Once the main risks to the systems have been identified, the company will need to develop a security policy for protecting the company's assets.

- A security policy consists of statements ranking information risks, identifying acceptable security goals, and identifying the mechanisms for achieving them.

- It drives policies determining acceptable use of the firm's information resources and which members of the company have access to those.

NPTEL

DRP/BCP

- Disaster recovery plan (DRP) devises strategies for the restoration of computing and communications services after they have been disrupted.
- DRP focuses primarily on the technical issues involved in keeping systems up and running, such as which files to back up and the maintenance of backup computer systems or disaster recovery services.
- Business continuity plan (BCP) focuses on how the company can restore business operations after a disaster strikes.
- The business continuity plan identifies critical business processes and determines action plans for handling mission-critical functions if systems go down.



NPTEL

Security Audit



- Organizations must conduct comprehensive and systematic security audits.
- Security audits review technologies, procedures, documentation, training, and personnel.
- A thorough audit will even simulate an attack or disaster to test the response of the technology, information systems staff, and business employees.

NPTEL

Legal and Regulatory Requirements

NPTEL

Cyber Laws

- Gramm-Leach-Bliley Act (1994): This act requires financial institutions to ensure the security and confidentiality of customer data.

- HIPAA (1996)

- Sarbanes-Oxley Act (2002): SOX is fundamentally about ensuring that internal controls are in place to govern the creation and documentation of information in financial statements. Because information systems (IS) are used to generate, store, and transport such data, the legislation requires firms to consider IS security controls required to ensure the integrity, confidentiality, and availability of their data.

NPTEL

Computer Forensics



- Computer forensics is the scientific collection, examination, authentication, preservation, and analysis of data held on or retrieved from computer storage media in such a way that the information can be used as evidence in a court of law.
- It deals with the following problems:
  - Recovering data from computers while preserving evidential integrity
  - Securely storing and handling of recovered electronic data
  - Finding significant information in a large volume of electronic data
  - Presenting the information to a court of law

NPTEL

Organizations that Promote Computer Security

- **CERT**
  - Responds to thousands of security incidents each year
  - Helps Internet users and companies become more knowledgeable about security risks
  - Posts alerts to inform the Internet community about security events
- **Microsoft Security Research Group**
  - Privately sponsored site that offers free information about computer security issues

NPTEL

References

- K. Laudon and J. Laudon (2016). Management Information Systems Publisher: Pearson. Edition 14e.

- R. De. (2018). MIS Managing Information Systems in Business, Government and Society. Publisher: Wiley. Second Edition.

NPTEL



NPTEL

THANK YOU !



NPTEL ONLINE CERTIFICATION COURSES

NPTEL

Management Information Systems

Prof. Surojit Mookherjee
VGSoM, IIT KHARAGPUR

Week 12: Managing Global Organizations and Global Projects

Lecture 01 : Managing Global Organization - Globalization (Part1)

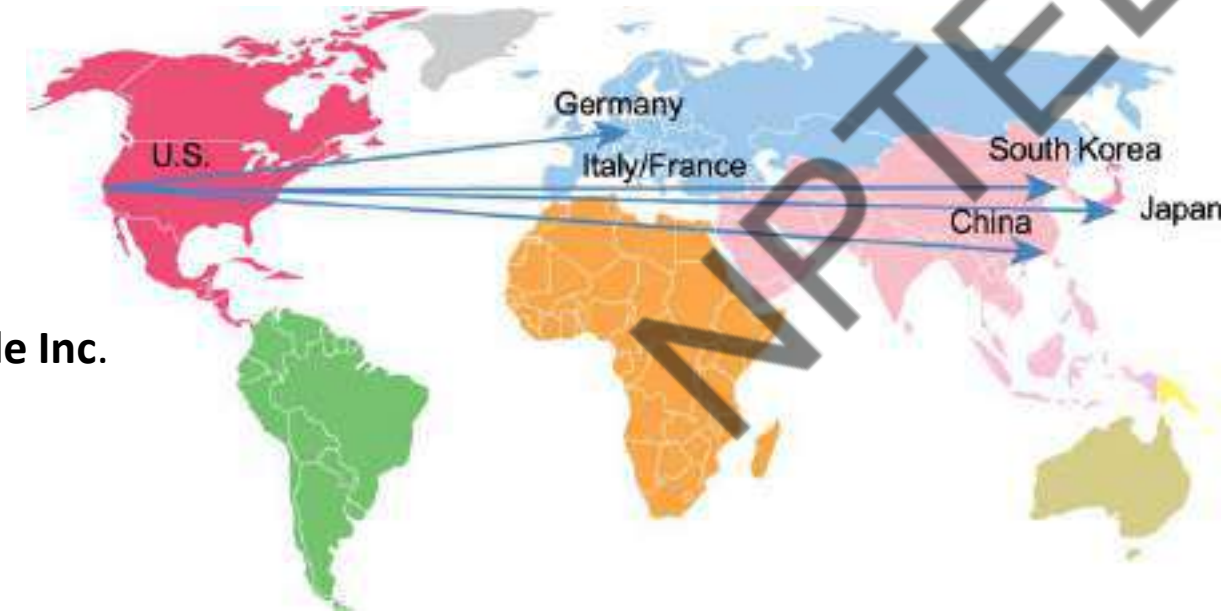
Learning Objectives

- What major factors are driving the internationalization of business?
- What are the alternative strategies for developing global businesses?
- What are the challenges posed by global information systems and management solutions for these challenges?
- What are the issues and technical alternatives to be considered when developing international information systems?

NPTEL

What is meant by a Global Organization (Globalization) ?

International business refers to the trade of goods, services, technology, capital and/or knowledge across national borders and at a **global** or transnational life. It involves cross-border transactions of goods and services between two or more countries.



NPTEL

Example - Apple Inc.

Characteristics of a Global Organization

- Large scale operations.
- Earns foreign exchange in multiple currencies
- Integrates economies.
- Large number of middlemen.
- High risk (e.g. US - China trade conflict)
- International restrictions.
- Highly sensitive nature towards local laws and regulations
- Global Political alignments / Trading Partnerships
- **Multi – Languages , Multi - Cultural**

NPTEL

What are the major Drivers (Factors) affecting International business

- Transport (Steam engine to Jet engine)
- Communication (Internet and Telekom)
- Political
- Economical
- Social
- Technological (Computing power , Cloud , Video conferencing , Analytics , IT Security)
- Legal

NPTEL

Globalization

Globalization* is the process of business integration among people, companies, and governments , across the world.

It is primarily an **economic** process in integration with **social** and **cultural** aspects of the countries involved.

So we have :

- Economic Globalization
- Cultural Globalization and
- Political Globalization (Belt and Road Initiative of China)

NPTEL

* Wikipedia

Internet users by region				
	2005	**2010**	**2017**	**2019**[a]
Africa	2%	10%	21.8%	28.2%
Americas	36%	49%	65.9%	77.2%
Arab States	8%	26%	43.7%	51.6%
Asia and Pacific	9%	23%	43.9%	48.4%
Commonwealth of Independent States	10%	34%	67.7%	72.2%
Europe	46%	67%	79.6%	82.5%
[a] Estimate. Source: International Telecommunication Union.[119]				

* Wikipedia

NPTEL

What Major Factors Are Driving the Internationalization of Business?

- Global economic system and global world order driven by advanced networks and information systems
- The growth of international trade has radically altered domestic economies around the globe
- For example, production of many high-end electronic products parceled out to multiple countries
  - For example: Apple iPhone's global supply chain
    Samsung's Mobile phone manufacturing

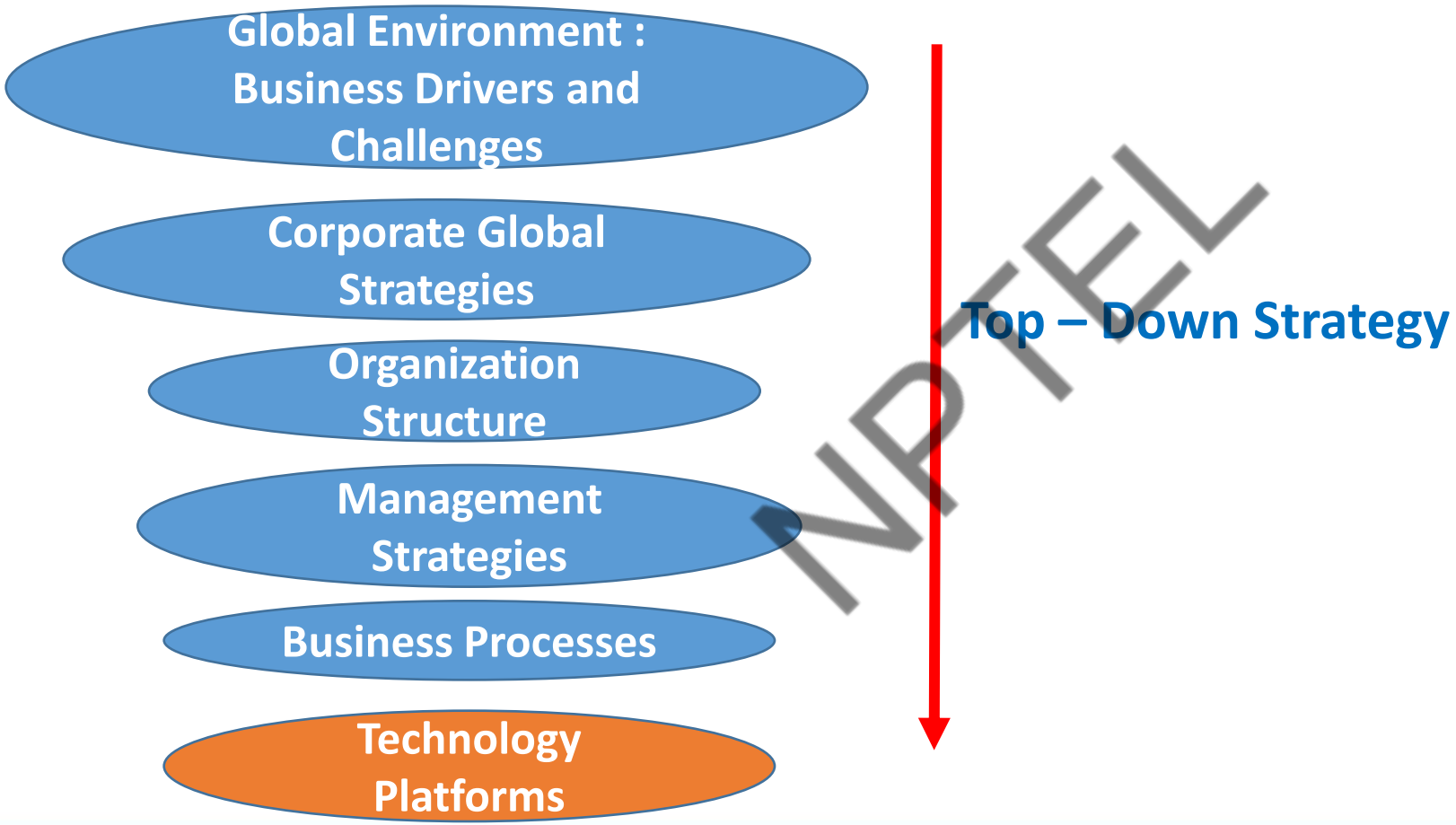
NPTEL

Strategy for developing an International Information Systems Architecture

- **Understand global environment**
  - Business drivers for global competition
  - Inhibitors creating management challenges
  - Multi – cultural aspects of employees and partners
- **Develop corporate strategy for global competition**
- **Develop organizational structure and division of labor**
- **Consider management issues**
  - Design of business procedures, reengineering, managing change
- **Consider technology platform**

NPTEL

International Information Systems Architecture



NPTEL

The Global Environment: Business Drivers and Challenges ….

- **General cultural challenges**
  - Cultural particularism
  - Social expectations
  - Political laws
- **Specific challenges**
  - Standards
  - Reliability
  - Speed
  - Personnel

NPTEL

The Global Environment: Business Drivers and Challenges ….

GENERAL CULTURAL FACTORS	SPECIFIC BUSINESS FACTORS
Global communication and transportation technologies	Global markets
Development of global culture	Global production and operations
Emergence of global social norms	Global coordination
Political stability	Global workforce
Global knowledge base	Global economies of scale

NPTEL

Challenges and Obstacles to Global Business Systems

GLOBAL	SPECIFIC
Cultural particularism: Regionalism, nationalism, language differences	Standards: Different Electronic Data Interchange (EDI), e-mail, telecommunications standards
Social expectations: Brand-name expectations, work hours	Reliability: Phone networks not uniformly reliable
Political laws: Transborder data and privacy laws, commercial regulations	Speed: Different data transfer speeds, many slower than United States
-	Personnel: Shortages of skilled consultants

NPTEL

State of the Art

- Most companies have inherited a patchwork international system using traditional batch-oriented reporting, manual data entry, legacy systems, and little online control.
- Significant difficulties in building appropriate international architectures
  - Planning a system appropriate to firm's global strategy
  - Structuring organization of systems and business units
  - Solving implementation issues
  - Choosing right technical platform

NPTEL

REFERENCES

- **Management Information Systems: Managing the Digital Firm - Kenneth C. Laudon & Jane P. Laudon – Ch 15**
- **The World is Flat : Thomas L Friedman**
- **Enterprise Resource Planning, Rajesh Ray, Tata McGraw-Hill**

NPTEL



NPTEL

THANK YOU !



NPTEL ONLINE CERTIFICATION COURSES

NPTEL

Management Information Systems

Prof. Surojit Mookherjee
VGSoM, IIT KHARAGPUR

Week 12: Managing Global Organizations and Global Projects

Lecture 02 : Managing Global Organizations – Part 2

Global Strategies and Business Organization….

- Three main kinds of organizational structure
  - Centralized: In the home country
  - Decentralized/dispersed: To local foreign units
  - Coordinated: All units participate as equals

NPTEL

Global Strategies and Business Organization

Four main global strategies

- Domestic exporter (e.g. Tata Motors)
- Multinational (e.g. IBM)
- Franchisers (e.g McDonalds)
- Transnational (e.g. Coca Cola)

NPTEL

Transnational corporations share many qualities with **multinational corporations**, with the difference being that **multinational corporations** consist of a centralized management structure, whereas **transnational corporations** generally are decentralized, having country specific controls.

Global Systems to Fit the Strategy

- Configuration, management, and development of systems tend to follow global strategy chosen
- Four main types of systems configuration
  - Centralized: Systems development and operation occur totally at domestic home base
  - Duplicated: Development occurs at home base but operations are handed over to autonomous units in foreign locations
  - Decentralized: Each foreign unit designs own solutions and systems
  - Networked: Development and operations occur in coordinated fashion across all units

NPTEL

Reorganizing the Business

To develop a global company and information systems support structure:

1. Organize value-adding activities along lines of comparative advantage
   - For example: Locate functions where they can best be performed, for least cost and maximum impact.
2. Develop and operate systems units at each level of corporate activity—regional, national, and international
3. Establish at world headquarters:
   - Single office responsible for development of international systems
   - Global CIO position

NPTEL

A Typical Scenario: Disorganization on a Global Scale

A traditional multinational consumer-goods company based in United States and operating in Europe would like to expand into Asia

- World headquarters and strategic management in United States
- Separate regional, national production and marketing centers
- Foreign divisions have separate IT systems
- E-mail systems are incompatible
- Each production facility uses different ERP system, different hardware and database platforms, and so on

NPTEL

Global Systems Strategy (1 of 3)

- Share only core systems
  - Core systems support functionality critical to firm (e.g. key business processes like Procurement , Pricing , Manufacturing , Inventory)
- Partially coordinate systems that share some key elements
  - Do not have to be totally common across national boundaries
  - Local variation desirable (e.g. HR , Finance)
- Peripheral systems
  - Need to suit local requirements only (e.g. Tax rules)

NPTEL

Global Systems Strategy (2 of 3)

- Define core business processes
- Identify core systems to coordinate centrally
- Communicate across organization leadership for Change Management
- Choose an approach
  - Piecemeal and grand design approaches tend to fail

NPTEL

Global Systems Strategy (3 of 3)

- Make benefits clear
  - Global flexibility
  - Gains in efficiency
  - Global markets and larger customer base unleash new economies of scale at production facilities
  - Optimizing corporate funds over much larger capital base

NPTEL

The Management Solution: Implementation (1 of 2)

- Agreeing on common user requirements
  - Short list of core business processes
  - Develop common language, understanding of common elements and unique local qualities
- Introducing changes in business processes
  - Success depends on legitimacy, authority, ability to involve users in change design process
- Coordinating applications development
  - Coordinate change through incremental steps
  - Reduce set of transnational systems to bare minimum

NPTEL

Issues and Technical Alternatives When Developing International Information Systems (1 of 3)

- Computing platforms and systems integration
  - How new core systems will fit in with existing suite of applications developed around globe by different divisions
  - Standardization: Data standards, interfaces, software, and so on
- Connectivity
  - Internet does not guarantee any level of service
  - Many firms use private networks and VPNs
  - Low penetration of PCs, outdated infrastructures in developing countries

NPTEL

Issues and Technical Alternatives When Developing International Information Systems (2 of 3)

- Software
  - Integrating new systems with old – Version and Upgrade management
  - Human interface design issues, languages
  - Commonality of softwares used amongst different countries.
- Software localization
  - Configuring to meet local requirements. Very important for ERP type enterprise systems (SCM , CRM etc.)

NPTEL

Issues and Technical Alternatives When Developing International Information Systems (3 of 3)

Most important software applications:

- Enterprise Resource Planning
- Business Warehousing and MIS
- Electronic Data Interchange (EDI)
- Enterprise systems like CRM , SCM , SRM , PLM
- Collaboration tools, e-mail, videoconferencing , Chat
- Servers and Operating Systems
- IT Security systems
- Data Analytics and Reporting systems.

NPTEL

REFERENCES

- **Management Information Systems: Managing the Digital Firm - Kenneth C. Laudon & Jane P. Laudon – Ch 15**
- **The World is Flat : Thomas L Friedman**
- **Enterprise Resource Planning, Rajesh Ray, Tata McGraw-Hill**

NPTEL



NPTEL

THANK YOU !



NPTEL ONLINE CERTIFICATION COURSES

NPTEL

Management Information Systems

Prof. Surojit Mookherjee
VGSoM, IIT KHARAGPUR

Week 12: Managing Global Organizations and Global Projects

Lecture 03 : Shared Services

What is Shared Services….

Defining shared services

Organizations share delivery of similar services across locations / countries to :

- ✓ reduce cost,
- ✓ enhance efficiency and
- ✓ improve quality

However, it requires careful design of the Organization's structure , governance and IT backbone for high quality Management Information System, to ensure that it achieves the level of cooperation required to implement an effective shared services model.

Seamless and real-time communication play a key role in the effectiveness of MIS for the success of the Shared Services.

NPTEL

Three Key things provided by Shared Services

- **external services** — services that Organization provide to its customers such as : consulting or advisory services , non-core service activity like Catering by Railways (IRCTC)
- **back office functions** — functions that support the Organizations functioning such as : Information Technology, Finance, Legal, Payroll, and Human Resources , Corporate Social Responsibility , Training , R&D etc;
- **procurement** — Centralised purchase of goods and services to achieve economies of scale , short lead time of delivery , maintaining lean inventory levels.

NPTEL

Major benefits of Shared Services ….

- **Cost reduction through economies of scale**
- Increased efficiency through sharing of assets and people
- Improved service delivery and consistency across regions
- Reduced duplication of processes and skilled human resources
- Improved quality of service through larger and more skilled resource pool
- Standardizing the business processes across locations and countries
- Common IT tools – lower costs

NPTEL

Major benefits of Shared Services

- Promotes the consolidation of scarce resources with specific technology skills that can be shared across business units
- Engages industry standards and best practices
- Flexible engagement models to satisfy clients' diverse needs
- Ownership and control remains with the Organization
- Keeps the Organization focused on core activities
- Support of local economies by sustaining local employment (Outsourcing)

NPTEL

Shared Services – Delivery Mechanisms

- **Resource sharing** – Multi National organizations share key resources (plant, equipment or personnel) of individual countries to deliver for Group companies.
- **Centralisation of services** – relocation of multiple delivery sites or services to one centre which then serves the whole organization. These could be located at one place for world-wide operation or have locations each at continent level (Americas , EU , Asia-Pacific etc.)
- **Joint venture** – establish stand-alone entity to share costs and risks of providing goods or services (e.g a separate company for procuring raw-material for the parent organization)
- **Outsourcing** – Key functions outsourced to provider organisations. Non-core functions can be outsourced to a third-party for efficiency and cost saving and agility.

All of these get delivered on a platform (back – bone) of Integrated IT systems and technology

NPTEL

Scope of Shared Services in a business enterprise……..

FINANCE

- Planning and Budgeting
- Treasury/Cash Management
- Internal Audit
- Tax
- Foreign Exchange
- Business Support Analysis

HR

- Benefits Administration
- Records Management
- Training & Development
- Relocation Services
- Recruitment
- Payroll

IT

- Hardware & Software
- Application Development
- Data Center Operations
- Standards
- Technology Planning
- Security
- Cloud
- ERP , CRM , SRM, KM etc. enterprise softwares

NPTEL

Scope of Shared Services in a business enterprise

Customer Service

- Call Centers
- Credit & Collections
- Order Processing
- Returns Processing

etc.

Legal

- Regulatory Compliance
- Public Affairs/Media Relations
- Litigation Support & Coordination
- Insurance

etc.

Procurement

- Transportation & Logistics
- Strategic Sourcing
- Warehousing
- Inventory Management
- Vendor Management

etc.

NPTEL

Challenges for Shared Services functioning …..

Not having common tools and approaches

Lack of Top Management support

Lack of common process management

Maintaining key employee talent

Inconsistent approach to customer service

Lack of a shared center infrastructure

Different Time Zones

Different Legal Requirements

Functional silos for processes execution

NPTEL

Challenges for Shared Services functioning – Change Management

- Biggest challenge is fear of reduced importance ,for the employee, when ‘business processes’ get shifted to a Shared Service Organization , which would usually be in a different country.
- Consolidation of common functions in a single location would result in staff reduction
- Staff relocation to different country with associated problems (family issues etc.)
- Reporting to managers from different country , different cultural background , habits etc.
- Need to learn new technologies
- Multiple Time Zones

NPTEL

How Technology is helping Shared Services

ERP

ERP provides the platform for building operational efficiency and generating MIS reports

Electronic Processing

Electronic Invoicing , Document Management , Automated payment , User Self service, Workflow etc.

Cloud Environment

Has enabled quick setting up , scaling up and scaling down as necessary without any Capex involvement

Social Media

Easier and effective collaboration with partners without any geographic limitation

NPTEL

Locational features impacting global shared service centers

- **Americas**: The Americas region, particularly in the US, have the most established shared services models
  - technology, common language, standardized policies , skilled man power.

- **Latin America**: Low cost manpower , skilled in IT and Spanish language support

NPTEL

Locational features impacting global shared service centers

- **Eastern Europe**:
  - EU process knowledge
  - Muti European language support
  - Moderately expensive labor
- **China**:
  - China is in its infancy stages of developing shared services organizations
  - Most large manufacturing companies have a base in China
  - Lack of English knowledge

NPTEL

Locational features impacting global shared service centers

- **India / Philippines**:
  - Cheap labor
  - Strong knowledge in English
  - Global leaders in business process outsourcing
  - Large technical pool
  - Very good IT and Telecom Infrastructure

NPTEL

REFERENCES

- Enterprise Resource Planning, Rajesh Ray, Tata McGraw-Hill
- Management Information Systems: Managing the Digital Firm - Kenneth C. Laudon & Jane P. Laudon
- Shared services Analysis report by KPMG
- Trends and Best Practices in global shared services – Chezey Partners

NPTEL



NPTEL

THANK YOU !



NPTEL ONLINE CERTIFICATION COURSES

NPTEL

Management Information Systems

Prof. Surojit Mookherjee
VGSoM, IIT KHARAGPUR

Week 12: Managing Global Organizations and Global Projects

Lecture 04 : Offshoring / Transitioning Global IT systems

Outsourcing / Offshoring of IT systems of an Organization

- The world has seen since the early 90's , major shift in policies of large corporates towards managing their IT systems.

- They decided to "Outsource" both 'support- maintenance' as well as 'new developments' to specialised IT organizations , whose business is IT development and maintenance (e.g – IBM , TCS, Wipro, Infosys, HCL-Tech , Tech-Mahindra , Cognizant etc.).

- Cost – saving being a major factor , all of these service delivery organizations developed in countries like India , Philippines , Brazil which were located far away from the Headquarters of the Clients. The new locations were thus referred to as "Off-shore Development Centers" .
- **Off-Shoring** thus became a common terminology it the IT domain.

NPTEL

Transition

Transition – is the process of transferring the **knowledge and management** of the IT systems from one organization to another.

Activities in Transition Cycle

Typical transition activities are as follows:

- ✓ **Knowledge transition:** This involves transferring knowledge from the implementation team to support team or from Client to new Service Provider
- ✓ **Project documentation transition**: Different documents created during the project need to be stored properly and to be handed over by the implementation team to support team.
- ✓ **Human resource transition:** Once the implementation is over, some people will go back to their old job, some will remain to support the application in future, new consulting company may take over the support work – planning need to be done for all these.

NPTEL

Transition Activities

Knowledge Transfer

- Client's business process knowledge
- Application Specific knowledge
- Logics for configuration, developments etc

Human Resource Transition

- Location transition / Transfers
- Role transition / New roles
- Compensation and
- Employment conditions

Documentation

- Blueprints
- Business Logics / Design documents
- Development Specifications
- Configuration documents
- Application Information Documents
- Test Scripts

NPTEL

Typical Support Activities for Transitioning / Outsourcing …..

System Administration

- ✓ Daily Database Administration
- ✓ System performance monitoring
- ✓ Managing Job Schedules
- ✓ Managing Technical Infrastructure

Change Management

- ✓ Develop and Implement Change Control Procedures
- ✓ Applying regular patches, notes etc
- ✓ Testing while implementing changes
- ✓ Transporting changes from one environment to another

NPTEL

Typical Support Activities for Transitioning / Outsourcing

End user application support

- ✓ Provide Ongoing Support to End Users
- ✓ Manage and Resolve problems and issues
- ✓ Root Cause Analysis of problem

Knowledge Management

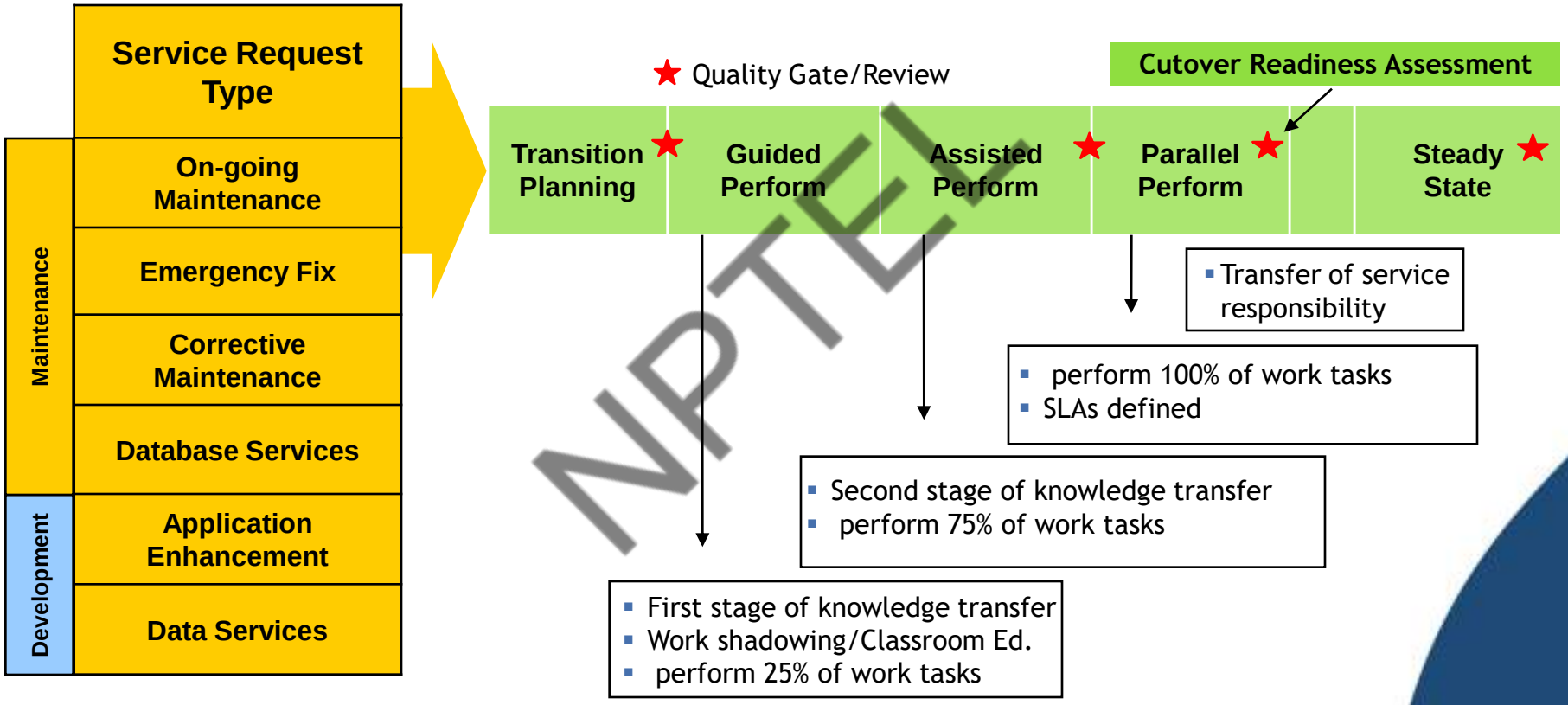
- ✓ Deliver continuous End User Training
- ✓ Evaluate learning effectiveness
- ✓ Document Lessons Learned

Continuous Improvement

- ✓ Plan for Continuous Improvement
- ✓ Value Realization
- ✓ Measuring Business Benefits

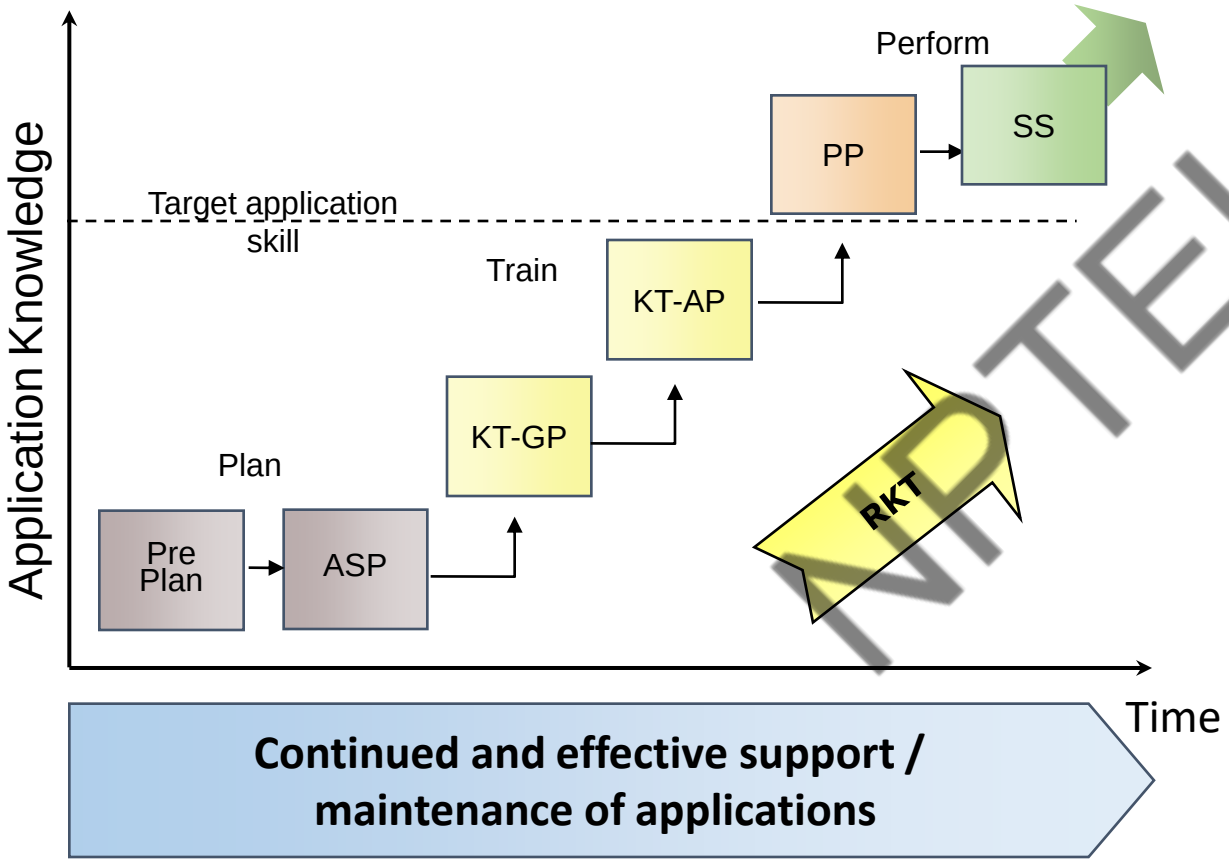
NPTEL

Transition activities manage a successful transfer of knowledge to global delivery support staff.



NPTEL

Incremental Progress towards the Goal

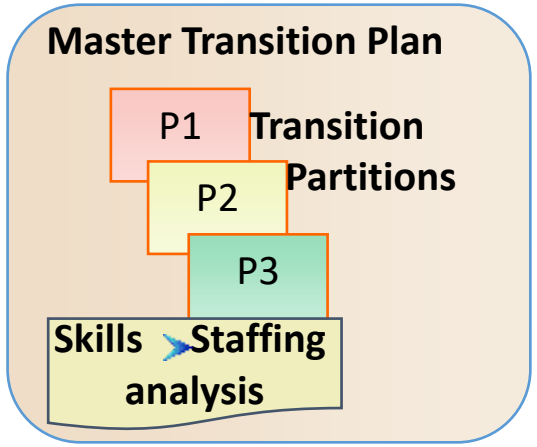


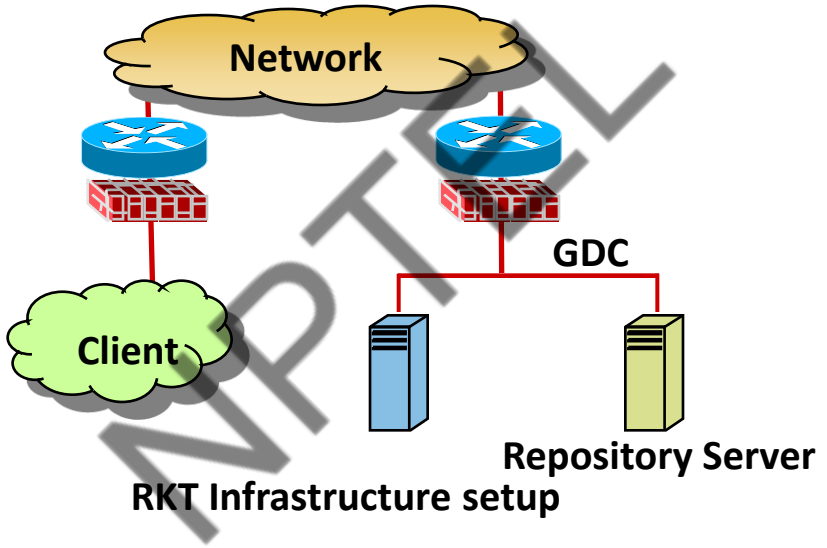
ASP	**Application Specific Planning**
KT -	**Knowledge Transfer Guided Perform**
KT -	**Knowledge Transfer Assisted Perform**
PP	**Parallel Perform**
SS	**Steady State**
RKT	**Remote Knowledge Transfer**

NPTEL

Plan - Transition Master Planning

Deliverables



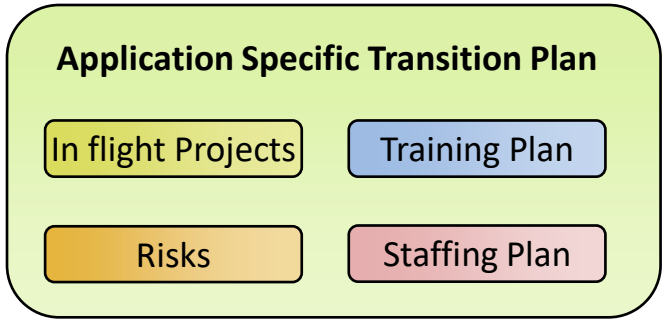


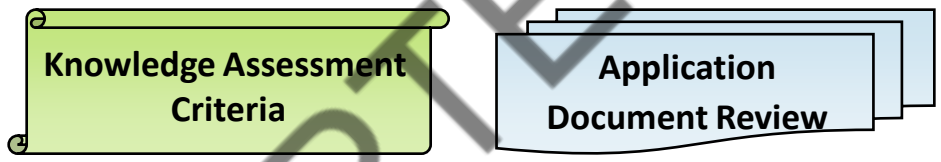
NPTEL

RKT – Remote Knowledge Transfer
<Virtual> – Desktop to Desktop
GDC – Global delivery Center

Plan - Application Specific Planning

Deliverables





NPTEL

AID – Application Information Document
ASCP – Application Support Control Plan

Application Specific Planning (ASP) is done for each logical application group to develop detailed Training Plans

Perform - Parallel Perform & Steady State

Deliverables

- ✓ Steady State Delivery Organization
- ✓ Service Levels
- ✓ Disaster Recovery (DR) / Business Continuity Plan (BCP)
- ✓ AID / ASCP updated in steady state
- ✓ Recording Repository of Knowledge Transfer Recordings

NPTEL

AID – Application Information Document
ASCP – Application Support Control Plan

Transition documentation……

Application	Document Repository
Steady State Organization	▪ Reporting Chain ▪ Management Functions ▪ Contact Information
Application Specific Transition Plan	▪ Transition Staffing, Schedule ▪ Training, Resources ▪ Risk Management
Steady State Governance Model	▪ Client/ Governance Model ▪ Metrics & Measurements ▪ Escalation Procedures

NPTEL

Transition documentation

Application	Document Repository
Application Information Document	Application Overview Technical Architecture Functions, Interfaces and Modules
<Virtual> Recordings (RKT Tool)	Business Specific Training Detailed Application Reviews Support Tasks Training
Global Delivery Center Disaster Recovery Plan	Risks and Mitigation Identification Key Resource Identification Recovery & Contingency Planning Travel Ready Planning

NPTEL

eCC Remote Knowledge Transfer Tool

- **eCollaboration Center**
  - Uses Industry standard Audio, Video & Data channels
  - Trainers (SMEs) and Trainees (Global Resources) conduct Knowledge Transfer from eCC equipped Conference Rooms

- **Recorded Training Sessions**
  - Training sessions are recorded in remote location (VCR recording)

- **Replay/reuse of recorded sessions**
  - Valuable knowledge base available resources in steady state for retraining, induction of new resources into team and cross training to create backups

- **Collaborative environment in Steady State**
  - The on-site and remote teams can collaborate for reviews, design and critical production problems ticket resolution



eCollaboration Center

NPTEL

REFERENCES

- **Enterprise Resource Planning, Rajesh Ray, Tata McGraw-Hill**
- **The World is Flat: A Brief History of the Twenty-first Century ; Friedman, Thomas. Picador/Varrar, 2005.**

NPTEL



NPTEL

THANK YOU !



NPTEL ONLINE CERTIFICATION COURSES

NPTEL

Management Information Systems

Prof. Surojit Mookherjee
VGSoM, IIT KHARAGPUR

Week 12: Managing Global Organizations and Global Projects

Lecture 05 : Globalization of Corporates in a Flat World

The Three Eras of Globalization

Era	Year /	Dynamic Force / Key Agent of Change
Globalization 1.0 (Country)	1492 (Columbus set sail) to 1800	How brawn, how much muscle. Colonization by countries
Globalization 2.0 (Company)	1800 to 2000	Breakthroughs in transportation and telecommunication costs. Railways , ships , Steam engines Telegraph, Telephone, PC, Fibre Optics, WWW.
Globalization 3.0 (Individual)	2000 to present	Connectivity , softwares and Collaboration

NPTEL

Ten Forces That Flattened The World

NPTEL

Ten Flattening Factors…..

- Berlin Wall Coming Down (collapse of Communism)
- Netscape IPO
- Work Flow Software
- Uploading
- Outsourcing

NPTEL

Ten FlatteningFactors

- Offshoring
- Supply-chaining
- Insourcing
- In-forming
- The Steroids



NPTEL

Flattener # 1

1989 -

- Berlin Wall comes down
- Tipping point for free market societies
- Moved away from central planning
- Empowered individuals

NPTEL

Flattener # 2

1995 -

- Netscape IPO -4 years after Berners-Lee invented the Web
- Universal usability, accessibility and interoperability of the Web
- Win 95 shipped
- Overabundance of fiber optic capacity
- Communication costs plummeted

NPTEL

Flattener # 3

Workflow Software

Have Your Application Talk to My Application

- Collaboration
- Rapid development
- Rapid deployment
- Cheapest sources for coding, testing, implementation
- Everyone can create and maintain digital content

NPTEL

Flattener # 4

Uploading – The Open Source Movement

Self-Organizing Collaborative Communities

- Open source
  - For production
  - Linux Operating Systems
  - Geeks making business decisions
- Blogs
- Wikipedia
- Not controlled by corporations or Government







Firefox



NPTEL

Flattener # 5

Outsourcing



- Y2K and India – Major outsourcing started with this
- Competent programmers and engineers who will deliver the same quality at a lower cost (coas-arbitrage)
- Economic imperative (cost saving , getting large volume of skilled resources)
- Outsource everything you possibly can (and build on your core competence)

NPTEL

Flattener # 6

Offshoring

Rather than outsource jobs :-

- Send the whole factory to India or China or Indonesia or Philippines or ……
- Not just manufacturing but also and not limited to :
  - Legal processing
  - Medical Transcription
  - Consulting Services
  - Accounting back-office

NPTEL

Flattener # 7

Supply-chaining

- Wal-Mart (one of the largest organization)
- JIT (Totally integrated)
- IT infrastructure as a key competitive advantage
- Suppliers as Partners
- RFID (Radio Frequency Identification)

NPTEL

Flattener # 8

Insourcing

UPS (one of the largest Courier service company)

- Intimate collaboration
- 3rd-party managed logistics
- "They" act as part of "Your" company
  - Fixing Toshiba laptops
  - Managing delivery of Papa John's Pizza supplies
  - Packaging Segrest Farms live tropical fish for delivery
  - Picking, inspecting, packing and delivering Nike shoes
  - Working with Plow and Hearth furniture suppliers to improve packaging and reduce breakage

NPTEL

Flattener # 9

In-forming

- Building and deploying your personal information supply-chain
- Google – now 1 billion searches per day
- Search Engines - Soon “everything” will be searchable
- Wikipedia (Knowledge sharing)

NPTEL

Flattener # 10

The Steroids….

- "*Making collaboration digital, mobile, personal, virtual*" – Carly Fiorina
  - Computing capabilities - including speed, I/O rate and storage capacity
  - Instant Messaging and File sharing – BitTorrent, Kazaa
  - VoIP – Skype
  - Video Conferencing – Google Hangout , Zoom , CISCO-Telepresence
  - Advanced graphics – from video games
  - Wireless – communicate with Anyone from Anywhere

NPTEL

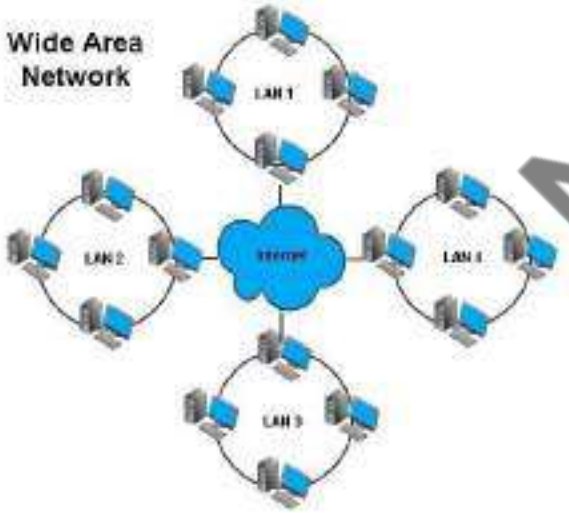
The Steroids













NPTEL

The Great Convergence

We are in a convergence of three powerful, technological forces:

- ✓ Low cost and powerful computing devices
- ✓ Low-cost, high bandwidth network connectivity
- ✓ Open standards

To Summarize :

We have computing everywhere and anywhere, anytime and all the time, with access to limitless amounts of information and services.

NPTEL

Emerging Regions – BRIC Countries



REFERENCES

- **The World is Flat: A Brief History of the Twenty-first Century ; Friedman, Thomas. Picador/Varrar, 2005.**
- **Andi S. Boediman ; Strategic Innovation Consultant ; www.ideonomics.com**

NPTEL



NPTEL

THANK YOU !